



Commodities Strategy | December 2017

Annual Commodities Market Outlook 2018

Sailing ahead into 2018 but will China or oil take the winds out?



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See Appendix A-1 for Analyst Certification, Important Disclosures and non-US research analyst disclosures

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Prepared for Rong (research email)

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Prepared for Rong Lu (research email)

Despite uncertainties ahead, 2018 offers plenty of commodity investor options*

Both YTD commodity performance and expected higher global growth point to potentially tighter balances across many commodities. But uncertainties about China's 'New Economy' and the lingering impacts of its commodity-intensive "Old Economy" create bumps in the road ahead. *Here are our main conviction calls for 2018.*

- **Brent: Sell time-spreads (2Q/4Q 2018).** Producer hedging and bullish prompt fundamentals have pushed up time spreads sharply. With high hedge cover for 2018, producers flows should abate; meanwhile we expect fundamentals to weaken, especially into 2H'18 as producer cuts come to an end.
- **WTI: Get short the 2Q/3Q'18 put-skews by buying put-spreads, delta-hedged.** We expect WTI prices to sustain above the high \$40s through 2018 and producers to continue to cover forward production. The hedges should roll into Cal'19, with producer flows abating the bid for downside vols on the Cal 18 strip, weighing on the put skew.
- **European natural gas: Sell NBP or TTF versus US Henry Hub prices for 2Q and 3Q'19.** Beyond this current winter, global natural gas and LNG production should rise in 2018 and beyond, lowering longer-term prices and loosening both LNG and European gas markets. US prices, meanwhile should remain firm with modest inventory. The current arbitrage is \$1.25/MMBtu while the NBP-HH spread is trading at more than double that level.
- **Gold: A cheap geopolitical and equity hedge.** Even as the rates and FX channels dominate the fundamental outlook for gold pricing, bullion is increasingly being used by investors as a risk-off and policy tail risk hedge. Event-driven bids for gold may be the new normal over the medium-term given 'vox populi' political movements and record central bank balance sheet unwinds. An equity market correction would also be supportive for gold prices for value-buyers. We like allocating gold to macro portfolios and getting long potential dips in 2H'18 towards \$1,200-1,225/oz.
- **Zinc: producers should consider selling long-dated (2H'19/'20) material over the next 6-9 months, as these prices are unlikely to last.** We believe zinc prices should rise by more than 10% to \$3,600 by 2Q'18, with the potential for a 25% increase to \$4000 as refined stocks draw down and Chinese mines fail to respond while induction furnace losses continue. Longer-term we believe this is zinc's last hurrah for a while given that the current price path should bring more supply to the market globally.

(Continued on next page)

*Note: These suggestions represent views in which we hold high conviction. However, we offer no recommendations on the specific structures used to express these views.

There is a rich array of investor options for commodities in 2018* (continued)

- **Iron Ore: Go long 1Q'18 iron ore, given asymmetric upside.** We expect iron ore to relay to average \$68/t (10%) during the next quarter and trading above \$70/t before moderating again, with risks weighted to the upside. The rally is based on an expected tightening of China's steel market, increasingly dependent on high-grade ore and with the supply growth passing its peak in the near term.
- **Platinum-Palladium: We expect the spread between the two PGM metals to widen further through 1H'18.** This view entails a sustained deficit for palladium and higher demand from gasoline vehicle growth globally and especially in Europe.
- **Sugar: We remain long 2H'18 ICE sugar.** We expect ICE sugar prices to trade 10-15% higher than the forward curve in 2H'18 and we recommend positioning for upside. Cheap volatility and flat skew also make outright calls and bullish risk reversals look to be appropriate. This view is supported by Brazilian ethanol dynamics and tightening balances.
- **Gasoline: Buy '1H'18 European gasoline cracks.** We expect margins to remain high and spare refining capacity to be tight. This means that Europe, as the world's marginal or incremental refining center, should be particularly tight. Reinforcing this is the expected deep refinery turnaround schedule in 1H'18 with clean product stocks low.
- **Diesel-HSFO: Buy Dec'19 or the Cal'20 strip for European gasoil-fuel oil.** The 2020IMP 0.5% sulfur cap on global marine gas oil should lead to a late rush by shippers to source diesel and ULSFO, leaving a glut of HSFO on the market. Shippers are running out of time in retrofitting scrubbers on vessels to enable them to run HSFO compliantly. Diesel demand could jump by over 1-m, b/d and HSFO demand could fall by over 1.5-m b/d.
- **US natural gas: basis differentials in 2019 should widen against Henry Hub.** Waha basis, reflecting natural gas prices in the Permian Basin, could widen substantially as oil and associated natural gas output grows and as the building of transportation pipelines out of the Permian lag production growth. Waha basis could wide from ~\$0.6 today to \$1 below Henry Hub. That level would justify new pipe construction as well as the sale of more gas to the Midwest to compete against low-cost gas from Western Canada and Appalachia.

*Note: These suggestions represent views in which we hold high conviction. However, we offer no recommendations on the specific structures used to express these views.

Source: Citi Research, NOTE: Futures trading involves substantial risk of loss

1. Macro Section: Commodities sailing ahead into 2018; But will China or oil take the winds out?

Commodities should start 2018 with a bang, but wildcards could disappoint

Commodities are performing well as 2017 comes to and end, with a mid-year burst for metals and bulks followed by energy's outperformance as the year comes to an end. But oil could reverse course through 2018 and a pending China slowdown could add to other wildcards to prevent a third year of commodity outperformance.

- **Citi expects sustained high commodity demand growth through next year and 2019, given expectations of robust 3.4% real GDP led by 4.8% growth in emerging markets (EMs) and 2.4% in advanced economies (AEs). With the broad-based recovery driven by investment and trade growth commodity demand growth should be well supported. But all of this is accompanied by huge uncertainties about Chinese policy and the Chinese economy.** Citi's China economics team indicates that the big EM bellwether, China, may be set to slow. Nevertheless, global investment-led growth should continue to pick up with rising capacity utilization, benign financial conditions, an absence of negative catalysts and some structural reforms (e.g. in India) paying dividends.
- **It's tempting to overstate the impact of demand growth on commodity fundamentals given that economic growth is steadily less commodity-intensive, diluting the impact of stronger growth.** What's at work are efficiency gains, the move up the value chain and a lack of successor to China as the commodity consuming juggernaut. While China should still partially rely on "old economy" growth drivers for now, no combination of countries seems big enough to pick up the potential slack in boosting commodity demand. China's increasingly strict environmental regulations and supply-side reforms (capacity rationalization) should limit the deflationary and bank balance sheet pressures of lower demand. This could even result in shortages should demand surprise to the upside given the collapse in Chinese mining FAI over the past two years).
- **Yet China's role in commodity price formation will remain critical in 2018, as it has been the for nearly two decades. Growth targets will no doubt nominally be met, partly because of the stated plan to double per capita income by 2020. However, Xi's now supreme authority means that the path of commodity intensity of growth is increasingly in one man's hands.** Added to the relatively high policy uncertainty in China is the potential for a PBOC policy shift. The current governor may extend his term and is increasingly hawkish, or a new governor with an unknown policy bias could be announced soon. Market risks are also rife given the potential consequences of capital controls (asset bubbles), supply-side reforms (shortages), and QE withdrawal (credit spreads and equity market volatility) loom large.
- **Nonetheless, our base case sees China-sensitive commodities posting positive demand growth in 2018, as the China policy put likely remains intact.** This demand growth is set against a likely continuation of higher environmental standards, pressure to continue to support bank balance sheets by eliminating overcapacity and reducing oversupply, which could result in some commodity market tightening notably in iron ore -where FAI in iron or mining in China is down 60% over the past 3 years, as well as zinc, and where Chinese domestic supply would simply not be able to respond.

Prepared for Rong Lu (research email)

Beyond China, oil also looks bearish, while currencies, gold could uplift

Prepared for Rong Lu (research email)

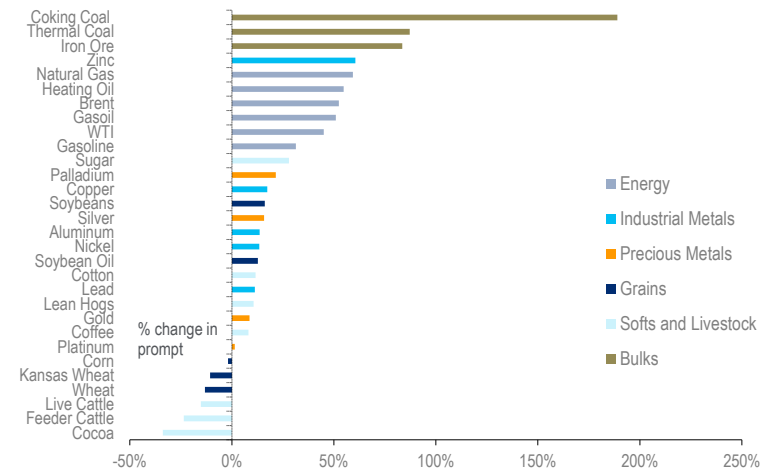
- **Even as crude oil markets continue to rebalance into 2018, by year-end Citi expects growing non-OPEC supply to meet virtually all incremental demand leaving little room for a return to higher production by OPEC/Russia putting downward pressure on prices into the second half of next year and 2019, damping exuberance for commodities.** Yet geopolitical risks in OPEC producers point to a greater likelihood of disruption risk, especially from weaker states (Iran, Iraq, Libya, Nigeria, etc) than a likelihood of higher supply going into next year.
- **Citi's forecasts for a weaker dollar also have implications for commodities, given that weak USD historically lifted commodity prices. But fundamentals appear to be more crucial to 2018 prices, as the correlation between commodity prices and USD has weakened markedly throughout 2017, even falling to around zero in 3Q. If there is a currency with implications for commodity prices it's more likely to be the renminbi than the USD.** Our China economists expect two-way fluctuations of the RMB's exchange rate in 2018, with a depreciation bias in 1H. Their USDRMB forecasts are 6.71, 6.74, 6.69, and 6.60 at the end of 1Q, 2Q, 3Q and 4Q, respectively. They forecast that a headline growth slowdown should undermine market sentiment in 1H18, and the large business deals signed with the US during Trump's visit to China showed that outward investment led by SOEs will likely resume early next year. By 2H'18, as economic growth gets back on track and financial sector opening-up starts to attract capital inflows, the RMB may then appreciate gradually.
- **Beyond oil fundamentals of some key commodities, instead of being at some steady state so that currencies could exert greater effects, could be at inflection points that amplify impacts of commodity supply-demand on prices.** For metals and bulks, China's partial reliance on "old-economy" growth vs the macro transitioning that drive capacity cuts and environmental regulations could support prices, while the battle between shale and OPEC could tip in the other direction. Yet, a more substantial depreciation of USD should give support to commodity prices, all else equal. At the same time the pending Trump tax cuts and the closing of output gaps could exert inflationary pressures with implications for commodities.
- **Even as the rates and FX channels dominate the fundamental outlook for gold pricing, gold is increasingly being used by investors as a policy and tail risk hedge.** 'Vox Populi' political movements, low oil prices, large fiscal and wage gaps and rapid technological change are just a few factors that should keep geopolitical risks structurally elevated over the medium-term, just as record central bank balance sheets begin to unwind, increasing the odds for policy error. We are likely to see wildcard event-driven rallies in gold prices on a more frequent basis.
- **Commodity volatility should rise in 2018, but the current low-vol/low-corr regime should persist, barring a sizable shock to global demand.** Asset price volatility should trend higher as core global central banks continue to tighten, pushing borrowing costs and risk premia higher. But the slow pace of the tightening cycle suggests volatility may be restored to the market gradually as well.

Commodities outperform in 4Q, providing second year of strong returns

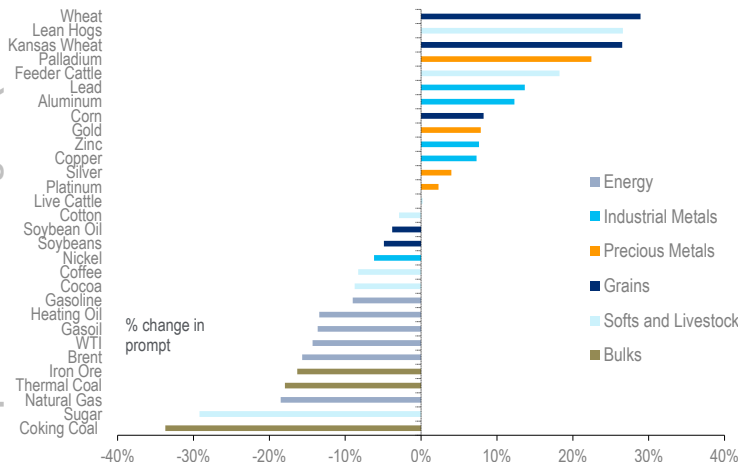
Will past be prologue? Commodities outperformed all other asset classes QTD with the BCOM TR index returning 2.9%, largely driven by the surge in the energy complex.

- Commodity markets performed well in 2017, with a late year surge in the energy complex following through on the outperformance in the precious metals and China-linked commodities in 1H and 3Q. Though bulk commodities were among the best performers in 3Q, the complex pulled backed drastically in 4Q to put all of the bulks complex in the red YTD joined by the soft agriculture commodities and natural gas. But all other commodities have seen positive returns YTD, with palladium the top performer.
- Strong returns across the commodities complex drove the BCOM TR index higher 2.9% QTD, the best quarterly performance for commodities since 2Q'16 and the best performance across all asset classes in 4Q, besting even the equities markets which had been outperforming all other asset classes in the first three quarters..

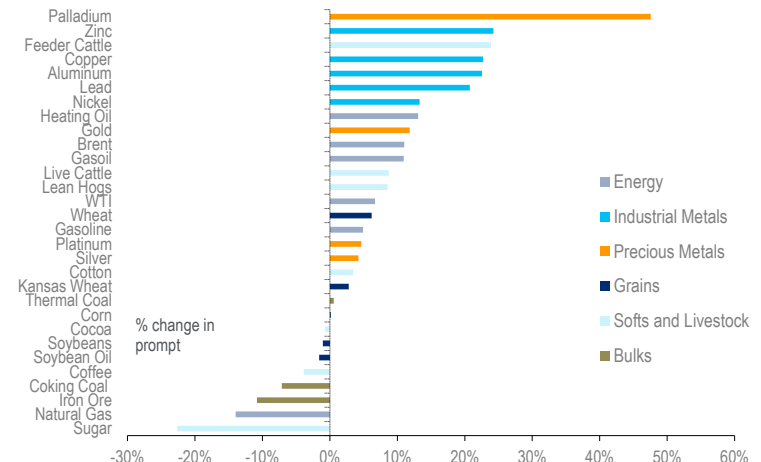
Commodities Price Changes 2016



Commodities Price Changes 1H'17



Commodities Price Changes 2017 YTD



Prepared for Rong Lu (research email)

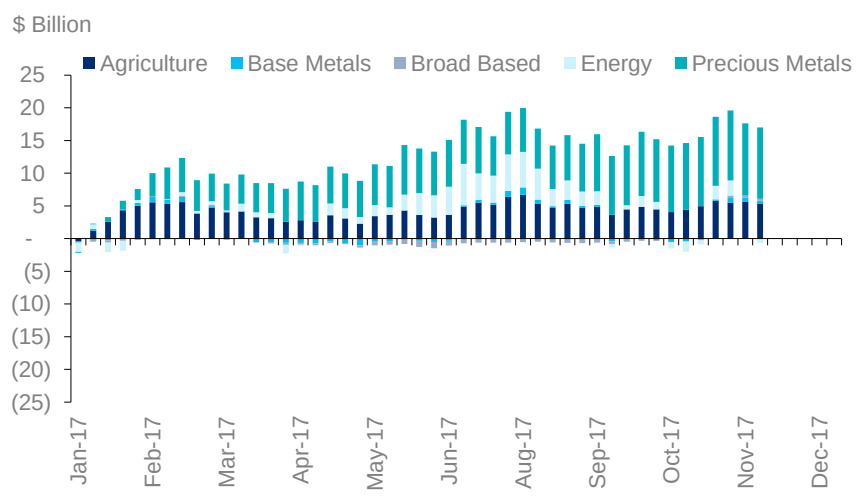
Investment inflows gathered pace again and look to sustain into 2018

Commodity AUM has continued to grow strongly in 2017, reaching the highest level since 2Q'14 before the pause in the YTD oil price rally. Strong curve term structure and a constructive macro backdrop should continue to buttress investor inflows in the short-term, although lower oil prices later in 2018 may challenge this growth.

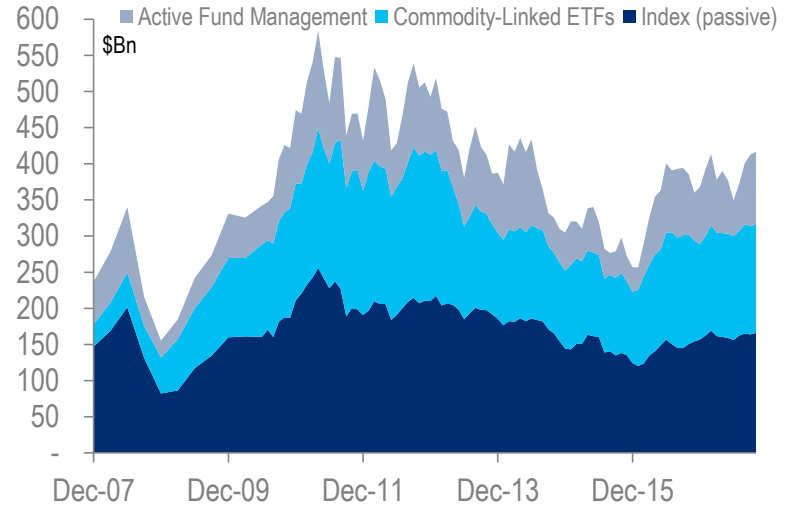
- **Citi estimates of commodity AUM** rebounded strongly in 2H, growing by almost 20% to ~\$417Bn through the end of October, the highest since June 2014.** This was primarily driven by the sharp rally in oil prices, with Brent flat price running up as much as 43% from the trough in June to early November, which helped lift index valuation to \$166Bn. Strong energy prices and Brent term structure have also reversed the bearish commodity sentiment that prevailed among the investor community in 2Q. Active fund length exceeded \$100Bn for first time in more than three years. ETF market caps, which are weighted towards precious metals, and physical gold ETF holdings in particular, edged slightly higher as well to \$150Bn.
- **Inflows into commodity-linked passive indices and ETFs have gathered pace again after stalling through 3Q, with YTD cumulative inflows hitting ~\$17Bn through mid-November 2017.** Precious metals ETFs have seen steady inflows throughout the year and contributed to approximately half of the total addition. Heightened geopolitical risks—uncertainty around the French elections in 1H and the rising tension between Washington and Pyongyang in 2H—boosted demand for gold and other safe havens, a trend we expect to continue in the medium term. The additional pick-up of investor flows was mostly led by additional investment in energy indices. Besides the bullish sentiment, stronger crude oil futures curves, with Brent and WTI in full backwardation, have also attracted passive investors who look to gain exposure via positive roll yields.

Prepared for Rong Lu (research email)

2017 YTD Cumulative Commodity Investment by Sector*



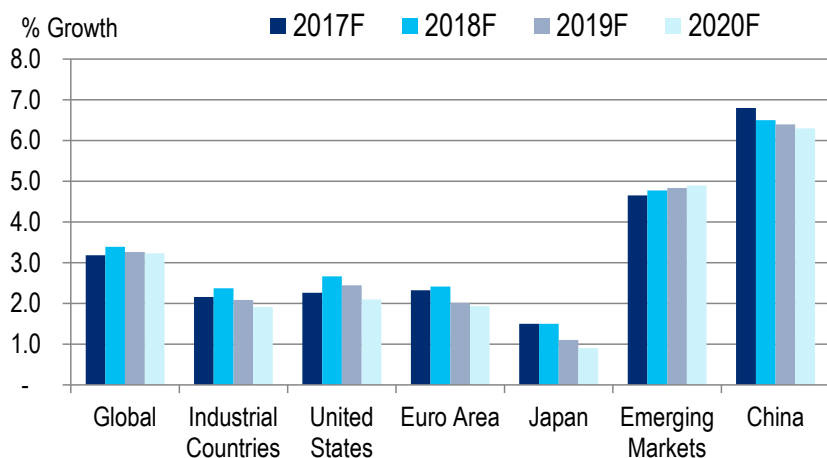
Institutional and Retail Commodity AUM**



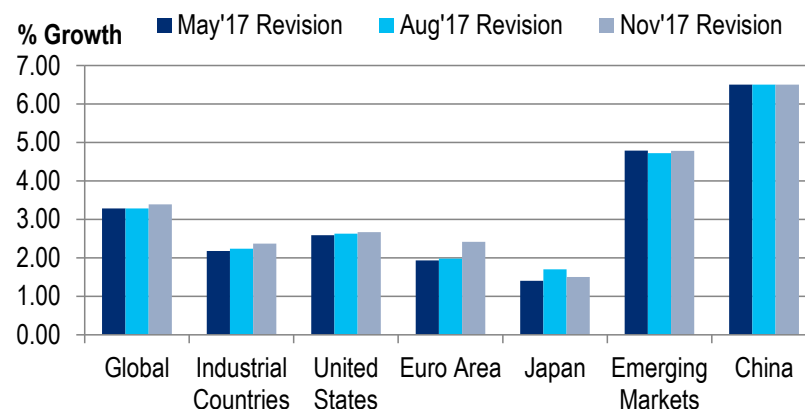
Source: CFTC, Bloomberg, Citi Research *subject to revision; passive index and commodity ETFs through 21 November 2017. **biased towards US trading activity and subject to revision

Global growth is expected to be robust and broad-based but be mindful of risks

Citi Economics Growth Forecast for Selected Countries/Regions



Citi Economics Forecast Revisions for Selected Countries and Regions (2018)



Citi expects robust, above-consensus economic growth pegging global real GDP growth in 2018 at 3.4% led by 4.8% growth in emerging markets (EMs) and 2.4% in advanced economies (AEs). The recovery is broad-based, with still accommodative monetary policy and potential fiscal stimulus (see US tax reform) leading the charge in the AEs, while a pick-up in investment and recoveries from recession/weakness, particularly in commodity-linked economies, supports EM growth.

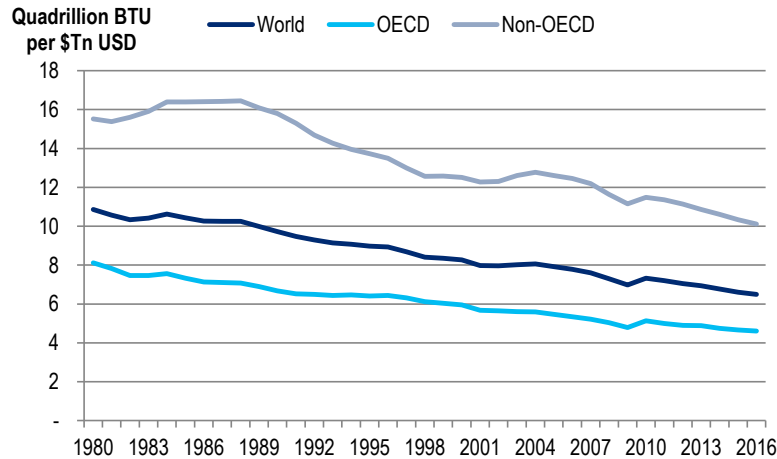
Investment-led growth, which is more closely correlated with commodity demand than consumer-led growth, should continue to pick up next year amid more robust activity, rising capacity utilization, benign financial conditions, an absence of negative catalysts and some structural reforms such as those in India paying dividends.

Trade growth should be strong, mostly in line with strong industrial activities and higher investment – another positive for commodity demand

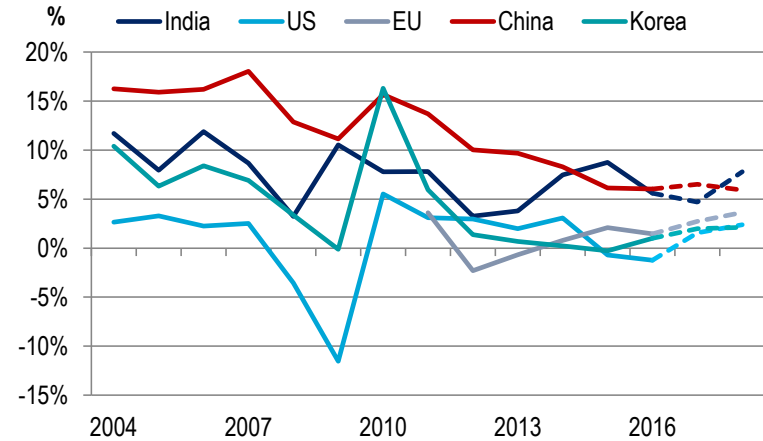
Risks remain, however. (1) Monetary policy begins to tighten across AEs. Although Citi's economics team expects only a moderate impact on real activity, there is risk of a correction in asset prices and heightened volatility, as financial markets tighten. (2) The uncertainty around US policies continues to pose risks. For one, Citi's above consensus call on US growth hinges on the passage of a sizeable tax cut in early 2018 but disappointment here could lead to slower than expected growth. How far the new administration will go with respect to trade protectionism is another, with open questions still surrounding NAFTA and KORUS. (3) Geopolitical risk also abounds, not least of which in North Korea.

Stronger macro should lift commodity demand yet demand intensity is easing

World Energy Intensity (1980-2016)



Global Industrial Production



- **Stronger global growth should help boost commodity demand in 2018, but economic growth is also becoming less commodity-intensive, which could dilute the impact of stronger growth on commodity demand.** For example, energy intensity, the total amount of energy consumed per unit of GDP, has been declining for many years in the OECD and non-OECD. This trend is not only apparent on an absolute basis, but also in percentage growth terms: in the 1980's, a 1% increase in global GDP led to ~1.5% increase in primary energy consumption, but since 2000 that has fallen to 1%. Why?
- **Efficiency:** Declining commodity-intensity of growth is partly a function of increasingly efficient industrial processes, sometimes spurred on by regulation, as well as improvements in energy efficiency. Energy efficiency regulations have been tightening globally but especially after the Paris accord, including efficiency standards for building and automobiles. China's supply cut policy due to environmental reasons and removal of excess, inefficient capacity, is another key driver.
- **Moving up the value-chain:** Developing economies are also deemphasizing from labor and energy-intensive industries. As economies become more developed, they tend to diversify away from commodity-intensive manufacturing processes and towards commodity-light consumer services. China, the powerhouse of commodity demand growth over the last decade, is now in the midst of just such a transition, with President Xi highlighting the need to further develop technology.
- **After China, who's next?** As China transitions its economy, which had also driven exporters to China to be commodity-intensive, no country or region big enough is emerging to pick up the slack in boosting commodity demand the way China did. Although China should still partially rely on the "old economy" growth drivers for now, environmental regulations and capacity rationalization should limit commodity demand growth. We examine the situation in China in our special section.

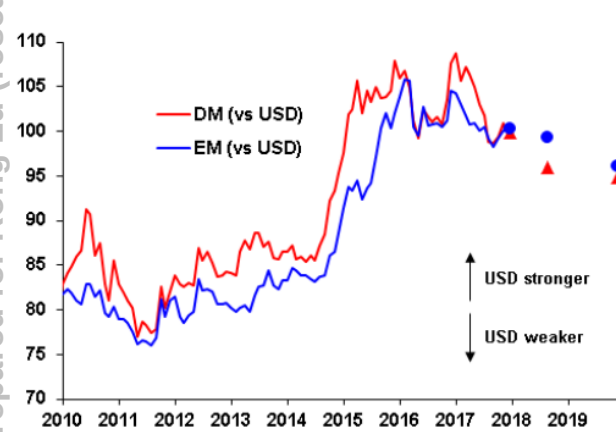
Weak USD should lift prices but fundamentals more crucial to 2018 prices

2017 has seen the lowest correlation between USD and commodity prices since early-2000s, as BCOM rebound to largely flat on the year while USD depreciating ~8% against other G10 currencies. We expect the muted relationship to continue into 2018 and see limited impacts on commodity pricing except metals.

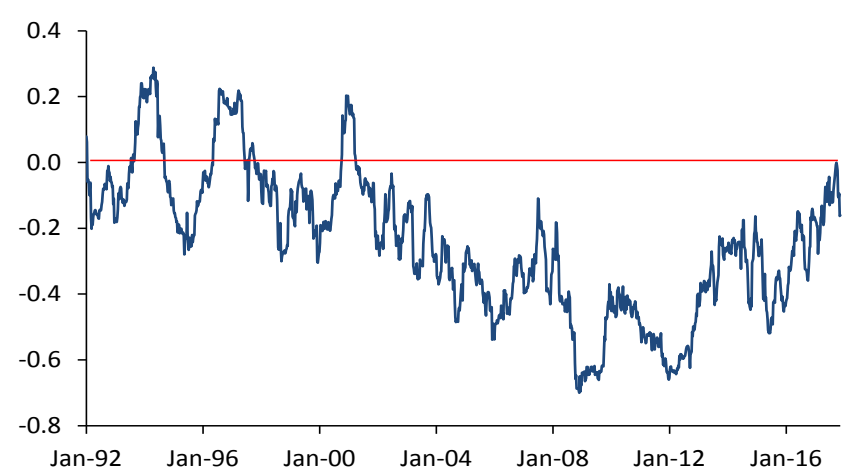
- **The correlation between commodity prices and USD has weakened markedly throughout 2017, even falling to around zero in 3Q. The correlation could stay relatively muted in 2018 for most commodities.** Note that the denomination effect is still key in metals pricing, especially gold, whose daily trading resembles a currency more than a traditional physical commodity.
- **Fundamentals of some key commodities, instead of being at some steady state so that currencies could exert greater effects, could be at their respective inflection points that amplify impacts of commodity supply-demand on prices:** for metals and bulks, China’s partial reliance on the “old-economy” growth model vs the macro transitioning that drive capacity cuts and environmental regulations, and for oil, the battle between shale and OPEC output cuts. Further, weather for natural gas and agriculture should outweigh USD’s denomination effect in short-term pricing.
- **However, a more substantial depreciation of USD should give support to commodity prices, all else equal.** Citi’s FX strategists are bearish USD, with the USD index dropping below 90 vs. G10 peers over a 6-12 month horizon. The strength in EURUSD comes from ECB’s taper in reducing the negative net cash requirement in the EUR bond market in 2018, so that Bund yields should rise and at a faster pace than UST yields. In EM, a friendly combination of high real interest rates (proxy for return) and current account surplus (proxy for risk) should keep a number of EM currencies strong vs. the USD.

Prepared for Rong Lu (research email)

DM and EM FX (vs. USD)*



BCOM Spot Index vs. USD Index rolling 1Y weekly correlation



Source: Bloomberg, Citi Research *Forecast as of mid-November, 2017

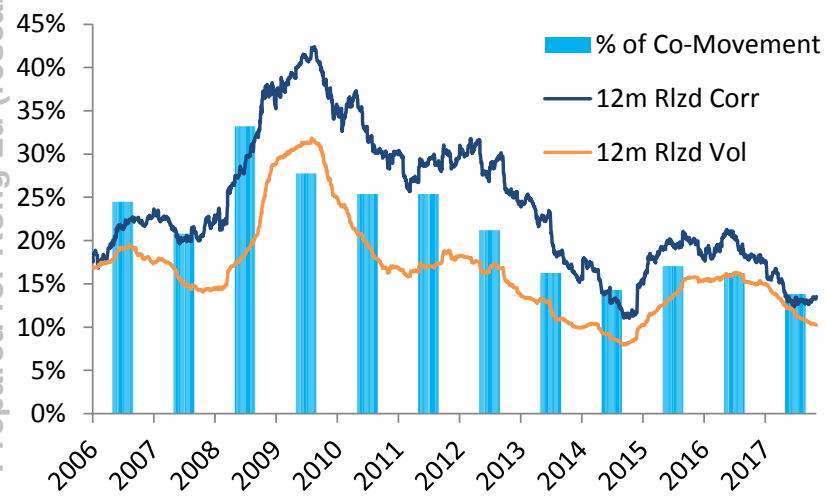
With fundamentals crucial, uncorrelated moves across sectors to limit index vol

A lack of co-movements among sectors should dampen the move in commodity index levels. The current low-vol/col-corr regime is likely to persist in 2018, barring a sizable shock to global demand.

- The subdued cross-commodity correlation, especially a lack of co-movements among sectors, in addition to the muted volatility at the single commodity level as explained in the next slide, should keep index volatility lower for longer. Only ~14% of the trading sessions YTD have seen a concerted rally or sell-off in all five major sectors (excluding livestock). This measure is the lowest since 2002, before index products became popular and commodities started being traded more widely as an asset class by investors. Although correlations between nearly all sector pairs are positive, they remain well below their long-term averages in the 30s corr pts. Notably, the correlation between precious and industrial metals sectors, both of which have seen decent positive returns this year, has been 17% vs. a 10-year average of 37%.
- Co-movement across sectors would not pick up unless and until a sizable and sustained shock to global demand, such as an abrupt deceleration in the economic growth in China or the US. This remains a tail risk in our view, as global macro seems to be gradually improving. In contrast, supply shocks may bring out enhanced price volatility of individual commodities or at a sector level, and they are unlikely to have a long-lasting and widespread effect across the board. The current low-vol/low-corr environment should persist in 2018, as each sector should trade largely on its own fundamentals.

Prepared for Rong Lu (research email)

BCOM 12M Realized Vol, Correlation and % of Co-Movements across Sectors*



6-Month Inter-Sector Correlations

Cross-Sector 126D		Sectors		
Correlation		Energy	Precious	Industrials
Sectors	Precious	10.7%	<-- Current 6M Correlation	
		24%	<-- 10-year Average 6M Corr	
		-0.7 25%	<-- 10-year z-score & %ile rank	
	Industrials	17.9%	16.8%	
		38%	37%	
		-1.1 19%	-1.2 13%	
Agriculture		21.7%	2.7%	-0.8%
		34%	25%	36%
		-0.7 30%	-1.5 7%	-2.2 1%

Source: Bloomberg, Citi Research *% of sector co-movements is the defined as the % of trading days where all five sectors - energy, industrial metals, precious metals, softs and grains – rally or sell off at the same time in that year. The 12M realized correlation is calculated as the weighted average pairwise 252-day correlation between the sub-indices of BCOM constituents.

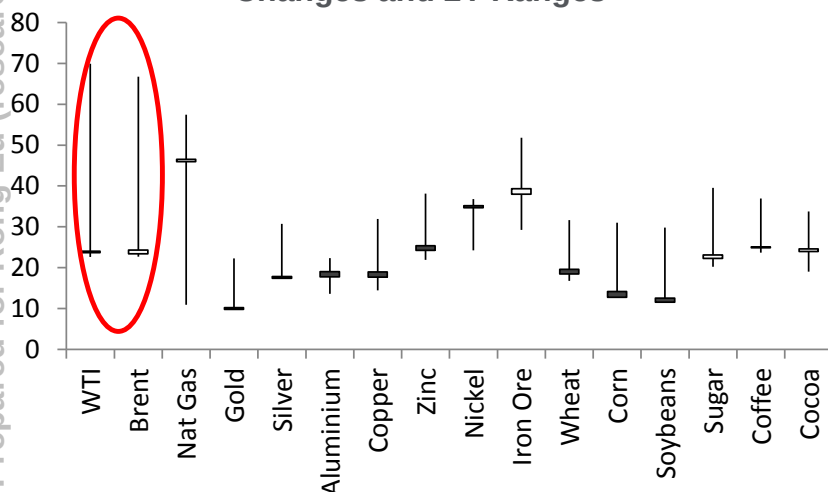
Volatility may rise in 2018, but only modestly

The commodity complex is likely to see a modest uptick in volatility next year, as less accommodative central bank policies should restore higher volatility to the broader market and as crude oil price volatility picks up.

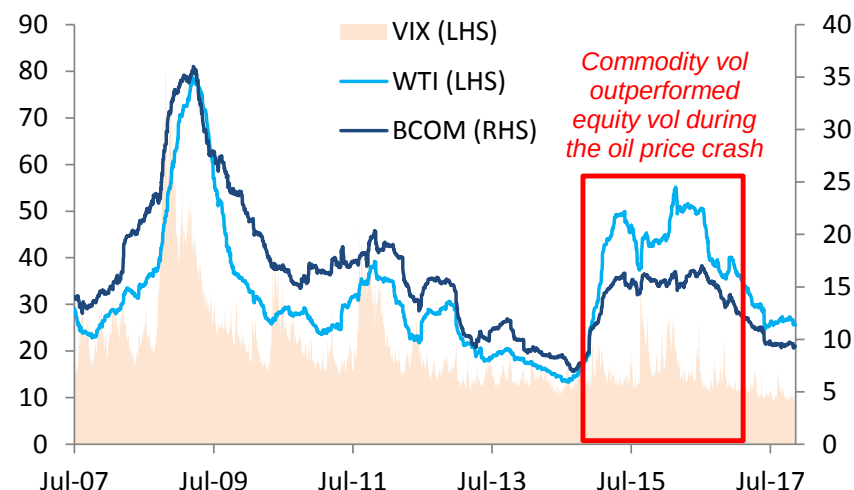
- **In the current low-vol environment that prevails all asset classes, muted volatility is seen across commodities.** As equity index vols continued to hover around historical lows, the volatility of commodity indices have declined to the lowest since 2014 as well. Precious metals and grains are seeing the most muted price moves in more than a decade, while commodities in the industrial metals and energy sectors have been trading a lot less volatile than in 2015-16. Accordingly, implied vols are also trading near the bottom of their 2-year range for a majority of commodities, pricing in only a little vol risk premium.
- **Broad market volatility could increase as global central banks continue to tighten, pushing rates and risk premia higher.** But the very slow pace of the tightening cycle suggests volatility may well be in check.
- **Crude oil vol should rise next year but should not spike to levels seen during the oil price crash in 2014-15.** The volatility in crude oil and the broader petroleum complex, which takes up the highest index weight (>20% in BCOM), should largely determine the trend in the index vol. A sharp sell-off in oil prices could send oil and commodity index vols significantly higher, outperforming the volatility of other assets, just as seen during 2H'14-2015. Given the strong negative relationship between oil vol and prices and the lower oil price path (see the Energy section), we expect oil vol to pick up slightly next year.

Prepared for Rong Lu (research email)

3M Implied Volatility across Commodities with 1W Changes and 2Y Ranges*



BCOM, WTI Crude Oil 6M Realized Volatility vs. VIX

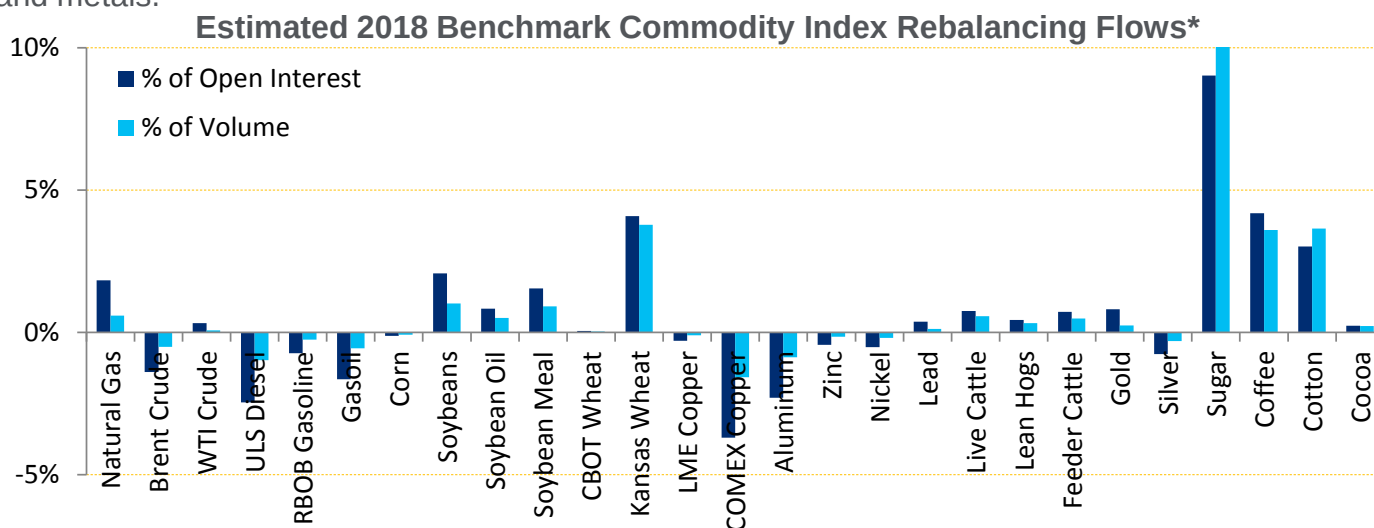


Source: Bloomberg, Citi Research *Levels are as of mid-November.

Annual index rebalancing should have muted impacts on prices

The two major commodity benchmark indices, BCOM and SPGSCI, are going to rebalance during the roll window from Jan 8 to Jan 12, 2018. We expect the annual rebalancing process to have muted impacts on the price of constituent contracts, assuming the current index weights stay stable into year-end.

- **With our estimates that approximately \$50Bn of AUM is invested in each of these two benchmarks, the energy and industrial metals sectors are expected to see a total reduction of ~\$3Bn in index AUM, while the agriculture sector is likely to see the largest inflows.** On a contract level, WTI and Brent crude oil will continue to have the largest index weights, although, all else equal, index investments in Brent are likely to drop by \$1.2Bn, primarily driven by the lower target weight in GSCI. While the adjustments to this year's target weights are small for most other commodities, we should see flows out of YTD outperformers, i.e. industrial metals, and into the underperformers (softs and grains), due to the shift in weights through the year.
- **Softs are likely to see the strongest price impact from rebalancing flows during the roll period.** We gauge the impacts by estimating the number of contracts to be bought or sold due to the re-weighting as a percentage of open interest and daily trading volume over the roll period for each commodity. In particular, we expect 11k ICE sugar contracts, which are down 23% and have underperformed all other commodities significantly this year, to be bought every day during the roll period (~10% of its average daily volume). Meanwhile, the added buying/selling pressure only accounts for less than 2% of the volume for energy commodities and metals.

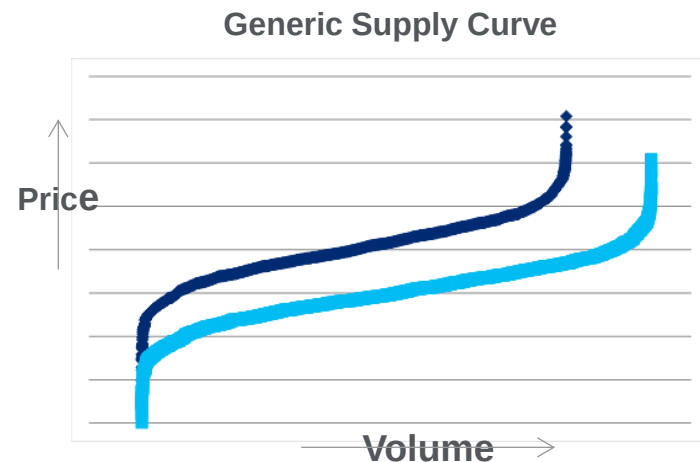


*Estimates are based on index weights and the first futures' settlement prices as of Nov 14, 2017. Rebalancing flows are measured as the number of contracts expected to be bought or sold at close during the Jan 2018 index rebalancing period, as a percentage of the average daily trading volume and as a percentage of the average open interest of short-dated contracts for each commodity (first 4 contracts for energy and base metals, and first 3 contracts for other sectors). The percentage of volume is computed using 20% of the estimated contracts to be bought or sold into the settlement over the roll period. We proxy each commodity's open interest and volumes using the average during the past 3 roll windows (Sep, Oct and Nov rolling periods).

The Big Differentiator: Longer-Term Costs and Longer-Term Prices

While short-term supply vs. demand balances have been largely responsible for the radical changes in prompt prices since 2014, cost structures are also playing an important role – sometimes helpful, sometimes misguided – in pointing to the medium and longer term outlook for commodity prices.

- The radical deflation in costs across most commodities since 2014 was both long overdue and bound to start settling following the even more radical increases of the last decade. While there is still considerable noise about the direction and magnitude of change ahead, patterns are emerging to guide assessments of supply curves for individual commodities for the medium- to longer-term. Virtually all extractive commodities' prices are determined by the marginal cost of incremental supply. If costs remain constant or change at a slow pace across an industry, then incremental demand will increasingly meet supply constraints unless more supply is brought into the market at a higher price, as is shown in the modeled cost curve on the right. A period of cost deflation can quickly and sharply flatten and lower cost curves.



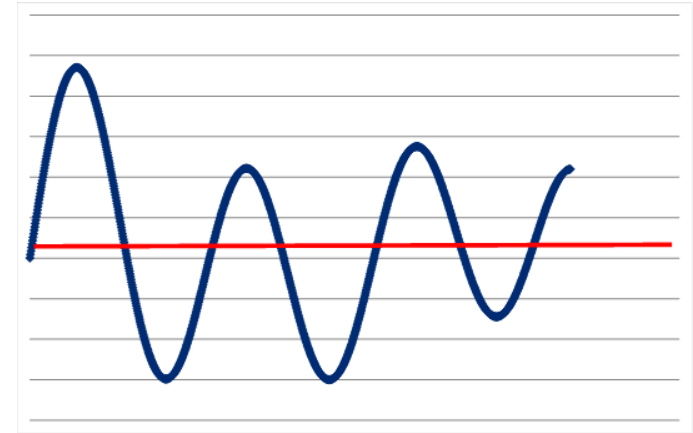
- Going forward we expect most of the cost deflation that has occurred in the supply curves for oil and gas to continue for several years to come, but in other sectors, we expect supply curves to both move up and to steepen. This reflects three different phenomenon. (1) Since all commodities are significantly energy intensive, lower energy costs should persist and put a drag on other cost curves. But (2) overcapacities in services that built up in the capital spending sprees of the last decade should taper with cost inflation replacing the non-energy portion of recent cost deflation. Moreover (3) lack of technological increases in productivity in most non-energy commodities (thermal coal and iron ore possibly being exceptions) should not only mean that incremental volumes required will come at a higher cost but should also mean cost curves for those commodities should start to rise again. China reductions in overcapacity and pollution are resulting in higher raw materials prices – most notably coal and steel, key energy and consumable inputs for the metals and mining and smelting industry. A large portion of this shift up in costs is likely to be permanent.
- There are yet other twists to energy costs and prices. Historically changes in cost structures have occurred slowly and where forward price curves have had significant liquidity in deferred pricing, long-dated contracts have shown remarkable price stability, effectively mean-reverting as WTI and Brent did at ~\$21 from 1987 to 2003. Lack of liquidity in deferred contracts has effectively eliminated mean reversion as a way to indicate where prices might be heading in the short-run.

But there's no straight line forward for any commodity – the process is cyclical

A major curve ball impeding projections is the fallacy that price changes are always linear. The fact is that price movements have been linear for only short-periods of time. Indeed cyclicality has been the main feature of commodity markets historically, with no indication that cyclicality will end any time soon.

- **Across commodities there appear to be three main factors lying behind cyclicality:** (1) **The supply function for commodities is characterized by sharp discontinuities**, particularly in the marshalling of capital for new supply but also in the depletion of older supply; (2) **There are meaningful lags in the way supply and demand functions interact with one another**, including impacts of growing demand on prices that can trigger sharply higher or lower prices and feed back into supply (e.g. projects under development continue to be developed even if less supply is needed; and a small increment in demand can trigger a significant price increase); and (3) **there are significant and lagging externalities to other markets and market sectors from overreactions to huge swings from commodity.** Over-reactions can even spill over into military conflict, as per Iraq's Kuwait invasion in 1990.

Generic Commodity Cycles



- **While the main factors assuring cyclicality are generally easy to understand, the specific factors change from cycle to cycle and no two cycles are the same, all for understandable reasons.** Global politics and global economics have had significant impacts on cyclicality. Long periods of recession can slow down the demand response to lower prices; producers cooperating to curtail production can prolong the duration of a period of higher prices
- **Cycles tend to have unexpected turning points that are sharp in the direction and magnitude of change. These can be mild or highly disruptive. The most recent cycle – the so called ‘super cycle’ of the last decade – was particularly brutal and disruptive, perhaps because the turning points were simultaneous across most commodities and over the last half century similar only to what occurred during the Malthusian 1970s. It was then that the so-called North-South split between commodity producers and consumers led to efforts at market power for producers via cartelization as well as efforts to create buffer stocks to moderate cyclicality.** In the oil markets, which along with copper and to some degree gold were most prominent in their level of increased prices last decade, the disruptive changes that occurred led to the exploitation of what were highly costly new sources of output, with production growing in huge quantities and triggering unexpected technological developments that brought costs down radically.

Unconventional oil supplies have re-set long-term prices

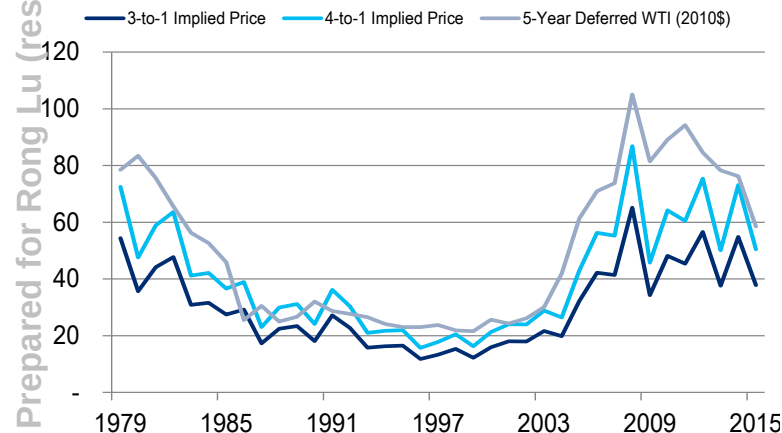
The rise in prices in the last decade pushed oil and natural gas exploration into what had been high cost areas: shale, deep water and oil sands, creating superabundant production growth, establishing short-cycle output, expandable infrastructure, depriving traditional producers of their pricing power.

Citi's Projected Crude Oil Prices Through 2020

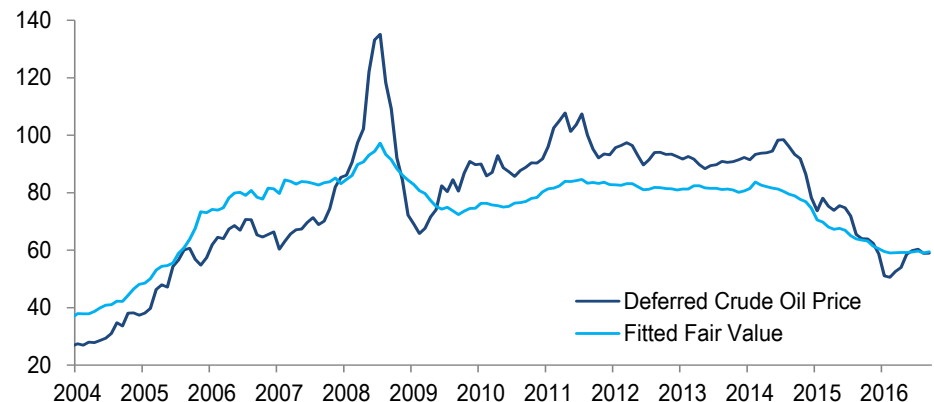
Base Case Price outlook (\$/bbl)	1Q17	2Q17	3Q17	4Q17E	1Q18E	2Q18E	3Q18E	4Q18E	1Q19E	2Q19E	3Q19E	4Q19E	2017E	2018E	2019E	2020E
Brent	55	51	52	61	57	54	54	52	48	47	49	50	55	54	49	56
WTI	52	48	48	55	52	49	50	48	43	42	43	44	51	50	43	48
Brent-WTI	3	3	4	6	5	5	4	4	5	5	6	6	4	5	6	8

- **Citi's longer term price projections are based on multiple different modeling techniques**, including those based on historical cost factors underlying the so-called Adelman method as well as Citi's own Fair Value Index projecting where longer term deferred prices will be.
- **Short-cycle crude oil creates a "call on US shale" as a market balancer**, which also puts a soft floor and soft ceiling on pricing, pointing to the low likely of oil exceeding \$70/bbl (Brent) or falling below \$40/bbl for any sustained period of time over the next five years.

5-year deferred WTI price vs implied price with 3:1 and 4:1 ratio to R&D costs



Citi's Fair Value Index Vs. Deferred Crude Oil Price



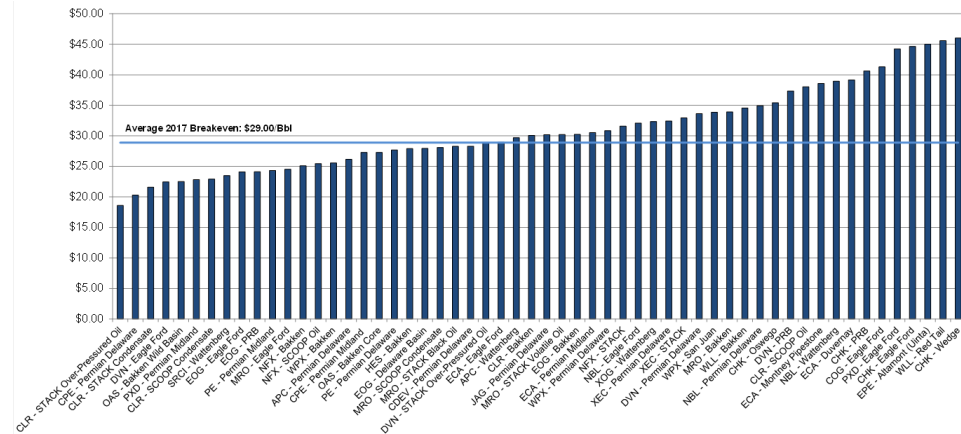
Cost factors are supportive of sustainable non-OPEC output growth

Though costs have inflated with the recovery in shale production, offsetting productivity gains and efficiency improvements have kept shale breakevens low; other drivers are pushing down costs for deepwater and oil sands.

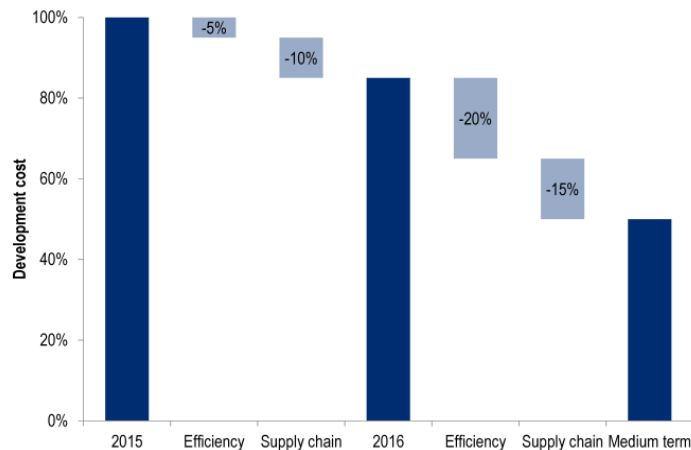
Average Shale Well Cost



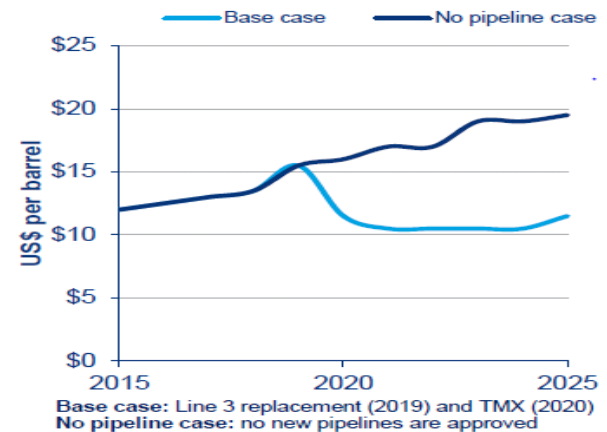
Citi E&P Oil Price Breakeven Analysis – Benchmarking by Sub-Basin or Play*



Deepwater Project Development Cost Outlook



New Pipelines Increase Oil Sands Netbacks

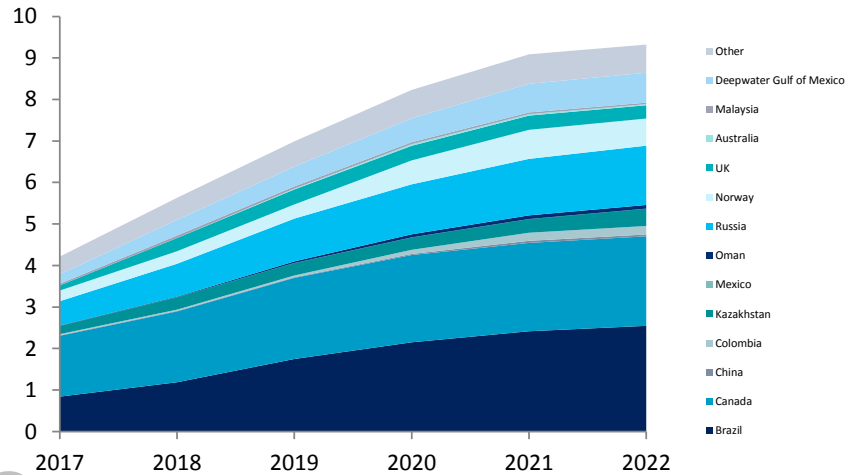


Source: Woodmac, EIA, Citi Research, *Based on each company's current modeled type curve (at \$3.00/Mmbtu flat natural gas price)

Global Supply, Cost Stability Should Cap Oil Prices Longer Term

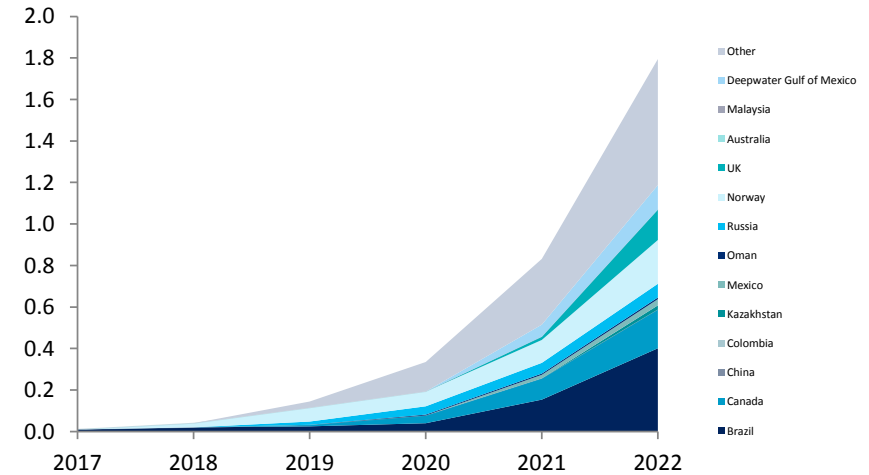
Non-OPEC non-shale production from FID projects*

Already-sanctioned projects could add over 5-m b/d through 2022



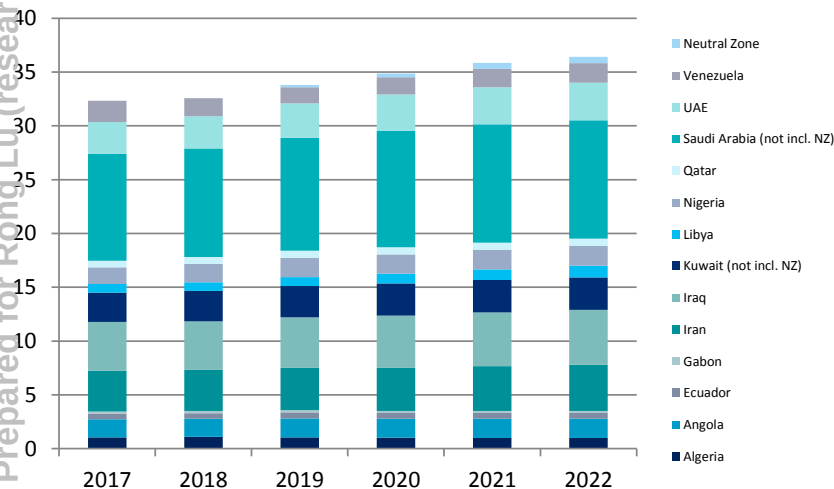
Non-OPEC non-shale production from non-FID projects*

Additional ~2-m b/d of growth possible from non-FID projects, with upside

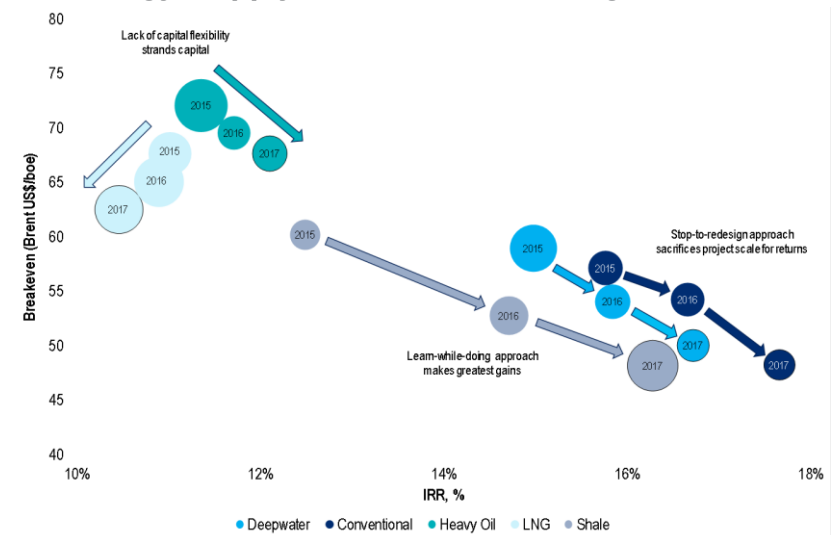


OPEC liquids production outlook

The return of OPEC barrels after the recent cuts, plus medium-term expansion plan, could add ~4-m b/d by 2022 (m b/d)



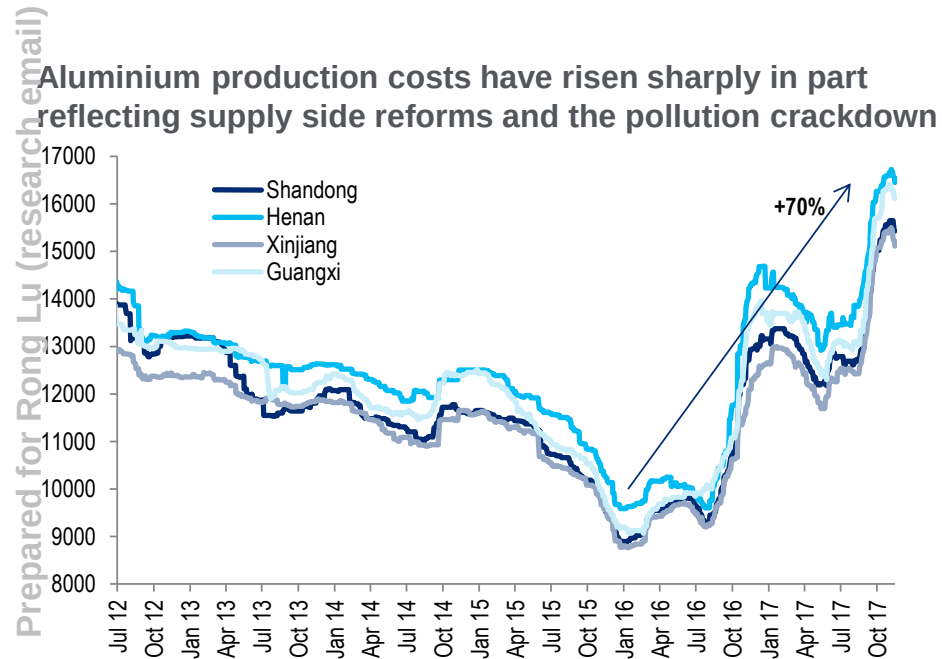
Global Supply Curve for Oil Getting Flatter, Lower due to technology, supply shocks and Evolving Production Costs



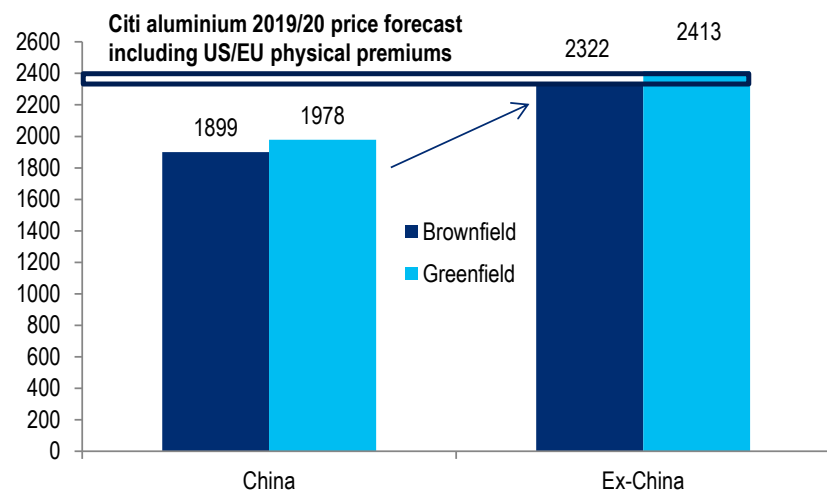
Aluminium production costs are set to remain higher for longer

Beyond the impact of the pollution crackdown on aluminium production costs, Chinese supply side reforms are likely to mean that ex-China will need to build aluminium smelters over the long term. This is set to shift incentive pricing from China to ex-China, where capital costs and hurdle rates are higher.

- China's crackdown on pollution (aimed at reducing pollution by cutting aluminium, alumina, carbon output) and supply side reforms (aimed at reducing overcapacity, affecting coal and alumina output) have contributed to a 70% increase in the cost of producing aluminium over the past two years. These policy dynamics are likely to persist indefinitely, and while lower cost production alternatives may be found, a large portion of the increase in costs is likely to continue.
- Perhaps even more critically, Chinese supply side reforms are likely to mean that incentive pricing shifts from China to ex-China, as growth in Chinese exports of aluminium (in all forms) slows down and potentially stops. The reduction in export growth would mean that ex-China would ultimately need to build aluminium smelters. Ex-China incentive prices are much higher than Chinese incentive prices, and this is likely to underpin higher long-dated aluminium prices for longer.



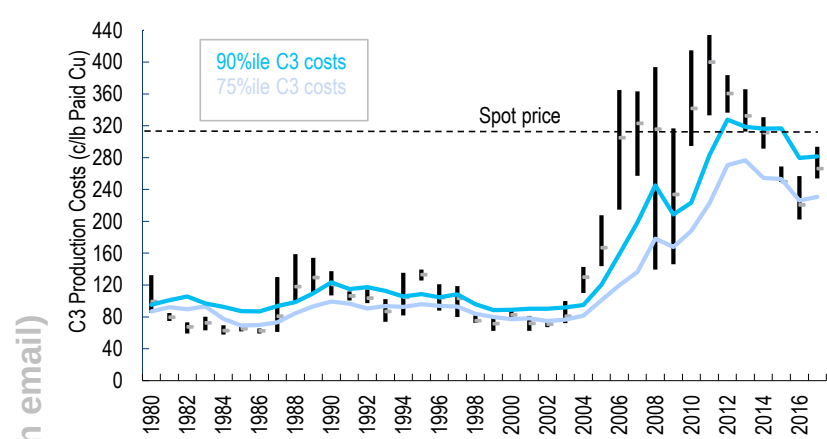
Aluminium smelters are increasingly likely to need to be built outside of China, where costs are higher



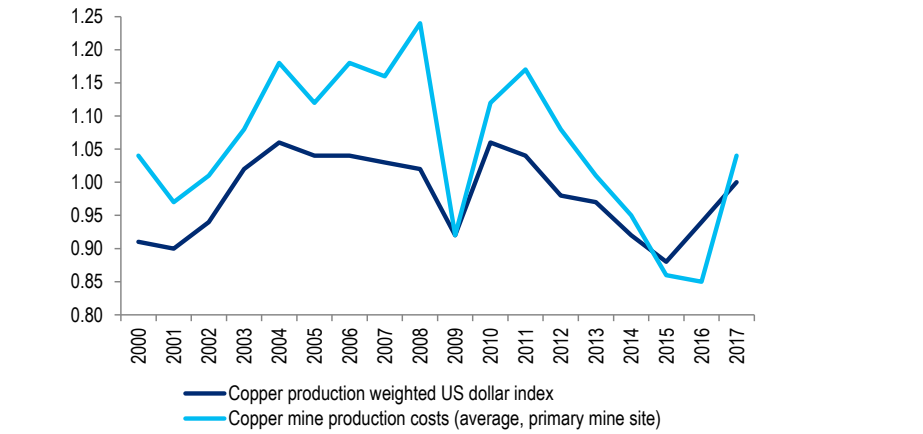
Copper costs of production have turned the corner

Copper production costs have begun to increase again on the back of a broad based rebound in costs across the labour, services, fuel, electricity, and stores (consumables) components, in turn reflecting the depreciation of the dollar against copper producing country currencies and higher oil, steel, and coal prices.

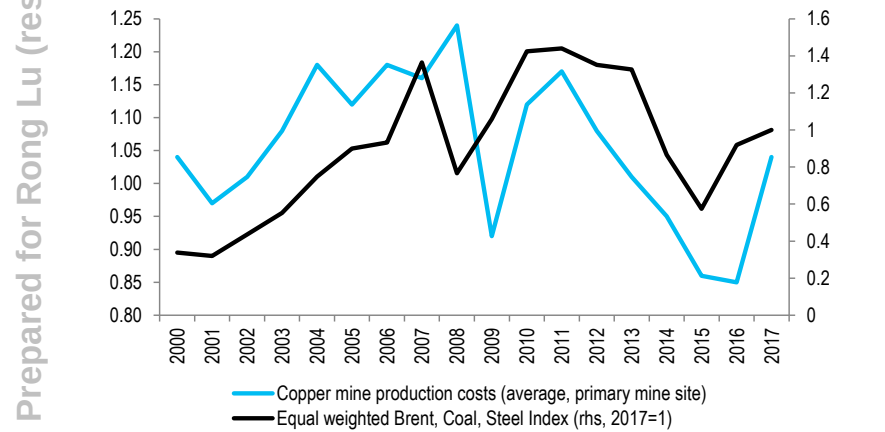
Annual average copper production costs have bottomed out, and are likely to rise during 2018



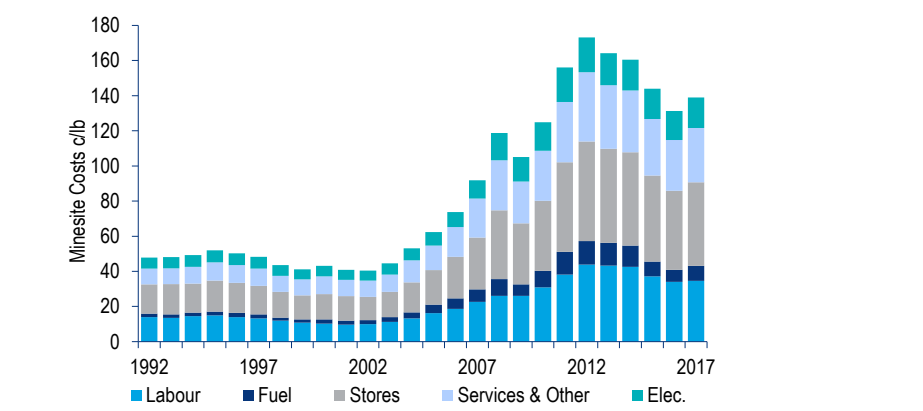
US dollar weakness against copper producing country currencies is placing upward pressure on costs



China's supply side reforms and pollution crackdown are driving up power (coal) and stores (steel, oil) costs



Cost increases have thus been broad based across the fuel, labour, services, stores and electricity components

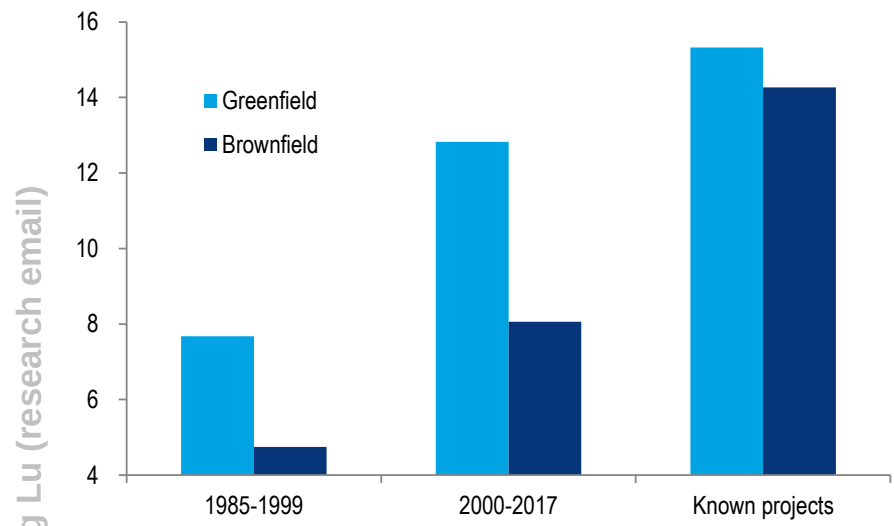


Source: Bloomberg, Wood Mackenzie, Citi Research

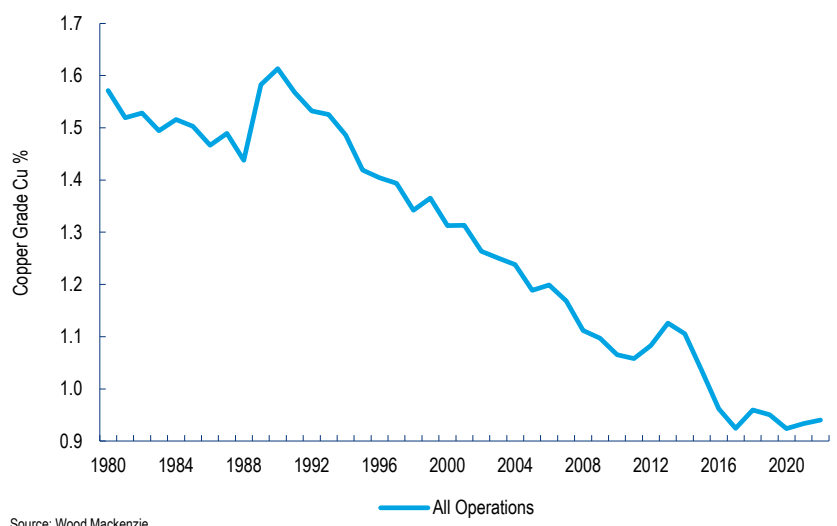
Rising capital intensity & falling grades underpin upward copper cost pressures

With Chinese supply side reforms and pollution controls are set to be maintained going forward, a large part of the increase in stores (rebar) and electricity (coal) costs are likely to be sustained. The trend of rising capital costs per tonne and low ore grades are also set to continue to place upward pressure on copper production costs.

Copper project capital costs have risen sharply, and continue to rise (\$/kt copper equivalent)



Ore grades are expected to continue to decline over the next 5 years, and could decline more if transitions to high grade underground mines are delayed



Source: Wood Mackenzie

Prepared for Rong Lu (research email)

Gold: an effective tail risk hedge in uncertain times

Increasingly, geopolitical events in a risky world have supported the tactical use of gold for short-term gain as well as the strategic use of gold as an inexpensive wildcard hedge. And gold is a particularly good hedge during financial crises and periods of asset market turmoil. We expect this to be the case in 2018.

- Even as the rates and FX channels dominate the fundamental outlook for gold pricing, gold is increasingly being used by investors as a policy and tail risk hedge. ‘Vox populi’ political movements, low oil prices, large fiscal and wage gaps and rapid technological change are just a few factors that should keep geopolitical risks structurally elevated over the medium-term, just as record central bank balance sheets begin to unwind, increasing the odds for policy error. What we’ve seen over the past two years—the sharp rally in gold prices after “Brexit”, ahead of the 2016 US presidential election and before the first round of the 2017 French elections—are not one-offs. Going forward, we are likely to see such wildcard event-driven rallies in gold prices on a more frequent basis. This is relevant for gold in a world of lowflation.

Gold has been trading increasingly sensitive to market risk sentiment and in a stronger inverse relationship with equity prices. We have not seen a meaningful equity market drawdown in over two years. If another crisis hits the market (within or outside of the US) and risk assets sell off, bullion prices should rally sharply and may sustain in a higher range. Indeed, gold has rallied almost 80% of the time after major market crises broke out over the past four decades, as shown on the right. And for crises that happened outside of the US, the price of the yellow metal has usually appreciated massively in local currencies. We also study the price impact from other types of geopolitical events—elections and political votes, and military strikes and attacks—in the Precious Metals section.

Gold price returns after major economic crises and market crashes

Event	Date	Returns in USD			Returns in Local Ccy		
		1D	1W	1M	1D	1W	1M
1979 Energy Crisis	3/26/1979	1.1%	-1.0%	-2.3%			
LatAm Sovereign Debt Crisis	8/20/1982	8.4%	16.9%	19.0%			
US Savings and Loans Crisis*	6/30/1986	0.2%	-0.3%	2.5%			
Financial Panic of '87	10/14/1987	0.3%	1.1%	1.3%			
Japan Asset Bubble Crash	12/29/1989	0.7%	-0.5%	5.1%	1.0%	-0.5%	4.9%
ERM UK Currency Crisis	9/15/1992	0.9%	1.7%	0.3%	2.4%	12.5%	11.0%
Asian Stock Market Crisis**	10/7/1997	-0.3%	-1.2%	-6.0%	-1.3%	-1.9%	2.3%
Russian Financial Crisis	8/17/1998	0.4%	0.3%	2.2%	2.9%	12.0%	86.2%
Mexico Peso Crisis	12/20/1994	0.6%	1.0%	1.0%	15.4%	45.0%	60.4%
Dotcom Bubble	3/10/2000	-0.8%	-1.6%	-3.2%			
the Great Financial Crisis	9/15/2008	2.8%	14.1%	8.8%			
European Sovereign Debt Crisis	1/14/2010	0.4%	-2.4%	-1.7%	0.5%	0.4%	3.6%
China Stock Market Crash	8/24/2015	-0.5%	-2.3%	-2.4%			
Median Returns		0.4%	0.4%	0.4%			
Average Returns		1.1%	2.0%	1.9%			
Positive Returns as % of Total		77%	46%	62%			

*The savings and loan crisis started from 1986 and lasted through 1995. We use mid-year 1986 as the start of the crisis. **The 1997 Asian financial market crisis impacted several Asian countries and regions, including Thailand, Malaysia, Singapore, Indonesia, the Philippines, and South Korea. We use the exchange rate of Thai Baht (THB) to compute gold's price returns in local currency.

Citi Commodities Price Forecasts*

		Point Prices														Annuals				
		0-3M	6-12M		Q4 2017E	Q1 2018E	Q2 2018E	Q3 2018E	Q4 2018E	Q1 2019E	Q2 2019E	Q3 2019E	Q4 2019E	2016	2017E	2018E	2019E	2020E		
Energy				5Y Cyclical																
ICE Brent	USD/bbl	57	53	62	61	57	54	54	52	48	47	49	50	45	55	54	49	56		
NYMEX WTI	USD/bbl	52	49	59	55	52	49	50	48	43	42	43	44	43	51	50	43	48		
Henry Hub Natural Gas	USD/MMBtu	3.2	3.0	3.0	3.2	3.2	3.1	3.1	3.0	3.1	3.0	2.9	2.9	2.6	3.0	3.1	3.0	3.0		
Base Metals				LT Price																
LME Aluminum	USD/MT	2,100	2,100	2,200	2,145	2,100	2,100	2,100	2,100	2,200	2,200	2,200	2,200	1,610	1,985	2,100	2,200	2,200		
LME Copper	USD/MT	7,000	7,100	7,000	6,900	7,000	7,200	7,200	7,000	7,000	7,000	7,000	7,000	4,869	6,210	7,100	7,000	6,600		
LME Lead	USD/MT	2,700	2,500	2,000	2,505	2,700	2,900	2,500	2,500	2,300	2,300	2,300	2,300	1,876	2,325	2,650	2,300	2,300		
LME Nickel	USD/MT	11,500	11,500	15,000	11,745	11,500	11,500	11,500	11,500	12,000	12,000	12,000	12,000	9,644	10,485	11,500	12,000	13,000		
LME Zinc	USD/MT	3,400	3,100	2,400	3,215	3,400	3,600	3,200	3,000	2,800	2,800	2,500	2,500	2,099	2,895	3,300	2,650	2,500		
Precious Metals				LT Price																
COMEX Gold	USD/T. oz	1,300	1,250	1,200	1,285	1,300	1,275	1,250	1,250	1,295	1,355	1,380	1,370	1,248	1,265	1,270	1,350	1,370		
Silver	USD/T. oz	16.0	17.8	16.5	17.0	17.0	16.8	16.5	16.5	17.8	17.8	17.8	17.8	17.1	17.2	16.7	17.8	18.0		
Platinum	USD/T. oz	925	980	1,150	925	925	925	950	950	975	1,000	1,000	1,025	988	950	940	1,000	1,150		
Palladium	USD/T. oz	1,000	900	780	975	1,000	1,100	1,200	1,100	1,050	1,000	1,000	950	614	865	1,100	1,000	900		
Bulk Commodities				5Y Cyclical																
Hard Coking Coal (Spot)	USD/MT	170	140	90	170	170	160	150	140	135	130	125	120	145	180	155	128	120		
Thermal Coal Asia (NEWC)	USD/MT	85	80	55	94	80	75	76	80	75	72	75	78	66	87	78	75	65		
Iron Ore Spot (TSI)	USD/MT	65	62	42	65	68	65	60	62	60	60	60	60	58	71	64	60	55		
Agriculture																				
CBOT Corn	USd/bu	360	370	N/A	345	360	365	370	365	380	390	390	400	358	360	365	390	425		
CBOT Soybeans	USd/bu	945	955	N/A	975	945	930	955	975	990	1015	1010	1025	989	975	951	1,010	1,050		
CBOT Wheat	USd/bu	468	485	N/A	445	468	480	485	505	510	505	500	525	436	442	485	510	550		
ICE Sugar	USd/lb	15.0	17.0	N/A	15.0	15.0	16.0	17.0	17.0	NA	NA	NA	NA	18.4	16.0	16.5	16.0	NA		
ICE Coffee	USd/lb	134	138	N/A	126	134	140	135	130	NA	NA	NA	NA	138	135	135	134	NA		
ICE Cocoa	USD/MT	2,070	2,200	N/A	2100	2070	2150	2200	2175	NA	NA	NA	NA	2,880	2005	2150	2200	NA		

Citi 2H'18 Commodities Market Outlook (November 2017)**

	Bullish	Neutral-to-bullish	Neutral	Neutral-to-bearish	Bearish
Energy			US Gas		EU Gas, Brent, WTI
Base Metals		Aluminium, Copper	Nickel	Zinc	
Precious Metals		Palladium	Platinum	Gold, Silver	
Bulks				Iron ore, Thermal coal	Coking coal
Agriculture	Sugar	Coffee, Ethanol	Corn, Wheat, Cocoa	Soy beans	

Citi Commodities Price Forecast Changes versus 4Q'17 outlook

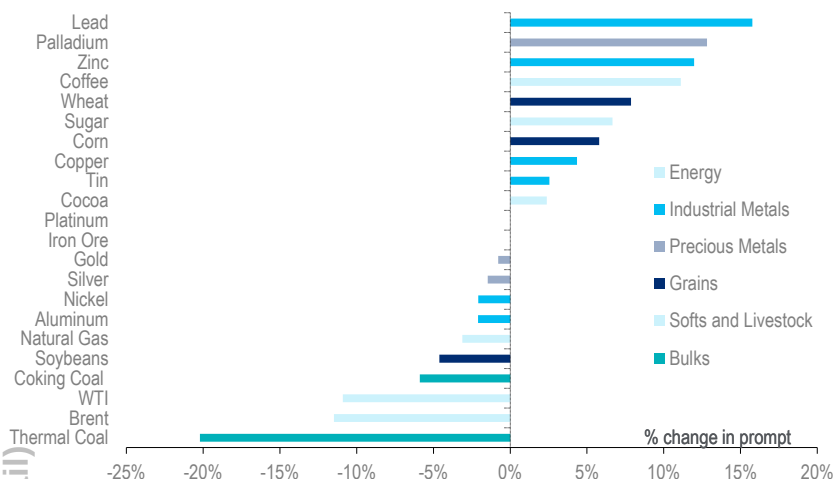
		Point Prices												Annals			
		0-3M	6-12M		Q4 2017E	Q1 2018E	Q2 2018E	Q3 2018E	Q4 2018E	Q1 2019E	Q2 2019E	Q3 2019E	Q4 2019E	2017E	2018E	2019E	2020E
Energy				5Y Cyclical													
ICE Brent	USD/bbl	-1	-1	0	3	2	2	-1	-1	NA	NA	NA	NA	1	0	NA	NA
NYMEX WTI	USD/bbl	-2	-1	0	1	1	1	-1	-1	NA	NA	NA	NA	1	0	NA	NA
Henry Hub Natural Gas	USD/MMBtu	-0.3	0.0	0.0	-0.3	0.0	0.1	0.2	0.0	NA	NA	NA	NA	-0.1	0.1	NA	NA
Base Metals				LT Price													
LME Aluminum	USD/MT	-	100	550	45	-100	100	155	155	NA	NA	NA	NA	-10	75	NA	NA
LME Copper	USD/MT	900	750	800	800	800	850	700	400	NA	NA	NA	NA	160	685	NA	NA
LME Lead	USD/MT	500	-	-200	5	500	650	225	200	NA	NA	NA	NA	5	395	NA	NA
LME Nickel	USD/MT	1,500	1,000	-6,000	1,545	1,400	1,300	1,250	1,200	NA	NA	NA	NA	395	1,285	NA	NA
LME Zinc	USD/MT	625	(100)	300	215	400	700	200	50	NA	NA	NA	NA	65	335	NA	NA
Precious Metals				LT Price													
COMEX Gold	USD/T. oz	(75)	-	150	-50	0	0	0	0	NA	NA	NA	NA	-10	0	NA	NA
Silver	USD/T. oz	-	-	0.0	-0.7	0.0	-0.8	-1.3	-1.5	NA	NA	NA	NA	-0.2	-0.9	NA	NA
Platinum	USD/T. oz	(25)	(20)	0	-55	-25	-30	0	0	NA	NA	NA	NA	-15	-10	NA	NA
Palladium	USD/T. oz	100	100	0	75	90	180	270	160	NA	NA	NA	NA	20	175	NA	NA
Bulk Commodities				5Y Cyclical													
Hard Coking Coal (Spot)	USD/MT	-100	25	0	20	30	25	20	15	NA	NA	NA	NA	5	23	NA	NA
Thermal Coal Asia (NEWC)	USD/MT	-10	10	0	14	5	1	4	9	NA	NA	NA	NA	4	5	NA	NA
Iron Ore Spot (TSI)	USD/MT	0	9	0	0	12	11	10	10	NA	NA	NA	NA	-1	11	NA	NA
Agriculture																	
CBOT Corn	USd/bu	5	(15)	N/A	-10	-15	-20	-25	-50	NA	NA	NA	NA	-2	-28	NA	NA
CBOT Soybeans	USd/bu	3	(35)	N/A	33	0	-60	-45	5	NA	NA	NA	NA	7	-24	NA	NA
CBOT Wheat	USd/bu	18	5	N/A	-5	3	0	0	20	NA	NA	NA	NA	-3	5	NA	NA
ICE Sugar	USD/lb	0.5	2.0	N/A	0.0	0.0	0.0	0.0	0.0	NA	NA	NA	NA	0.0	0.0	NA	NA
ICE Coffee	USD/lb	(1)	(7)	N/A	-9	-5	-5	-10	-20	NA	NA	NA	NA	0	-10	NA	NA
ICE Cocoa	USD/MT	145	50	N/A	175	0	0	0	0	NA	NA	NA	NA	0	0	NA	NA

Citi Commodities Forecast Heat Map

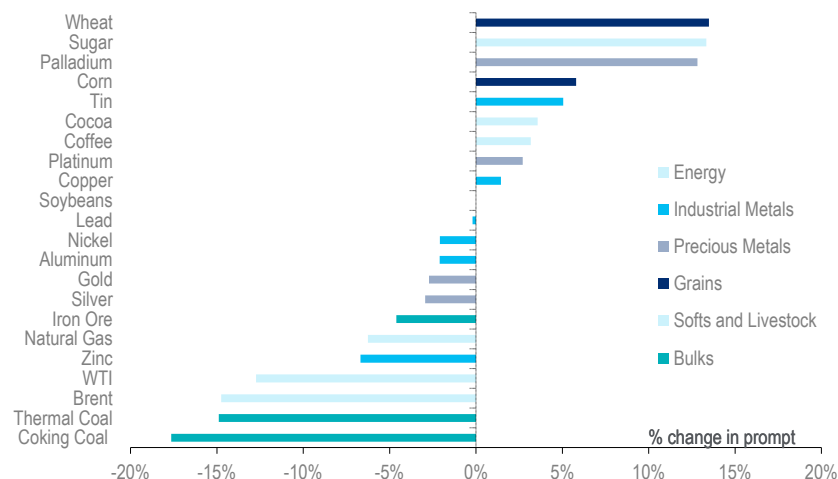
Average Annual Return (%)		2008	2009	2010	2011	2012	2013	2014	2015	2016	2017E	2018E	2019E	2020E
Energy														
ICE Brent	USD/bbl	36%	-36%	28%	38%	1%	-3%	-9%	-46%	-16%	22%	-2%	-9%	14%
NYMEX WTI	USD/bbl	38%	-38%	28%	19%	-1%	4%	-5%	-48%	-11%	17%	-2%	-14%	12%
Henry Hub Natural Gas	USD/MMBtu	25%	-53%	5%	-8%	-30%	32%	14%	-38%	-3%	17%	3%	-3%	0%
Base Metals														
LME Aluminum	USD/MT	-2%	-35%	29%	10%	-15%	-8%	1%	-11%	-4%	23%	6%	5%	0%
LME Copper	USD/MT	-3%	-25%	45%	17%	-10%	-8%	-7%	-20%	-11%	27%	14%	-1%	-6%
LME Lead	USD/MT	-19%	-17%	25%	10%	-13%	4%	-2%	-15%	4%	24%	14%	-13%	0%
LME Nickel	USD/MT	-42%	-30%	48%	5%	-23%	-14%	12%	-30%	-19%	9%	10%	4%	8%
LME Zinc	USD/MT	-42%	-11%	29%	1%	-11%	-1%	12%	-11%	8%	38%	14%	-20%	-6%
Precious Metals														
COMEX Gold	USD/T. oz	25%	12%	26%	28%	6%	-16%	-10%	-8%	8%	1%	0%	6%	1%
Silver	USD/T. oz	12%	-2%	38%	74%	-12%	-24%	-20%	-18%	9%	0%	-3%	7%	1%
Platinum	USD/T. oz	20%	-23%	33%	7%	-10%	-4%	-7%	-24%	-6%	-4%	-1%	6%	15%
Palladium	USD/T. oz	-1%	-25%	99%	38%	-12%	13%	11%	-14%	-11%	41%	27%	-9%	-10%
Bulk Commodities														
Hard Coking Coal (Spot)	USD/MT					-35%	-22%	-23%	-23%	67%	22%	-14%	-18%	-6%
Thermal Coal Asia (NEWC)	USD/MT	96%	-33%	38%	23%	-21%	-12%	-17%	-17%	11%	33%	-11%	-4%	-13%
Iron Ore Spot (TSI)	USD/MT	25%	-44%	69%	16%	-23%	5%	-29%	-43%	5%	21%	-10%	-6%	-8%
Agriculture														
CBOT Corn	USd/lb	41%	-29%	15%	59%	2%	-17%	-28%	-9%	-5%	0%	1%	7%	9%
CBOT Soybeans	USd/lb	43%	-16%	2%	26%	11%	-4%	-12%	-24%	5%	-1%	-2%	6%	4%
CBOT Wheat	USd/lb	25%	-33%	10%	22%	6%	-9%	-14%	-14%	-14%	1%	10%	5%	8%
ICE Sugar	USd/lb	22%	49%	24%	21%	-20%	-19%	-6%	-20%	39%	-12%	3%	-3%	NA
ICE Coffee	USd/lb	13%	-5%	31%	55%	-31%	-28%	41%	-26%	3%	-1%	0%	-1%	NA
ICE Cocoa	USD/MT	36%	9%	5%	-1%	-20%	2%	25%	3%	-8%	-30%	7%	2%	NA

Citi Commodities Price Forecasts

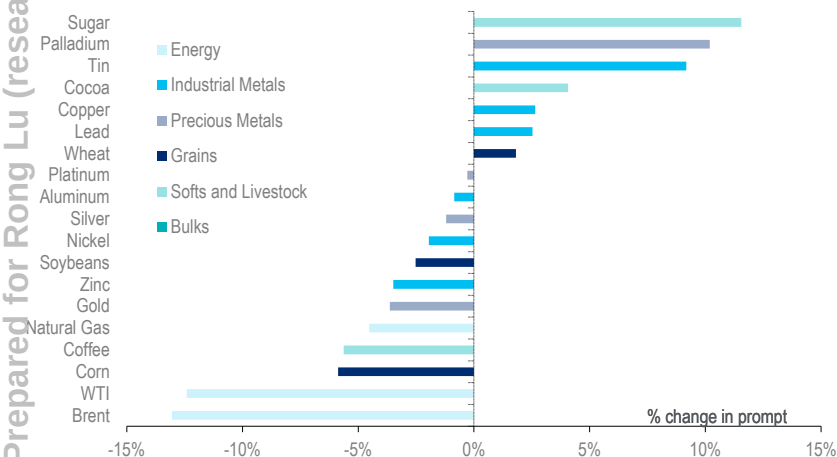
**Citi 1H'18 Commodity Returns Forecasts
(Citi 2Q'18 versus Citi 4Q'17 prices)**



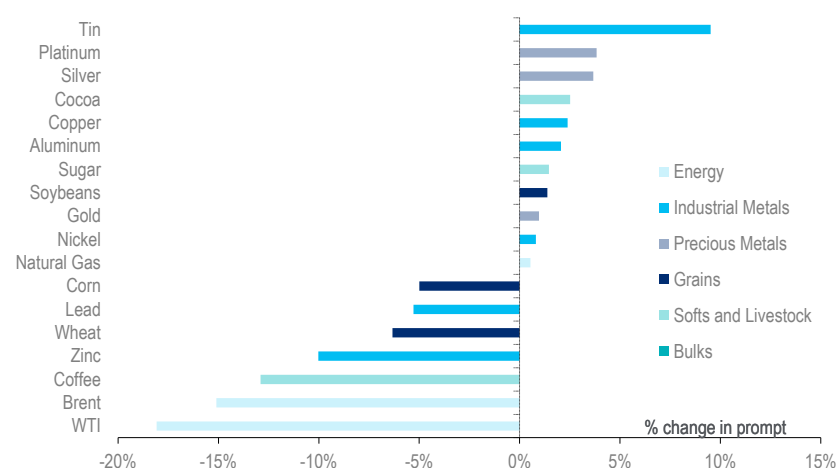
**Citi 2018 Commodity Returns Forecasts
(Citi 4Q'18 versus 4Q'17 prices)**



**Citi End-2018 Prices vs Forward Curve
(Citi 4Q'18 price forecasts versus Dec'18 futures*)**



**Citi End-2019 Prices vs Forward Curve
(Citi 2019 price forecasts versus Dec'19 futures*)**



2. Special Focus: As goes China's 'new era', so go commodities

As goes China's 'new era' so go commodities

China is critical to the commodities outlook, both in terms of driving demand and industry cost structure, and the outlook for China for 2018 is particularly uncertain as the dust settles from the party congress.

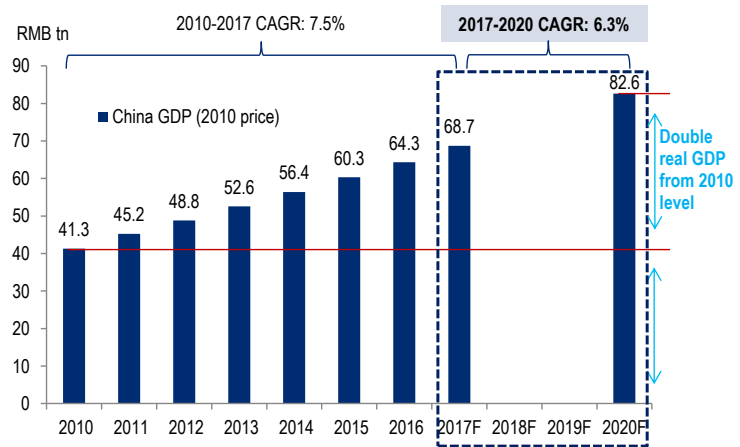
- **As growth slows the policy response function – both its timing and its quantum – will be critical.** Citi economists expect growth to slow notably, and we are already seeing signs of this slowdown in property sales and FAI. While the policy response function has always been difficult to predict, Xi's now supreme authority means that China's future growth path, including the commodity intensity of the growth path, as well as how China responds to shocks, are now ultimately in the hands of one man. Further policy uncertainty comes from a potential PBOC policy shift - the current governor may extend his term and is increasingly hawkish in tone, or a new governor with an unknown policy bias could be announced as soon as December. Other substantial risks to the Chinese growth outlook for 2018 include the potential consequences of capital controls (asset bubbles), supply side reforms (shortages), and QE withdrawal (credit spreads and equity market volatility) loom large.
- **Having acknowledged these risks, our base case is that China sensitive commodities see positive demand growth during 2018, as the China policy put is likely to remain intact.** The composition of this growth will be critical to metals, but less so to crude and natural gas. 'Old-economy' growth is already slowing down, and the policy response function is expected to kick in during 1Q'18.
- **Higher environmental standards and supply side reforms will significantly impact commodity markets in 2018, and are likely to be permanent.** A solid demand growth backdrop is set to coincide with a continuation of high environmental standards, and sustained pressure to support bank balance sheets by eliminating overcapacity and reducing oversupply. These measures are set to result in higher seasonal volatility in Chinese commodity demand and supply, higher costs of commodity production. Potential shortages in the face of positive demand shocks are likely to ensue.
- **As China moves away from being a low value exporter (or, a pollution importer), ex-China will increasingly need to meet its own demand with its own supply, requiring ex-China prices to move to incentive levels.** Ex-China incentive prices have already been reached in coal, are in the process of being reached in steel, and one day, in our view will be reached in aluminium.
- **A collapse in mining FAI in China means that there is limited scope for Chinese domestic supply to respond to stronger demand.** China has traditionally been a swing supplier of commodities such as zinc, coal and iron ore. However fixed asset investment in Chinese mining of iron ore, coal, and metals has fallen by 20-30% y/y, and 50%-2/3rds since mid 2014. When/if demand reaccelerates a domestic supply response will be both difficult and at best delayed.

Navigating China's new era – 2020 targets require ‘old economy’ growth

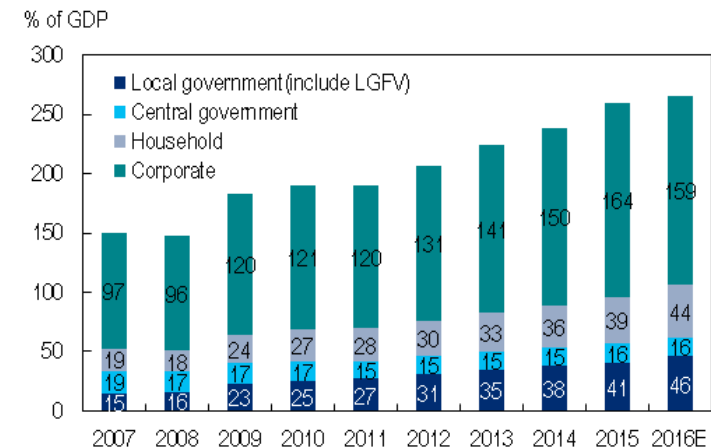
Our base case is that Chinese growth will remain solid through to 2020, with Xi pledging to double per capita income over the period.

- We believe growth target for China post 19 Party Congress will still remain around 6.5% until 2020. The next five years are critical for whether China can stride over the middle-income trap or not. The CPC has affirmed its commitment to doubling China's 2010 GDP and per capita income by 2020, which we estimate would require at least 6.3% annual growth. It also reiterated its plan to lift all of the poor (43.35mn as of 2016) out of poverty, known as “2020 poverty relief target”.
- The top leadership during the 19th Party Congress reiterated the importance of “quality” over “quantity” growth. In the transition from rapid growth to high-quality development, China should improve its economic structure and foster new growth drivers, via 1) advanced manufacture – initiatives such as “Made in China 2025” should receive continued policy support; 2) rural vitalization strategy – the *Political Report* extended current rural land leasing contracts by another 30 years upon expiration; 3) coordinated regional development – the second round urbanization should support infrastructure investment and drive economic growth; 4) financial reforms – China will strengthen the macro control framework and macro prudential assessment; 5) SOE reform – the key to resolve China's debt problem; 6) “Belt-and-Road” strategy and opening up.

An average growth rate of only 6.3% is needed in the next 3 years to achieve the doubling real GDP goal



Corporate debt, especially the SOE debt, is the key for debt issue in China



Source: CEIC, Citi Research

Navigating China's new era (2) – though challenges point to rising volatility

‘Crossing the river’ is set to be as challenging as ever for China. Major risks through 2020 are the potential impact of a tightening in global financial conditions as output gaps close in the US and Europe, gradually building capital outflow pressures, and the risks associated with reforming and transitioning the growth model.

- **China, and President Xi specifically plan to “cross the river by feeling the stones”.** We do not doubt that the river will be crossed – GDP per capita targets through 2020, ostensibly or otherwise, will ultimately be met - but **how volatile will the crossing be? How commodity intensive will the crossing will be?**
- **This depends on both on how Xi chooses to execute his transition to a higher quality of growth new era, and the way in which Xi responds to the inevitable challenges that China will face.** Specifically:
 - How will China react to a tightening in global financial conditions as interest rates rise and QE is drawn (potentially)? The attempt to withdraw QE may provide larger than anticipated challenges to global growth and China's economy.
 - How will Xi manage one of the largest and most rapid debt build ups in history? No large country has navigated either such a rapid debt build up nor a transition to a consumer driven economy without suffering recession. China certainly can re-write the history books, but this transition will by no means and there is no successful precedent to follow.
 - Following the deflationary oil and dollar shocks of 2015, China closed up its capital account. Every year that passes domestic assets build relative to foreign reserves. How does China open up its capital account over the medium term without risking major devaluation?
- **Though a reduction in Chinese commodity intensity is a concern for all commodity producers and investors, it is crucial to note that China’s increasingly strict environmental regulations and supply side reforms (including capacity rationalization) should limit the deflationary and bank balance sheet pressures of lower demand.** Specifically, for those commodities where China is long, such as coal, steel, aluminium, chemicals, China is likely to attempt to protect producer margins over the course of the cycle by acting as a swing supplier, and raising capacity utilisation. Should demand surprise to the upside, the long lead time between investment and production may result in commodity shortages (we note that Chinese mining FAI has collapsed over the past two years).

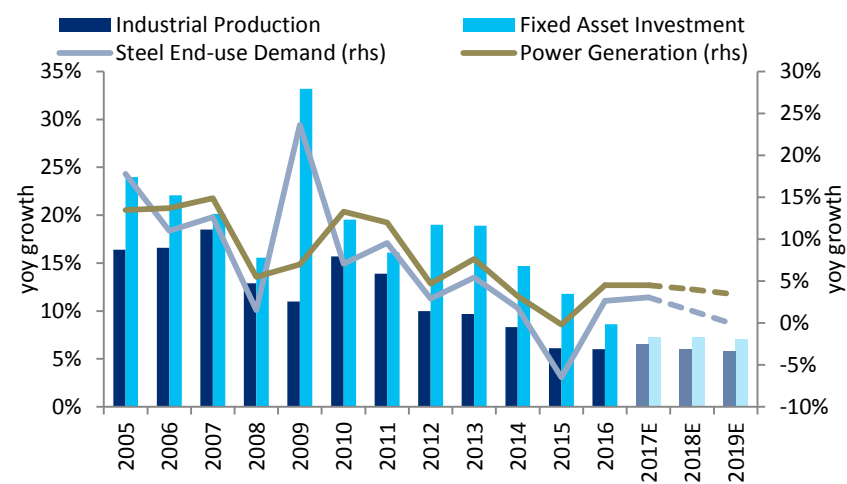
Prepared for Rong Lu (research email)

Chinese FAI/IP growth forecast to slow, recent developments warrant caution

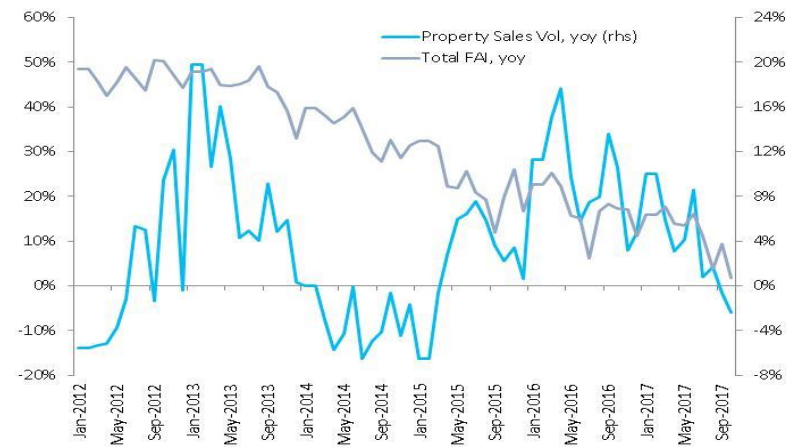
The latest data imply Chinese FAI and property sales will contract by c.1% and c.10% y/y during 1H'18, respectively, should recent run rates be maintained. However, China has substantial room to maneuver to shore up 'old economy' growth, and we expect it to leverage some of these tools during 1Q'18.

- Our economists forecast FAI growth to slow modestly from 8.6% to 7.0%, and IP growth to slow from 6.0% to 5.8% over the next 3 years.
- Most recently, Chinese FAI and property sales have been **weakening**. Unlike during 2014 when property sales were last negative, FAI is weak, and growing at its slowest pace for over 5 years, should the monthly data be accurate. On the more bullish side, manufacturing activity both in China and globally is growing at a solid clip. Overall the latest data leaves us cautious China-sensitive commodities, and looking to buy any dips in zinc and iron ore as we head in to the new year.
- Property sales and FAI weakness of late is at least in part by design, and as such there is substantial room for policy easing.
 - There is in our view substantial **pent up demand to purchase property**.
 - Nationwide property sales have been above completions, with inventories falling and prices rising nationwide, and land sales strong, pointing to **pent up demand for new property starts**.
 - Relaxation of restrictions on either front are possible if things continue to deteriorate. Easing of restrictions on either would lead to an **improvement in Property FAI**.
 - **Manufacturing FAI stimulus** for higher value add goods production could be expanded.
 - **Infrastructure FAI** could be improved by **accelerating project development**.

Chinese FAI and IP are projected to slow down



Near term China's 'old economy' data has been weak, though this is in part by design

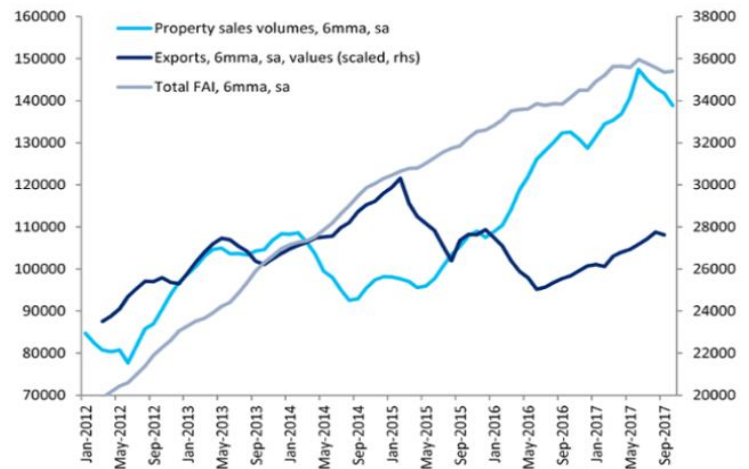


Source: NBS, Caixin, Bloomberg, PBoC, Citi Research

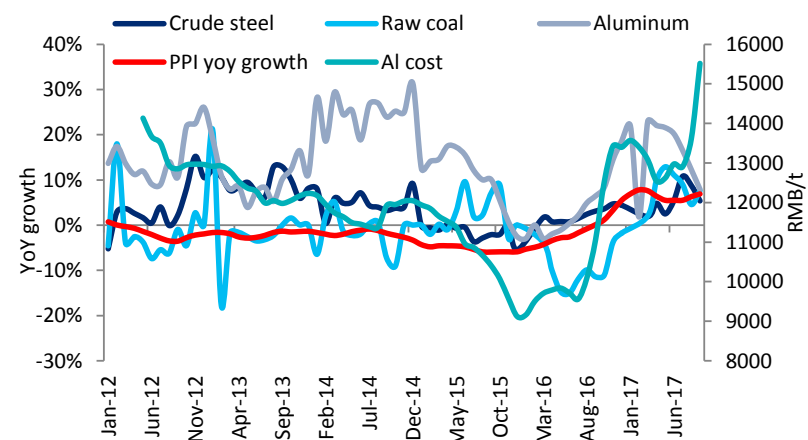
China's 'old-economy' in charts – slower activity growth is already upon us

Property sales and Total FAI have both weakened in recent months on a sequential basis, and the total credit 'impulse' has been negative for some time. Chinese manufacturing is the bright spot, with PMIs around 5-year highs on the back of the prior housing and infrastructure stimulus and solid exports growth

The last time property sales weakened (1H'14), FAI and exports were growing solidly, now it is only exports that are supportive

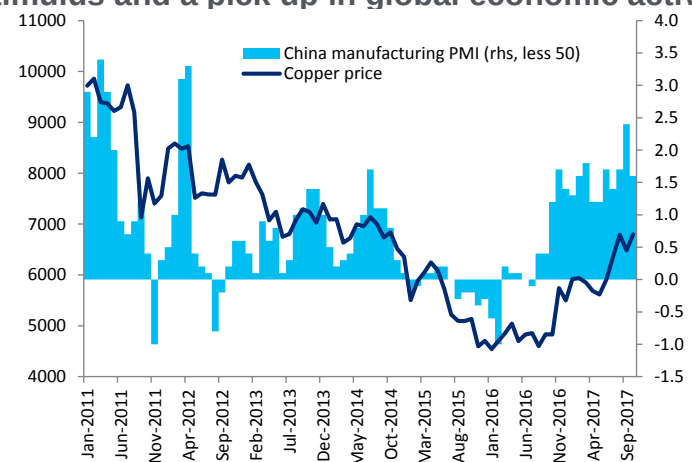


Supply growth has slowed and PPI and costs of production have risen

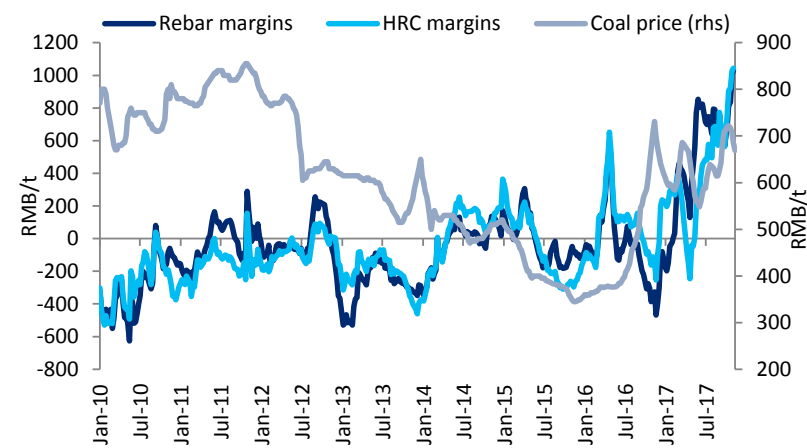


Prepared for Rong Lu (research email)

Chinese manufacturing is the bright spot, on the back of prior stimulus and a pick up in global economic activity



In steel, supply cuts and demand strength triggered price hikes and improved margins



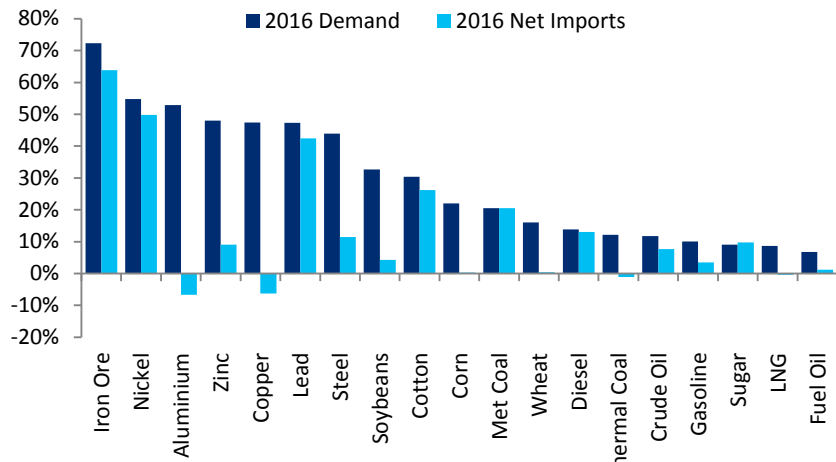
Source: NBS, Wind, Citi Research

But slower growth is set against restricted commodity capacity growth...

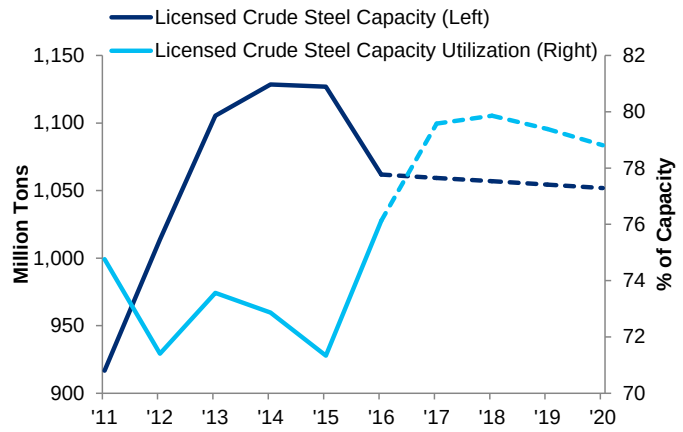
China's move to reduce overcapacity and pollution is set to result in lower domestic commodity output growth, higher utilization rates, solid domestic producer margins, and higher costs of commodity production.

- **Supply-side reforms were stressed in Xi's Political Report and should continue to be carried out strictly in the next few years, although the approaches could become more moderate over time.** The five major targets of the supply-side reforms include cutting industrial capacity, reducing housing inventory, cutting leverage, lowering corporate costs, and improving weak economic links. Capacity curtailments allowed enhanced utilization rates and margins in the commodity producing sectors, but at the same time rising Chinese domestic commodity prices might trigger higher PPI inflation domestically and worsened Chinese terms of trade.
- **The ongoing Chinese supply-side reforms and enhanced environmental regulations should yield profound implications on Chinese import and export dynamics.** In particular, China's role as the world's major commodities processor may weaken over time as the country is set to export fewer semi-processed manufacturing goods. The goal is to significantly lower exports of energy-intensive and pollution-intensive manufacturing goods and to reduce unnecessary emissions of waste gas, water and solid domestically.
- **We therefore anticipate lower net exports of steel products, aluminum products, petroleum products, coke, carbon anode etc. in the next few years.** The rest of the world should cope with the aftermath of the Chinese policies by producing more high-quality raw materials and building new processing capacity to tackle frequent price spikes created by Chinese production cutbacks.

China accounts for the bulk of global commodities demand



Chinese capacity utilisation is set to rise in steel, aluminium, and other 'old economy' industries

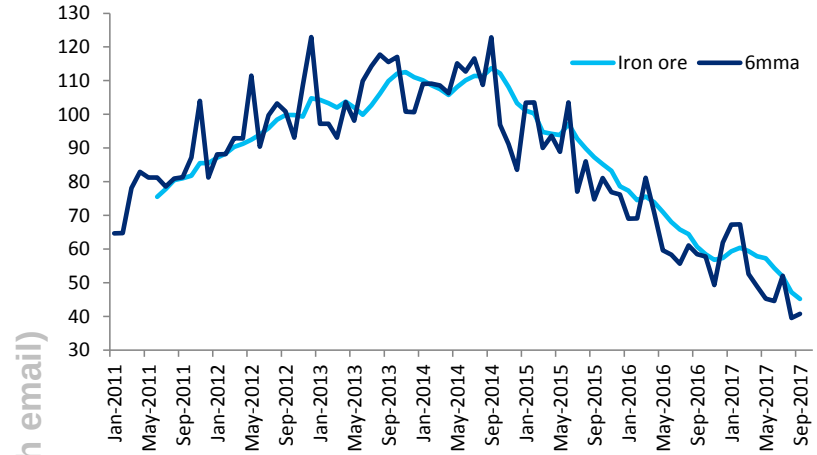


Source: NBS, Caixin, Bloomberg, PBoC, Citi Research

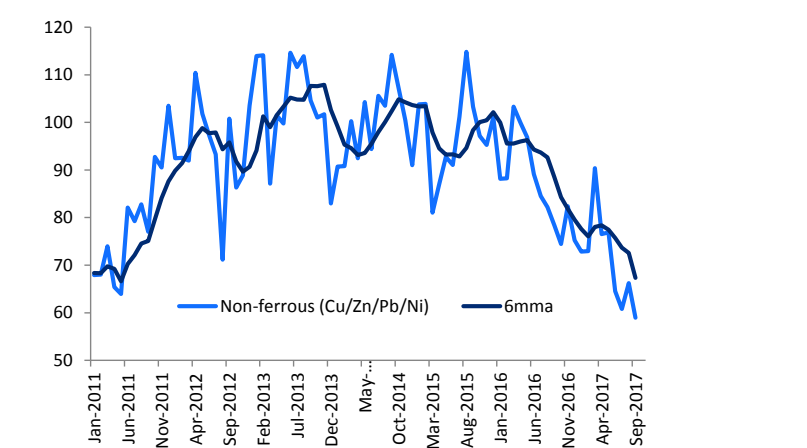
...and China's FAI in commodity supply is collapsing

FAI in mining iron ore, coal, and metals has fallen by 20-30% y/y, and 50%-2/3rds since mid 2014. Should Chinese demand reaccelerate a domestic supply response will be both difficult and delayed.

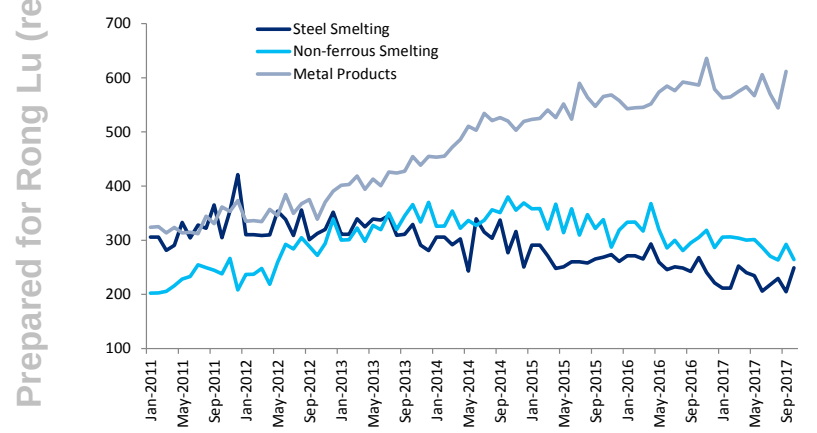
FAI in iron ore mining has fallen 30% y/y and 67% from mid 2014 levels



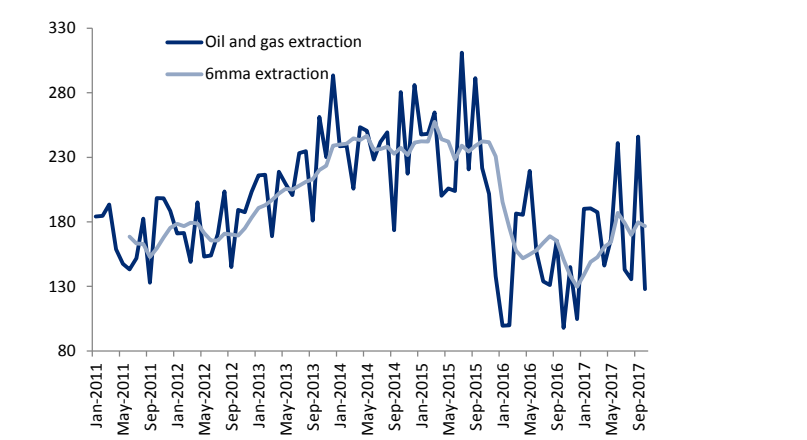
FAI in mining base metals (indicative for zinc) is down 20% y/y and 50% since its peak



China is still adding to downstream capacity in steel/metals but smelting FAI still trending lower



China oil and gas extraction FAI has recovered somewhat but remains down 30% from its peak

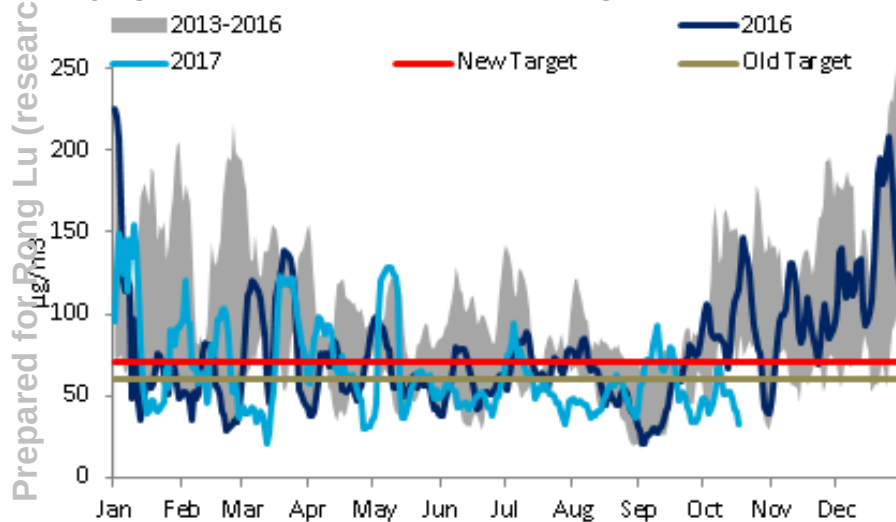


Source: CEIC, Bloomberg, Citi Research

China's "war on pollution" looms large

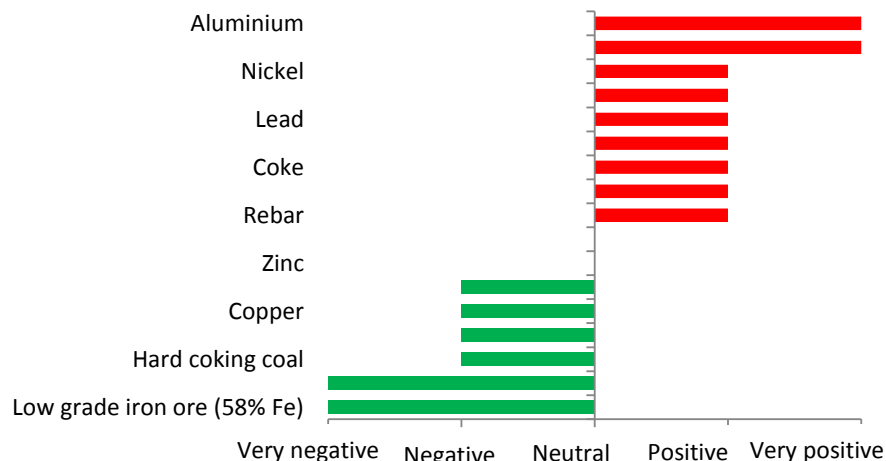
- **Environmental protection is embedded in the Chinese government's pursuit of "quality" over "quantity",** which was emphasized in the 13th Five Year Plan (2015-2020) and President Xi's Work Report to the 19th Party Congress.
- **We expect strict enforcement of the winter supply cuts policies to guarantee planned reductions in pollution.** According to the State Council's *Air Pollution Control Action Plan* published in 2013, Beijing's PM2.5 concentration is expected to be reduced to 60 $\mu\text{g}/\text{m}^3$ by the end of 2017.
- **The State Council plans to send 120 inspection teams during the winter to Hebei, Shandong, Shanxi, Henan and particularly to the two municipalities and 26 cities (2+26 cities).** Seven out of the 2+26 cities volunteered to cut more and/or earlier than required by the MEP's guidelines.

Beijing PM 2.5 concentration and targets

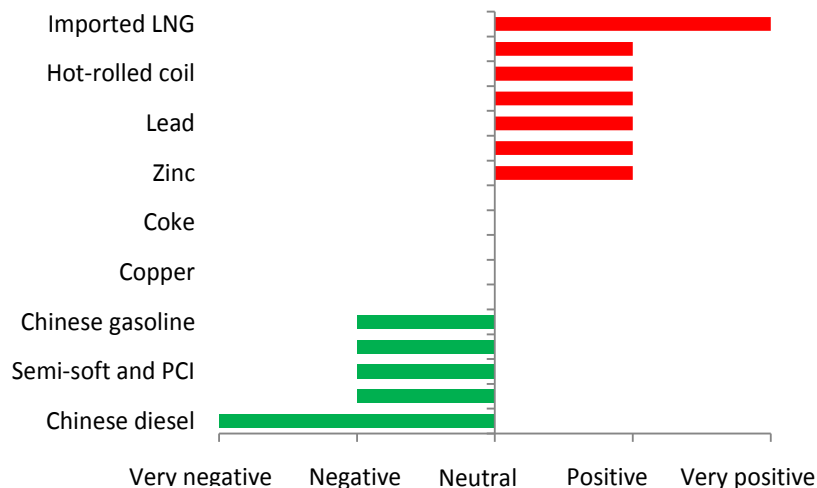


Source: Wind, Citi Research

Aluminum, flat steel products and nickel are the three largest beneficiaries of the 2017 winter cuts in terms of support to prices...



...For 2018 and beyond, extreme price moves on the back of winter cuts are less likely as the industries should be better prepared



Commodities output is expected to be hit by winter environmental crackdowns

Coke and carbon anode are expected to be affected the most by the winter cuts policies, followed by alumina, aluminum, steel and nickel pig iron on a gross output cut basis (not considering restarts or ramp-ups elsewhere).

Types of Commodity	Cities subject to supply cut policies	Total operating capacity (Mt/y)	% required cut against nameplate capacity	Total output losses (Mt)	% losses against 2016 winter output (Nov.16-Feb.17)	% losses against 2016 annual output	Comments
Steel	Tangshan, Handan, Shijiazhuang, Anyang, Tianjin, Xingtai, Changzhi, Jincheng, Zibo, Zhengzhou, Linfen, Jinzhong, Xuzhou, Weifang, Binzhou	308.8	< 50%	25-35	9.6%-13.5%	3.1%-4.3%	Measured by steel capacity; monitored via power consumption. Coking processing time extended from an average of 24 hours to 36-72 hours.
Coke	2+26 cities, Weifang	166.4	25-75%	16.6	11.3%	3.7%	Measured by number of aluminum electrolytic cells; the biggest swing factor is when Weiqiao starts to cut.
Aluminum	2+26 cities	11.7	> 30%	1.0	8.9%	3.2%	Measured by number of alumina producing lines.
Alumina	2+26 cities, Luoyang, Pingdingshan and Sanmenxia	37.6	30%	2.5-3.0	10.8%-12.9%	4.1%-4.9%	Plants that fail to meet emission standards are required to fully shut down.
Carbon Anode	2+26 cities	11.6	> 50%	1.5	25%-30%	8%-10%	The cut might fail to happen because NPI is not listed as one of the key winter curtailment target sectors.
Nickel Pig Iron	Linyi	0.114 (Ni)	50%	0.017 (Ni)	12.1%**	4.4%**	

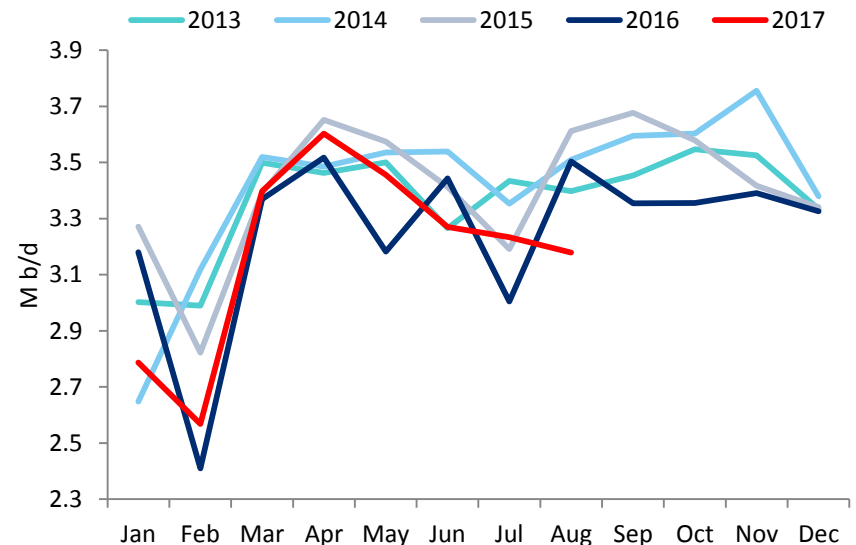
There could be two sets of direct impacts of the winter air-quality control measures on commodities markets: 1) upstream industries will likely be hurt more severely than downstream industries, i.e. commodities supply should generally come in tighter than demand; 2) flight to quality could occur across the space, and therefore price differentials of ores/raw materials with different qualities should widen. Given China's importance in the global commodities markets, the rest of the world should also cope with the aftermath of the Chinese policies by producing more high-quality raw materials and allowing relocation of some of China's polluting industries to other countries in the long run.

The impact of winter cuts on demand losses could be limited

- **Construction, transportation and power consumption could also take a hit from the winter environmental measures, but overall commodities supply cuts should outweigh demand cuts into the winter.** A few northern Chinese cities including Beijing banned ground construction work over the winter, and this could hit real estate and infrastructure buildouts. Yet, the actual impact on commodity demand should be limited, as winter is a typical low season for construction. So far only a few key cities have planned to suspend ground construction work and key infrastructure and housing projects for livelihood could be exempted from the cuts. Demand for copper and other commodities used in the late stage of housing construction may be less affected as internal wiring and decoration work is technically allowed.
- **Heavy polluting fabricators and processors, including zinc galvanizers, might be asked to shut down in the upcoming winter. Truck loading and transportation could be restricted for some industrial plants, ports and urban areas.**
- **Power consumption and demand for diesel should get hit as China is expected to curtail industrial activities across the space.** Production cuts in aluminium and other industrial sectors could reduce China's power demand growth to low single digits during the winter heating season. Curtailments in mining and truck transportation activities could create headwinds for diesel, the apparent demand of which has already fallen 150-k b/d so far this year.



Chinese diesel apparent demand

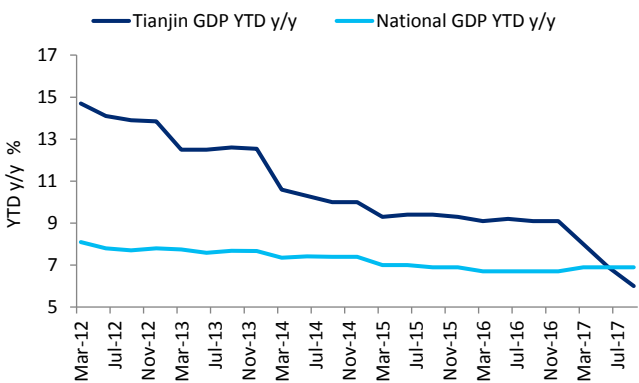


Unintended consequences of China's policies?

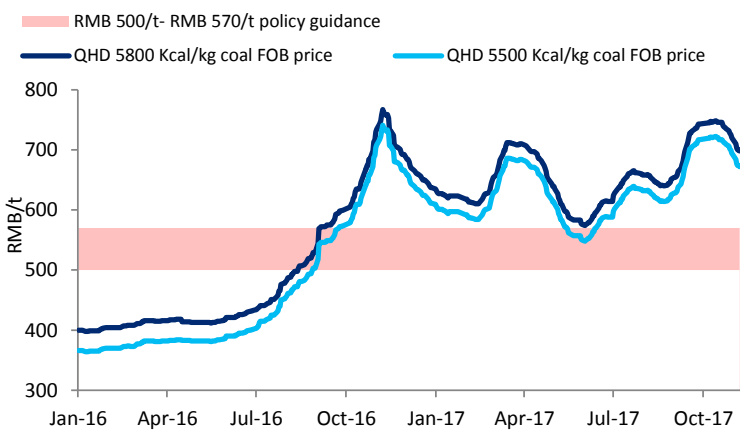
Supply side reforms and higher environmental standards involve trade-offs and risks, including lower growth and higher inflation and unemployment, and potential shortages, potential long term ex-China supply responses, and generally higher commodity price volatility (should policy mistakes occur).

- **Sustainability of the current winter supply cut measures may depend on the trade-offs between actual air quality improvement and a series of unintended consequences**, including a potential slowdown of economic growth, and higher inflation and unemployment. If the production cutback policies continue over the coming years, the market should expect weak winter industrial production across different sectors and more profound seasonality of Chinese economic activities.
- **Beyond the winter cuts, heightened policy interventions in the coal market resulted in price spikes, though the NDRC has been dedicated to keep prices in a much lower range.** The failure to control coal spot prices was due in part to the pursuit of other policy priorities including safety initiatives and environmental protection, which effectively curbed mines' run rates. We expect the NDRC will ultimately encourage enough supply to be brought online in coal.
- **Given that the winter cut policies will likely continue in the next few years, generally higher commodity price volatility could be expected.** This is partly because commodity consumers, in particular the construction sector and manufacturers with spare capacity, could generally be more flexible in adjusting their operations according to winter cut schedules. Yet a lack of spare capacity could prevent commodity producers from the same practice. Such a re-occurring mismatch between supply and demand could keep prices volatile.

The city of Tianjin, which experienced intense environmental measures, saw more rapid GDP slowdown than national average



Chinese spot thermal coal prices stayed above the government guidance for most of this year



Chinese policies could drive a supply response ex-China in some commodities

Supply side reforms and higher environmental standards have contributed to the recent rally in global metals and bulks prices. In some cases, prices are already at incentive price levels. For other commodities, we are not there yet and so a continuation of Chinese reforms could see further price ex-China price upside.

Should current spot prices be maintained, most commodities with the exception of iron ore and aluminium should see greenfield projects be funded

		China	Ex-China	Ex-China
		Incentive price	Incentive price	Incentive price
	Spot		Restarts/Brownfield	Greenfield
Thermal Coal	\$90-100/t			
Coking coal	\$190-200/t			
Zinc	\$3,200/t			
Alumina	\$480/t			
Steel (rebar)	\$460/t			
Aluminium	\$2,100/t			
Iron ore	\$62/t			
At incentive price				
At incentive price but restricitons on new capacity additions				
Not at incentive price (evidenced by either incentive price data or capex decisions)				

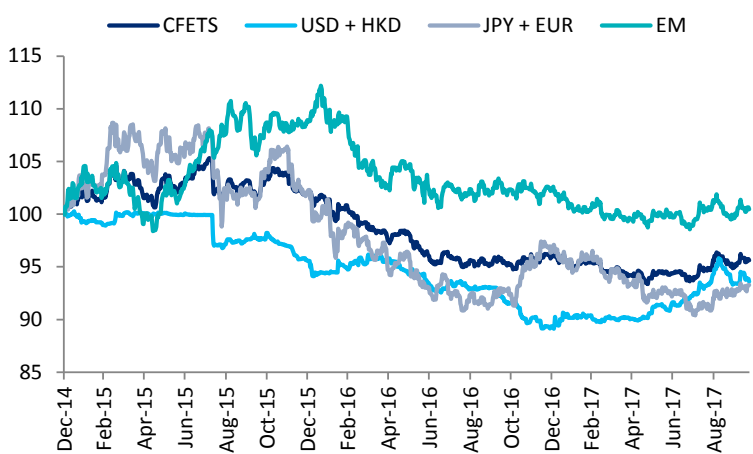
Prepared for Rong Lu (research email)

Depreciation pressure looms for RMB in 1H'2018

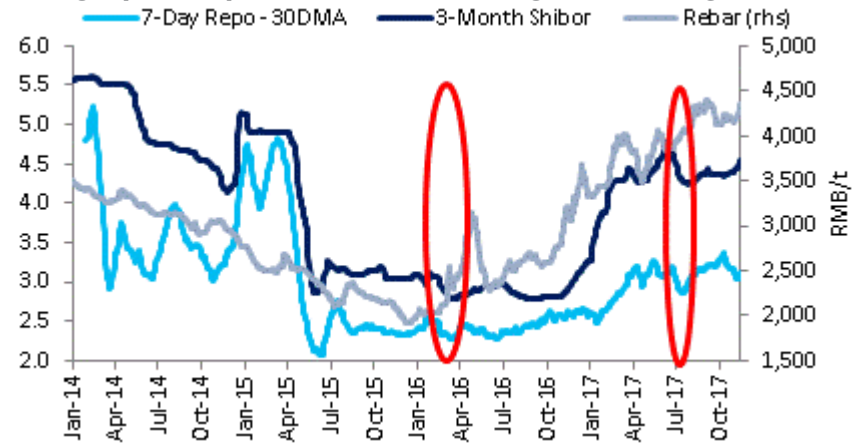
Citi expects two-way fluctuations of the RMB in 2018 with a depreciation bias in 1H'18 owing to the growth momentum softening in 2H'17 and 1H'18. Should growth get back on track and policy measures to attract capital inflow via financial sector liberalization start to take effect, RMB may appreciate gradually from 2H'18 onwards.

- **Citi's China economics team expects USDCNY to average 6.69 in 2018 and 6.29 in 2019 compared with 6.75 in 2017.** As discussed by multiple PBoC senior officials during the 19th Party Congress, widening CNY trading band is "not a current focus," and there remain possibility of "non-routine" intervention to the currency. Maintaining currency stability appears to be on PBOC's mind in its effort to reduce onshore FX risk reserves, after persistent appreciation of CNY since early this year.
- **RMB's appreciation against the USD had a positive impact in lifting prices of most metals and bulks,** though it might have been partially offset by the reduced need to import commodities as a storage of value when RMB was depreciating earlier.
- **Liquidity conditions could largely stay neutral for now, but a proposed targeted RRR cut is set to take effect in January 2018 to boost liquidity.** That said, we believe financial regulatory overhaul is a still half-finished project. Despite some forbearance that we have seen, we expect it to resume after the 19th Party Congress, which may put some pressures on market liquidity without the PBoC's offsetting injection.
- **The commodities market could continue to react to liquidity signals next year, with the market having likely priced in the proposed targeted RRR cut.** Some market participants also worry that the government might consider to "save bullets" of liquidity easing for future growth challenges.

CNY has remained stable in recent months



Commodities price hikes in 2017 have largely moved in tandem with Chinese rate hikes and deleveraging, though price spikes occurred alongside easing

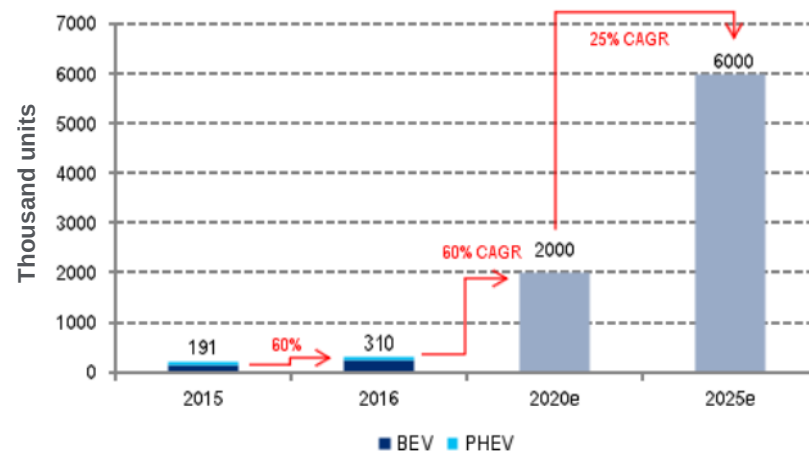


China is critical in driving EV output and battery metals demand

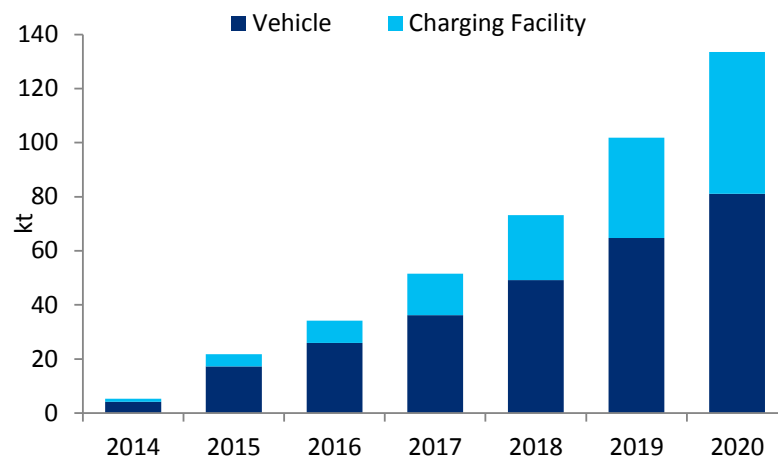
Battery demand will have a profound impact on metals such as lithium, cobalt, vanadium and nickel over the coming decade, principally driven by Electric Vehicles (EV). China being the world's largest passenger vehicle market is also the world's largest EV market. China will be essential in driving battery metal demand globally.

- **There is a huge capital pool in China that is being directed by the government to invest in potentially disruptive EV and EV battery development.** China's closed capital account, a domestic high savings rate and supportive policies mean that the probability of disruptive technological innovation in EV batteries in China is very high. China unveiled a new policy in September this year to encourage NEV production based on a credit-score system that will take effect in 2019. Automakers that sell above 30,000 cars annually must produce and sell enough NEVs that reach the credit threshold equivalent to 10% of the passenger vehicles annual sales and the level will rise to 12% for 2020.
- **China has seen incredible growth in electric vehicles since first implementing NEV (New Energy Vehicles) subsidies in 2009, with the market growing 5x between 2014 and 2016.** Our China metals team expects 2017E total NEV growth to slightly slow to 45% y/y, from 55% y/y in 2016. But China is in the right trajectory to deliver 30-40% y/y growth in NEV production volume and eventually reach annual production of 2.1 mn units in 2020,
- **Implications to metal demand are two-fold in terms of cathode materials as well as charging infrastructure.** We estimate industrial metals nickel and copper and to a lesser extent aluminium will have a key part to play as demand for cathode materials increases to service the growing share of EVs as a percentage of the total car fleet globally. Specifically, cumulative nickel demand from the EV battery sector may total ~150kt by 2020, however we note risks to the upside as the dominant battery chemistry in China remains the swing factor.

China's ambitious EV penetration growth targets



Chinese copper demand from EV



3. Energy: Short-Term bullishness can persist
But a more bearish end-2018 looms...

Prepared for Rong Lu (research email)

Energy Top Calls

- **Crude:** Bearish view medium-term, but neutral near-term, with view the balance of risks likely to the upside. Producer hedging and bullish prompt fundamentals sharply pushed up timespreads. With producer flow for 2018 likely to abate with hedge cover already quite high, crude becoming oversupplied in 2018, and OPEC and Russian crude coming back to the market by end-year and before consensus expectations then **selling Brent timespreads (Jun/Dec-18), spreads** can capture these themes.
- **Crude vol:** WTI longer-dated put skews have been trading in tandem with prices, as US shale producers adding to downside hedges whenever prices rally, steepening the put skew; e.g. Jun18 15-delta put skew rallied to ~5 vol pts (18%) of the ATM vol, the highest since 1Q'17. We expect WTI prices to sustain above the high-\$40s through 2018 and producers to continue covering their production. The hedges should be rolled into Cal19, with producer flows likely to abate the bid for downside vols on the cal18 strip, weighing on the put skew. We like **getting short Jun-18 put skew by buying put spreads, delta-hedged**.
- **Refined products:** Refinery capacity utilization is likely to be high and spare capacity is likely to be tight. Hence overall margins, especially in Europe (the world's marginal refining center), are likely going to stay high. Potentially getting additional upside would be to focus on 1H'18 given a deep turnaround schedule and clean product stocks likely start 2018 low as a result of 2017 dynamics. Given that there is little ability to flex up gasoline yields materially on a global basis (whereas there is for distillates) then combining all these themes means that our conviction call would be to **buy 1H'18 European gasoline cracks**.
- **Diesel-HSFO:** The 2020 IMO 0.5% sulfur cap on global marine bunker fuels is expected to lead to a late rush by shippers to source diesel and ULSFO, leaving a glut of HSFO on the market. Shippers have been slow to retrofit scrubbers on vessels enabling them to run HSFO compliantly. Diesel demand could jump by over 1-m b/d and HSFO demand could fall by over 1.5-m b/d. Refiners look constrained to be able to raise diesel and lower FO yields enough, while raising utilization would exacerbate the FO glut. We like **buying Dec-19 or Cal20 strip European gasoil-fuel oil**, currently at \$32-33/bbl (~\$280-290/mt) which could widen to \$60/bbl (~\$400/mt) or more.
- **Global natural gas:** Although global gas prices should climb this winter as multiple heating fuel markets confront tight fundamentals, global production should strengthen in 2018 and beyond as new supply comes online, lowering prices longer term. The long-anticipated strong LNG production growth in the next few years, first from Australia then the US, should quickly loosen up the LNG and European gas markets. Meanwhile, very modest inventory expected in 2018 and 2019 should keep US prices supported. Hence, our top call is to **sell NBP or TTF vs. US Henry Hub for summer 2019 (2Q and 3Q'19)**. The current economical arbitrage is ~\$1.25/MMBtu while the NBP-HH spread is currently trading at ~\$2.90/MMBtu.
- **US gas basis:** Waha basis (gas prices in the Permian) could widen substantially as oil and associated gas output grows. Unlike oil/NGLs, which can be shipped by pipes truck or rail, gas can be shipped only on pipes. The congestion expected to develop could **widen 2019 basis differentials to \$1 below Henry Hub from ~\$0.6 now**. These levels are needed to justify new pipe and sending more gas to the Midwest to compete against low-cost Marcellus/Utica and Western Canadian gas

2018 is likely to be a transition year for oil; OPEC timing is key

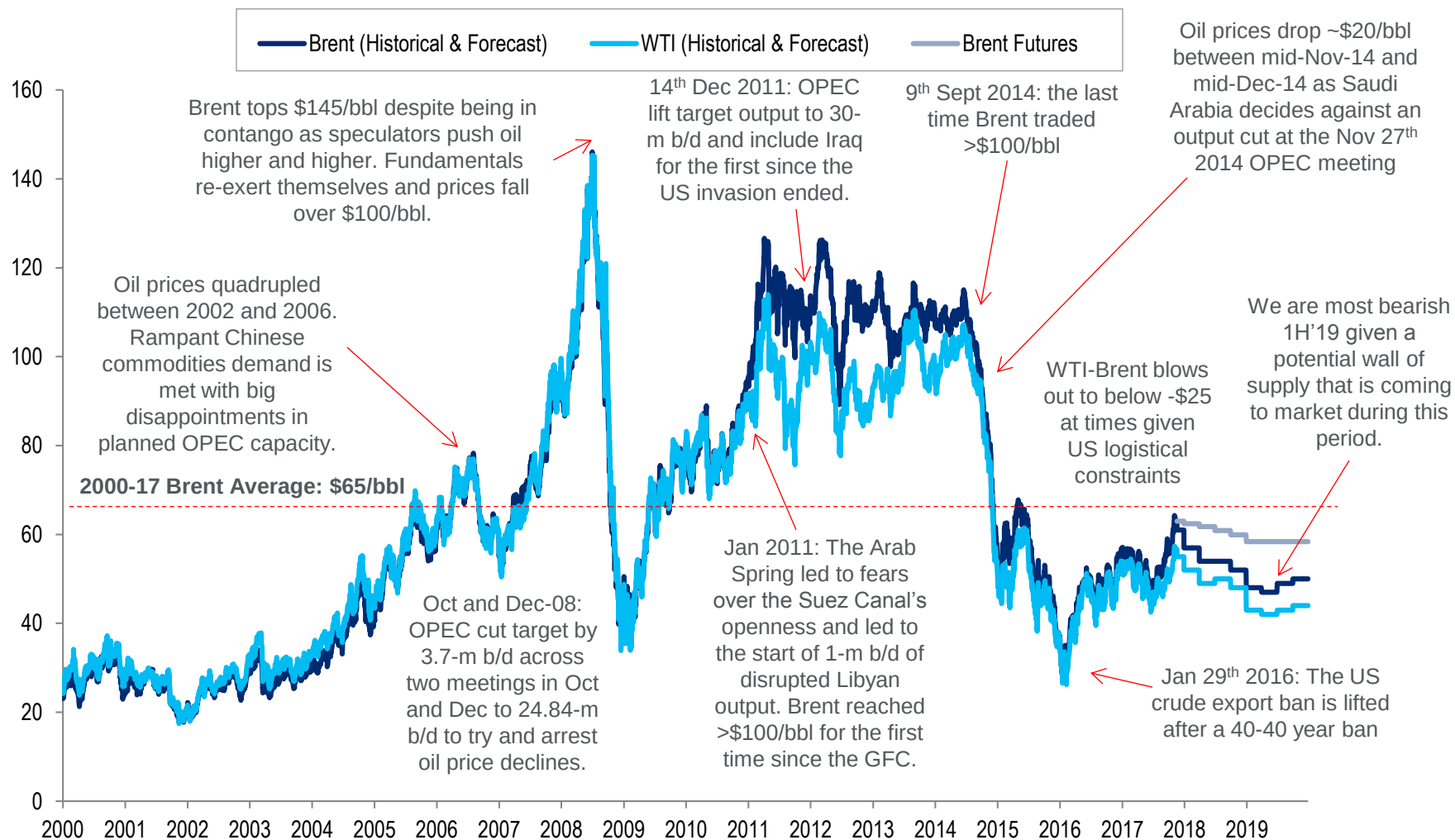
Base Case Price outlook (\$/bbl)	1Q17	2Q17	3Q17	4Q17E	1Q18E	2Q18E	3Q18E	4Q18E	1Q19E	2Q19E	3Q19E	4Q19E	2017E	2018E	2019E	2020E
Brent	55	51	52	61	57	54	54	52	48	47	49	50	55	54	49	56
WTI	52	48	48	55	52	49	50	48	43	42	43	44	51	50	43	48
Brent-WTI	3	3	4	6	5	5	4	4	5	5	6	6	4	5	6	8

Base Case Price outlook (\$/bbl)	1Q17	2Q17	3Q17	4Q17E	1Q18E	2Q18E	3Q18E	4Q18E	1Q19E	2Q19E	3Q19E	4Q19E	2017E	2018E	2019E	2020E
Brent	-	-	-	3	2	2	-1	-1	-	-	-	-	1	-	-	-
WTI	-	-	-	1	1	1	-1	-1	-2	-2	-2	-2	-	-	-2	-4
Brent-WTI	-	-	-	2	1	1	-	-	1	1	2	2	-	1	2	4

- **Our annual oil price expectations are essentially unchanged, but tweaked a bit for end-2017 and through 2018.** The bullish supply/demand dynamic that we have been talking about over the last 1-2 years looks like it is running out of steam. For YE2017 and 1H'18, there is however, likely to remain a constructive backdrop of strong oil demand growth, OPEC supply restraint, and heightened oil supply disruption risk, that non-OPEC supply growth is unlikely to be able to match immediately. To this end, we are pushing our year-end 2017 and 1H'18 oil price expectations up by \$2-3/bbl.
- **Oil prices are expected to end the year somewhat lower than the current spot Brent price of ~\$64/bbl.** As the market had overwhelmingly expected, OPEC and Russia extended their production cuts the full 9 months beyond 1Q'18 when producers met on Nov 30th. But hanging over the market are unknowns about what OPEC might consider when it meets again in mid-year, the level of Iraq's production, and how much production will be coming out of non-OPEC producers, especially the US. In addition, there are questions about the sustainability of the net length in today's market and how much geopolitical risk now priced into oil will sell-off, especially in January when an upsurge in drilling in the US as well as a rise in crude oil inventories could spook markets.
- **The oil market should move from deficit in 2017 to balanced in 2018.** Oil demand continues to be buoyed by improving global economic growth and oil-specific macro indicators (e.g. manufacturing PMIs and trade growth). Supply should grow more robustly, with the US leading the way with an expected ~1.6-m b/d of y/y growth (incl. NGLs) and possibly 400-k b/d more, given hedges. Brazil and Canada likely add ~0.5-m b/d, leaving only a small amount of room for OPEC and Russia to fill, making the timing and pace of their return crucial, and for now we expect a phased return in 2H'18, perhaps only during 4Q'18, hence our bearish YE2018 Brent forecast of ~\$52/bbl. Further clouding the picture are probably strong refinery margins even as crude stocks build due to tight refining capacity. This would be bearish curve structure with an unclear impact on flat price.
- **We remain bearish 2019 with a forecast of ~\$49/bbl for Brent as OPEC, US, Canada, Brazil and Russia look to add material volumes of supply (likely >2-m b/d) which would likely be at a faster growth rate than demand.** This returns the oil market to a sizeable surplus, and hence a lower oil price would be needed to rein in US crude production in following years.

A Brief Recent History of the Oil Price

Prepared for Rong Lu (research email)



Source: Bloomberg, Citi Research estimates

Oil balances shift from deficit to surplus by 2019*

Citi Global Oil Supply/Demand Balance (m b/d)

Demand	1Q17	2Q17	3Q17	4Q17	1Q18	2Q18	3Q18	4Q18	2017	2018	2019	17'	18'	19'
OECD Demand	46.9	47.0	47.4	47.6	47.0	46.8	47.7	47.7	47.2	47.3	47.3	0.30	0.07	0.05
Non-OECD Demand	49.7	50.5	50.9	50.7	51.0	51.8	52.3	52.5	50.5	51.9	53.3	1.26	1.43	1.44
Total Demand	96.6	97.5	98.3	98.3	98.0	98.6	100.0	100.2	97.7	99.2	100.7	1.56	1.51	1.49
Supply	1Q17	2Q17	3Q17	4Q17	1Q18	2Q18	3Q18	4Q18	2017	2018	2019	17'	18'	19'
US	9.0	9.1	9.2	9.4	9.8	10.1	10.3	10.6	9.2	10.2	11.0	0.34	1.01	0.80
Canada	4.0	3.6	3.8	4.0	4.2	3.9	4.0	4.2	3.8	4.1	4.3	0.23	0.25	0.23
Mexico	2.0	2.0	1.9	1.9	1.8	1.8	1.8	1.8	1.9	1.8	1.7	-0.21	-0.14	-0.08
Brazil	2.6	2.6	2.6	2.8	2.8	2.8	3.0	3.0	2.7	2.9	3.2	0.15	0.22	0.34
Norway	1.7	1.7	1.6	1.7	1.7	1.7	1.6	1.7	1.7	1.7	1.6	0.01	0.02	-0.04
UK	0.9	0.9	0.9	1.0	1.0	1.0	1.0	1.1	0.9	1.0	1.0	0.01	0.10	-0.08
Russia	11.1	11.0	10.9	11.0	11.0	11.0	11.1	11.3	11.0	11.1	11.4	0.03	0.09	0.34
China	3.9	3.9	3.8	3.8	3.8	3.8	3.7	3.7	3.8	3.7	3.7	-0.16	-0.11	-0.09
Kazakhstan	1.8	1.8	1.8	1.8	1.9	1.9	1.9	1.9	1.8	1.9	2.0	0.16	0.10	0.06
Non-OPEC Crude + NGLS	53.1	52.6	52.5	53.4	54.3	54.1	54.5	55.4	52.9	54.6	56.1	0.48	1.70	1.49
Iraq	4.6	4.6	4.6	4.4	4.5	4.5	4.6	4.6	4.5	4.6	4.6	0.09	0.03	0.04
Iran	3.8	3.7	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8	0.22	0.02	0.00
Kuwait	2.7	2.7	2.7	2.7	2.8	2.8	2.8	3.0	2.7	2.8	3.1	-0.17	0.12	0.21
Saudi	9.9	10.0	10.0	9.9	9.9	9.9	10.4	10.2	9.9	10.1	10.4	-0.48	0.17	0.24
U.A.E	3.0	2.9	2.9	2.9	3.0	3.0	3.0	3.1	2.9	3.0	3.2	-0.11	0.05	0.21
Venezuela	2.1	2.0	2.0	1.9	1.8	1.7	1.6	1.6	2.0	1.7	1.4	-0.26	-0.30	-0.27
OPEC Crude	32.2	32.3	32.7	32.4	32.4	32.4	33.0	33.0	32.4	32.7	33.1	-0.44	0.29	0.39
OPEC Condensate & NGLs	6.7	6.7	6.7	6.7	6.8	6.8	6.8	6.8	6.7	6.8	6.9	0.16	0.12	0.09
Total Supply	96.8	97.0	97.8	98.1	98.6	98.9	100.3	100.9	97.4	99.7	101.9	0.30	2.28	2.22
Inventory adjustments	-1.2	-1.3	-1.1	-0.8	-1.2	-1.1	-1.0	-0.7	-1.1	-1.0	-1.0	-	-	-
Error Term/Seasonality Adjustment	2.0	1.2	0.4	0.2	1.4	0.1	0.3	0.2	1.0	0.5	0.4	-	-	-
Total Stock Change	0.9	-0.6	-1.2	-0.9	0.8	-0.6	-0.4	0.2	-0.4	0.0	0.6	-	-	-
Call on US Crude	8.1	9.7	10.4	10.3	9.0	10.7	10.6	10.4	9.6	10.2	10.4	-	-	-
Call on OPEC crude	31.2	32.9	34.0	33.3	31.6	33.0	33.4	32.8	32.8	32.7	32.5	-	-	-

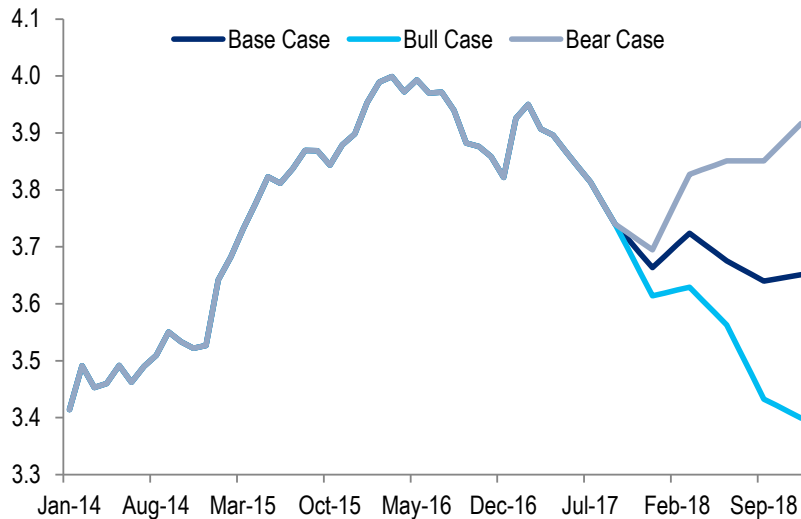
Source: IEA, FGE, JODI, PPAC, Chinese Customs, PEMEX, NEB, EIA, ANP, NPD, OPEC, Petrologistics, ICIS, Lloyds List, Citi Research estimates; *Additional hedging by US producers could add some 300- to 400-k b/d to US production growth in 2018.

S/D uncertainty remains high; 2018 oil range could be >\$30/bbl

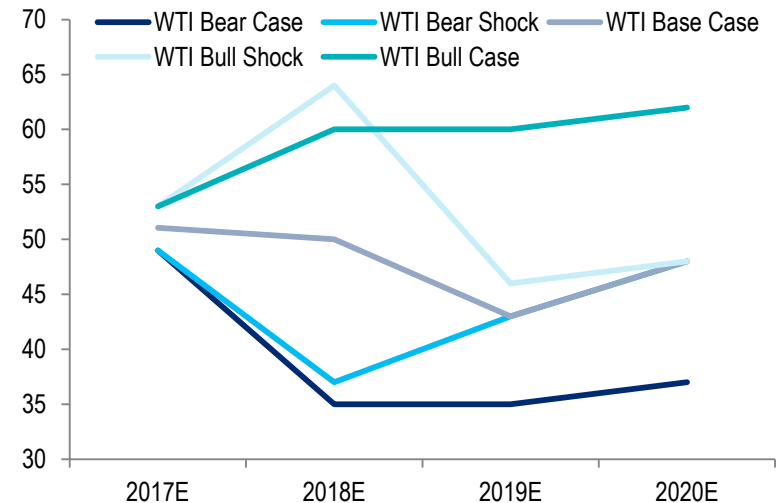
• Our oil price and inventory scenarios are:

- **The bull case** is driven by faster oil demand growth, delays to OPEC expansion plans, additional geopolitical supply disruptions, and higher shale cost inflation (but eventually leading to stronger US crude production given higher prices).
- **The bear case** is driven by upside to non-OPEC supply growth, slower oil demand growth, some recovery of geopolitical supply disruptions, but most likely from more robust and earlier OPEC supply expansions.
- **The bull shock case** shows our estimate of the price impact of a sustained 1-m b/d geopolitical supply loss (LatAM, Libya/Nigeria).
- **The bear shock case** for a sustained 0.5-m b/d return of currently disrupted supply.

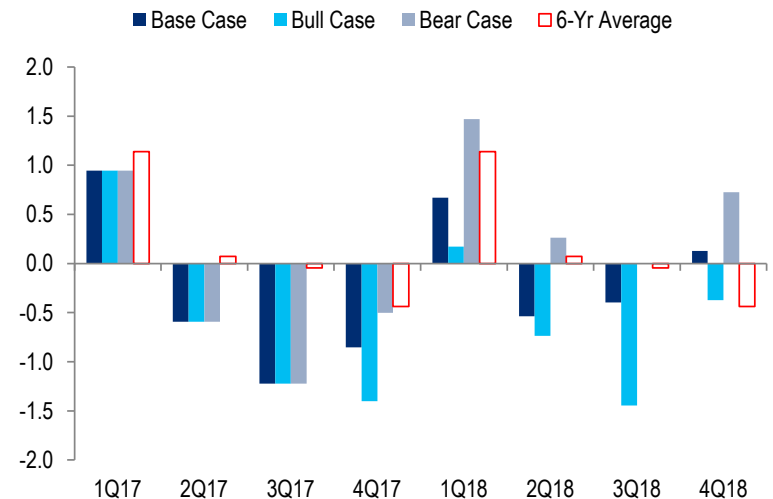
Scenario Analysis for Global Oil Stocks (bln bbls)



WTI Price Scenarios (\$/bbl)



Scenario Analysis for Global Oil Stock Q/Q Changes (m b/d)

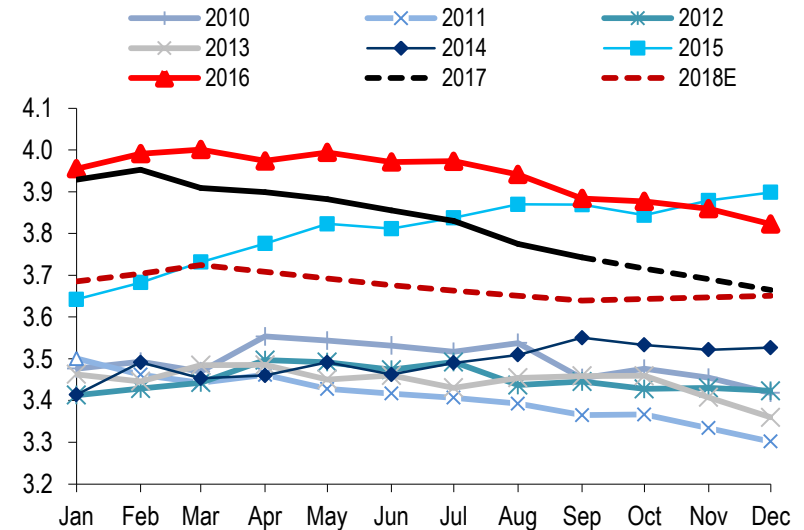


Source: IEA, EIA, METI, KNOC, Euroil, Xinhua, SCI99, Lloyds, JODI, Bloomberg, Citi Research estimates

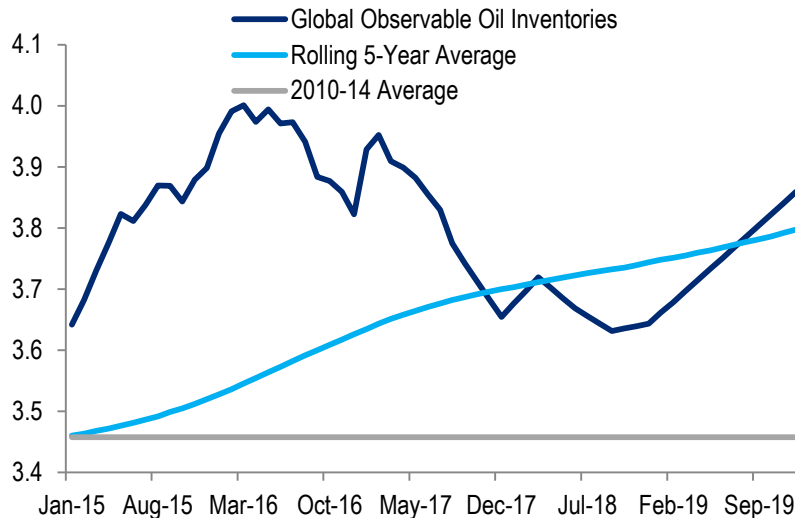
Rapid stock draws have supported prices but builds are coming

- **Global oil stocks are expected to reach their 5YA in absolute level and days of cover terms during 1H'18.** This has proven to be the most bullish catalyst for oil prices in 2017 and despite a very low chance that oil inventories will fall below the static 5YA level of 2010-14, current levels of oil in tank point to oil prices at >\$60/bbl, based on simple recent historical relationships...
- **However, higher prices and lower inventories now are self correcting; by end-2018 we expect the oil market to return to surplus and inventories to start rising again.** Our current market imbalance is around +0.6-m b/d in 2019 which would push oil inventories back above their rolling 5YA. This is clearly a negative for oil pricing and our simple oil inventory vs. oil price model indicates that this level of inventory would warrant a Brent crude price of below \$50/bbl.

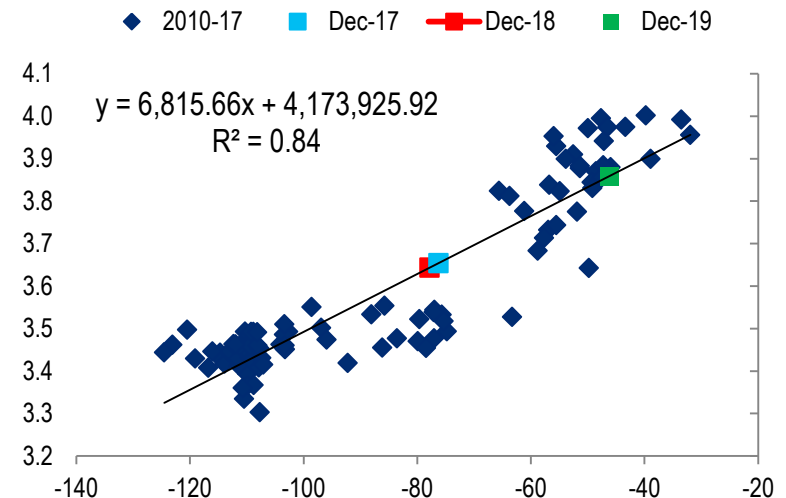
Global Oil Stocks (bln bbls)



Global Oil Stocks vs. 5-Year Averages (bln bbls)



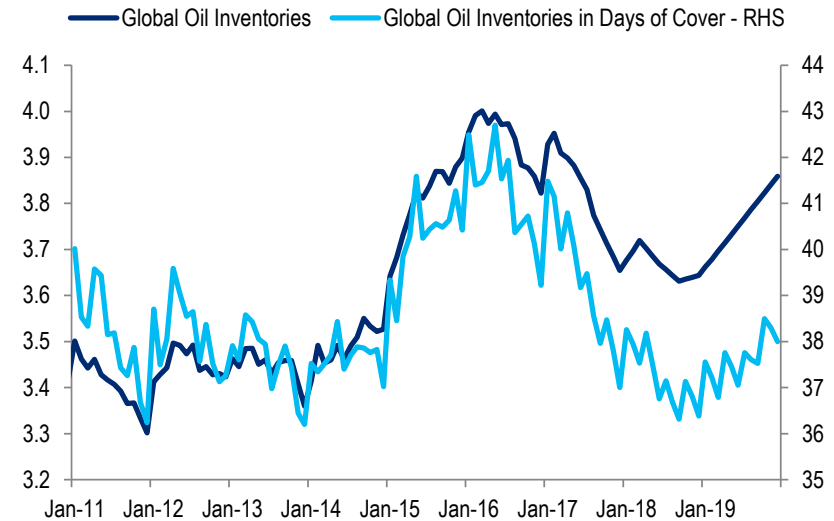
Global Oil Stocks (bln bbls) vs. Inverse Brent (\$/bbl)



OPEC's targeting of five-year absolute stocks is misguided

- OPEC's keen focus on targeting 5YA average absolute stocks levels seems wrong to us for many reasons:
 - **Rolling 5YA levels will naturally mean revert** without action.
 - **Demand cover is a more appropriate metric** as the purpose of holding stocks is to cover demand excess or supply shortfalls.
 - **Absolute US crude stock surplus measurements may be misleading** given the changing dynamics of the US crude network. The on-shoring of US crude imports has reduced oil-at-sea heading to the US which was de facto storage. More crude is also needed in stocks now to cover growing production, exports and crude runs. The chart below shows the “actual” surplus by end-3Q'17 may have been closer to ~25-m bbls compared to 2012. And this could well be too high still given that Cushing likely now holds material volumes of blended crude (DSW type) that is less desirable to domestic refiners.

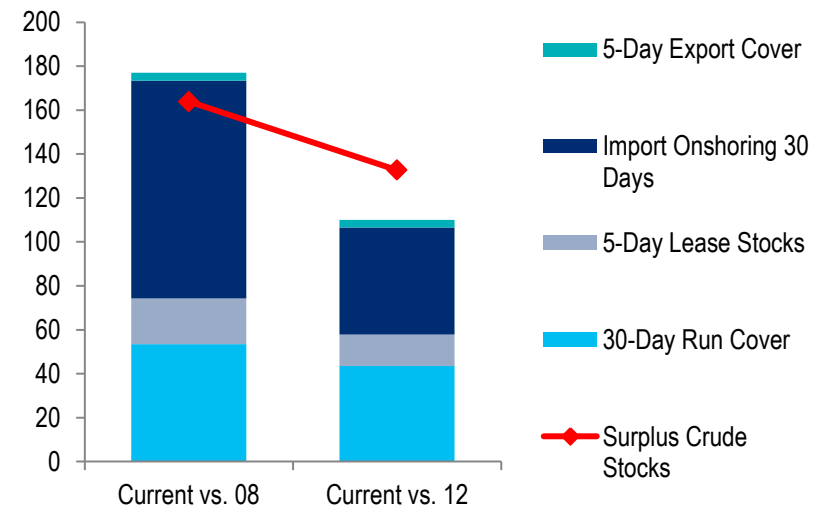
Global Oil Stocks (bln bbls, Days of Cover)



Cushing Crude Stocks (bln bbls)



Surplus US Crude Stocks (mln bbls)



OPEC + Russia's return to market likely defines 2018/2019 prices

OPEC and Russian Crude Production (m b/d)



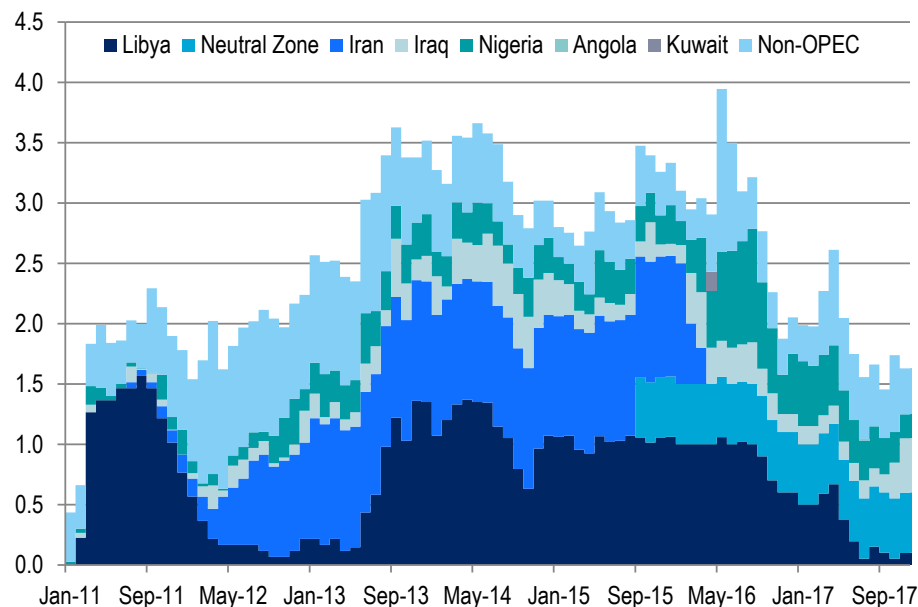
US shale production, non-OPEC production and oil demand are going to grow next year—the question is just by how much. With US producer hedging, momentum from already sanctioned greenfield projects and longer-term demand trends, a good portion of 2018 growth rates are “locked in”, though to varying degrees. The most uncertain aspect of late 2018 balances, for both physical and financial flows, is when will OPEC and Russia return to the oil market, by how much and how quickly.

By when? Citi stands out in its expectation of OPEC's return to the market as we expect the current deal to last until mid-year or end-3Q'18, but not end-2018, given wiggle room in their November 30th agreement. The drive by Russia to see the current deal through all of 2018 seems lacking. At the same time, the likelihood that Saudi Arabia would go it alone is low and as such offline OPEC and Russian barrels are likely to be phased back in by YE'18 when producers meet again in June 2018.

By how much? It is likely that non-GCC OPEC output will not be able to increase much, with Iraq the exception if they fix the government take of their upstream contracts. This leaves the GCC and Russia as the key producers to watch, and while output has been restricted, capacity has been expanded by Russia, Kuwait and the UAE. There is now likely ~1-m b/d of spare capacity in those three countries, and Russia could be back at its max capacity of 11.4-11.5-m b/d by early 2019. Kuwait and the UAE may show some restraint, in line with Saudi Arabia that is unlikely to rush back to the market if IPO deliberations are still ongoing. But it is going to be a difficult balancing act between supporting the oil price, and regaining market share.

Weaker OPEC states could pose 2-m b/d of supply disruption risk

Global Oil Supply Disruptions (m b/d)



Global oil supply disruptions reached a five year low of ~1.5-m b/d in September 2017 but the heightened level of geopolitical risk in the MENA region leads us to believe that this does not last and the level of disrupted oil is likely to rise, perhaps materially. Weaker OPEC countries have recently seen disruption risks increase,

Libya: The GNA's mandate expires on Dec 17th and if no political breakthrough is made then the risk of conflict likely escalates as General Haftar's LNA could seek to take control which could put Western Libyan oil production at risk (300-400-k b/d).

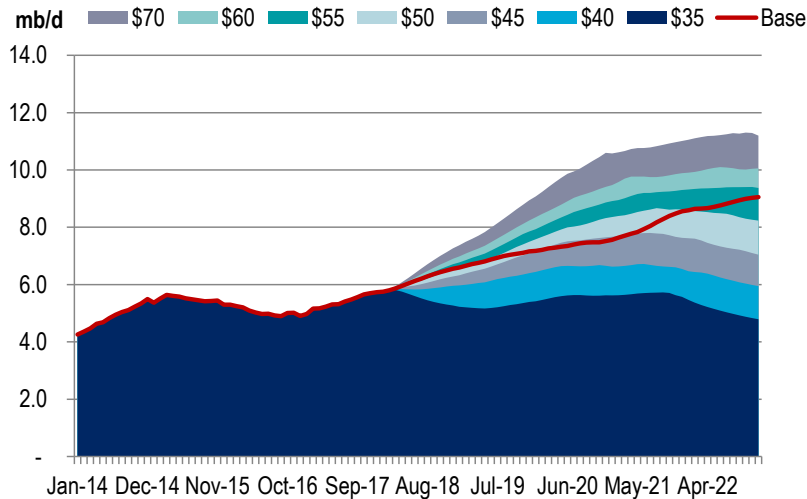
Iran: President's Trump's decertification of the Iran deal likely means less FDI in Iran and could elicit a response from the Iranian regime. There is also some probability, though low, that President Trump does not sign the sanctions waiver on Jan 13th 2018.

Iraq: Kurdish crude flows have been cut in half (300-k b/d) but could stop entirely if Turkey or the KRG cut flows.

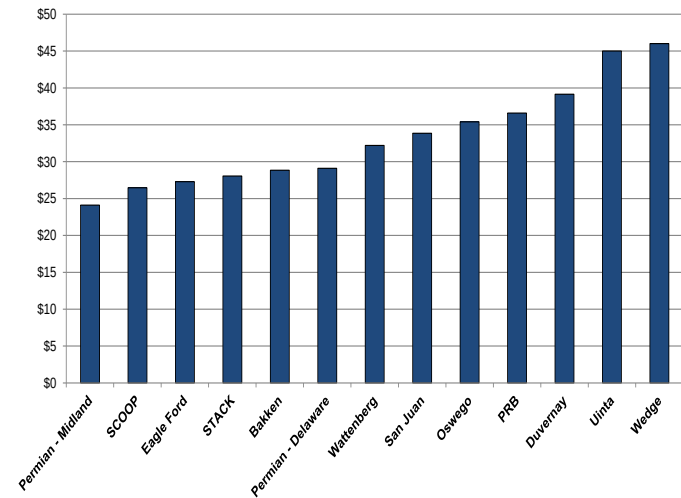
Nigeria: The Niger Delta Avengers have recently reneged on the ceasefire and Presidential elections (Feb-19) often cause increased disruptions.

Robust shale output expected in 2018, buttressed by >\$50 WTI

US Shale Production Forecasts



Citi E&P sector oil price breakeven analysis* (\$/bbl)



We expect momentum in the shale recovery to continue in earnest next year, with an expected ~830-k b/d of growth Dec'18/Dec'17, only slightly lower than the expected 850-k b/d of growth Dec'17/Dec'16. Half-cycle breakeven costs remain very low, with Citi's E&P equity team putting the average breakeven cost for their coverage group below \$50/bbl in all basins. Thus, with the recent move in deferred WTI to above \$55/bbl giving producers an opportunity to lock in profits for next year, there seems little reason to expect a pull-back from the current growth trajectory. We expect rig counts, which had begun to decline slightly since mid-summer, to resume an upward trajectory this month as stronger deferred WTI pricing likely holds, for now. Already US oil rig counts are up 9 rigs from their "recent bottom" of 729 rigs in early November.

Several factors which were supposed to hinder growth seem likely to overstated.

- Projections for our US E&P coverage group indicate that 2018 cash flows should be able to cover additional CAPEX spend as well as return shareholder capital through dividends and/or potential share buybacks.
- Congestion issues and cost inflation clearly pose a concern, but also tend to be relieved over time. For example, worries over tightness in the sand market have given way to substantial new in-basin mining capacity in the Permian.
- Expectations that increased capital restraint going forward could hurt drilling activity are probably overstated. With the recent acreage-grab coming to an end, E&Ps have more cash-flows to return to shareholders without jeopardizing production.
- Well productivity is not falling as some measures would suggest, and technological improvements continue.

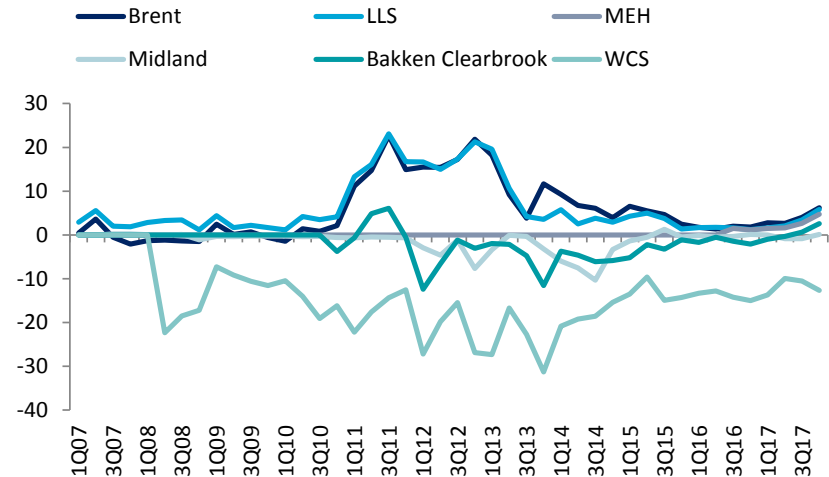
Brent-WTI diff reflecting US pipe/export infrastructure challenges

- Since the “shale revolution,” the Brent-WTI diff has reflected the fitful dynamics of US crude production growth, and the need for pipeline and export infrastructure to evacuate it at low cost, centered on Cushing, OK. Crude-by-rail has filled in the gap, added destination optionality, access to East/West Coast refiners.
- The recent widening of Brent-WTI spreads from <\$3 to >\$6 since late August echoes the infrastructure problems in 2011-13, when the differential blew out to \$15-30.
- While storm impacts exacerbated the problem, shutting in US refineries for a prolonged period, maxed out pipelines out of the Permian Basin and Cushing pointing to the role of outflow constraints. Ultimately, low refinery runs and limited ability to reach the Gulf Coast led to bloated inventories at Cushing. Crude quality concerns over Domestic Sweet Blend may have played a part too.

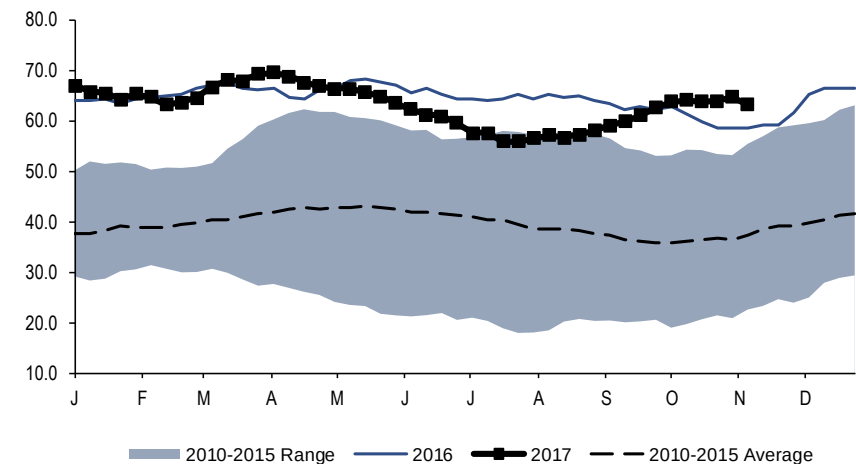
Prepared for Rong Lu (research email)

- Main sources of supply growth are from Canada, Permian Basin in west Texas, Bakken in ND, Niobrara in Rockies region, Eagle Ford in south Texas.
- After serving local refinery demand, crude oil needs to reach the Gulf Coast or East Coast for export.
- Given the pipeline system, the following price points are important: WTI Cushing, WTI Midland (Permian Basin), Magellan East Houston (MEH), Louisiana Light Sweet (LLS, St James, LA), West Canadian Select (WCS, Canada), and others.

Selected NAM crude diffs vs. WTI Cushing
(\$/bbl, 2007-2017, quarterly averages)



Crude oil inventories at Cushing, OK
(m bbls, 2010-15 range, 2010-15 average, 2016, 2017)



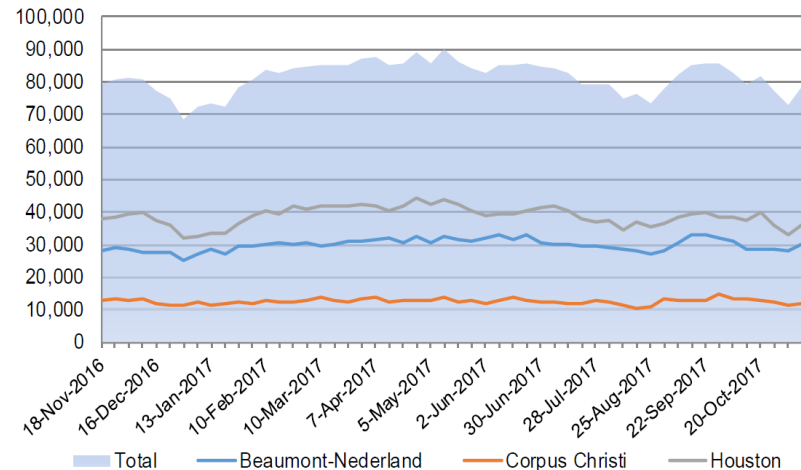
Takeaway constraints in the Cushing/Permian/Houston triangle

- **Cushing** receives growing supply from Canada, Bakken Niobrara, Oklahoma, with more inflow pipes over time; outflow pipe capacity surged but plateaued end-2014 with Seaway Twin; outflows are close to maxed out to USGC (hence wide MEH-WTI). **Cushing is pricing even below Midland too** to discourage barrels flowing into Cushing. Weak WTI also incentivizes Canadian, Bakken barrels to bypass Cushing, railed to the coasts.
 - However, the new **200-k b/d Diamond pipeline**, with line fill this past November and expected start-up in December, should add ~700-k bbls per week of additional outflows, and helps narrow MEH-WTI, move Cushing to a premium to Midland again.

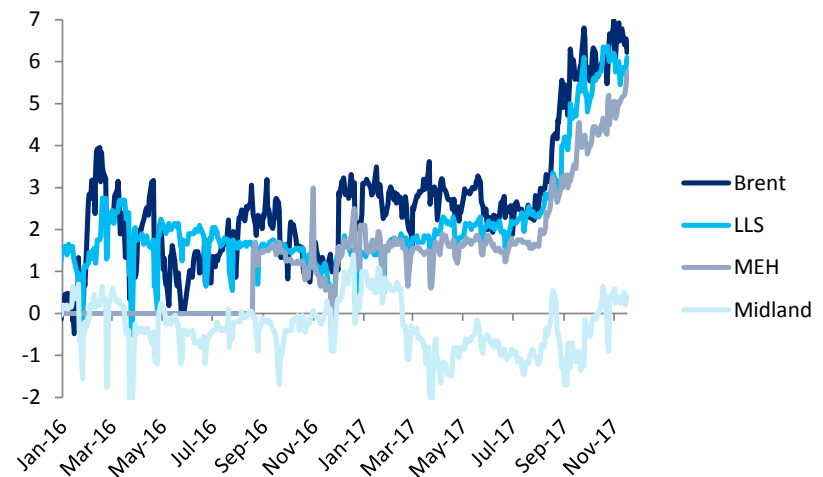
Midland sees strong Permian supply growth, needs to discount to send marginal barrels to Cushing and Houston by pipe; Midland crude stocks have risen and stayed high as pipe capacity to USGC looks close to maxed out (hence wide MEH-Midland). The **450-k b/d Midland-Sealy pipeline** expected 1H'18, should narrow MEH-Midland, and strengthen Midland vs. Cushing, reducing Basin flows into Cushing – perhaps sharply, but only for a short period.

- **Houston needs to grow export capacity over time;** after serving local refineries, some barrels move to St James, LA via the Zydeco (formerly “Ho-Ho”) pipeline, but the growing remainder needs to be exported. LLS, on the eastern USGC, has stayed fairly close to Brent, save for a Harvey disruption, though MEH has priced below it, pointing to export constraints too. Nevertheless, the US made record export volumes recently, over 2-m b/d.

Gulf Coast crude oil storage
(k bbls)



Selected NAM crude diffs vs. WTI Cushing
(\$/bbl, 2016-17, daily)

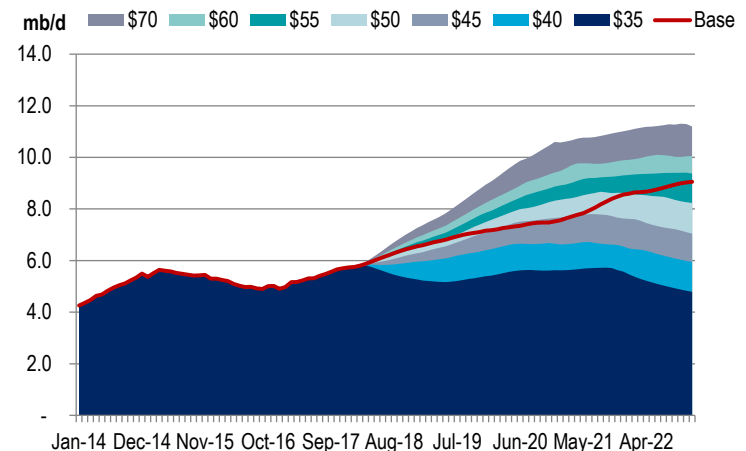


Pipe/export capacity narrows the diff, but more needed in 2019

- **Cushing stocks are already drawing down again**
 - 200-k b/d Diamond pipeline started line fill in November
 - Refinery runs rising after fall maintenance
 - 1Q'18 could see 450-k b/d Midland-Sealy pipeline
 - But after that, Cushing/Patoka needs more outflow capacity (e.g. reversal of 1.2-m b/d Capline, 1H'22?), and more Permian-to-USGC capacity (440-k b/d TexStar/Castleton/Ironwood pipeline in 1Q'19?). This does not look timely enough, so WTI could fall to wide discounts to MEH, LLS, Brent again in 2019...
- **Canadian supply grows over 0.2-m b/d y/y, some +0.7-m b/d from end-2017 to end-2020**, adding to rail flows into the US (at 130-k b/d 2017 YTD) with WCS weaker vs. WTI and USGC grades, before gaining with +370-k b/d Line 3 in 2019/20, +830-k b/d Keystone XL in 2020/21...
 - ...but the TransMountain pipe (2020?) should add +540-k b/d capacity to Pacific, diverting rail and pipe flows from the US, could narrow Brent-WTI and Maya-WCS.
- **Meanwhile, can the US Gulf Coast export enough?**
 - According to FGE, Texas has export capacity of ~2.2-m b/d now: 0.8-m b/d in Houston, 1-m b/d at Corpus Christi, 0.4-m b/d in Nederland. 2018 could see +0.75-m b/d to ~3-m b/d. 2019 could see +1-m b/d to ~4-m b/d. However, this alone may not be enough to evacuate US supply. LOOP could start sending VLCCs (+2-m b/d capacity by 2020?) but looks underutilized without a Capline reversal—and without Capline+LOOP, midcon pipes/storage tanks could become a bottleneck again.

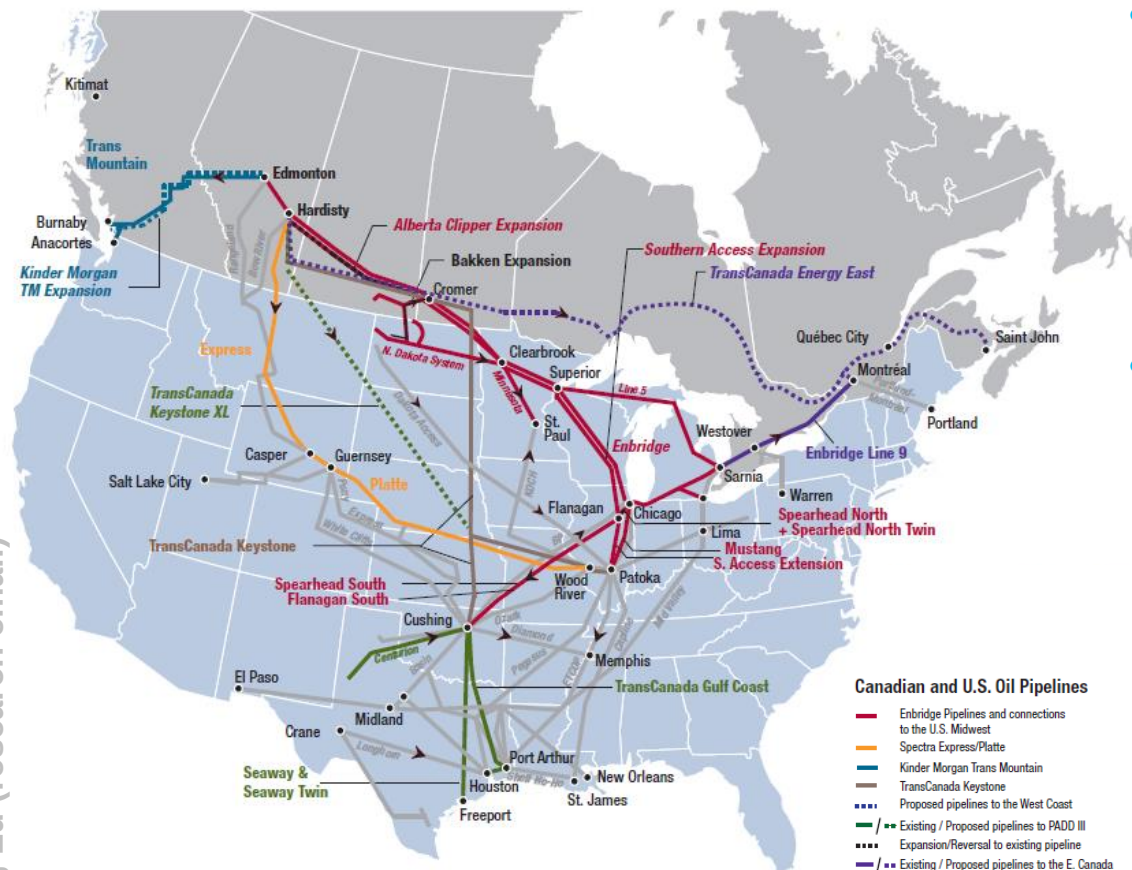
- **How much US oil does the world want?** Global balances have limited room in 2019/20 for shale to grow without stock builds. If global balances become very weighty, Brent-MEH could (more than) close, not just to restrain shale, but also to store surplus barrels in US storage tanks, reminiscent of early 2016.
- **The 2-3-m b/d of US+Canadian supply growth to 2020 could be covered by the +3-4-m b/d of additional export capacity** on the USGC (though incl. LOOP), and then Canadian flows into the US could drop by up to 0.5-m b/d after 2020, diverted to the Pacific coast of Canada for export.
- **But Cushing, Patoka, Permian takeaway capacity to the USGC still look challenged.** Brent-MEH could go to parity in 2019, but MEH-WTI could widen again at the same time, without additional pipelines built to evacuate crude from the midcontinent.

US shale oil production forecasts at various WTI prices
(m b/d, 2014-2022E)



Brent-WTI can widen 2019 onwards, without new midcon pipes

Prepared for Rong Lu (research email)



- Canadian crude: WCS stays weak as oil sands growth in 2018-19 calls for more crude-by-rail**, before falling to minimal levels as Line 3 (+370-k b/d), Keystone XL (830-k b/d), and TransMountain (540-k b/d) come online—there could be 1-m b/d of empty Canadian takeaway pipes in 2020/21, boosting WCS, though the risk is that this is offset by the IMO sulfur rule hitting sour.
- WTI Cushing and Midland: Permian output is hitting constraints** ahead of 450-k b/d Midland-Sealy coming on, but when it does (1H'18?) Midland/Cushing rally vs. USGC grades. But by 2019, Permian growth challenges pipes again. In Cushing, outflows are maxed out, but just adding 200-k b/d outflow capacity (Diamond). Midland-Sealy diverts barrels away from Cushing, reducing Basin flows (currently >0.3-m b/d) for ~0.5-m b/d tightening of Cushing balances. 2019 might see this reverse with Permian growth of 0.6-m b/d in 2018, 0.7-m b/d in 2019. Line 3 (2019) adds 370-k b/d of Canadian oil to the midcon, thus glutted by >0.1-m b/d in 2018, >1-m b/d in 2019, without a Capline reversal (currently planned for 2022). It might take another Brent-WTI diff blowout after Midland-Sealy before midcon pipeline build-out accelerates again. TransMountain (2020) could lessen the US glut by ~0.5-m b/d, but this is not enough alone.

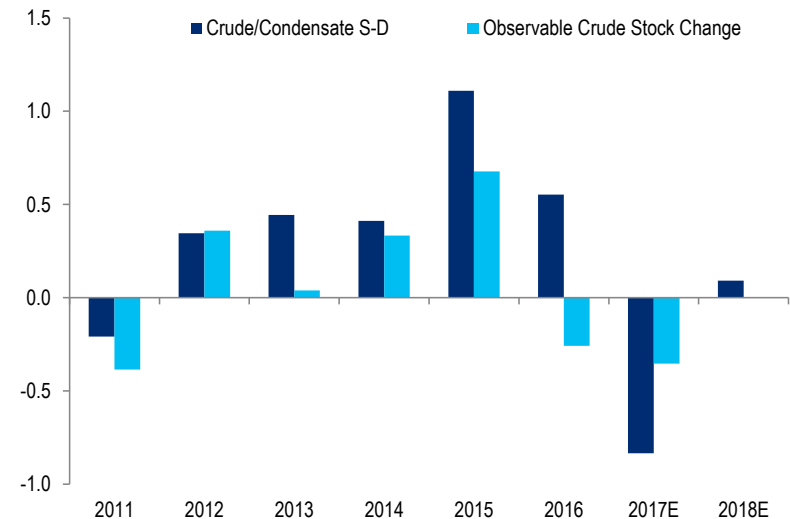
	Brent	LLS	MEH	WTI	Midland	WCS	Brent-ME	MEH-WTI	Brent-WTI	WTI-WCS
1Q17	55	54	54	52	52	39	1	1	3	14
2Q17	51	50	50	48	48	38	1	2	3	10
3Q17	52	51	51	48	47	38	1	3	4	10
4Q17	61	61	60	55	55	47	1	5	6	8
1Q18	57	57	56	52	52	40	1	4	5	12
2Q18	54	54	53	49	50	36	1	4	5	13
3Q18	54	54	53	50	51	37	1	3	4	13
4Q18	52	52	51	48	48	34	1	3	4	14
2019	49	48	49	43	42	30	0	6	6	13
2020	56	56	55	48	46	39	1	7	8	9

Source: CAPP, Citi Research estimates * The formerly proposed Energy East line has been canceled.

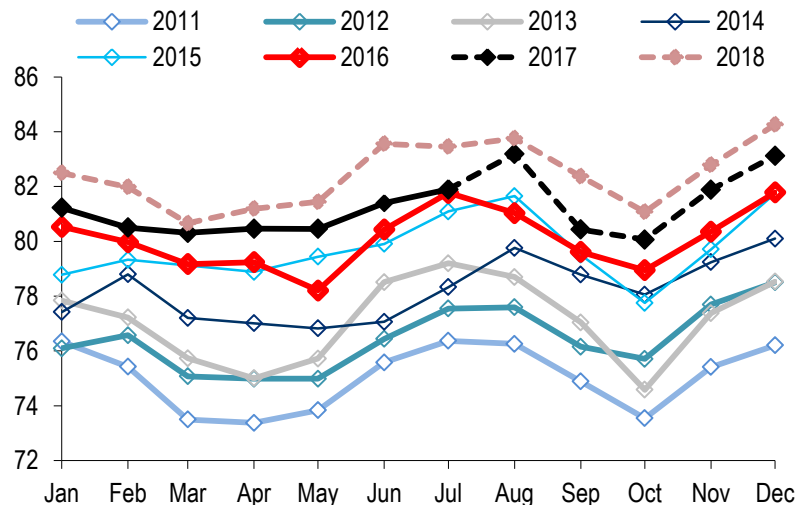
Crude dynamics may lean more bearish than balances suggest

- Crude specific dynamics do lean a bit more bearish than the headline global oil S/D suggests in 2018 as crude supply is expected to grow ~1.7-m b/d y/y while crude runs are only expected to increase by ~1.2-m b/d y/y. The crude market was in deficit this year however by an estimated 0.3-m b/d, so the implied crude stockbuild for next year is 0.1-0.2-m b/d.
- In addition to this:
 - 1H'18 TARs are hefty, especially in the Middle East, meaning crude exports could be higher than expected.
 - US SPR sales are likely to average ~50-k b/d in 2018
 - Chinese “overpurchasing” of crude is likely to slow Y/Y as refinery site stocks likely absorbed a lot of crude in 1H'17. There is a lot of uncertainty here however and may only ease slightly from 0.9-m b/d in 2017 to 0.8-m b/d in 2018.

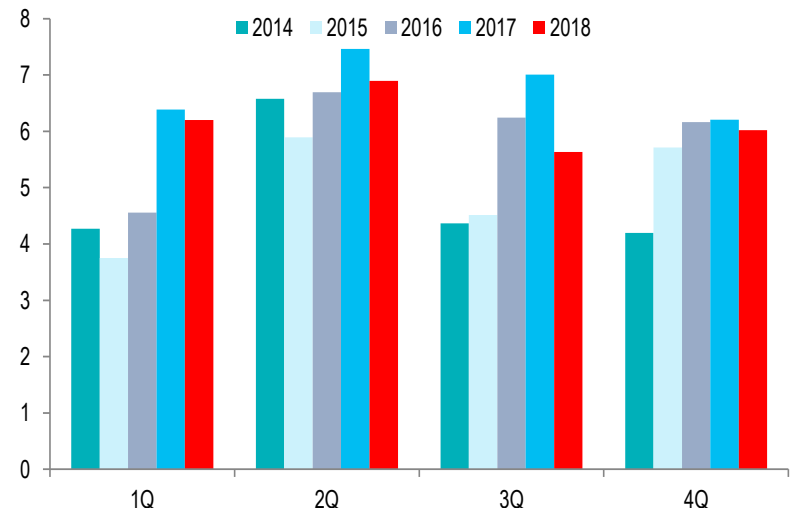
Crude & Condensate Supply/Demand Balance (m b/d)



Global Crude Runs (m b/d)



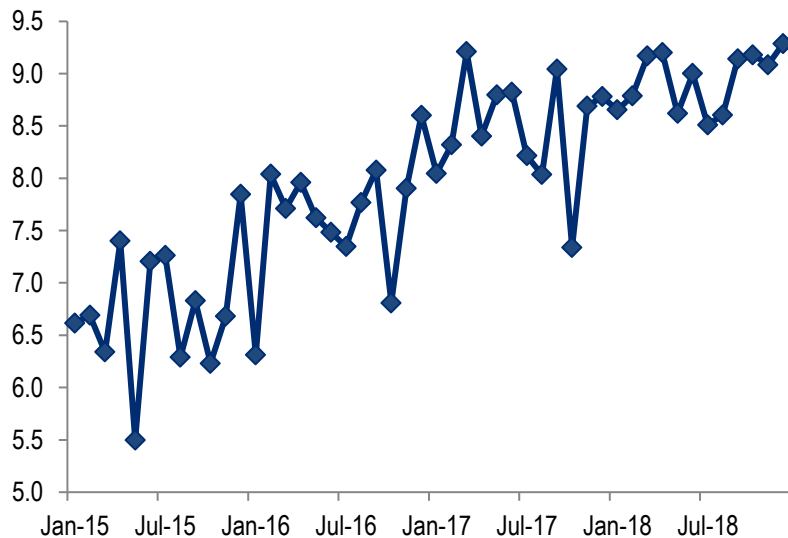
Global TAR & Unplanned CDU Outages (m b/d)



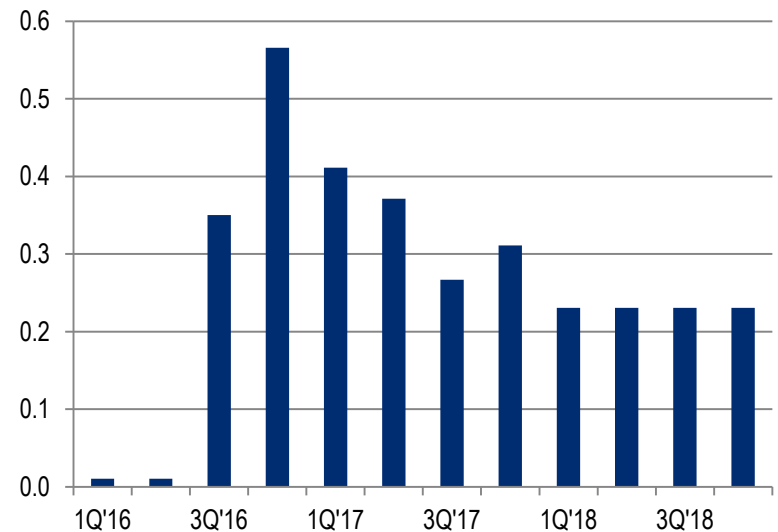
Chinese crude import growth to slow; policy is the key unknown

- Chinese crude imports are expected to grow another ~0.5-m b/d y/y in 2018 to ~8.9-m b/d as rising refinery runs, falling crude output and stockpiling demand increase China's crude deficit. This is a small slowdown from the +0.8 b/d crude import growth rate estimated for 2017 given the smaller pipeline of SPR and private storage capacity additions next year (122-m bbls in 2017 and 84-m bbls in 2018). There is upside to this number however as spare refining capacity in China could be utilised if product export quotas are increased in 2018. This would allow SOE refiners (or potentially independents if they are granted quotas) to increase refinery run rates even further and service ex-Chinese petroleum product demand.
- These expectations are amongst the most uncertain however as Chinese policy remains unclear for next year. 2017 saw a more restrictive approach on product exports, with independent refiners receiving no quotas, and the level of export quotas is a key determinant for crude runs. 2018 product export quotas are still a mystery, but Independent refiners have already been issued 2.8-m b/d of crude import quotas and 2-m b/d of crude usage quotas for 2018, which if fully utilized would be materially more bullish for crude demand near-term.
- There is also the potential that if Chinese product export quotas are too restrictive and global product markets are tight, then SOE refiners will lobby hard to export additional petroleum products to capture the additional margin.

Chinese Crude Imports (m b/d)



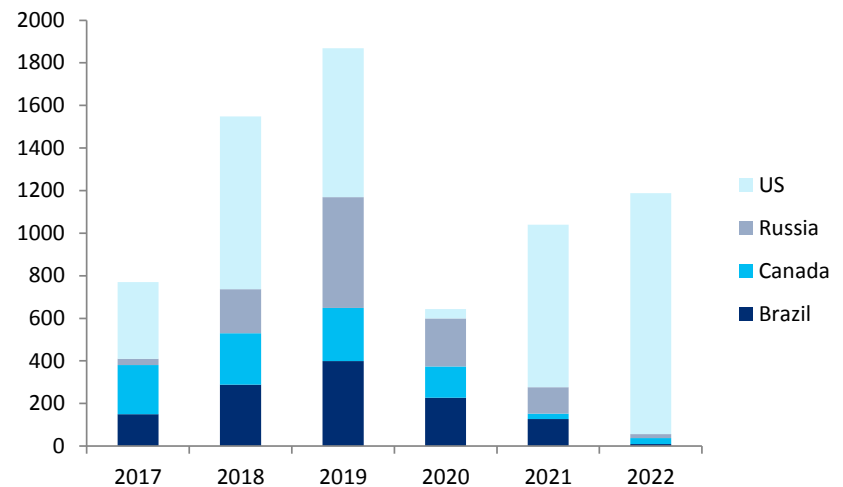
China SPR and Private Storage Capacity Additions (m b/d)



Upside to non-OPEC non-shale supply

- **Brazil's oil supply could grow by 1- to 1.5-m b/d to reach 3.8-4.3-m b/d by 2022**, with IOC investment coming in with an easing of local content requirements / minimum ownership stakes.
- **Canada could add over 0.7-m b/d to reach 5.3-m b/d or over by 2022**, helped by "Canadianisation" of the sector, pipeline build-out.
- **Russia could see 2-3% annual growth**, with 2018-19 likely seeing a fast return to market after the joint OPEC/non-OPEC cuts; combined with an expected surging of production (see [From Russia With Love](#)) this could mean growing some 1-m b/d to 2020, and likely more after that, beyond visible horizon of upcoming projects.
- **All of this means that the period to 2020 in particular sees limited space for OPEC expansions**, and Russian and non-OPEC supply growth; with the US as swing supplier, incentive prices needed to keep US shale flat to down late 2018-2019 could be in the \$40s for WTI.
- However, geopolitical supply disruptions remain a bullish risk, though shale can still cap spikes.

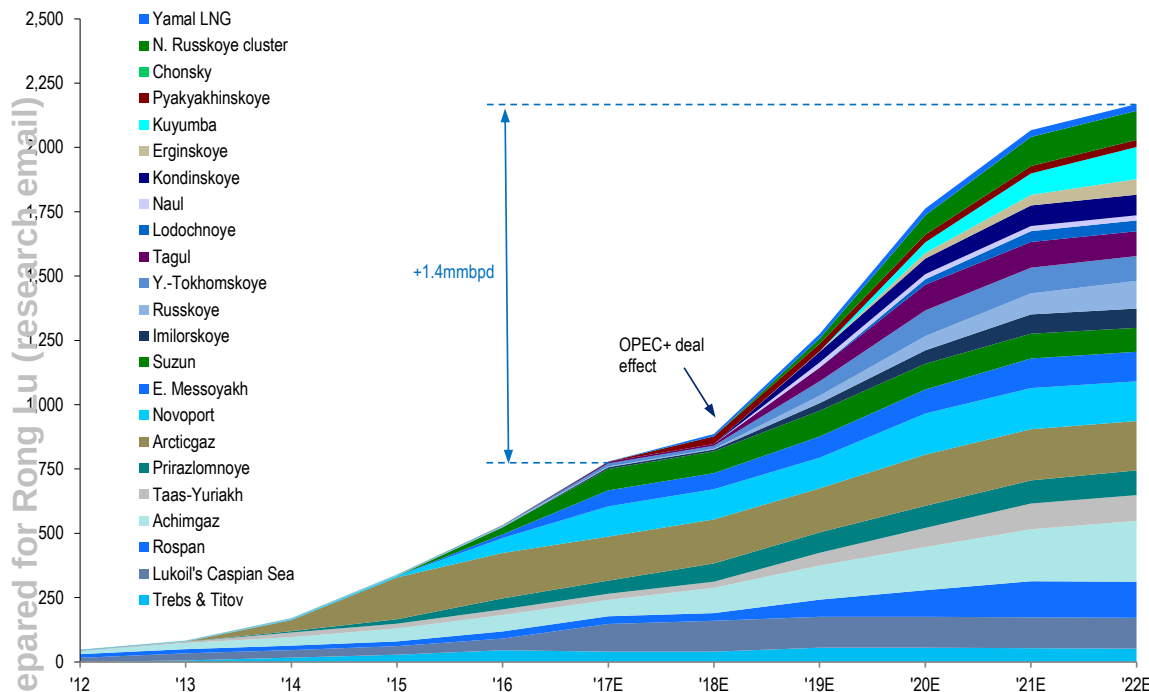
Potential oil supply growth—Brazil, Canada, Russia, US
(Year-on-year growth of annual average, 2017-2022E, k b/d)



Russia is set for 1-m b/d of supply growth to 2020

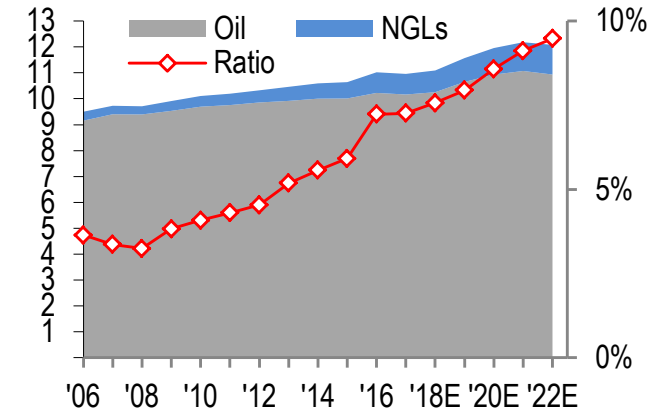
- **Russia can grow its crude oil production by 1-m b/d by 2020, and probably more** beyond the current visible horizon of upstream projects, for 2-3% y/y growth.
 - It can restart most of the capacity idled under the OPEC/non-OPEC production agreement
 - There are more than 23+ greenfield projects in the works adding 1-m b/d by 2020 and 1.4-m b/d by 2022, with several new fields in deeper, tighter plays
 - **Decline rates have been contained.**

Russia's greenfield list, kbd – Adding a 1.4mmbpd tailwind



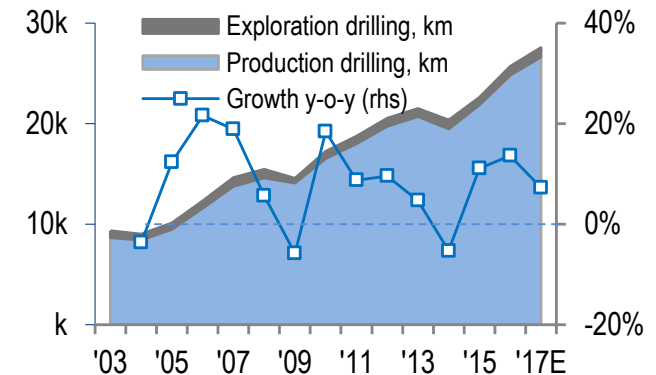
Source: Rosneft, Novatek, Gazpromneft, Lukoil, Interfax, CDU TEK, WoodMac, and Citi Research estimates

Russian oil production, mmbpd



Source: Interfax, CDU TEK, Citi Research

Russian drilling dynamics

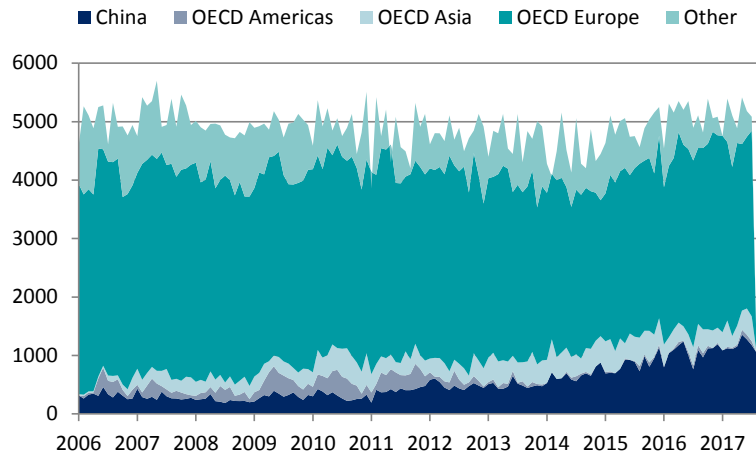


Source: Interfax, Citi Research

Russia look to line up supply destinations in China, worldwide

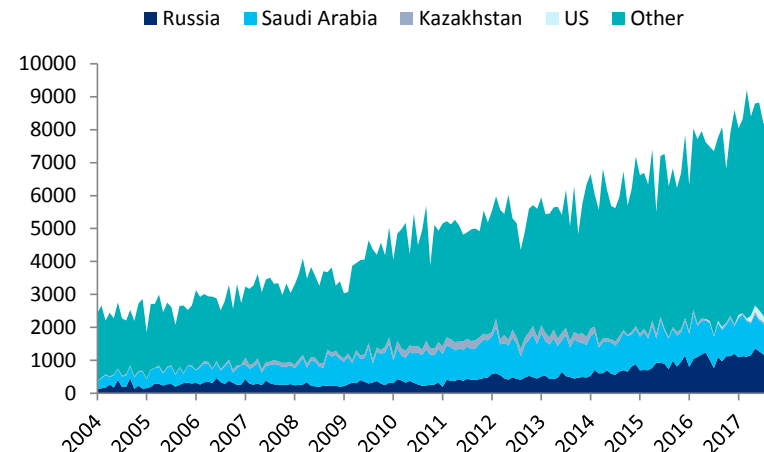
- **Russia has been expanding its ability to export crude oil to Asia and the Pacific Basin, with the development of the ESPO pipeline, supply via Kazakhstan, and in the development of Pacific upstream projects like those on and around Sakhalin island since the 2000s.** Russia now routinely exports over 1-m b/d of crude oil to China, hitting a high of 1.55-m b/d in Sept 2017, with a year-to-date average of 1.2-m b/d; capacity to export higher volumes that can be added quickly—in particular, the ESPO China spur to Daqing is set to expand by 300-k b/d at the beginning of 2018.
- **Russia, like the Saudis, has been investing in refinery capacity worldwide too.** Rosneft has stakes in three refineries in Germany, three refineries in the US (though stakes are loan collateral), and recently bought a 49% stake in Essar, acquiring the Vadinar refinery in India. JV refineries in Indonesia and China could bear fruit post-2020. Lukoil has stakes in four refineries in Europe. Gazprom has a refinery in Belarus, and is planning one in Uzbekistan. Private Russian companies have proposed a JV refinery in Angola.

Russian crude oil exports by destination (k b/d)



Source: Russian Federal Service of State Statistics, China Customs, IEA, Citi Research

Chinese crude oil imports by origin (k b/d)

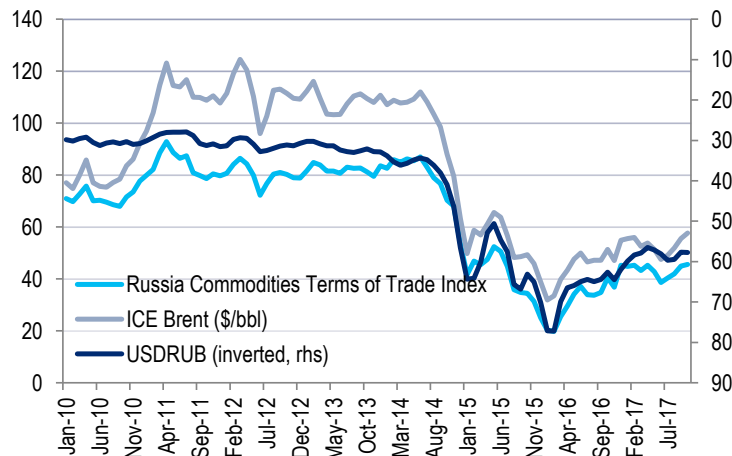


Source: China Customs, Citi Research

Will the Russian central bank have a say and hold some sway re: oil policy?

Higher global crude oil prices have led to appreciation of the Russian ruble against the wishes of the Finance Ministry and central bank. USDRUB doubled from 2014 to 2016, and even today, oil prices are down only 10% in ruble terms versus a 50% drop in USD terms. These FX dynamics could weigh on Russia-Saudi cooperation.

ICE Brent (\$/bbl) versus ruble cross and Russia Commodities Terms of Trade index (2010 – October 2017)



Crude oil prices in Ruble-terms are down only ~10% from their recent 2014 peak versus a ~50% drop in USD terms



USDRUB has declined from ~64-65 in 4Q'16 to ~58 in 4Q'17 amid higher oil prices. But ruble appreciation is becoming a concern for Russian policymakers and could become a point of contention in the Saudi-OPEC-Russia oil production agreement. Indeed, the Finance Ministry has been purchasing US dollars for the past several months trying to stem RUB gains as crude prices marched to the mid-\$60s/bbl. In advance of national elections at the end of 1Q'18, we also think the Kremlin may focus more on the internal political economy, and the competitiveness a weak ruble provides for foreign trade. Our EM strategists believe USDRUB at ~54-55 might be a key threshold to watch for more aggressive policy measures. Lower crude oil prices would likely allow the ruble to depreciate, easing fiscal concerns such as the budget performance.

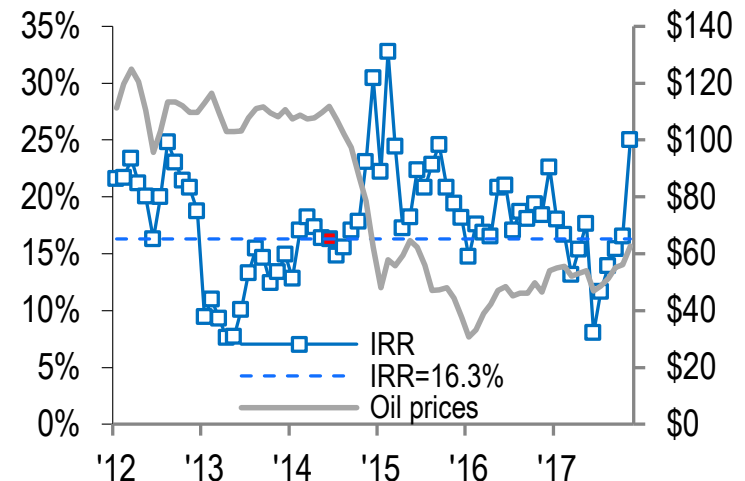
With Russia exiting recession this year with modest ~2% GDP growth, forecast by Citi at a still anemic ~2.3% in 2018, the CRB should persist in its disciplined approach to monetary policymaking—seeking to limit RUB appreciation. Indeed, local officials have already been talking down the ruble and this is at least partly driven by the Finance Ministry view that oil prices will decline significantly during 2018. In addition, Russia may be also guarding against the possibility of US sanctions on its sovereign debt in 2018. Given the strong positive correlation between crude oil prices and ruble relative value, we think the local currency dynamic pits several Russian policymakers at odds for extending crude oil supply cuts.

New Russian well economics stable, even counterintuitive

Between an automatically-adjusting tax regime and a USDRUB exchange rate that is highly correlated with the price of oil, new well economics are not only surprisingly stable, they can even move against oil prices in certain situations. Let's introduce the Citi equity research Marginal West Siberian oil well model, which has been designed to test the changes on well IRRs brought on by swings in oil prices, the USDRUB cross, and base tax regime changes.

- Our base model is expected to yield a 16.3% return in June of 2014 (when oil was c\$112/bbl and the rouble was at c34/USD)
- For most of the time since the oil crash, this same well would have seen expected returns generally well above that level since the oil price crash.
 - Note marginal well economics may be better now than in 2013 when oil prices were >\$100/bbl
 - Also note that a strong rouble drove down economics this summer for c4 months
- Three reasons for the robust well returns:
 - ① The Russian government absorbs much (79%) of the blow from lower oil prices via substantially lower taxes and duties...
 - ② A weaker ruble cuts operating costs...
 - ③ ...and that weaker ruble has a critical effect on up-front well CAPEX

Marginal oil well IRR dynamics



Source: Citi Research

Note: The model also takes into account changes to the oil tax regime as the so-called 'Tax Maneuver' was executed, which blurs somewhat the effect of changes in oil prices and the RUB/USD rate on well economics.

The math behind an expiry of a deal with OPEC for Russia

With ~1-m b/d of extra production either idle or coming online in the next 2-3 years, and with the extra cushion a floating USDRUB rate gives both producers and the Russian government's own budget, the math indicates the Russian budget is relatively insulated from an expiry of the OPEC+ agreement.

2020 Russian budget scenario analysis by oil price, production

	Base	First: Price fall	Second: Volume rise	Third: More production
Brent	\$55.0	\$45.0	\$45.0	\$47.5
RUB/USD	59.1	67.0	67.0	65.0
Russian production, mmbpd	11.0	11.0	12.0	12.0
o/w exported	5.1	5.1	6.1	6.1
Tax impact				
Oil MET revenues (est), \$bn/yr	\$57	\$43	\$47	\$51
Oil export duty revenues (est), \$bn/yr	\$19	\$14	\$17	\$19
Refined product export duty rev, \$bn/yr	\$6	\$4	\$4	\$5
Total government oil revenues, \$bn/yr	\$81	\$61	\$68	\$74
Total government oil revenues, RUBbn/yr	4,800	4,110	4,590	4,810
Statutory tax rates				
MET, \$/bbl	\$19.1	\$14.3	\$14.3	\$15.5
Export duty, \$/bbl	\$12.9	\$9.9	\$9.9	\$10.6
Typical integrated oil company				
Production, kbd (assumed)	100	100	109	109
Integrated EBITDA, \$/bbl	\$8.9	\$8.0	\$8.0	\$8.3
EBITDA, \$mn	\$324	\$291	\$318	\$329
EBITDA, RUBbn	19.1	19.5	21.3	21.4

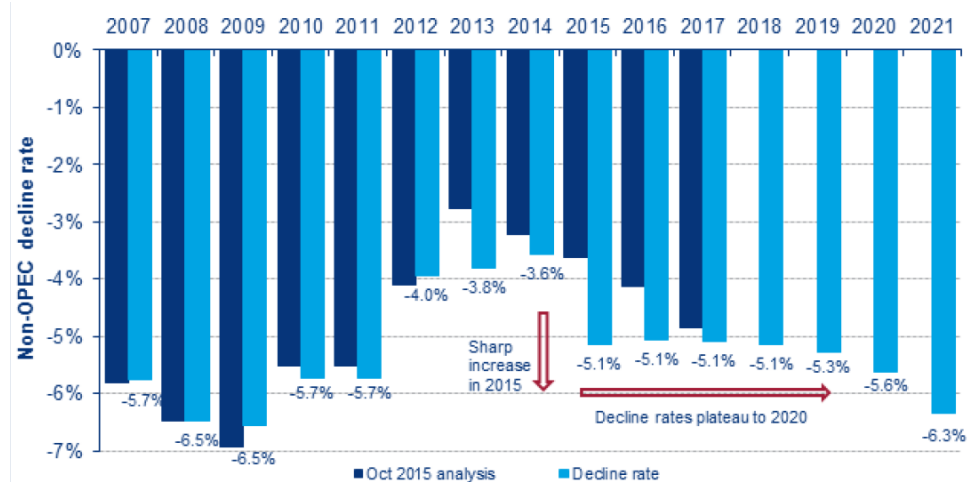
Source: Citi Research estimates based on Citi's Simple Russian Oil Model

Non-OPEC decline rates are stabilizing, not accelerating

- Most visibly within the non-OPEC producers, **China, Colombia, Mexico are set to see significant declines** going forward, though Mexico could reverse this by 2020.
- Elsewhere, decline rates do not look as bad as feared. Decline rates did worsen since 2014—perhaps from around 3.5% to about 5% now—but is expected to stabilize rather than continue accelerating, due to:
 - Operating efficiencies to minimize downtime from producing wells
 - Rethinking maintenance schedules
 - A shift in spending toward short-cycle supply boosts—infill drilling, near-field step-out drilling, and satellite tie-backs
 - Falling service sector costs
- Only a subset of all global supply is in this declining portion—importantly, oil sands are “zero-decline” resources, while there has recently been a shift toward a larger portion of global production coming from early-life assets, which are characterized by lower decline rates, though this likely ages again over the medium-term.

Annual average non-OPEC decline rates (ex-shale)

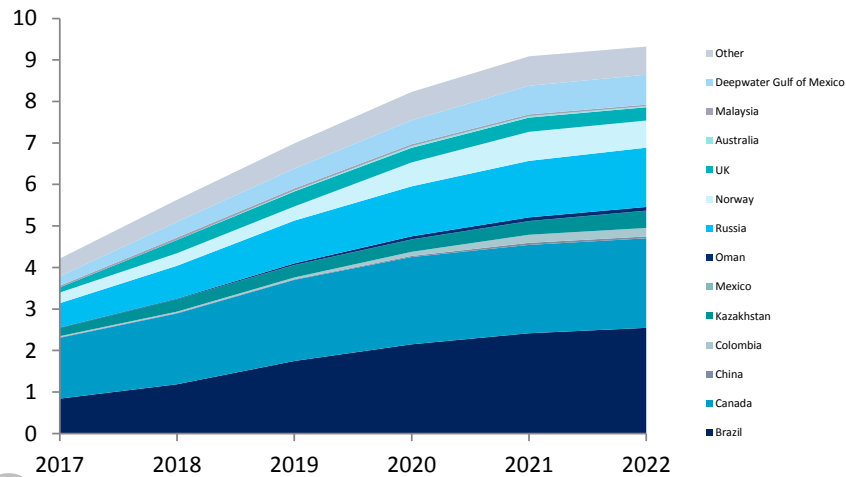
...have risen since 2014, but may hold steady this decade



Non-OPEC FID and non-FID projects vs. OPEC expansions

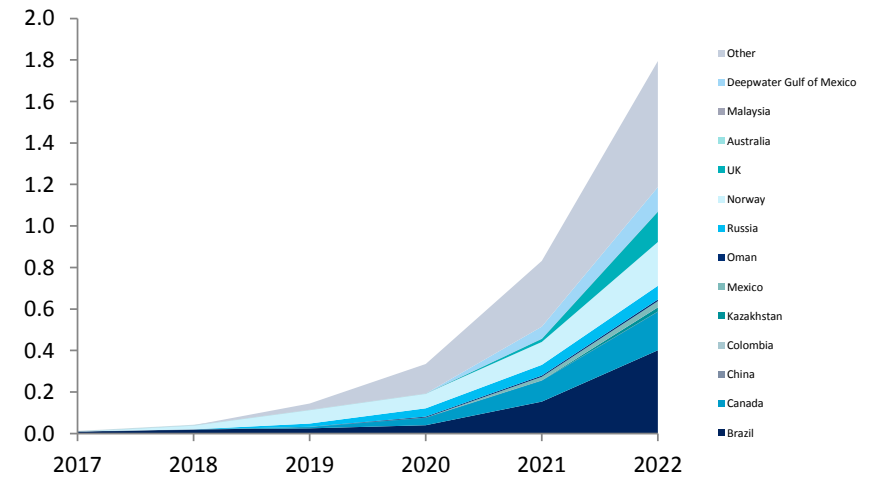
Non-OPEC non-shale production from FID projects*

Already-sanctioned projects could add over 5-m b/d through 2022



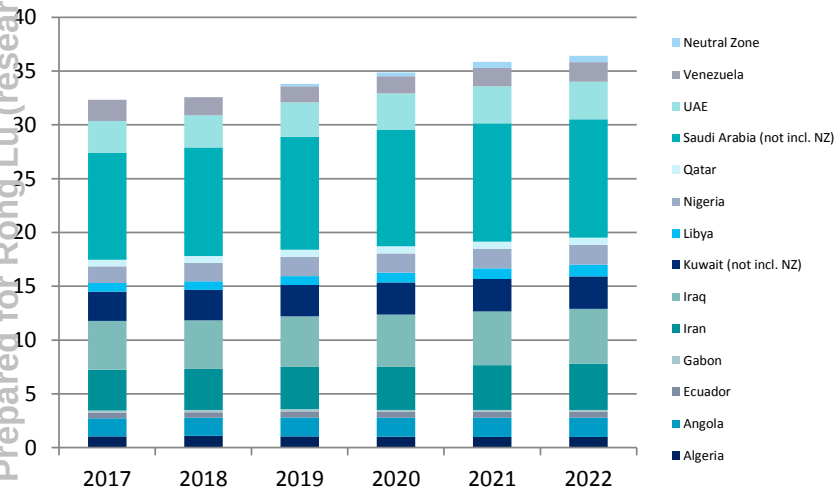
Non-OPEC non-shale production from non-FID projects*

Additional ~2-m b/d of growth possible from non-FID projects, with upside



OPEC liquids production outlook

The return of OPEC barrels after the recent cuts, plus medium-term expansion plan, could add ~4-m b/d by 2022 (m b/d)



OPEC liquids production outlook

The largest growth would come from the GCC countries, Iraq, Iran and possibly Saudi Arabia (k b/d)

	2017	2018	2019	2020	2021	2022 5YR CHG
Algeria	1060	1090	1040	1010	990	-70
Angola	1660	1680	1780	1760	1800	140
Ecuador	530	530	550	540	530	0
Gabon	200	210	200	195	190	-15
Iran	3790	3800	3933	4000	4150	510
Iraq	4530	4530	4700	4850	5000	570
Kuwait (not incl. NZ)	2720	2830	2900	3000	3000	280
Libya	800	800	850	900	1000	300
Nigeria	1550	1700	1775	1800	1825	300
Qatar	610	630	660	660	660	50
Saudi Arabia (not incl. NZ)	9950	10100	10500	10800	11000	1050
UAE	2940	2990	3200	3400	3450	560
Venezuela	1980	1680	1500	1600	1700	-180
Neutral Zone	0	0	200	350	550	600
OPEC NGLs, GTLs	6680	6790	6800	6875	6950	320

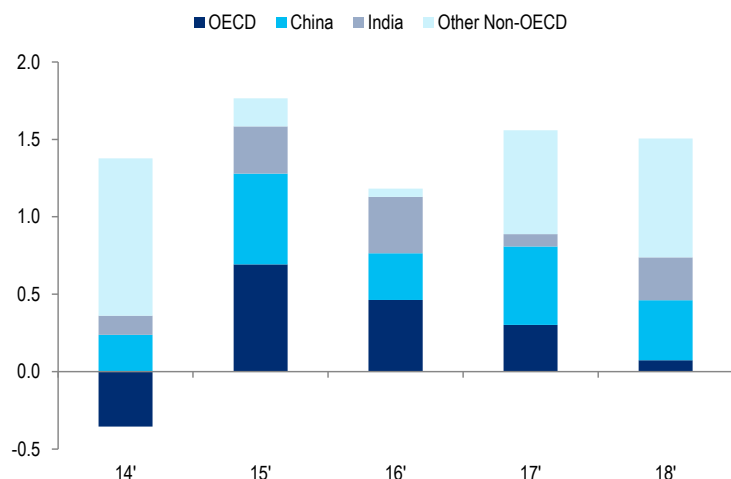
Oil demand growth likely to remain well above 1-m b/d

- Oil demand growth has averaged ~2-m b/d (2% y/y) over the last 6 months, and although this fast growth rate is unlikely to be sustained, oil specific macroeconomic drivers point to healthy demand growth through 2018. Global GDP growth is expected to increase 0.3% y/y to 3.4%; the OECD and non-OECD regions are both expected to grow faster next year than this year; Many of the major oil consuming nations are expected to see a healthy pick-up in headline GDP growth (US, Russia, Brazil, Saudi, India etc.); And Chinese growth is set to moderate, but not decline markedly. Manufacturing PMIs are also growing at their fastest rate since early 2011 yet recent price increases are likely to have a dampening impact on demand although this is likely to be focused on LPG, jet and gas.

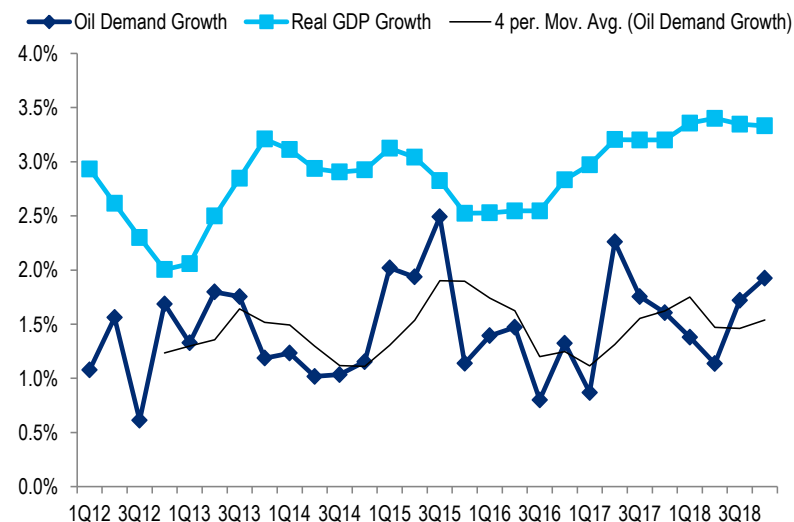
- Our expectation is for oil demand to average ~1.5-m b/d again in 2018 and 2019, with risks potentially to the upside in 2018.

Petchem demand remains robust; although there has been some switching towards naphtha from LPG given higher prices. This likely persists in 2018, but new US crackers will add specific LPG demand too.

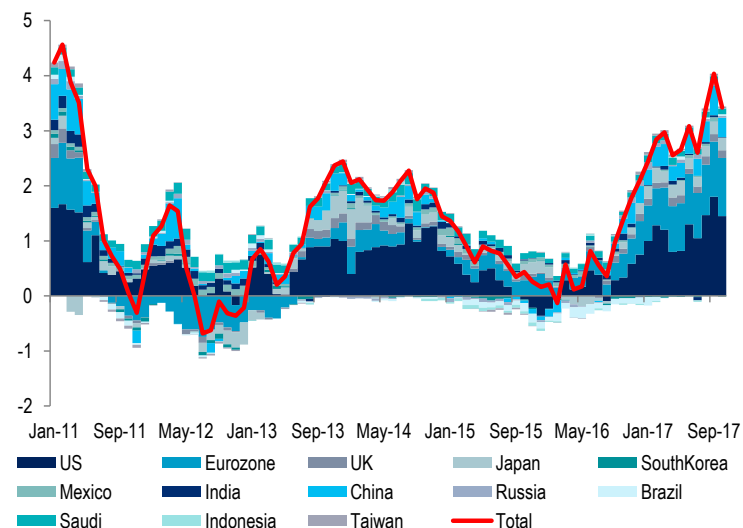
Oil Demand Growth by Region (m b/d)



GDP vs. Oil Demand Growth (m b/d)



Global Manufacturing PMI Index*



China demand: investment slows but manufacturing grows

- **Signs of stronger oil demand growth in 2017 of ~0.5-m b/d y/y—after a period in the doldrums—could ease to ~0.4-m b/d y/y (~3.5% y/y), but there remain questions:** how much are China's weakening "old economy" drivers of oil demand offset by infrastructure investment, road network development, an improving global outlook? What about headwinds from environmental policies, winter supply cuts, higher oil prices?
- **China's GDP growth is expected to trend down to 6.5% and below**, from >7% before 2015; IP and FAI growth is expected to ease to an even slower pace, just over 5%, versus double-digit growth in the first half of the 2010s.

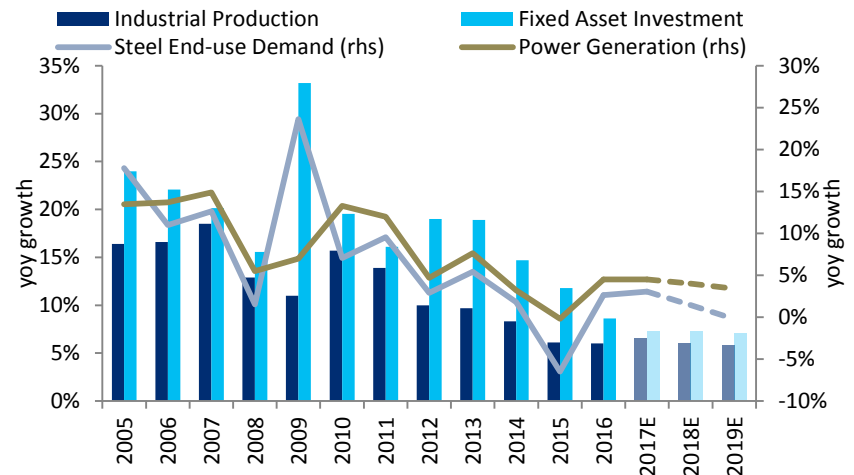
• **Weaker IP and FAI growth had slowed diesel demand to a standstill—even moving into reverse at times—since 2011** as freight traffic slowed from double-digit growth to the low single digits.

• **However, an improved global economic outlook has seen PMIs, freight traffic, container traffic recover significantly, helping to lift diesel demand back to ~2% growth over 2Q and 3Q'17.** September saw a spike in apparent oil demand across the board, but was likely due to pre-holiday stockpiling. 4Q'17 could see a slowdown due to winter supply cuts, October trucking restrictions ahead of the National Congress. Diesel, gasoline, naphtha could see faster growth in 2018, while LPG, jet and others could see slower growth.

• **Upsides to demand could boost y/y growth to 0.5-m b/d**, though slowing economic growth, policy drivers, higher oil prices likely stymie this.

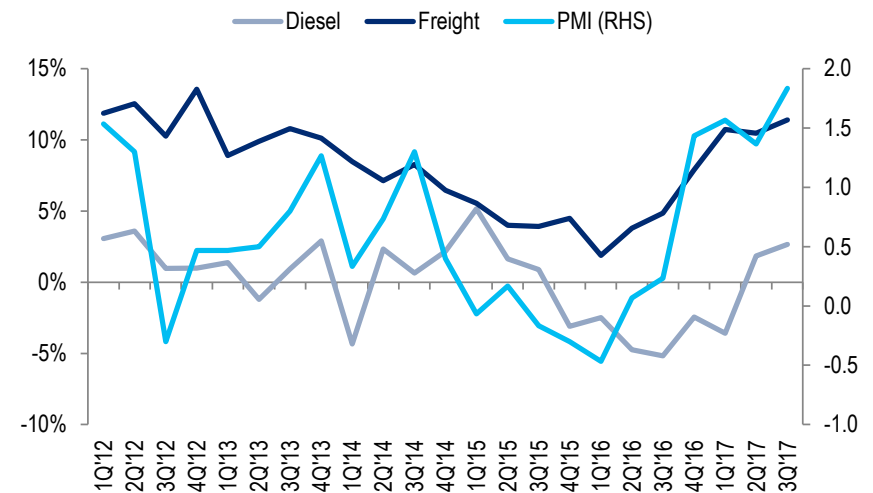
China's FAI and IP are slowing down...

(fAI, IP, steel demand, power gen, 2005-19E)



...but manufacturing and freight traffic are rising again, supporting stronger oil demand growth, incl. diesel

(% y/y on LHS, PMI less 50 on RHS, quarterly 2012-17)



Steady, not stellar, China gasoline demand as car sales slow

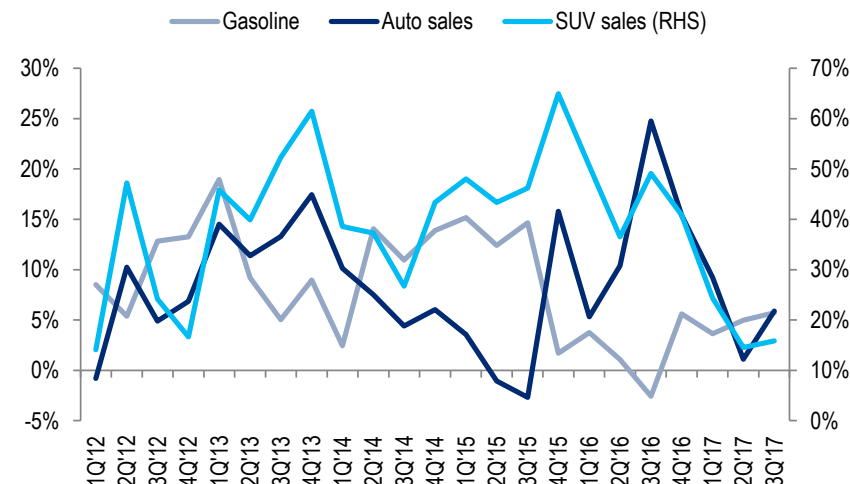
- China's gasoline demand could sustain ~6% growth going forward, or some 150-200-k b/d per year in the next few years, though there could be headwinds from sustained higher oil prices and falling car sales growth; after a 2014-15 demand surge (to over 10% y/y, or 200-250-k b/d y/y) on lower prices, gasoline consumption was flattish over 2015-16.

- Low oil prices had boosted auto sales, particularly SUV sales in 2014-15, which should lock in some lower efficiency gains for a period, but these sales numbers have come off sharply in 2017, perhaps partially as oil prices have recovered.
- EV penetration in China's vehicle fleet matters more in medium-term, but the directional trend is clear, and a headwind for transportation oil demand over time. For instance, China has set up an scoring system that means a portion of carmakers' vehicle sales should be electric, hybrids, hydrogen; the 12% score target for 2020 implies that "new energy vehicles" need to account for 4-5% of vehicle sales that year.

- Jet fuel demand, which has grown 11% YTD, should continue rising, though at a slower rate. YTD China domestic traffic has seen some 13% y/y growth, but has slowed quite somewhat in the past few months. Routes to/from Asia continue to grow at a healthy 5-10% clip for various regions.

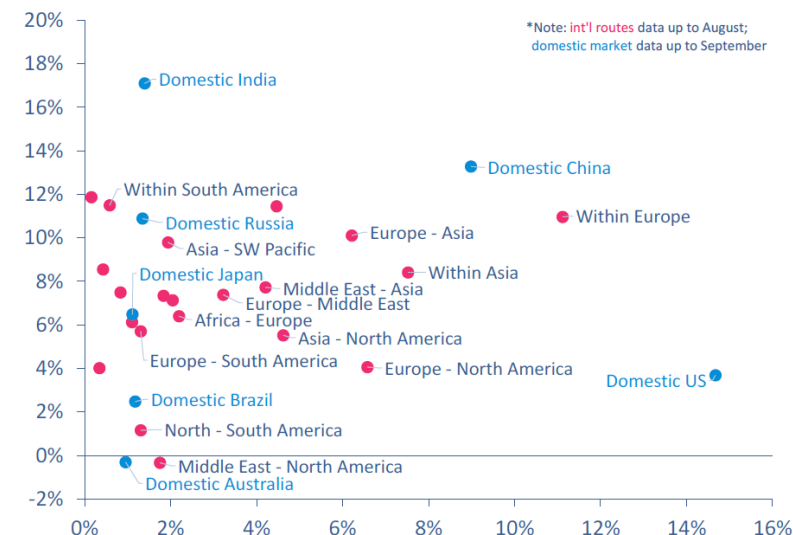
China gasoline demand and auto and SUV sales

(% y/y, quarterly 2012-17)



Passenger traffic growth by int'l route or market, 2017 YTD

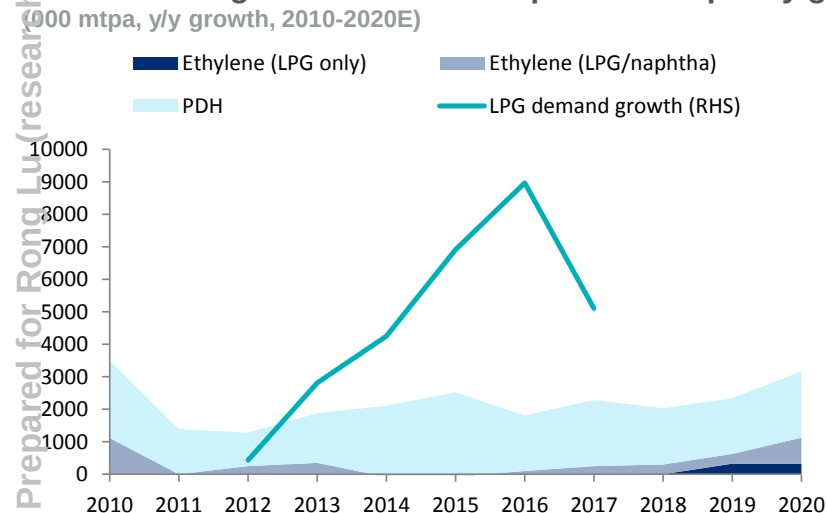
(Revenue Passenger Kilometers (RPK) % y/y)



Strong China petchem demand, but more naphtha than LPG

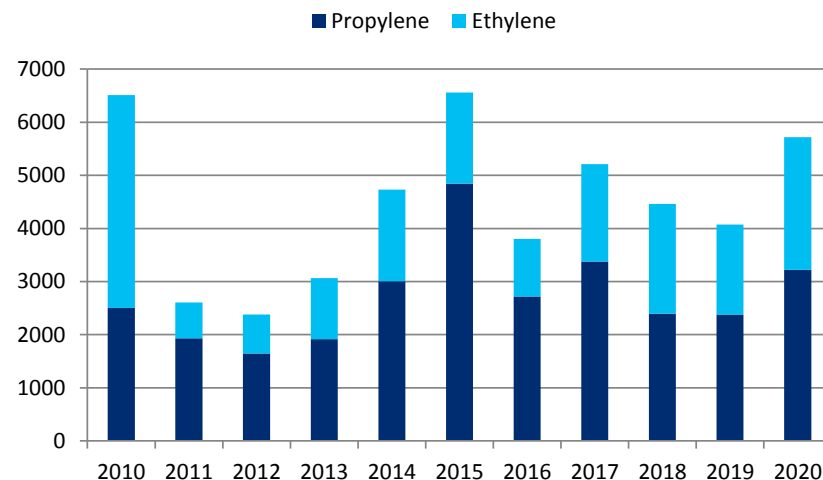
- **LPG demand for petchem feedstock has grown robustly, up 24% y/y in 2016 and 11% in 2017**, but looks set to slow with fewer ethylene or PDH capacity additions in 2018 before picking up again in 2020. **Naphtha demand could stay supported as 2018-19 ethylene cracker capacity opens** that can use this feedstock.
- **Fuel oil demand trends point to tepid growth:** (1) fuel oil feedstocks for teapot refiners remain low, having switched to crude, (2) bunker fuel demand growth somewhat offset by efficiency gains, (3) complex refinery capacity build-out has slowed.
- **Pet coke, asphalt, and other demand** can continue at ~0.1-m b/d of growth as seen in 2017, supported by infrastructure investment. "Other" includes aromatics for blending, which could indicate some underreported gasoline/diesel demand.

LPG demand growth vs. relevant petchem capacity growth



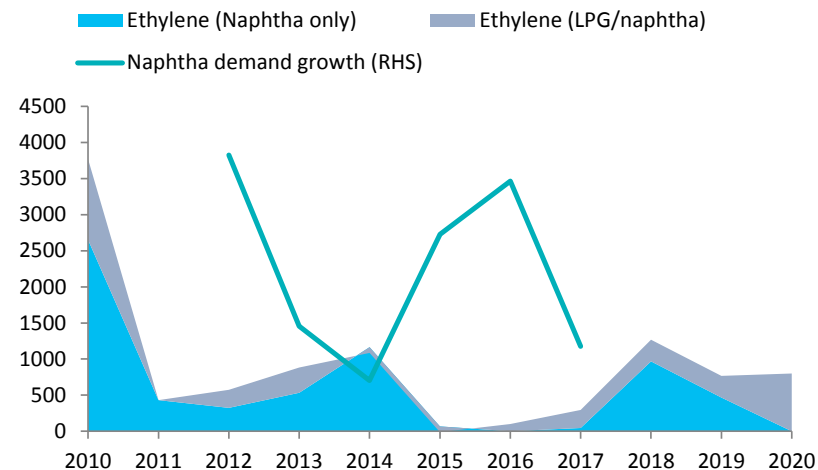
Ethylene and propylene output capacity additions

(000 metric tons)



Naphtha demand growth vs. petchem capacity growth

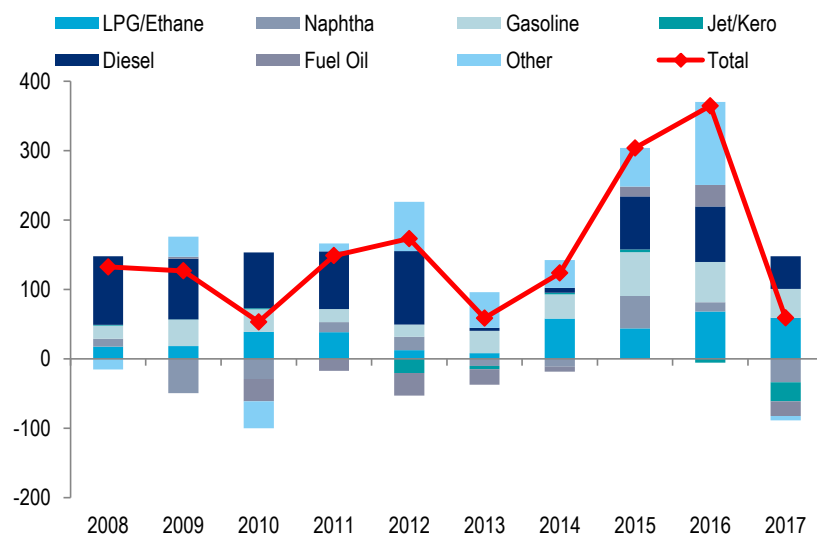
(000 mtpa, y/y growth, 2010-2020E)



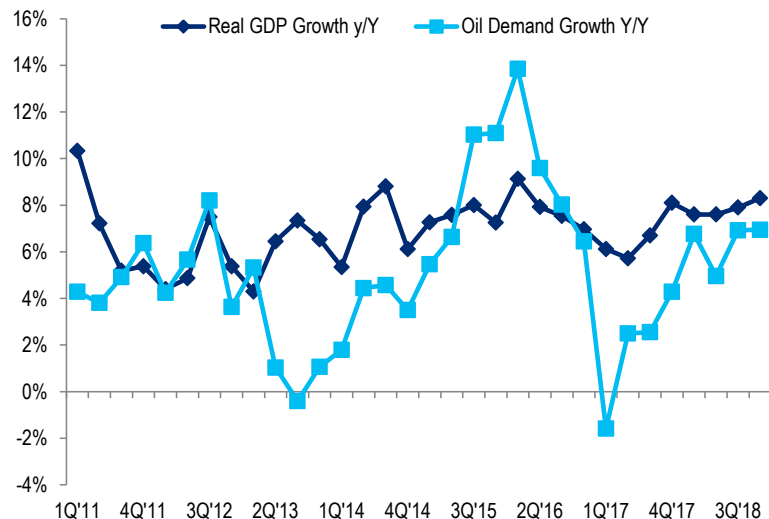
Stronger Indian oil demand growth to return in 2018

- Oil demand in India grew at its slowest y/y % growth rate (1.5%) since 2001, as its demonetization had a material impact on economic and oil demand growth. After surging to 300-350-k b/d (9-10%) in 2015-16, the Indian oil economy was rocked by this change in policy as well as more restricted pet coke usage and growth is likely to be <100-k b/d this year.
- Indian oil demand is expected to grow ~0.3-m b/d in 2018 but growth could potentially be higher. Headline GDP growth (Indian fiscal year) is set to grow at 7.6% next year and industrial production is expected to rise from 4.7% this year to 7.8% next year. Strong passenger and commercial vehicle sales are likely to lead to stronger gasoline and diesel consumption next year and the recent "7th Pay Commission" could see disposable income increases and hence provide more of a support to gasoline demand. A higher base oil price does work against demand though.

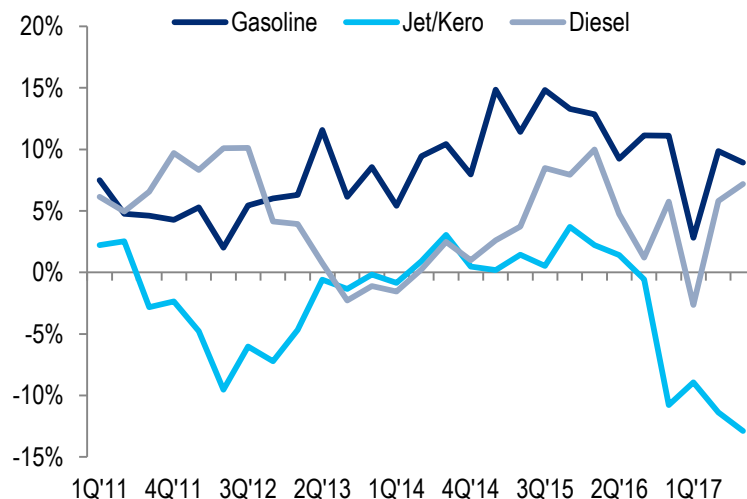
Indian Oil Demand Growth by Product (k b/d)



India Oil Demand vs. GDP Y/Y Growth (%)



Key Indian Oil Demand Growth Rates (%)

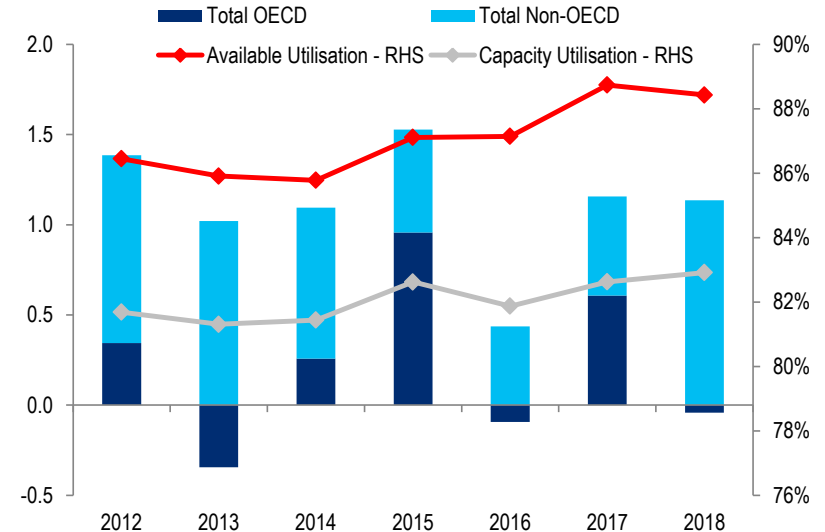


Refining margins likely to remain strong to support utilization

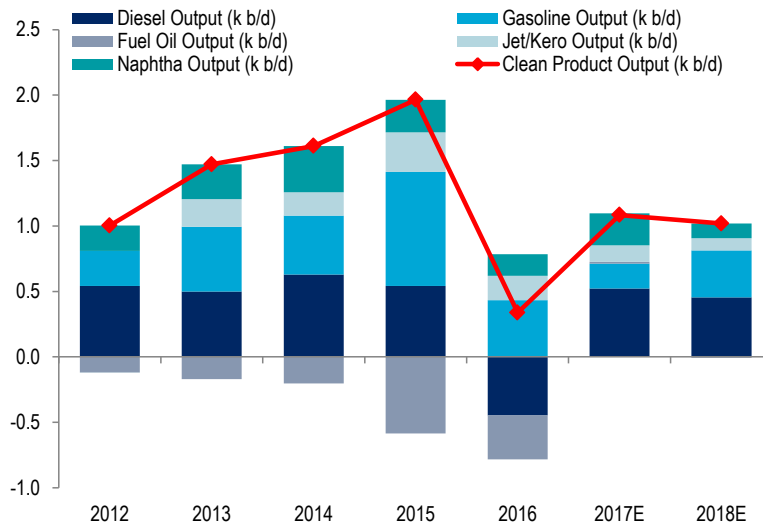
2018 refinery margins should be at or above current forward margins:

- In most scenarios oil and clean product demand growth is likely to outpace refinery run rate growth.
- Global refinery utilization rates are already very high, meaning a rally in margins/cracks may not bring much additional supply.
- OECD utilization rates are at-or-around max and 2018 run rate growth is entirely in the non-OECD region where the risk skew is likely to the downside.
- Spare refining capacity is largely located within China but 2018 product export quotas remain a mystery.
- All product markets have seen stocks draw and markets tighten making yield switching more difficult.
- CDU additions expectations are for 0.9-m b/d (Dec-18 vs. 17)

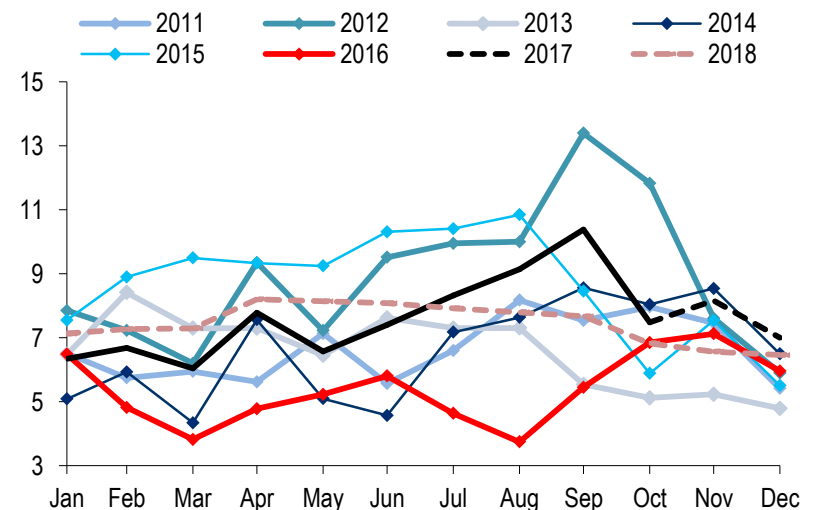
Crude Run Rate Growth (m b/d) vs. Utilization (%)



Global Refined Product Output Y/Y Growth (m b/d)



N.W Europe DTD Brent Cracking Margin (\$/bbl)

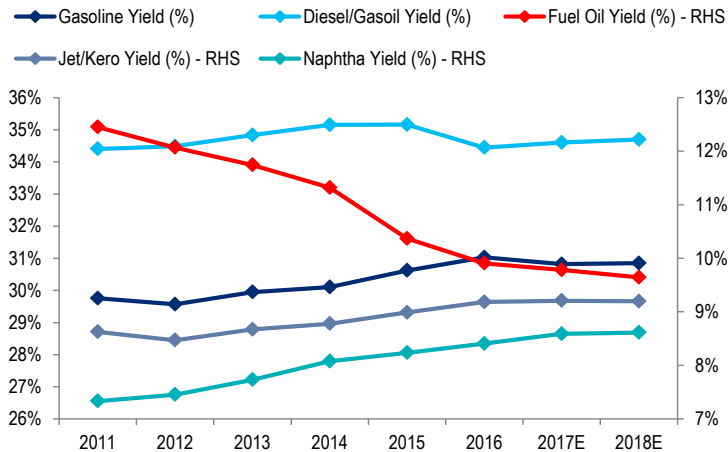


Source: ICIS, EIA, IEA, FGE, JODI, Chinese Customs, KNOC, METI, IIR, company reports, Bloomberg, Citi Research estimates

Clean product cracks will have to compete for additional yield

Clean product yields should continue to increase at the expense of fuel oil yields due to secondary unit upgrading but to see additional increases of diesel/yields, it will likely have to be at the expense of jet/kero or gasoline yields, yet these markets aren't oversupplied. Chinese runs will also be key, due to falling gasoil yields there.

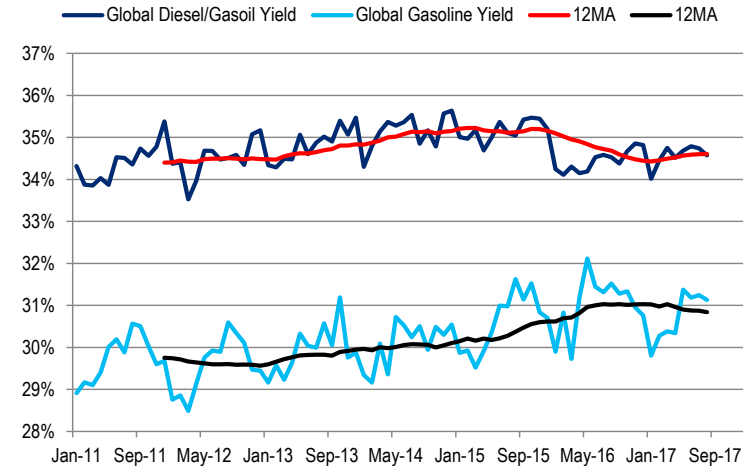
Global Refined Product Yields (%)



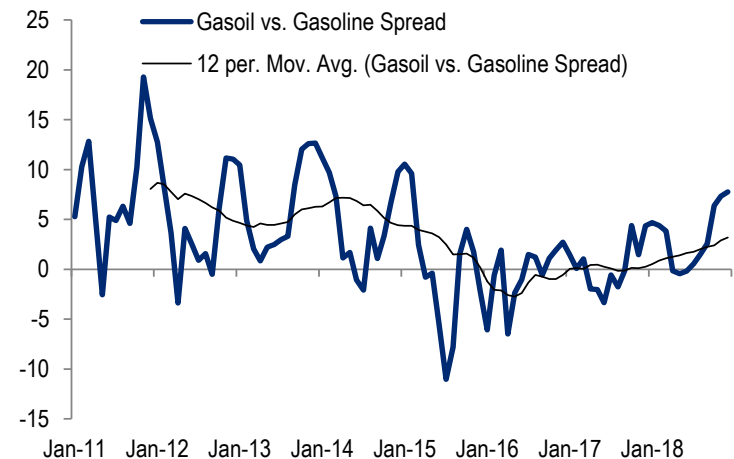
Refinery Utilization by Region (%)

Utilisation	2014	2015	2016	2017E	2018E
US	93.5%	93.6%	92.3%	95.0%	95.5%
OECD Europe	83.2%	88.5%	87.6%	90.3%	90.0%
Japan	88.6%	88.6%	88.4%	91.2%	90.6%
Korea	81.2%	88.5%	91.6%	92.2%	92.5%
Total OECD	87.6%	90.0%	89.5%	92.1%	91.9%
FSU	92.8%	93.4%	89.9%	90.2%	89.0%
China	73.9%	75.7%	78.7%	80.3%	80.5%
Other Asia	93.9%	93.7%	96.2%	96.5%	96.4%
Middle East	91.5%	92.0%	91.7%	92.7%	95.8%
Total Non-OECD	84.2%	84.7%	85.2%	85.9%	85.7%
Global	85.8%	87.1%	87.1%	88.7%	88.5%

Global Diesel/Gasoil vs. Gasoline Yields (%)



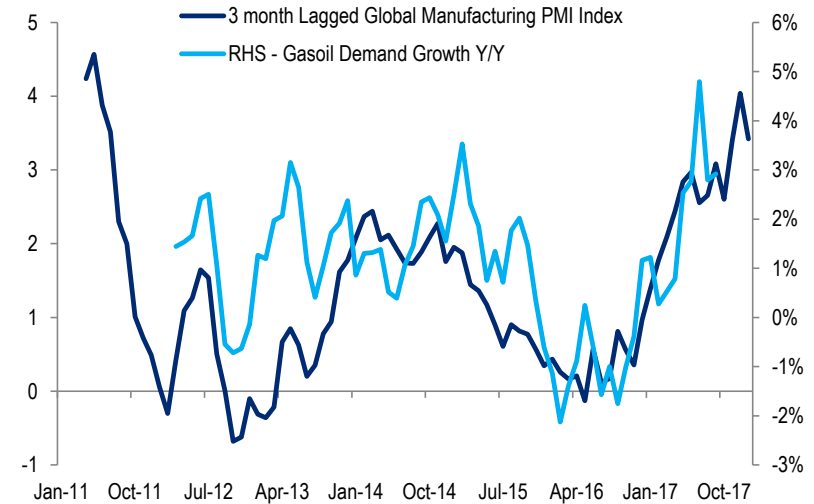
Europe Gasoil vs. Gasoline Spread (\$/bbl)



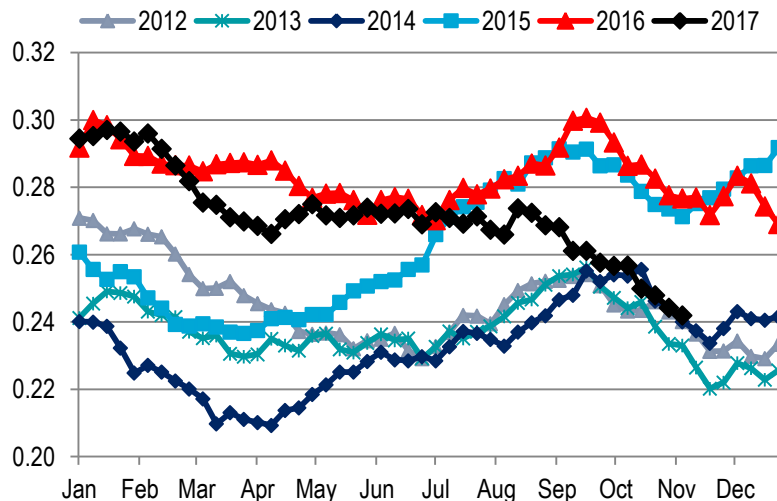
Distillate cracks look fairly priced as there is yield flexibility

- ULSD and gasoil cracks returned to levels not seen since 2015 as restricted refinery supply and resurgent demand growth have led to a sharp draw-down in stocks.
- ULSD and gasoil cracks and timespreads have come off since their September highs but current forward cracks look around fair value as:
 - Using current yields, diesel/gasoil supply is expected to increase by ~450-k b/d next, which is 50-100-k b/d below 2017E demand
 - There is upside flex however in global diesel yields but it will have to be at the expense of jet or gasoline, which are both tight.
 - Manufacturing PMIs are a leading indicator of diesel demand growth and they remain very strong globally.

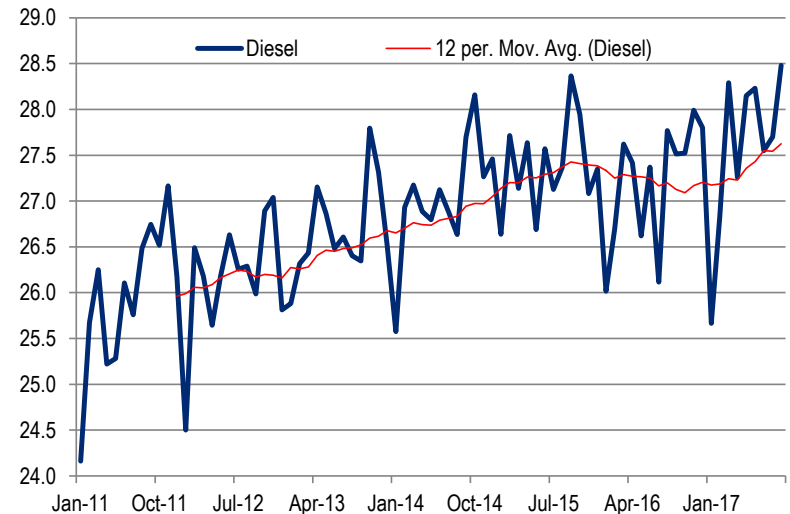
Manufacturing PMI vs. Global Diesel Demand Growth



High Frequency Middle Distillate Stocks (bln bbls)



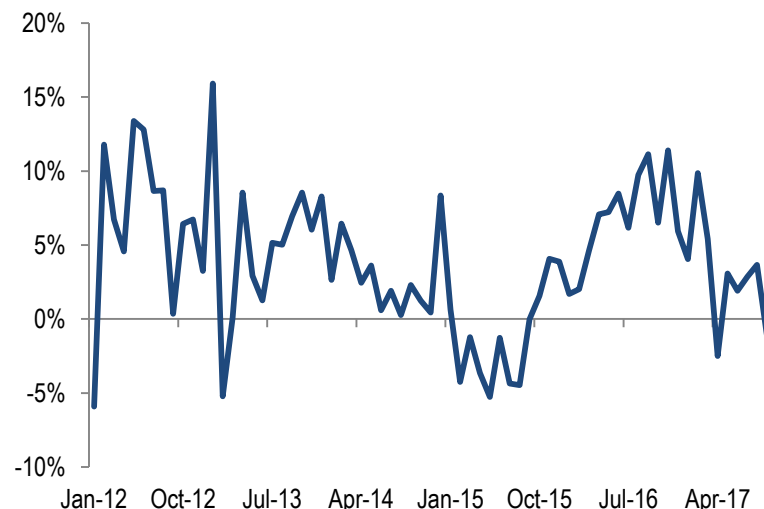
Global Diesel/Gasoil Demand (m b/d)



There is a stronger likelihood of gasoline crack upside

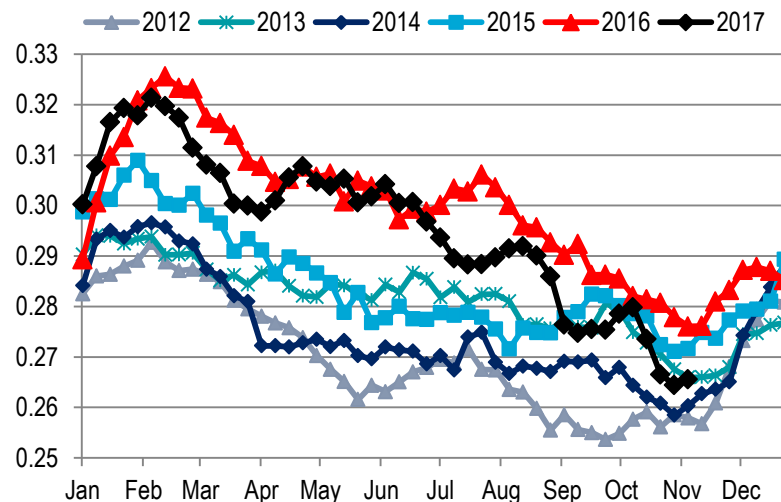
- After rampant growth in 2015 (+0.7-m b/d), gasoline demand growth eased in 2016 (+0.5-m b/d) and again in 2017E (0.4-m b/d). Slowing global auto sales growth is likely a contributing factor and a higher outright gasoline price in 2017/18 is likely to damp gasoline demand growth in 2018.
- Gasoline cracks have a potential upside bias however as refinery supply is expected at 0.3-0.4-m b/d in 2018 and there is little yield or utilization (especially ex China) to boost this. This implies that gasoline demand growth would need to ease again in 2018 in order to balance the gasoline market...
- ...but if oil demand is stronger than expected in 2018 then there is likely to be a higher chance of gasoline cracks spiking than diesel/gasoil cracks.

Global Auto Sales Y/Y Growth (%)

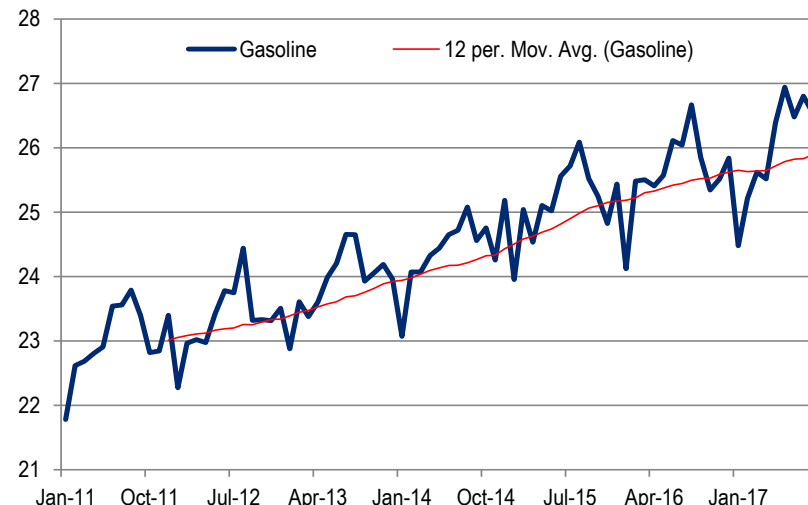


Prepared for Rong Lu (research email)

High Frequency Middle Gasoline/Naphtha Stocks (bln bbls)



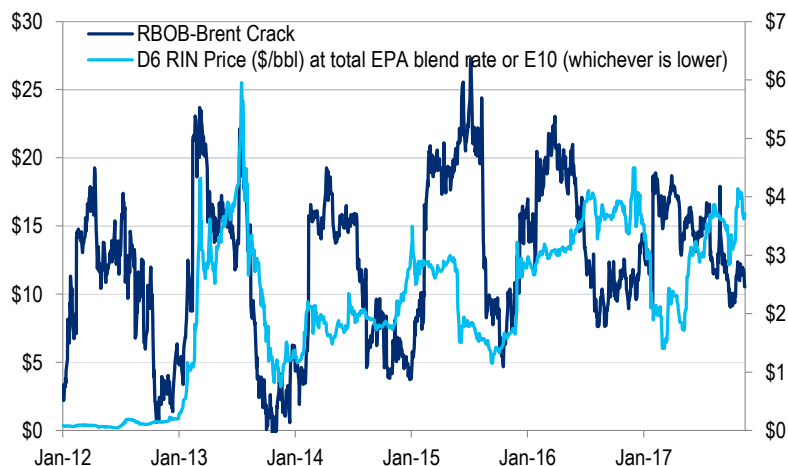
Global Gasoline Demand (m b/d)



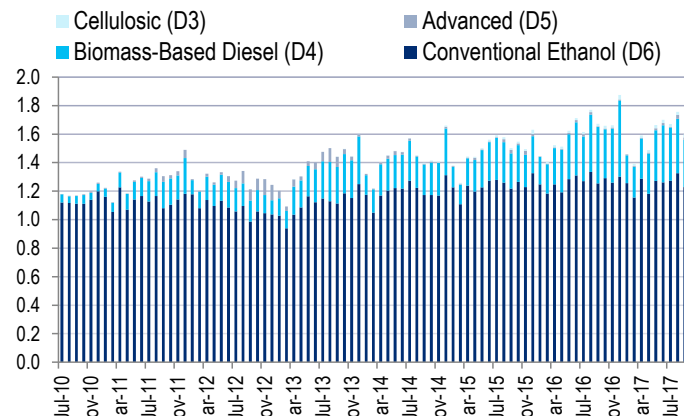
High RIN prices may support wholesale gasoline prices all-else being equal

Renewable Identification Numbers or RINs, regulated environmental credits that trade in US fuel markets are viewed as a 'cost' for refiners under EPA biofuel blending policy. They generally have a positive correlation with RBOB cracks*. Volatile RIN prices can feed into NYH and USGC gasoline and basis markets.

Implied RIN cost component in prompt RBOB-Brent crack (\$/bbl)



Major category gross RIN generation and annual growth rates versus growth in RVO mandates



	2014	2015	2016	2017F	2018F*
D3, D4, D5, D6 RIN generation growth (%)	3.6%	3.5%	8.7%	1.9%	0.0%
EPA Renewable Fuels Standard growth (%)	-1.6%	4.0%	7.0%	6.5%	-0.2%
*proposed EPA volumes					

Some 35% of the nearby NYH RBOB-Brent crack can be attributed to the RIN cost assuming full pass-through and max ethanol blending. Prior to 2013, RIN prices were a few pennies and credits were in surplus. But since 2013 RIN prices have averaged >\$0.60/gal as EPA mandates to mix biofuels into the finished US gasoline pool. They have generally collided with the E10 (10% ethanol) blend wall, requiring higher-blended volumes to fulfill the mandate despite difficulties in marketing such fuels. As such, RIN credit balances have tightened from over 2.6bn gal surplus in 2013 to ~1-1.3bn gal for 2018. Policy uncertainty has also reigned supreme in this area, under both Obama and Trump—sparking heightened RIN market volatility.

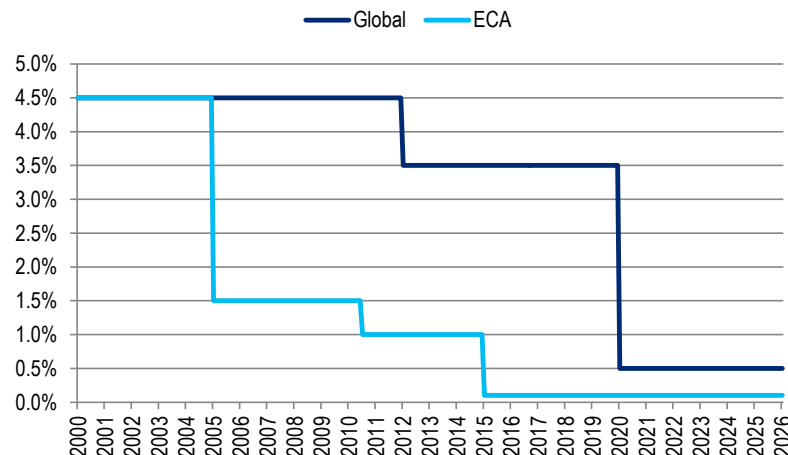
In recent years, EPA renewable volume obligations for refiners/importers of gasoline have exceeded the growth rate of organic RIN generation. It further seems that the Trump EPA is unlikely to meaningfully relax this policy anytime soon. RIN prices have averaged ~\$0.85/gal in 2H'17 and are likely to stay high in 2018 without policy relief. All-else equal, high RIN prices may continue to support gasoline prices and spreads, in an otherwise sluggish environment—even during refinery TAs.

Source: Bloomberg, Broker markets, EPA, Citi Research, *assumes full pass-through and max annual blend or cap at E10

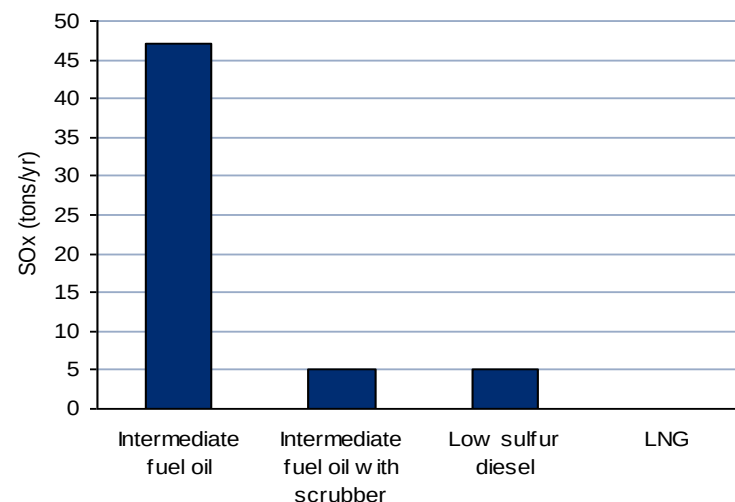
IMO 2020 bunker fuel sulfur rule is bullish diesel, bearish fuel oil

- The IMO's 0.5% sulfur cap on global marine fuels, starting January 1st, 2020, is **bullish for diesel**, **bearish for HSFO**. More stringent sulfur specs for global marine bunker fuel—from 3.5% to 0.5%—means that shippers need to consider how to comply with the new regulations.
- They can switch to low-sulfur fuels (MGO, LSFO, LNG, LPG, methanol) or install scrubbers on-board, allowing them to continue running HSFO. However, a financially strapped shipping sector has been reluctant to invest in retrofitting vessels with scrubbers, esp. as futures curves are not yet fully reflecting likely wider spreads come 2020.
- Refiners would need to raise diesel supply and sharply curtail HSFO, through raising already-high utilization, or shifting diesel and HSFO yields. This may be constrained, pointing to sharp widening of diesel-HSFO diffs, and correspondingly sweet-sour crude diffs as refiners change crude slate.

IMO sulfur spec for marine bunker fuels over time
(% sulfur, 2000-26F, globally, and for emission control areas)

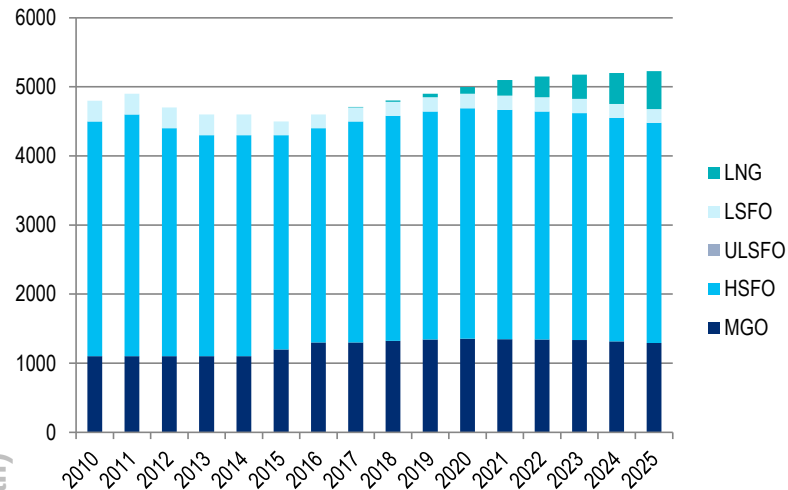


SOx emissions for selected marine bunker fuels
(ton/year)

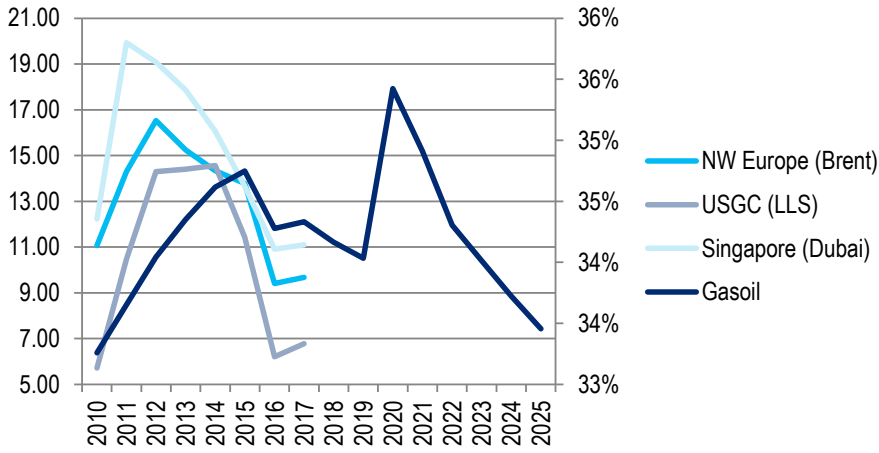


Impact on marine fuel mix, diesel and HSFO cracks

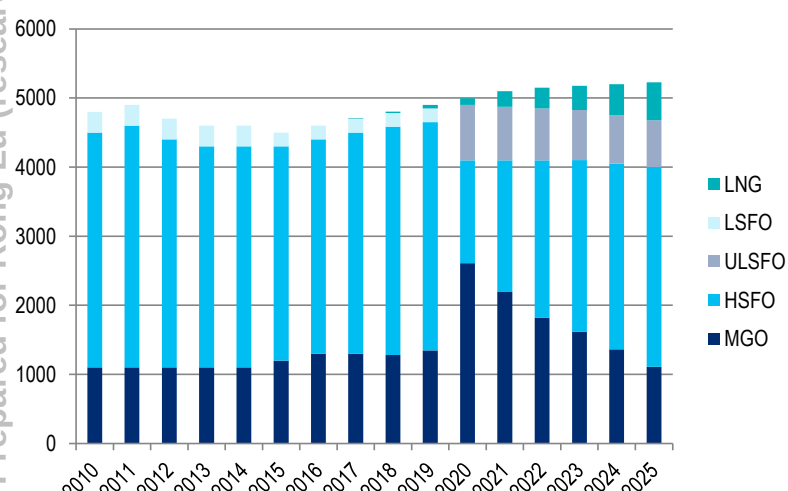
Global bunker fuel market, baseline w/o sulfur spec change
(k b/d, 2010-25F)



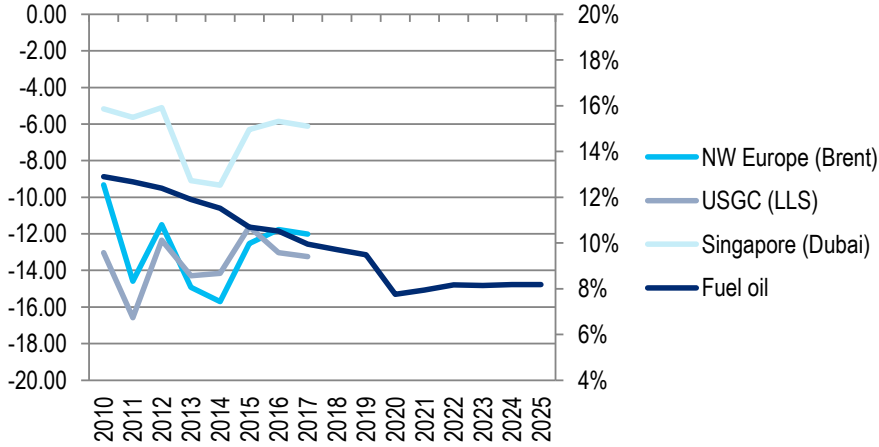
Projected gasoil yield (RHS) versus selected cracks (LHS),
with sulfur spec change
(\$/bbl, % yield, 2010-2025F)



Global bunker fuel market, *with* sulfur spec change
(k b/d, 2010-25F)



Projected fuel oil yield (RHS) versus selected cracks (LHS),
with sulfur spec change
(\$/bbl, % yield, 2010-2025F)



Source: FGE, IEA, Wood Mackenzie, Clarkson, Citi Research

Key marine bunker fuel market scenarios in 2020

2020 snapshot—global marine bunker fuel market scenarios and illustrative prices and cracks (base, bull, bear) with the IMO sulfur spec change, versus estimated baseline without IMO sulfur spec change

(m b/d, \$/bbl)

	MGO (m b/d)		HSFO (m b/d)		Diesel (\$/bbl)		HSFO (\$/bbl)		Brent (\$/bbl)
	vs.		vs.						
	Demand	<i>baseline</i>	Demand	<i>baseline</i>	Price	Crack	Price	Crack	Price
<i>Baseline</i>	1.4		3.3		64	8	44	-12	56
Base case	2.6	1.3	1.5	-1.8	86	30	26	-30	56
Bull case	3.4	2.1	1.3	-2.1	96	40	16	-40	56
Bear case	1.8	0.4	2.2	-1.1	66	10	46	-10	56

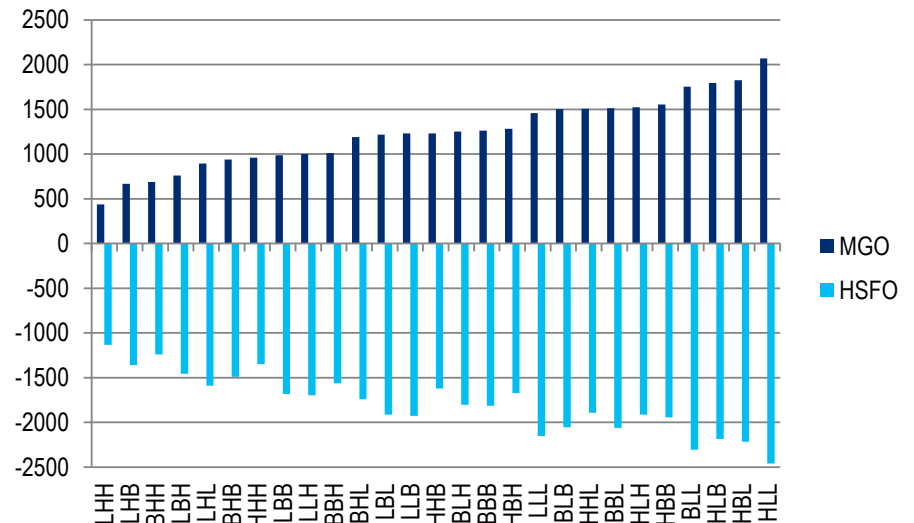
- ch email)
- **Scenarios for the jump in MGO demand range from +0.5-m b/d to +2-m b/d; corresponding to drops in HSFO demand of 1-m b/d to 2.5-m b/d.** The range shown on the right depends on (1) bunker market demand growth, (2) scrubber installations, (3) non-compliance. Other factors make a difference too, including the flexibility of ULSFO blending to make more 0.5% sulfur fuel oil.

Further, the pace of diesel demand growth outside of the marine fuel market, if faster than expected, could place additional strain on the refinery sector to supply this new mix of fuels.

Overall, directionally this strengthen diesel cracks and weakens HSFO cracks. This could be wider if refineries turn out to be less flexible, requiring raising utilization and running more sweet crude, which would produce more diesel and ULSFO, but would also likely lead to a larger glut of HSFO. The buyer of last resort may end up being the power generation sector, in such an extreme scenario.

Scenario analysis: boost to MGO and drop in HSFO demand*

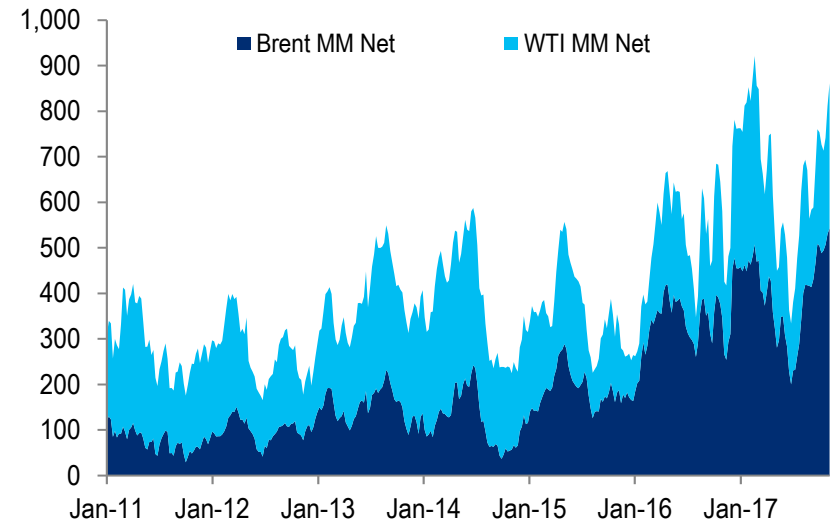
(k b/d, 2020 base case with IMO sulfur spec change, versus 2020 baseline without spec change)



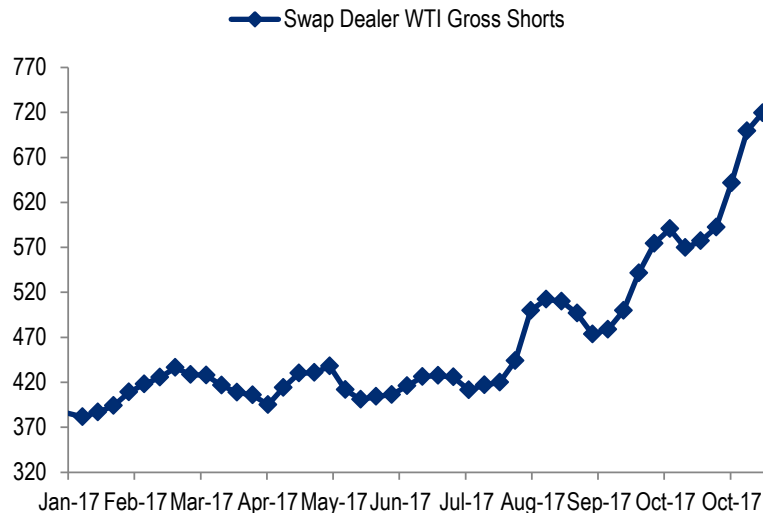
Yet recent investor and corporate flows could set oil up for a fall

- Financial flows are having an increasing impact on crude pricing, both corporate and investor, and current financial flows could be setting crude up for a more bearish turn.
- Investors held their highest all-time net long position on Brent in mid-November, which are likely to unwind following the Nov 30th OPEC meeting. In the weeks between early December and late January, heightened increases in US rig utilization and the expected crude inventory build in January are likely to trigger a sell-off, reducing price support.
- Hedging activity has sharply picked up in 3Q and 4Q'17 which all but locks in US shale barrels for next year. The sharp pick-up in swap dealer net length on WTI is clear evidence of this and as long as WTI prices for 2018 and 2019 stay in the \$52-55 corridor then additional hedging is likely.

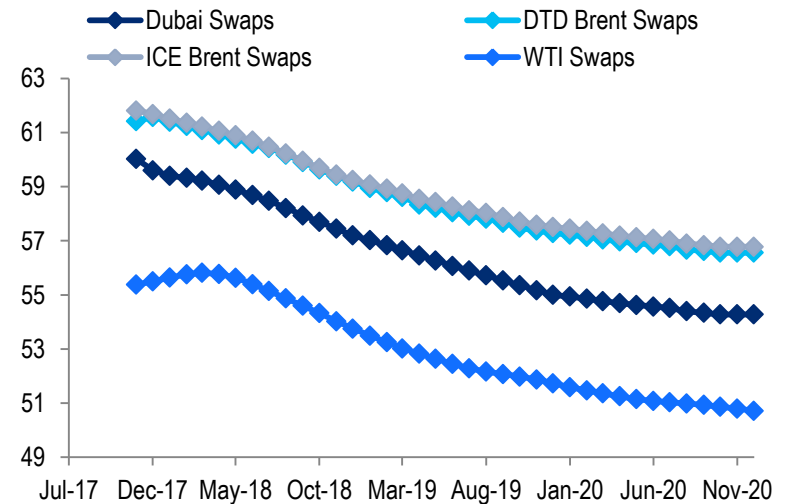
Crude Money Manager Net Length (mln bbls)



Swap Dealer Gross Shorts on WTI (mln bbls)



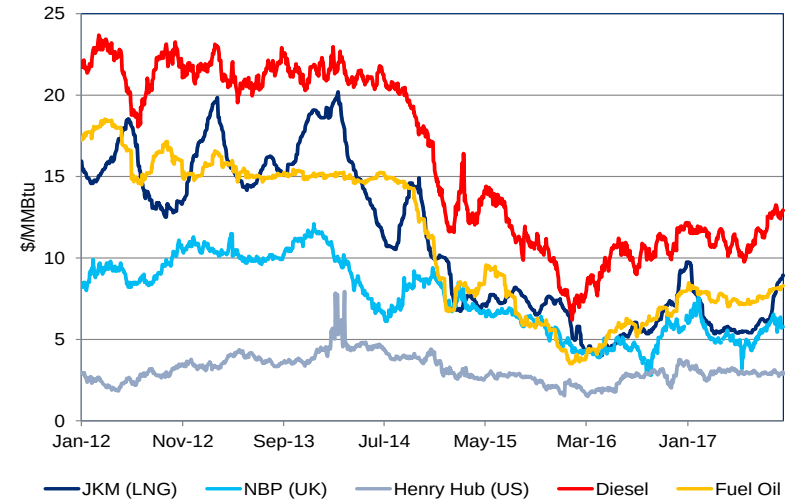
Crude Swaps Curve (\$/bbl)



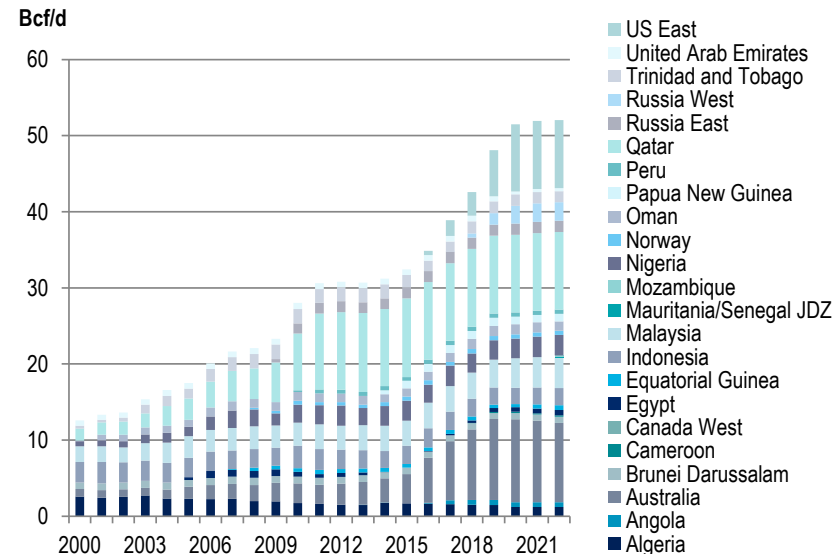
Natural Gas: Price hikes possible in winter but supply to strengthen in 2018

- Global natural gas prices should climb this winter as multiple heating fuel markets confront tight fundamentals, but global production should strengthen in 2018 and beyond as supply projects come online, lowering prices longer term.
- North American prices look skewed to the upside as a modest inventory outlook for 2018 should more than offset impacts of robust production growth. Results from our weather scenario analysis show that a one-standard deviation colder-than-normal winter could lead to an \$0.8 increase in the rolling 12-month strip price vs. a \$0.4 decline in mild winter case. Production should be robust, with new pipes debottlenecking the Northeast and associated gas production rising, but the rise should fall short of turning the market into oversupply. Low 2019 oil prices could reduce associated gas production growth and support gas prices. However, production growth should continue in the years to come, keeping HH prices at the \$3 or below range.
- LNG and European prices should rise because of seasonality (winter demand much stronger than supply) and relative tightness in the global LPG, heating oil and coal markets, thereby supporting prices of various heating fuels. The shutting of the UK's Rough gas storage facility, caps on Groningen output, maturing domestic gas fields and repeated issues with France's nuclear power plant fleet are leaving Western Europe with an increasing gas deficit.
- But the long-anticipated strong LNG production growth in the next few years, first from Australia then the US, should quickly loosen up the LNG and European gas markets. We are also bearish oil and coal prices post-winter. Looking ahead, spot LNG prices outside of winter could fall to <\$4 but winter price spikes could lift annual average prices to the mid-\$5 level.

Global natural gas prices are rising



Global LNG supply should finally surge, with new projects from Australia and the US



US natural gas prices supported due to moderate storage expected in 2018

US Natural Gas Prices Forecasts

Prices	2017	2018	2019	2020	1Q17	2Q17	3Q17	4Q17	1Q18	2Q18	3Q18	4Q18	1Q19	2Q19	3Q19	4Q19
Base (50%)	3.0	3.1	3.0	3.0	3.0	3.0	2.9	3.2	3.2	3.1	3.1	3.0	3.1	3.0	2.9	2.9
Weather normalized cases																
High (20%)	3.1	3.6	3.4	3.2	3.0	3.0	2.9	3.3	3.8	3.5	3.5	3.4	3.5	3.4	3.4	3.3
Low (30%)	3.0	2.5	2.5	2.7	3.0	3.0	2.9	3.2	3.0	2.3	2.3	2.4	2.6	2.3	2.4	2.6
Weather cases																
One-SD colder winter 17-18	3.1	3.7	3.0	3.0	3.0	3.0	2.9	3.4	4.2	3.9	3.5	3.0	3.1	3.0	2.9	2.9
One-SD milder winter 17-18	3.0	2.6	3.0	3.0	3.0	3.0	2.9	3.0	2.4	2.3	2.5	3.0	3.1	3.0	2.9	2.9

- **Base case (50% Likelihood):** Henry Hub prices should strengthen this winter after a mild autumn. Citi is revising 2018 prices higher from \$3.0/MMBtu to \$3.1, while maintaining 2019 prices at \$3.0.

The market should become more constructive toward the end of this year or the beginning of next year, as projected inventories remain modest at ~1.5-Tcf for Mar'18 and at ~3.65-Tcf for Oct'18. Hence, 2018 prices should still average \$3.1 despite robust production growth, because of strong demand growth from LNG exports, pipeline gas exports to Mexico and industrials, particularly as major facilities such as petchem plants are coming online. With lower oil price expectations in 2019, associated gas production growth could slow, which would help support natural gas prices in 2019 at \$3.0. However, strong production should persist beyond 2019, thereby keeping prices at \$3 or below in the future.

Bull Case (20% Likelihood): Prices should average ~\$3.6 in 2018 and \$3.4 in 2019. Low oil prices could slow associated gas production in 2018 and 2019. Gas demand for power generation could rise more strongly because of stronger electricity demand growth, nuclear retirements and changes in the US tax law that depress renewables growth.

Bear Case (30% Likelihood): Prices should average ~\$2.5 in 2018 and 2019. 2018 production growth could be strong, perhaps reaching 7.2-Bcf/d y/y, even after accounting for a reduction in ethane rejection. Global LNG demand growth could disappoint, leading to slower growth in US LNG exports. Gas demand for power generation could be weak due to rising renewable generation (unaffected by changes in US tax law), greater energy efficiency and weak electricity demand.

Winter cases: The price response to winter weather should be asymmetric to the upside. Prices could average ~\$3.7 in 2018 if the 2017-18 winter were one standard deviation colder than normal, but a one standard deviation milder winter could lower average 2018 prices to ~\$2.6, all assuming that prices would normalize in 4Q'18. It is harder to raise production quickly in a low inventory environment, than it is to curtail production and induce more coal-to-gas switching in a low price environment.

2018 storage level on the low end of recent range should support prices

	Jan-17	Feb-17	Mar-17	Apr-17	May-17	Jun-17	Jul-17	Aug-17	Sep-17	Oct-17	Nov-17	Dec-17	Y/Y Change	Annual
Total Supply	70.5	70.8	71.0	70.6	70.7	71.4	72.5	72.8	72.7	72.3	74.0	72.5	-1.5	71.8
Prod	69.9	70.8	70.7	70.9	71.0	71.8	72.8	72.7	73.0	73.0	74.8	74.7	0.8	72.2
LNG Imports	0.4	0.3	0.2	0.2	0.2	0.2	0.2	0.3	0.2	0.2	0.2	0.2	(0.0)	0.2
Exports to Mexico	(3.9)	(4.1)	(4.2)	(3.9)	(4.2)	(4.5)	(4.4)	(4.4)	(4.4)	(4.3)	(4.1)	(4.7)	(0.5)	(4.2)
Imports from Canada	5.8	5.6	5.7	5.1	5.6	5.6	6.0	5.6	5.6	5.8	5.5	5.4	(0.2)	5.6
LNG Exports	(1.7)	(1.9)	(1.4)	(1.7)	(2.0)	(1.7)	(1.7)	(1.5)	(1.7)	(2.5)	(2.5)	(3.1)	(1.4)	(1.9)
Total Demand	93.0	82.7	80.9	63.7	60.8	62.9	68.9	67.0	64.2	65.9	77.9	92.1	(1.6)	73.3
IND	23.6	23.0	22.3	20.8	20.1	20.4	20.1	20.5	20.1	21.1	22.3	22.9	0.3	21.4
ResComm	42.2	33.5	30.6	16.4	12.2	8.7	7.8	7.9	9.0	13.4	26.9	40.5	0.3	20.7
EG	20.6	19.6	21.5	20.4	22.5	27.8	33.9	32.4	29.1	25.3	22.4	21.9	(2.4)	24.8
Pipe Use	2.4	2.1	2.1	1.6	1.5	1.6	1.7	1.7	1.6	1.6	1.8	2.2	(0.0)	1.8
Lease and Plant Fuel	4.2	4.3	4.3	4.3	4.3	4.4	4.4	4.4	4.4	4.4	4.4	4.4	0.1	4.4
Transport	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.1
Inventory (Bcf)	2,601	2,317	2,044	2,272	2,597	2,876	3,021	3,213	3,523	3,774	3,652	3,061		
Coal-Gas Switching (Bcf/d)	5.92	6.82	6.60	6.18	6.03	6.53	6.70	6.82	6.56	6.76	6.26	6.08	(0.96)	6.44

	Jan-18	Feb-18	Mar-18	Apr-18	May-18	Jun-18	Jul-18	Aug-18	Sep-18	Oct-18	Nov-18	Dec-18	Y/Y Change	Annual
Total Supply	73.0	73.7	73.2	74.4	75.4	75.7	77.2	77.1	77.0	77.0	76.2	76.6	3.7	75.5
Prod	75.0	75.5	76.5	77.0	78.0	78.0	79.0	80.0	80.5	80.5	80.5	80.5	6.2	78.4
LNG Imports	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	(0.1)	0.1
Exports to Mexico	(4.9)	(5.1)	(5.3)	(5.4)	(5.4)	(5.4)	(5.4)	(5.4)	(5.4)	(5.4)	(5.4)	(5.4)	(1.1)	(5.3)
Imports from Canada	5.8	6.2	4.9	5.8	5.8	6.1	6.8	6.4	5.8	5.8	5.0	5.4	0.2	5.8
LNG Exports	(3.1)	(3.1)	(3.1)	(3.1)	(3.1)	(3.1)	(3.3)	(4.0)	(4.0)	(4.0)	(4.0)	(4.0)	(1.5)	(3.5)
Total Demand	97.7	95.7	81.2	70.5	64.2	64.7	66.6	66.8	64.3	66.5	75.7	92.3	2.2	75.5
IND	23.4	24.5	21.9	21.5	20.5	20.6	20.1	20.1	20.8	21.0	22.3	23.2	0.2	21.6
ResComm	45.5	42.5	30.1	19.3	12.5	8.9	7.4	7.5	9.1	15.3	25.2	40.5	1.2	22.0
EG	21.9	22.1	22.6	23.6	25.2	29.2	32.9	33.1	28.4	24.1	21.9	21.9	0.8	25.6
Pipe Use	2.4	2.0	2.0	1.6	1.5	1.5	1.6	1.5	1.4	1.5	1.6	2.0	(0.1)	1.7
Lease and Plant Fuel	4.4	4.4	4.4	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	0.1	4.5
Transport	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.1
Inventory (Bcf)	2,305	1,716	1,483	1,606	1,958	2,300	2,622	2,946	3,342	3,644	3,644	3,174		
Coal-Gas Switching (Bcf/d)	6.03	6.08	6.42	6.72	6.72	6.66	6.63	6.64	6.64	6.57	6.37	6.13	0.03	6.47

	Jan-19	Feb-19	Mar-19	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19	Nov-19	Dec-19	Y/Y Change	Annual
Total Supply	76.2	75.1	73.7	75.1	75.1	75.4	77.0	77.7	77.1	77.1	77.3	77.7	0.7	76.2
Prod	80.5	80.5	80.5	81.0	81.0	81.0	82.0	83.0	83.0	83.0	84.0	84.0	3.5	82.0
LNG Imports	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(0.1)	-
Exports to Mexico	(5.5)	(5.8)	(5.8)	(5.8)	(5.8)	(5.8)	(5.8)	(5.8)	(5.8)	(5.8)	(5.8)	(5.8)	(0.4)	(5.8)
Imports from Canada	5.8	6.2	4.9	5.8	5.8	6.1	6.8	6.4	5.8	5.8	5.0	5.4	-	5.8
LNG Exports	(4.6)	(5.8)	(5.8)	(5.9)	(5.9)	(5.9)	(5.9)	(5.9)	(5.9)	(5.9)	(5.9)	(5.9)	(2.3)	(5.8)
Total Demand	98.2	96.1	82.2	71.4	65.5	66.1	68.1	68.2	65.4	67.8	76.7	92.9	1.0	76.6
IND	23.7	24.8	22.3	21.8	20.8	20.9	20.4	20.4	21.1	21.3	22.6	23.6	0.3	22.0
ResComm	45.5	42.5	30.1	19.3	12.5	8.9	7.4	7.5	9.1	15.3	25.2	40.5	-	22.0
EG	22.1	22.2	23.3	24.2	26.2	30.2	34.1	34.1	29.2	25.1	22.6	22.2	0.7	26.3
Pipe Use	2.4	2.0	2.0	1.6	1.5	1.5	1.6	1.5	1.4	1.5	1.6	2.0	-	1.7
Lease and Plant Fuel	4.4	4.4	4.4	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	-	4.5
Transport	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	-	0.1
Inventory (Bcf)	2,501	1,936	1,691	1,809	2,111	2,402	2,673	2,971	3,336	3,598	3,597	3,142		
Coal-Gas Switching (Bcf/d)	6.08	6.21	6.71	7.15	7.18	7.13	7.11	7.13	7.12	7.03	6.80	6.52	0.38	6.85

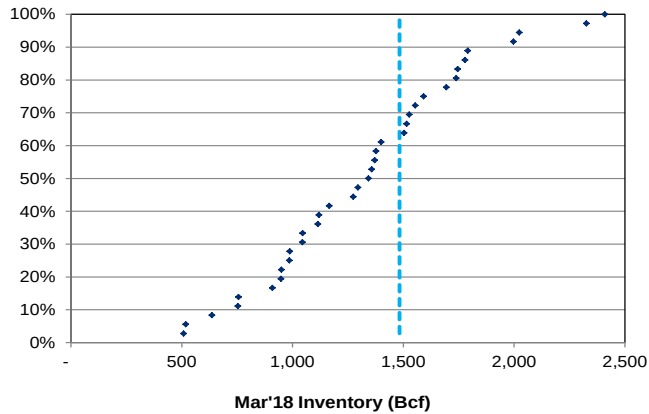
...But 1.5-Tcf in Mar'18 storage is relatively low, which could fall to less than 1-Tcf if the winter is 1-SD colder than normal...

...even with 6.2-Bcf/d of production growth y/y, having 3.6-Tcf in Oct'18 storage is barely enough. More extreme weather in winter or summer could lower this substantially.

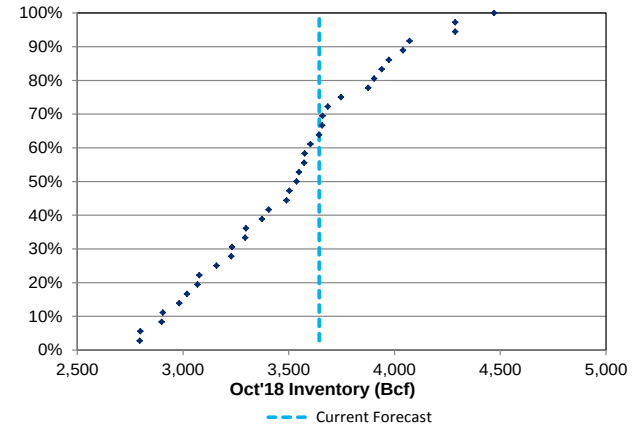
Weather scenarios also show asymmetric upside to prices this winter

Results from running multiple winter weather scenarios show that a one-standard deviation colder-than-normal winter could lead to an \$0.8 increase in the rolling 12-month strip price vs. a \$0.4 decline in a one standard deviation milder-than-normal winter scenario, assuming the market targets a 3.7-Tcf end-Oct'18 inventory level.

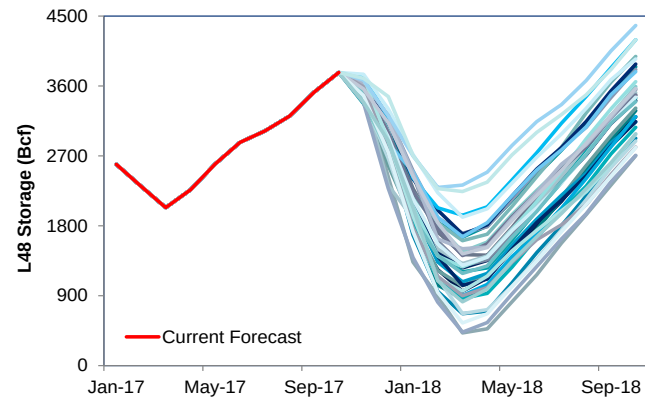
Cumulative distribution curve of possible Mar'18 gas inventories after running 36 years of weather scenarios starting 1980



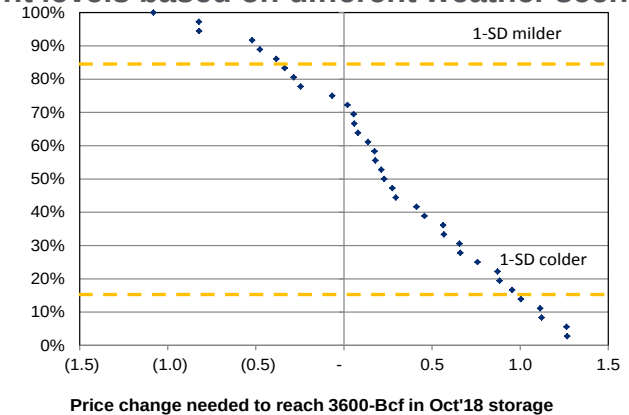
Cumulative distribution curve of possible Oct'18 gas inventories after running 36 years of weather scenarios starting 1980



Different realizations of gas inventories based on 36 years of different weather scenarios

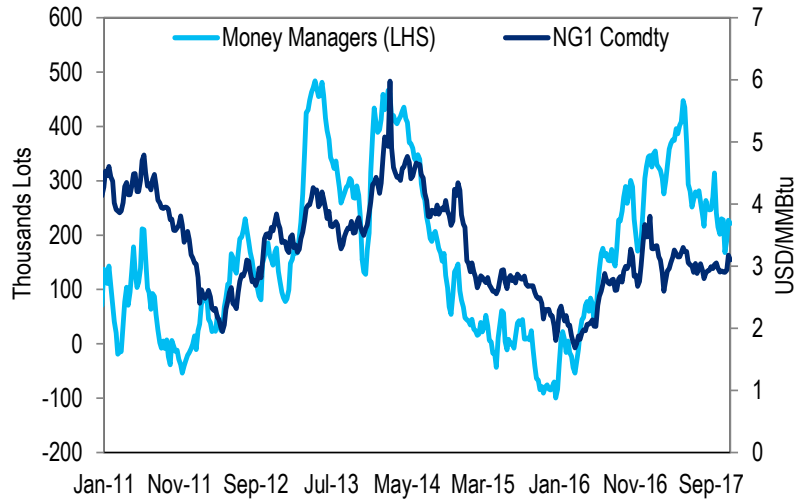


Cumulative distribution curve of possible winter strip price from current levels based on different weather scenarios

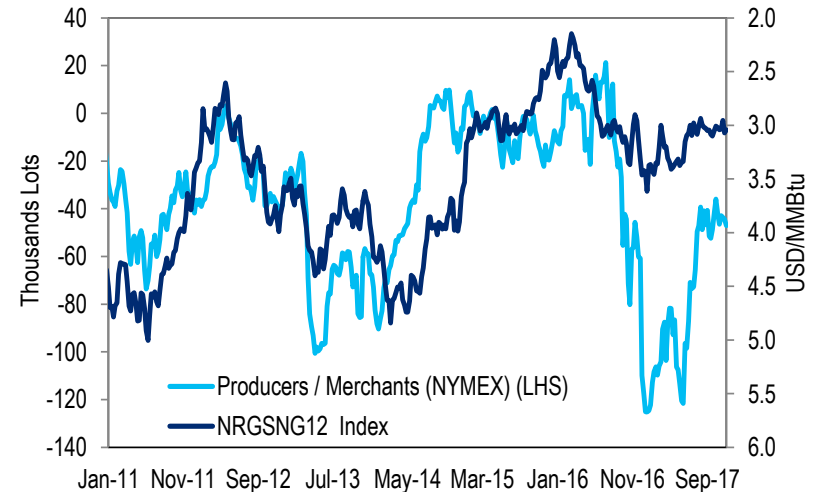


But mild weather, stagnant pricing so far kept funds from going bullish for now

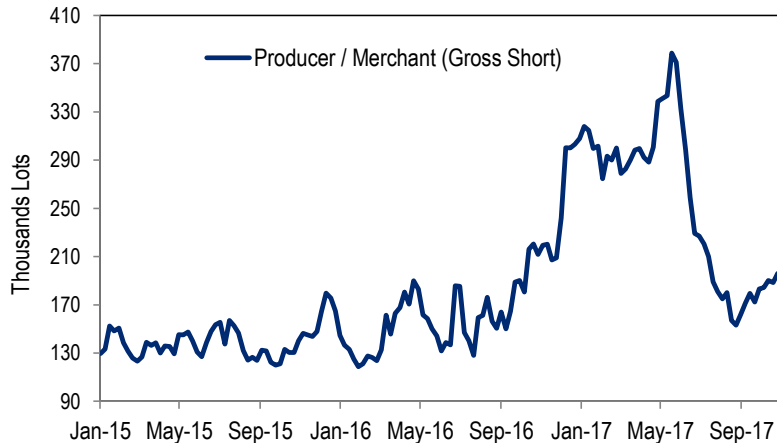
Managed money reduced their net length, as short-term mild fall weather trumped low projected storage in 2018



Producers/users/merchants trimmed their net short positioning, as some consumers came in to hedge



Producers/users/merchants also have a smaller gross short positioning, perhaps as producer hedges roll off amid sub-\$3 pricing, leaving some room for price upside



- Managed money net positioning is somewhat correlated with prompt month gas price movements. Managed money reduced their net length, as short-term mild fall weather trumped low projected storage in 2018
- Producer/user/merchant has reduced net short positioning (NYMEX only) significantly.
- Producer/user/merchant gross shorts are pulled back from the surge vs. mid-May, potentially because of hedges rolling off, producers not hedging future production as much due to lower futures prices, and possibly more hedging from consumers

Demand to rise due to industrial growth, rising exports, power market dynamics

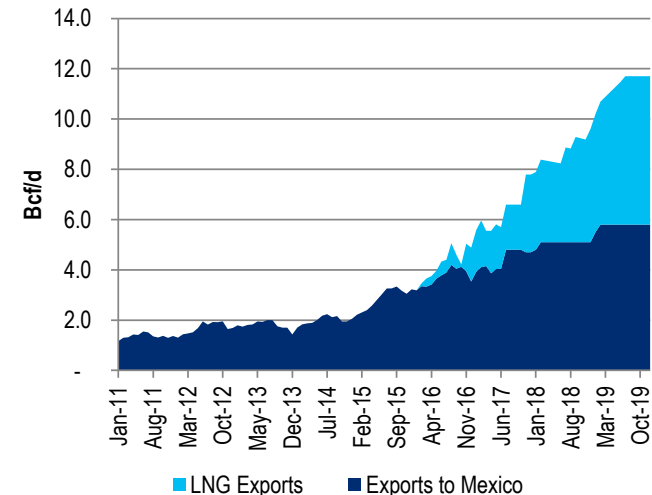
- **Natural gas exports should surge, as new LNG export facilities come online and pipeline developments in Mexico debottleneck much of country.** With minimal capex on upstream oil and gas production, Mexican gas production should keep falling over the next couple of years before recovering, though demand should continue to strengthen as gas gains shares in both power generation and industrial applications. LNG exports are also rising, with train 4 at Sabine Pass LNG and the Cove Point terminal coming online in 4Q'17. A bigger wave should come in 2019 or as early as late 2018.
- **Industrial gas demand should grow** as major industrial facilities come online, including methanol, ammonia, urea and ethylene, with additions in 2017 affecting the full-year 2018 balance.
- **Gas demand for power generation could get a lift due to strong economic growth, accelerated “denuclearization” and, longer term, slower growth rate of renewables.**

Although power demand was perceived to have decoupled with economic growth in the past few years, what actually happened was that, besides the rise of distributed energy resources reducing the demand for electricity generated by major power plants, the subdued economic growth was more than offset by gains in energy efficiency. But the robust improvements in unemployment rate and GDP growth could start to more than offset efficiency gains.

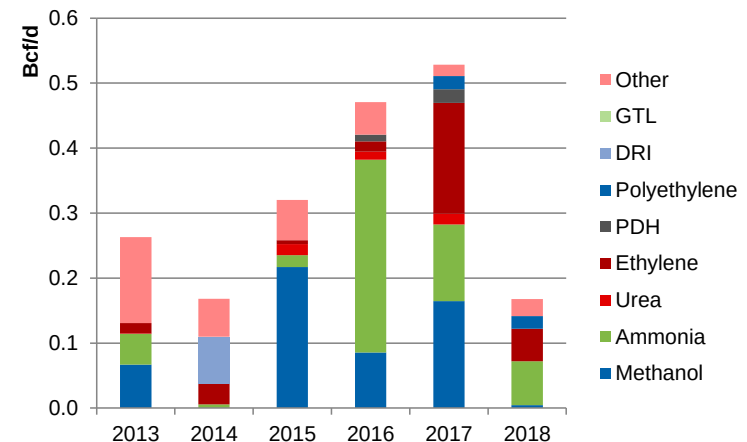
In a low natural gas and low power price environment, a number of nuclear reactors have announced their retirements, which should help support gas demand for power generation

The growth of renewables could slow if an import tariff on solar panels were imposed, making the cost of solar installation higher, and if proposed changes in US tax laws, if passed, sharply reduce the economics of tax equity and inflation adjustments.

Exports are key to lifting gas “demand,” with both LNG exports and pipeline exports to Mexico rising



Industrial gas demand addition very substantial in 2017, with the full effect showing up in 2018



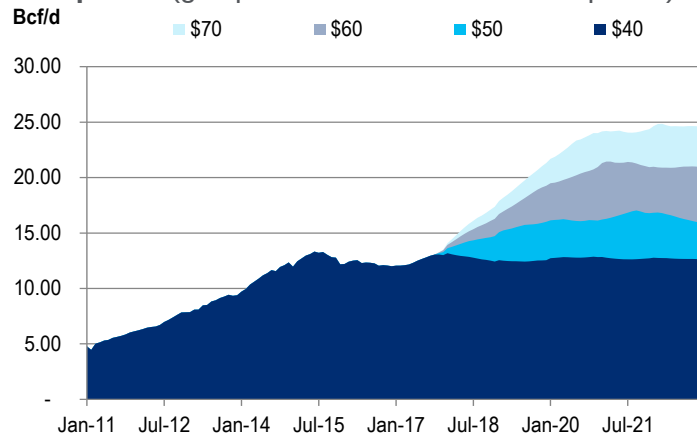
Politics looms large in affecting North American market fundamentals, pricing

- **North American Free Trade Agreement on US energy exports and imports – gas price negative**
 - We do not expect NAFTA renegotiations to impact the energy sector, given Trump's enthusiastic support for the industry and the importance of energy trade to the US economy. But recent renegotiations marked a change in tone from the US and considerably heightened the risk that NAFTA negotiations could breakdown.
 - The US has long run a trade surplus with Mexico on natural gas. Recently this surplus has grown significantly as exports have ballooned with new cross-border export capacity and the growth of US shale gas production, growing from 400-MMcf/d in 2001 to 3.8-Bcf/d in 2016 and 4.6-Bcf/d YTD in 2017. Strong gas exports to Mexico are key to supporting US gas prices.
 - Without NAFTA, guarantees of zero-tariff trade of natural gas would vanish. Overall, this would be primarily a negative for US producers. This may also result in Mexico and Canada ceasing to qualify as Free Trade Agreement (FTA) countries in the US.
 - Given that NAFTA's dissolution would have severe consequences for the US energy industry, free trade in energy should be preserved even in this worst case scenario. There is also little incentive for Canada or Mexico to impose tariffs on US energy.
- **The US Department of Energy rule-making proposal and FERC's independence – gas price negative**
 - Although the Energy Department (DOE) has asked FERC to consider new rules to essentially give support to coal and nuclear power plants for their so-called resiliency benefits to the grid, there are numerous downsides and obstacles to implementation.
 - The net loser from a successful implementation of this rule would be natural gas-fired power plants at the power sector level, as coal and nuclear could get help.
 - But this proposal looks likely to upset more parties than it pleases. This uncommonly broad opposition means there will likely be significant hurdles to implementation including challenges to the DOE's aggressive timeline and litigation risks. The independence of FERC is also at stake if FERC, now with a Republican majority, largely adheres to DOE's directive.
- **Import tariffs on solar panels – gas price positive**
 - If imposed, a tariff on low-cost solar panels imported from China would raise the cost of solar PV in the US, which should slow the growth of solar installations. This would be a positive for grid-connected power demand and gas demand for power generation. However, much of the solar industry, including developers and installers, are opposed to this tariff.
- **Changes in US tax laws on the economics of renewable energy development – gas price positive**
 - Proposed changes to US tax laws, if passed, would lower corporate tax rates (which affects tax equity), reduce interest deductibility (which affects the appeal of debt) and possibly eliminate accelerated depreciation and PTC's inflation adjustment, with the latter crucial to the economics of many wind projects. A slowdown in renewables growth should benefit gas demand.

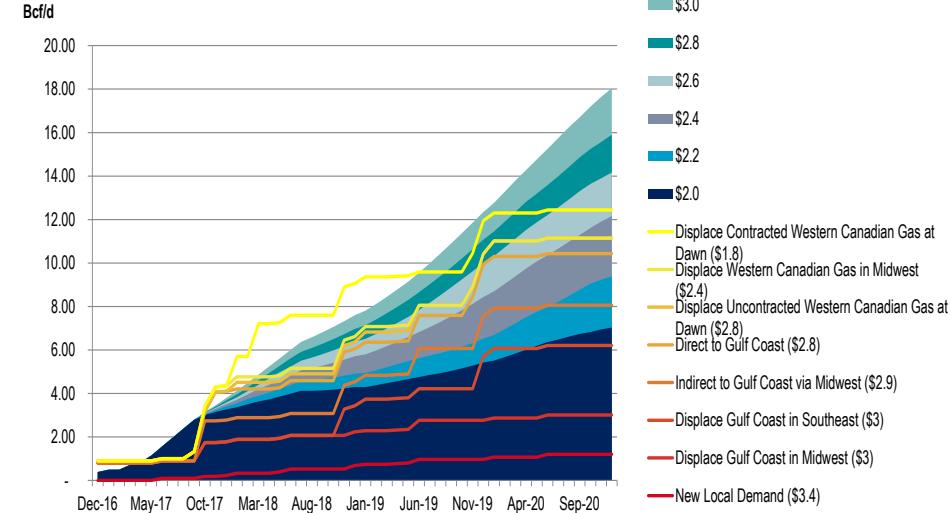
While production should soar, constraints and capital discipline could limit gains

- **Production growth in 2018 and 2019 should be strong**, with 2018 production up by ~6.2-Bcf/d y/y. It could be more than 7-Bcf/d if not for the reduction in ethane rejection, as ethane would be exported or used in new petchem plants. Such growth is helped by new pipelines debottlenecking the Northeast and oil wells possibly producing more associated gas than expected due to higher GOR (gas-oil ratios).
- **US Northeast production has been rising, with new pipelines coming online debottlenecking the region.** But environmental and regulatory challenges could still delay them. The Atlantic Sunrise pipe was almost delayed by a court challenge, while the Leach XPress has been delayed by two months.
- **Associated gas production from oil plays, such as the Permian, should also surge**
- **But producer capital discipline could limit output growth**

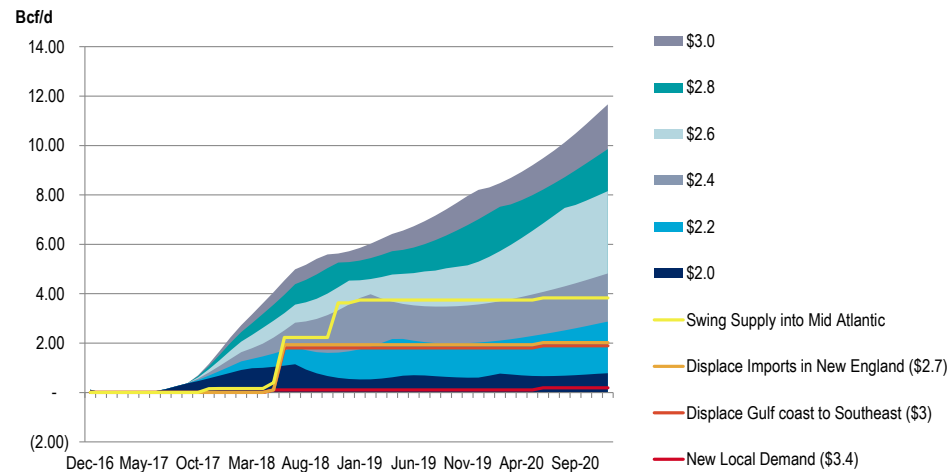
How much associated gas output would grow depends on oil prices (gas production at different WTI prices)



Gas-on-gas competition to limited how much Utica/SW Marcellus gas could grow (production at different flat Dominion South prices)



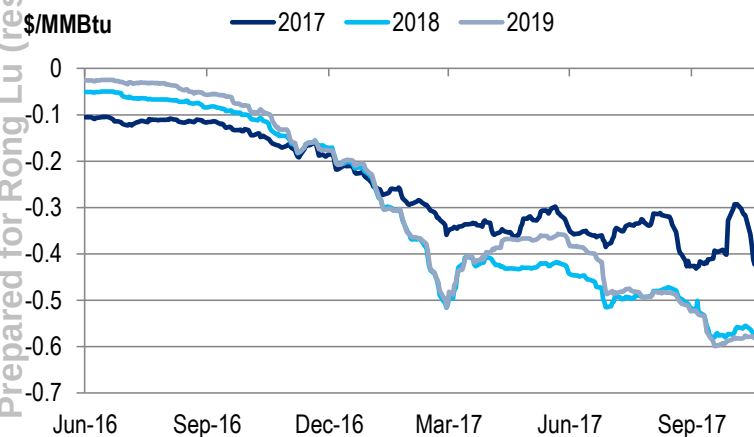
NE Appalachia production has great potential for growth, but limited by pipe constraints and gas-on-gas competition elsewhere. (production at different flat Transco Leidy prices)



Basis should see wild swings on strong production but lumpy pipe development

- **Waha basis, representative of gas prices in the Permian basin, should start to widen substantially as oil and associated gas production growth continues.** For every 1-mb/d of tight oil production growth, associated gas production could rise by ~2-Bcf/d. Unlike oil and NGLs, which could be put on a truck or railcars, gas must be transported on pipes. The congestion expected to develop could widen basis differentials to perhaps ~70c or \$1 below Henry Hub. These levels are needed to justify new pipe and sending more gas to the Midwest to compete against low-cost Marcellus/Utica and Western Canadian gas.
- **Waha basis moves could drive basis in the Rockies and Midcontinent.** Rockies gas has to price low enough to compete with Waha gas in California, Midcon and the Midwest. Midcon, propelled by SCOOP/STACK growth, also directly competes with Waha gas

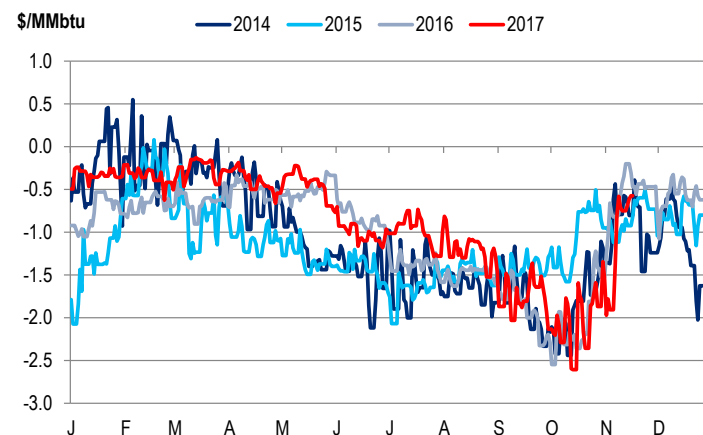
Forward Waha basis strip has declined but should fall to at least \$0.7 or \$1/MMBtu below Henry Hub



Source: Platts, Bloomberg, Citi Research

- **Northeast basis differentials to Henry Hub should narrow as new pipelines come online.** For Dominion South Point, representative of SW Marcellus/Utica, basis should narrow this winter as new pipes enter services and cold weather in winter boosts demand. For Transco Leidy, representative of NE Marcellus, basis should narrow starting in mid-2018, when the Atlantic Sunrise pipeline should begin operation.
- **But seasonality could still substantially widen basis in fall.** As seen in the past few years, basis could widen to more than \$2 below Henry Hub post-summer. Steady production growth bumps up against falling demand during fall, but new pipelines generally don't come online until winter, thereby creating massive production gluts that depress prices.
- **Western Canada gas, with AECO basis likely at \$1 below Henry Hub, should intensify the gas-on-gas competition with US Northeast gas,** as the latter tries to capture greater shares in the Eastern Canada and US Midwest markets.

Northeast prices (Dominion SP pictured) tend to exhibit strong seasonality and wild swings, particularly between fall and winter



High global prices keep US LNG export arbitrage wide-open

With US prices hovering around the \$3 range in 2018...

US Henry Hub Forecasts (\$/MMBtu)	1Q'18	2Q'18	3Q'18	4Q'18	1Q'19	2Q'19	3Q'19	4Q'19	2017	2018	2019
Base	3.2	3.1	3.1	3.0	3.1	3.0	2.9	2.9	3.0	3.1	3.0
Liquefaction fuel %											
LNG cost components (\$/MMBtu)											
Liquefaction fuel at 15% of gas cost	0.5	0.5	0.5	0.5	0.5	0.5	0.4	0.4	0.5	0.5	0.4
Transport											
Europe	0.6	0.6	0.6	0.6	0.6	0.5	0.5	0.5	0.5	0.6	0.6
Asia	1.2	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Regas	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Delivered gas cost to											
Europe (with regas)	4.5	4.3	4.3	4.3	4.3	4.3	4.1	4.1	4.3	4.4	4.2
Europe (without regas)	4.3	4.1	4.1	4.0	4.1	4.0	3.9	3.9	4.0	4.1	
Asia (without regas)	4.9	4.7	4.6	4.6	4.6	4.6	4.4	4.4	4.5	4.7	4.5
Current NBP forward	7.3	6.1	5.9	6.7	7.3	5.9	5.7	6.5	5.9	6.5	6.3
Implied Russian gas at German border											
Oil price: forward curve	5.8	6.0	5.9	5.9	5.8	5.7	5.7	5.6	5.0	5.9	5.7
Oil price: Citi forecast	5.5	5.4	5.2	5.1	4.9	4.6	4.5	4.6	5.0	5.3	4.7
LNG prices (Brent-implied)											
Oil price: forward curve	6.9	6.9	6.8	6.7	6.7	6.6	6.6	6.5	6.0	6.9	6.6
Oil price: Citi forecast	6.5	6.1	5.9	5.8	5.5	5.2	5.3	5.4	5.9	6.1	5.4
Oil prices (\$/bbl)											
Forward curve	63	62	62	61	60	60	59	59	56	62	60
Citi forecast	57	54	54	52	48	47	49	50	55	54	49
Coal-to-gas switching (Europe)											
Coal price (Newcastle \$/ton)	80	75	76	80	75	72	75	78	87	78	75
Generation cost: fuel + transport only (\$/MWh)	30	28	29	31	29	28	29	31	34	30	29

...Delivered costs of US LNG (variable cost only) should be in the low to mid-\$4 range...

...Which are much below NBP forwards. This keeps US LNG export arbitrage open. But the long-awaited surge in global LNG supply should drag NBP prices lower.

Oil-indexed LNG prices look higher than US LNG delivered costs, but the 2018 ramp-up of global LNG supply could flood market sufficiently to further weaken oil-indexation, lowering global LNG prices.

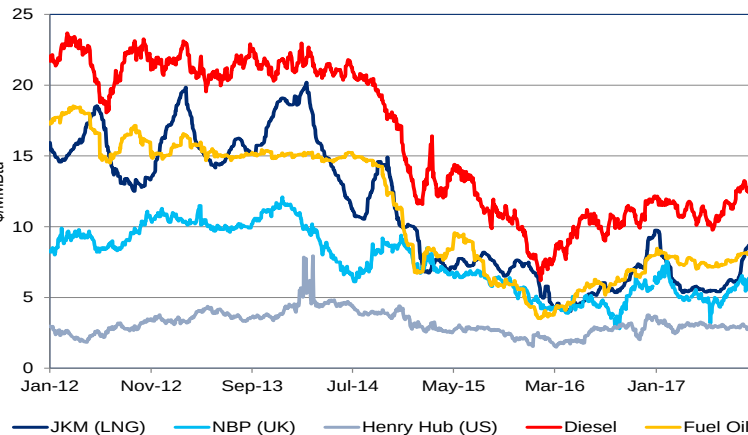
Between now and the end of 2017, multiple terminals are coming online, including Yamal in Russia, Sabine Pass's train 4 and Cove Point in the US, and Wheatstone in Australia.

Other Australian projects would come online in 2018, with a second wave of US projects entering service in 2019

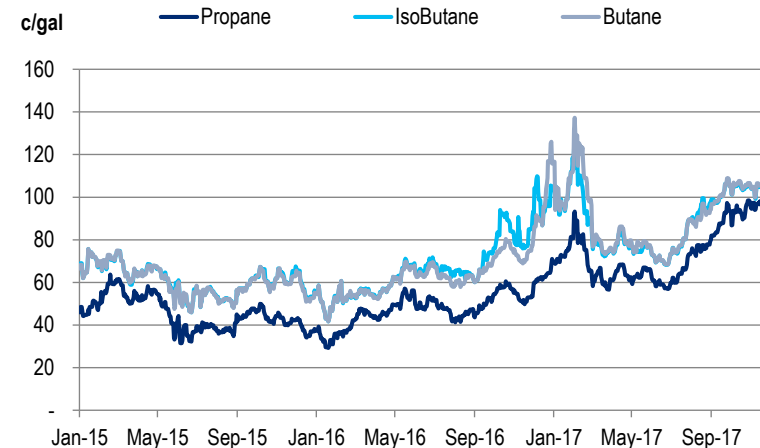
Global prices could spike in winter amid tightness in many heating fuel markets

- **LNG prices have begun their year-end surge early, having risen to nearly ~\$10/MMBtu, vs the mid-\$5 range seen this past summer.** LNG prices are highly seasonal because of large demand differences between winter and late spring, insufficient storage and a lack of spare capacity in supply. **But US LNG exports are not large enough yet to sway global prices.**
- **Impact from oil/diesel/fuel oil:** LNG prices could rise to \$12/MMBtu or higher this winter, a ceiling set by diesel (heating oil) prices in power generation terms (not oil-indexed terms). High prices of fuel oil, as a competing fuel, also help to lift LNG prices. With low middle distillate stocks globally amid a relatively tight crude oil market, heating oil prices could spike if the winter in Europe or Asia gets cold. Higher heating oil/gasoil prices could raise the LNG price ceiling further.
- **Impact from coal:** Prices of thermal coal act as a price floor giving support to LNG and European gas prices, partly through coal-to-gas switching. Although fundamentals of lower quality seaborne coal have loosened up already, just as China is making a concerted effort to raise production and limit price increases, prices of high quality coal (Newcastle and API2), which is generally consumed in markets that burn LNG as well, have stayed in the \$90/t range as fundamentals remain tight there.
- **Impact from LPG:** Prices of LPG, also a heating fuel, are elevated, with US Mont Belvieu prices surging from a low 50% ratio to Brent to a high 60% ratio from late summer to Nov'17. This likely reflects strong demand from petchem as well as residential/commercial sectors, particularly out of Asia. Although LPG prices could stabilize from here, an upside surprise to winter demand could lead to sharp price spikes, with spillover effects to other heating fuel markets. A tight naphtha market also gives support to LPG prices from the petchem side

LNG: prices spike almost every year, close to the marginal cost of diesel-fired generation



LPG: prices at Mont Belvieu and Japan have both surged



In particular, UK/EU gas has upside winter risk; 2019 is much more bearish

\$/MMBtu	1Q17	2Q17	3Q17E	4Q17E	1Q18E	2Q18E	3Q18E	4Q18E	1Q19E	2Q19E	3Q19E	4Q19E	2017	2018	2019
NBP Forecast	6.0	4.8	5.4	6.9	6.5	5.5	5.3	5.8	5.6	4.7	4.8	5.5	5.8	5.8	5.2
NBP Month Ahead Futures	6.0	4.8	5.4	6.9	6.9	5.8	5.9	7.0	6.8	5.6	5.8	6.8	5.8	6.4	6.2
Russian Oil Linked Price	5.8	5.0	5.3	5.7	5.7	5.9	5.8	5.6	5.4	5.1	5.0	5.0	5.5	5.7	5.1
US LNG in Europe	4.3	4.4	4.2	4.4	4.6	4.5	4.5	4.4	4.5	4.3	4.1	4.1	4.3	4.5	4.2
Average EU Coal-to-Gas Switching	4.9	4.8	5.5	6.0	5.7	5.6	5.4	5.4	5.3	5.3	5.3	5.2	5.3	5.5	5.3

- **Strength in European gas prices is likely to linger through winter 17/18 given the current tightness in oil, coal, LNG and European power markets.** Western European gas storage started winter at a healthy 90% capacity fill level (77-bcm) but UK and European hub gas prices are very much “price taking” markets and winter risks remain skewed to the upside.

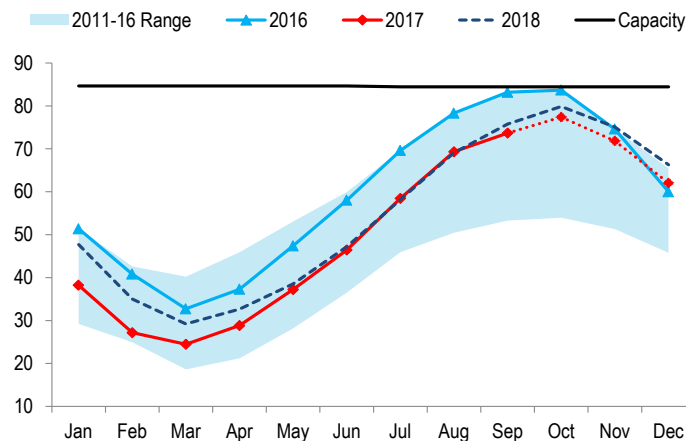
● **The shutting of the UK’s Rough gas storage facility, caps on Groningen output, maturing domestic gas fields and repeated issues with France’s nuclear power plant fleet are leaving Western Europe with an increasing gas deficit.** For 2017, it is expected to reach 800-mcm/d (~28-bcf/d). This makes relative energy costs extremely important and entering winter with Asian LNG at ~\$10/MMBtu, Brent crude over \$60/bbl and API2 coal over \$90/ton and with this tightness likely to persist, then European gas prices are likely to remain at-or-above current futures pricing this winter.

● **Tightness in near-term gas markets is expected to fade in 2018 and in an even more material way in 2019.** NBP and TTF futures for 2H’18 and 2019 seem too strong in our view as LNG and Russian gas exports are very much in-the-money and with Russian gas flows having upside flex and an additional 80-bcm (220-mcm/d) of global LNG supply arriving by 2019, then European balances will likely turn more bearish. In addition to this, the pipeline of European wind and solar capacity additions implies further reductions in thermal power generation while we are bearish oil and coal prices (vs. current futures) in 2018 and 2019.

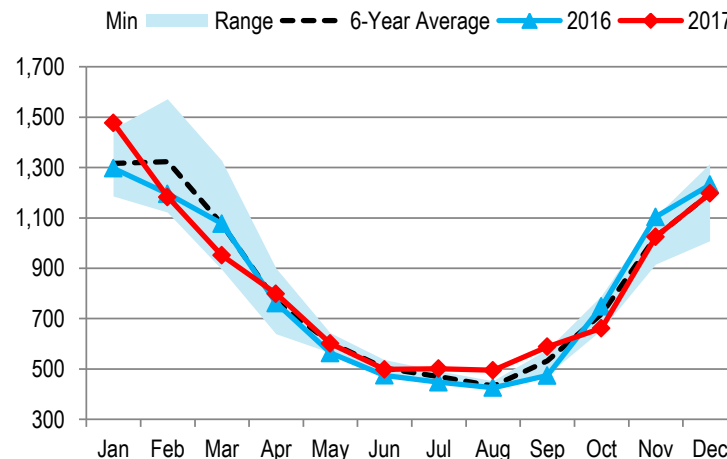
Near term demand driven by coal-gas switching; long-term still on a decline

Gas balances will likely tighten into 1Q'18 but if temperatures are equivalent to Jan-17 (+200-mcm/d of additional demand) then storage will be significantly lower. 2018 thermal power demand will very likely fall and if current coal fleet utilization holds than gas production from power should be flat y/y in 2018 at 240-mcm/d.

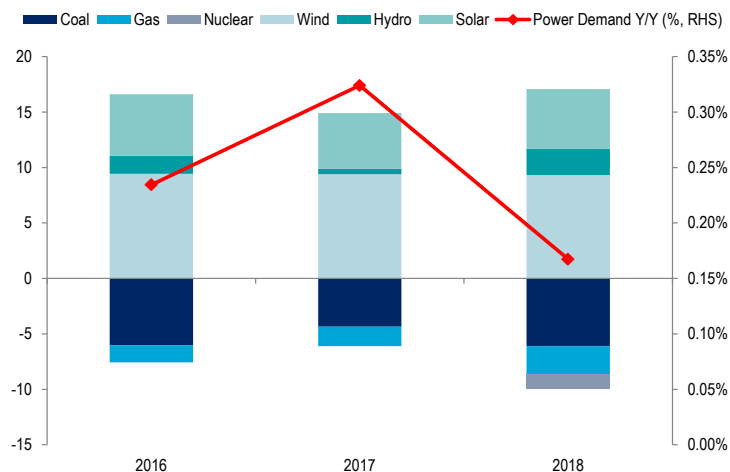
Western Europe Gas Balance (bcm)



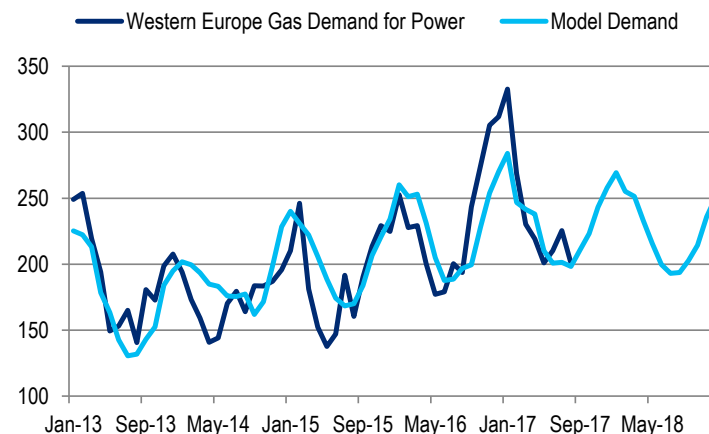
Western Europe Non-Power Gas Demand (mcm/d)



Western Europe Capacity Additions by Type (GW)



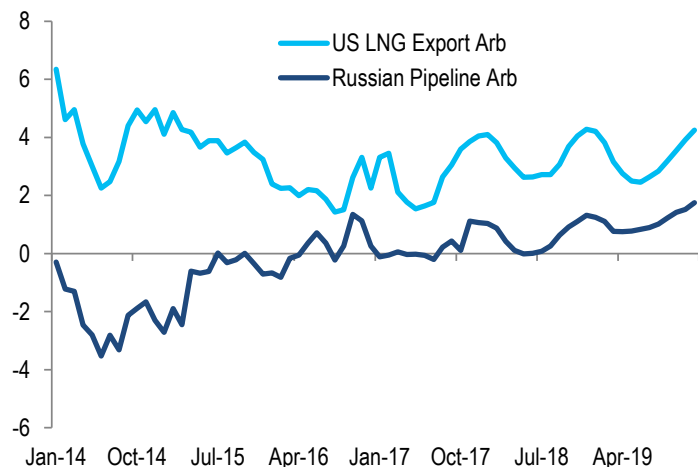
Western Europe Gas Demand for Power based on current coal/gas futures (mcm/d)



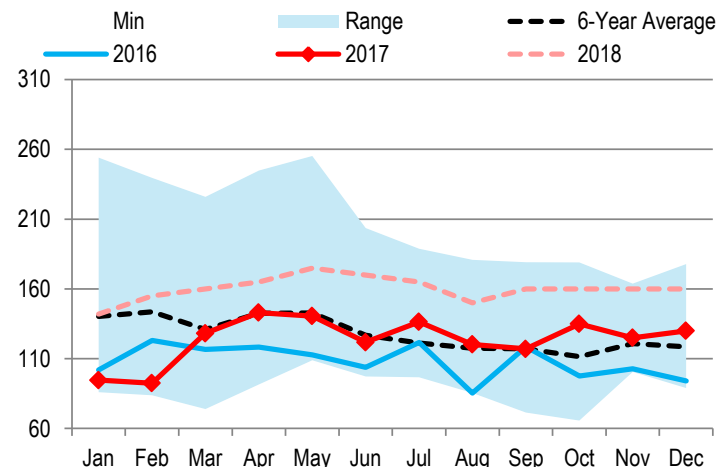
LNG and Russian exports to Europe look too high at current pricing

In 2018 and 2019, there is unlikely to be enough demand or natural supply declines that compensate for Russian gas volumes staying at current levels as well as LNG flows into Europe increasing as new projects come on stream. With oil-linked prices expected to fall into 2019, hub prices will likely have to follow to deter gas flows.

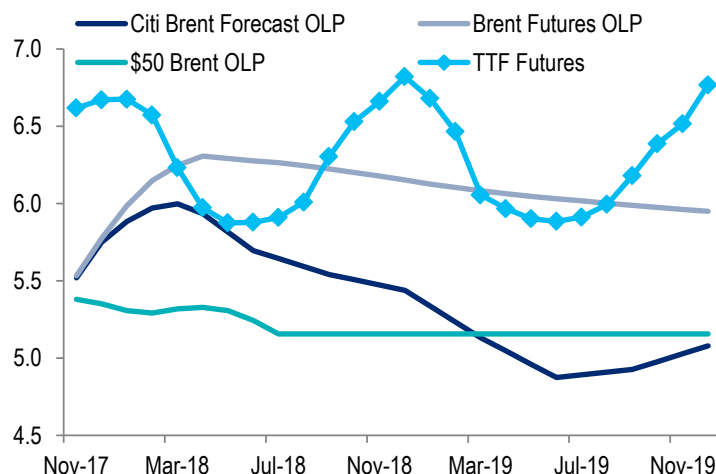
European Gas Import Arbs (\$/MMBtu)



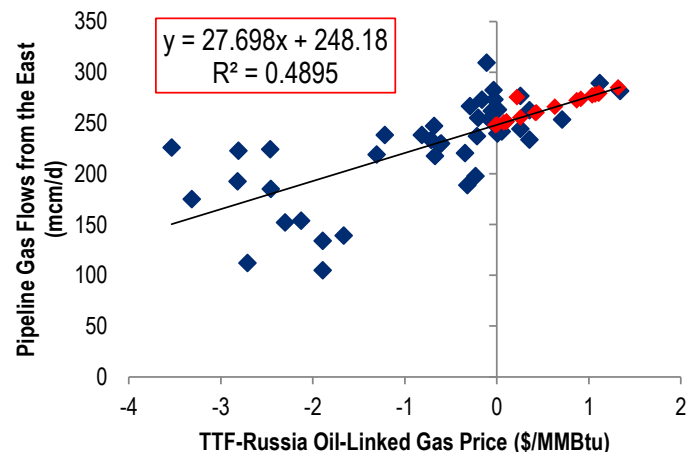
Western Europe LNG Imports (mcm/d)



Russian Oil Linked Prices at Different Brent Prices (\$/MMBtu)



European Gas Imports from the East vs. Price

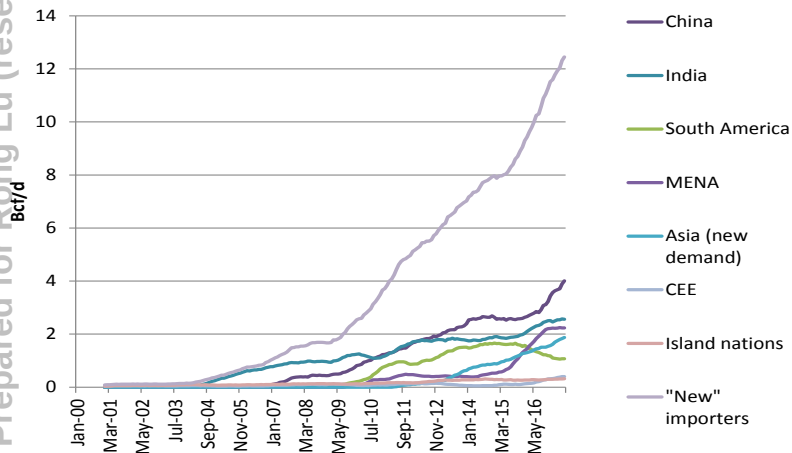


Bull-bear price cases depend on demand growth, Russia price war, other fuels

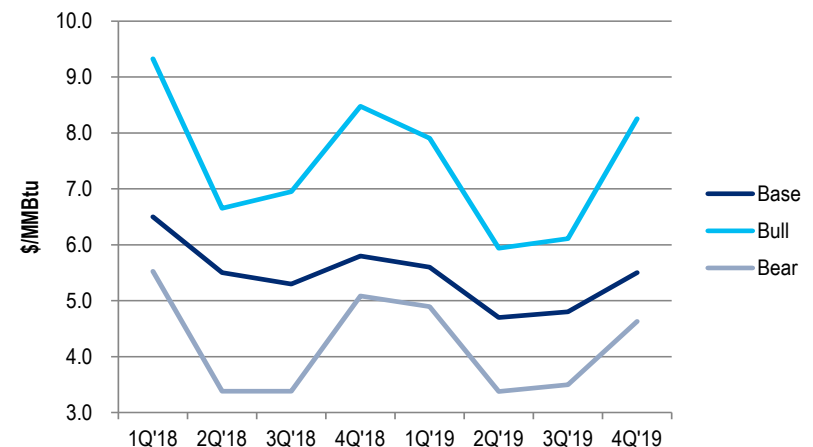
- Global LNG price scenarios hinge mainly on the pace of demand growth, marginal LNG costs and diesel, fuel oil prices:
 - If the pace of LNG demand growth from “new” importers outside of the “Original 6”^{*} accelerates, then the expected loosening of the market may not happen. The pace has quickened, with about 4 Bcf/d of growth in the last 2 years alone. Note that LNG liquefaction capacity should grow by more than 12-Bcf/d between now and 2020, or 4-Bcf/d per year.
 - Fuel substitution, including Japanese nuclear restarts, the closure of coal plants some due to environmental rules, are key
 - Oil prices matter, from the oil-indexation of contracted LNG, to fuel-switching from gas to fuel oil or diesel in power generation.
 - US LNG looks to be the marginal supply, while geopolitical disruptions (e.g. Yemen), new terminal delays would limit supply
 - **Bull-bear scenarios:** Prices in 2018 and 2019 could average \$6.1 and \$5.4 in the base case, with prices in the bull case averaging \$8.5 and \$7.7, while prices in the bear case averaging \$5.0 and \$4.7.
- European prices depend mainly on Russia’s supply and pricing strategy, the power market and prices of other fuels:
 - A price war between LNG and Russia pipe gas could start especially if a lot more excess LNG enters Europe
 - If some nuclear reactors shut down due to safety issues, then the power market could quickly tighten and raise gas demand
 - Coal prices falling due to a loosening of tight fundamentals should lower the price floors of gas through coal-gas switching
 - **Bull-bear scenarios** : Prices in 2018 and 2019 could average \$5.8 and \$5.2 in the base case, with prices in the bull case averaging \$7.9 and \$7.1, with prices in the bear case averaging \$4.3 and \$4.1.

Prepared for Rong Lu (research email)

Recent LNG demand growth driven by new importers



NBP bull-base-bear price scenarios



^{*} The Original 6 include: Japan, South Korea, Taiwan, Western Europe, North America and Turkey, as they were LNG importers even back in the 1990s.

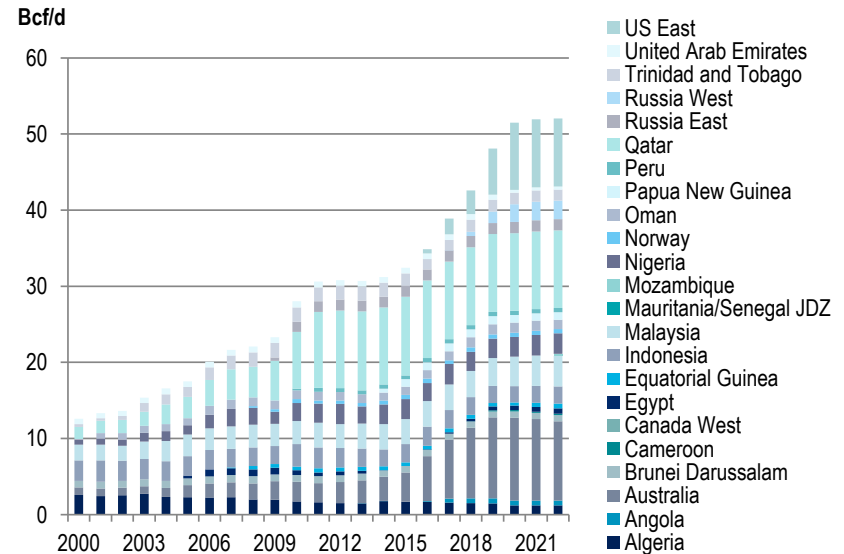
Source: Woodmac, Bloomberg, Platts, Citi Research

Profound change occurring in the global market, with new supply, pricing, hub

LNG supply is set to increase sharply...

- **Capacity should rise from 39-Bcf/d (300-mtpa) now to 52-Bcf/d (400-mtpa) by 2020**, with an increase of 3.7-Bcf/d (or ~28-mtpa) in 2018. Australia will lead the increase, up over 1.5-Bcf/d y/y, followed by the US in 2019 with nearly 3-Bcf/d.
- **Qatar, Russia and the US are likely to dominate the next wave of supply beyond 2022, with each country potentially producing 100 mtpa.** They all have vast reserves, experience developing export facilities, government support and aren't subject to domestic market tightness, unlike Australia or other greenfield facilities elsewhere.
- **Prices outside of winter could fall to \$4 or lower but winter prices could still spike, so that annual average prices could average around mid-\$5.**
- **With spot trading taking off displacing oil-indexation...**
- **Asia:** The finding in Japan that destination clauses are anti-competitive could unleash 30-mtpa of LNG to the spot market, encouraging anticompetitive probes elsewhere.
- **Middle East:** Qatar would raise exports by 30% to 100-mtpa in 4 years to recapture market shares.
- **Europe:** The UK becomes dependent on LNG imports, after the closure of Rough storage, and could foster spot LNG trading in the Atlantic basin, especially in winter.
- **North America:** Gas-on-gas competition gets more intense and make US LNG more competitive globally, connecting the North American market with global LNG fundamentals.

Global LNG supply should finally surge, with new projects from Australia and the US



...And pricing hubs forming to facilitate price discovery

- **Price markers are proliferating:** The Japan-Korea Marker is the most liquid. It has swaps assessed out three quarters. Prices are also assessed in key LNG importing areas. China, Japan and Singapore look to establish actual trading hubs.
- **The US Gulf Coast (USGC) could be well-suited for an LNG trading hub to develop:** It has breadth, as US LNG has reached South America, Europe, Asia and the Middle East. It has depth, as its location allows various arbitrage and hedging strategies with the world's most liquid onshore gas and power markets. Pricing is based on gas-on-gas competition. It is poised for growth, with lower capex for new builds. North America is a gigantic reserve, with midstream fully developed.

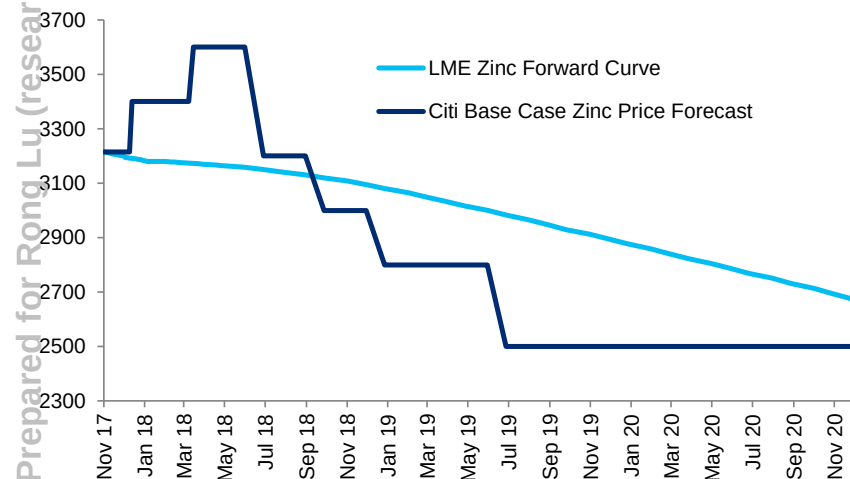
4. Base Metals: Zincs 'last hurrah' and quantifying the \$8,000/t copper bull case

Base metals – top trade: Zincs ‘last hurrah’

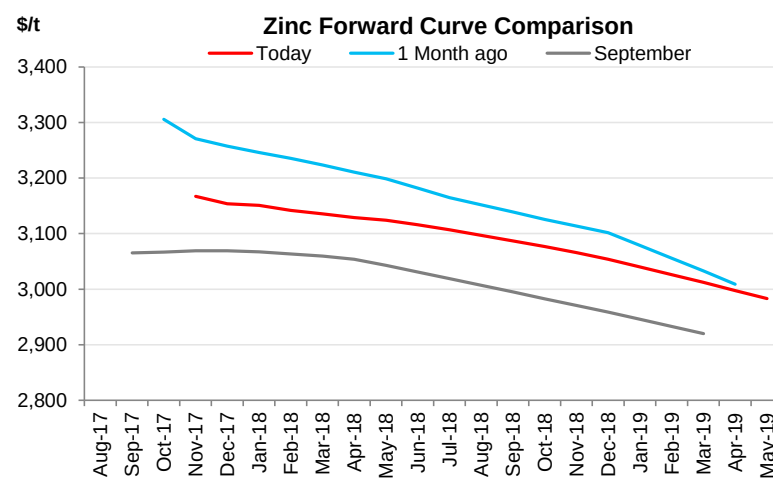
Zinc is set to continue its two year bull run over the next 6-9 months - we see prices rising by >10%/25% in our base/bull cases by 2Q'18. 2018 zinc rally or otherwise, we recommend producers consider selling long-dated zinc (2H'19/2020) over the next 6-9 months, as these long dated prices are very unlikely to last.

- **Our base case forecast is that prices rise by >10% to \$3,600/t during 2Q'18, and our bull case a 25% increase to \$4,000/t by 4Q'18.** Our zinc price forecast is based on our work that points to a) refined stocks drawing down to levels historically consistent with a much larger backwardation (zinc spreads are set to tighten) b) a larger than consensus deficit for this year c) Chinese zinc mine supply being unlikely to respond to high prices during 1H'18 d) c.150-200kt of zinc being lost from induction furnace losses in China post-June 2017, and finally e) our view that Glencore will only gradually ramp up starting from 2H'18.
- **Longer term, we have equally high conviction that 2018 will be zincs ‘last hurrah’ for this price cycle.** On the zinc supply side, c.80-90% of zinc projects work (on paper) at current zinc forward prices. We also identify nearly 10% mine supply growth during 2019/2020. On the demand side, absent a large take up of zinc as a new battery metal, zinc is the oldest of ‘old—economy’ exposed metals, with Chinese property and infrastructure accounting for >50% of demand. As China transitions away from construction heavy FAI in to its ‘new era’, the outlook is for a notable slowing in Chinese zinc demand growth.

Our base case forecast is that spot prices rise by >10% to \$3,600/t during 2Q'18, and then fall to \$2,500/t by end 2019



Zinc is the only base metal exhibiting genuine tightness as evidenced by a notable backwardation across the curve

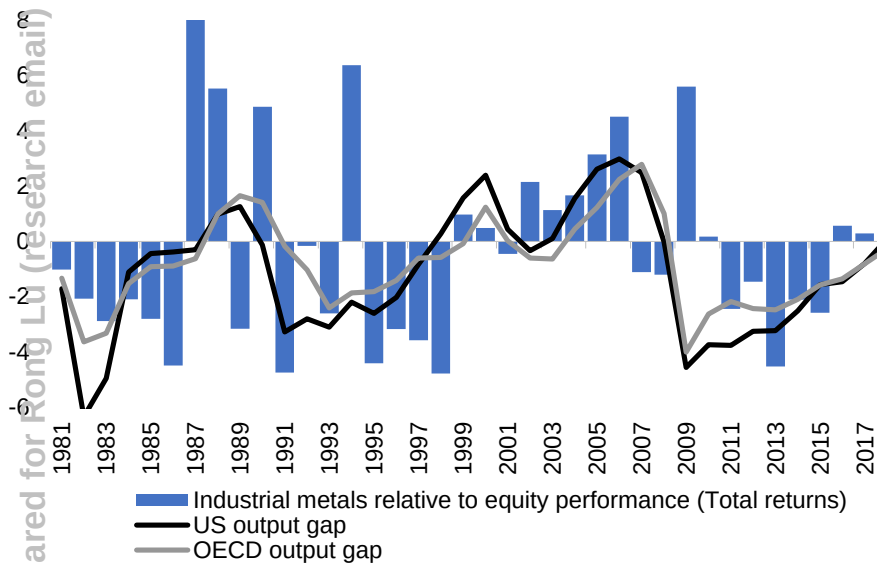


Base metals – the macroeconomic backdrop is set to be supportive in 2018

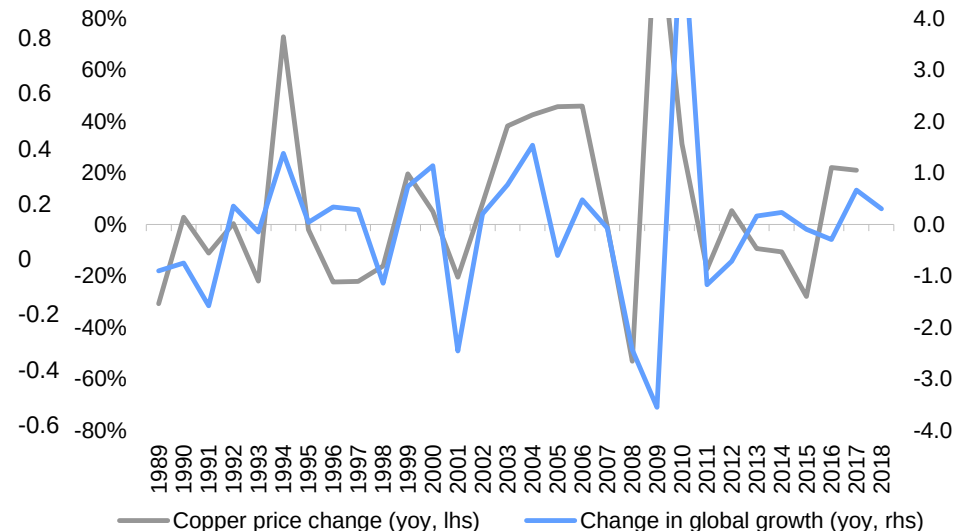
Accelerating of global growth and a closing of US and OECD output gaps have traditionally been supportive for metals returns, both in absolute terms and relative to equities. Our economists expect real global GDP growth to accelerate from 3.1% to 3.4% in 2018.

- A continuation of solid global growth is set to result in output gaps turning positive, and inflationary and rate hike pressures building. These periods have tended to see metals outperform equities. Increased awareness of this dynamic could see large inflows in to commodities over the next 12 months.
- Though the market may already be pricing an acceleration in global growth, there are always downside risks to growth being discounted. In this way, potential upside to metals prices comes from the lack of materialisation of these risks, or even stronger than anticipated growth.

Output gaps and metals relative to equity performance



Change in global GDP growth vs change in copper prices



Copper – moderately bullish outlook with risks evenly balanced

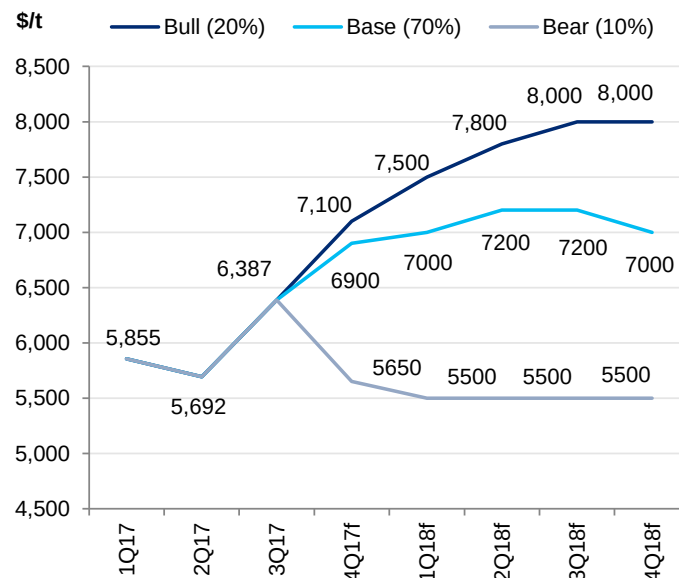
We forecast a small deficit copper market during 2018 and 2019, and we expect prices to trade above \$7,000/t for most of 2018, underpinned by a solid macroeconomic environment and anticipated higher than average mine supply disruptions. The main downside risk is that Chinese demand softens more than anticipated.

- **Base Forecast (70%):** Our base case is that supply disruptions are higher than average, at 6% of mine supply, with nearly 30 labour contract negotiations set to take place over the next 12 months against a backdrop of high prices, potentially affecting 25% of global mine supply. We expect copper demand growth will remain solid on the back of stronger global growth, and that China will support its 'old-economy' growth engine with a notable 1Q'18 credit impulse. Overall we see risks to the downside on the 'macro' front (back end downside) as broadly offset by risks to the upside on the 'micro' (spread upside / tightening) front.
- **Bull Case (20%):** In our bull case we see \$8,000/t copper by 2H'18 on the back of major outages of both scrap and mine supply, further strong global demand growth, and more bullish long term demand and supply expectations. The latter could come on the back of a reduction in fears surrounding the long term prospects for Chinese copper demand growth.
- **Bear Case (10%):** LME prices fall to \$5,500/t as supply growth is delivered without significant disruption and Chinese growth slows down materially – partly policy induced, partly structural.

WORLD COPPER Supply Demand Balance

kt	2016	2017F	2018F	2019F	2020F
Mine Production (conc.)	16,314	16,057	16,339	16,634	16,746
Mine Production (SxEw)	3,812	3,670	3,638	3,648	3,516
Total Mine Production	20,126	19,727	19,977	20,282	20,262
Mined % Change	-0.1%	-2.0%	1.3%	1.5%	-0.1%
Including Total Disr. Allowance (t)	0	199	1,261	1,507	1,737
Including Total Disr. Allowance (%)	0.0%	1.0%	6.0%	7.0%	8.0%
Total Refined Production	22,660	23,048	23,531	24,033	24,787
Refined % Change	3.8%	1.7%	2.1%	2.1%	3.1%
Consumption/Demand	22,537	23,027	23,607	24,137	24,634
Demand % Change	2.1%	2.2%	2.5%	2.2%	2.1%
Surplus/Deficit	123	21	-75	-104	153
Stock Change	87	80	-75	-104	153
Visible Stocks (incl bonded)	1,014	1,095	1,019	916	1,069
Stock:Consumption Ratio (wks)	2.3	2.5	2.2	2.0	2.3
Price (US\$/lb)	2.2	2.8	3.2	3.2	3.0
Price (US\$/t)	4,869	6,210	7,100	7,000	6,600

Copper price bull-bear scenarios

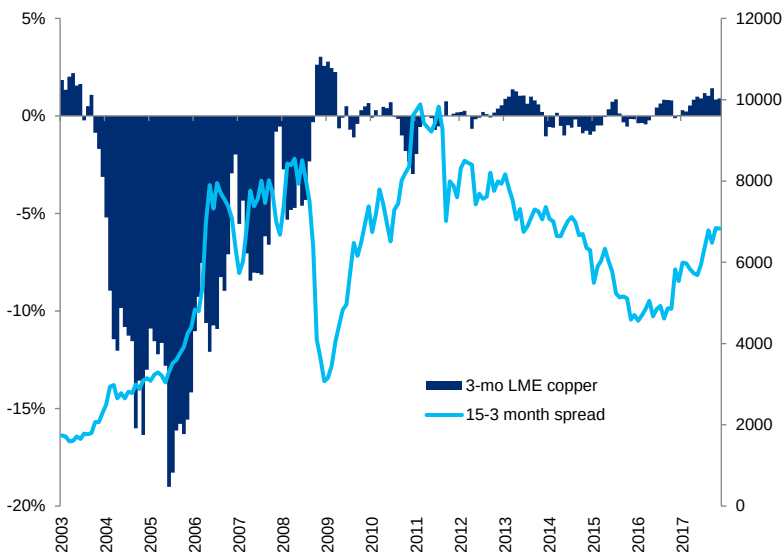


Copper – shifts in demand expectations have been critical to the rally

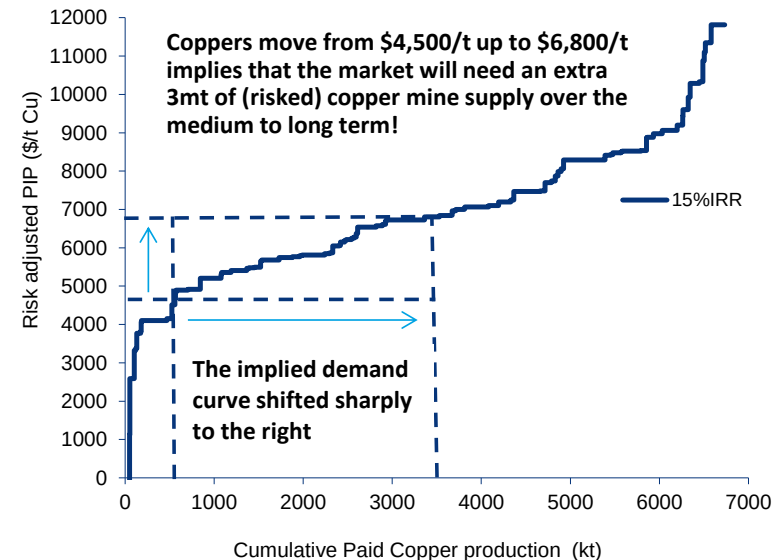
Copper's move higher both over the past 3 months and over the past 2 years has occurred despite a lack of fundamental tightness. The move was in our view driven by an increase in Chinese and global growth expectations. The market is now pricing that it needs an additional 3mt copper supply vs 2 years ago.

- **There is scant evidence of physical copper market tightness over the past two years.** The 50% price rally has occurred while SHFE and LME spreads have remained in contango and overall visible copper stocks have risen over the period.
- **So what drove the rally?** An improvement in current and future copper demand growth expectations, mainly driven by China. 2 years ago the market was pricing that Chinese and global copper demand would grow only negligibly, now it is pricing that the market will need an extra 3mt of copper supply over the long term (~2% incremental p.a. demand growth). The implied long term demand curve has shifted to the right. The supply curve has also shifted upwards over the past two years as the improvement in current and future demand expectations drove costs higher and contributed to a weaker dollar.
- **Looking ahead, we see the risks surrounding growth expectations as skewed to the downside, albeit from high levels.** Offsetting this is a likely tightening in the global copper balance which is expected to support prices around \$7,000/t.

LME and SHFE copper have been in contango as copper rallied 50% or \$2,300/t from \$4,500/t to \$6,800/t



The move higher was driven by an increase in demand expectations, as well as higher output costs

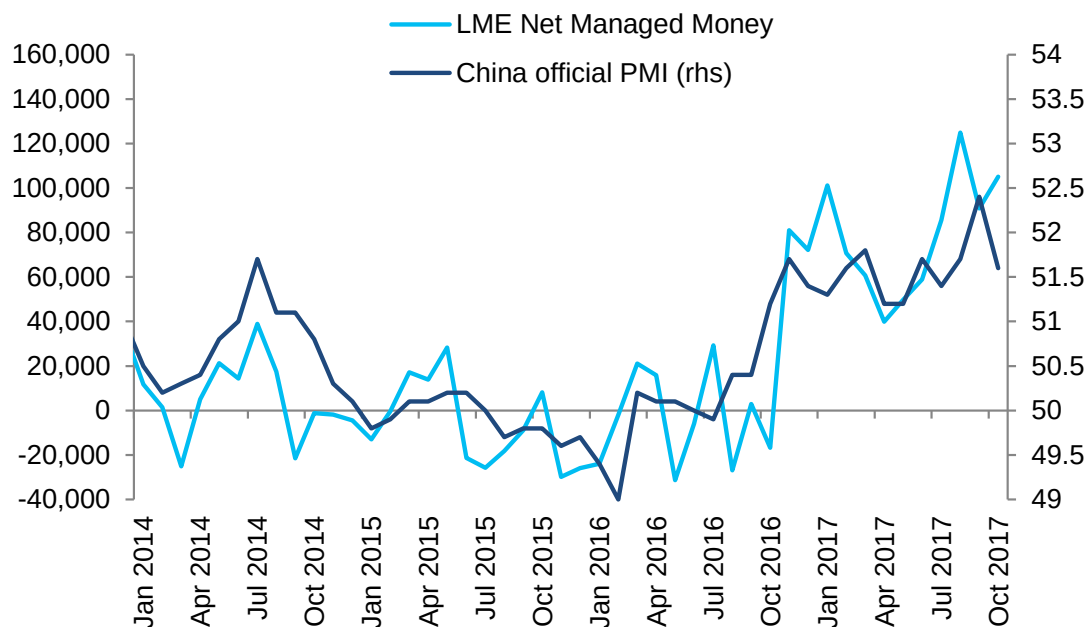


Copper – higher demand expectations have been consistent with the China PMI

The improvement in longer term demand expectations has in large part reflected a major improvement in current demand growth in China, and more broader, globally. Net speculative positions have increased in line with the improvement in near term Chinese growth, despite a lack of fundamental copper tightness.

- As we have seen over the past two years, with copper prices moving thousands of dollars higher despite rising inventories, the outlook for Chinese growth is critical to short and longer term growth expectations (the long term demand curve), as well as costs of production and the dollar (the long term supply curve).
- Expectations of future growth are often in part a reflection of current growth. Indeed, copper's rally has been associated with a major increase in the official China PMI, and a contemporaneous increase in net speculative positions.
- While growth expectations are high at present, and net speculative positioning is elevated, our base case is that these expectations are broadly met during 2018.

The official China PMI and LME copper Money Manager positions have been strongly positively correlated over the past 2 years



Source: Bloomberg, Wood Mackenzie, Citi Research

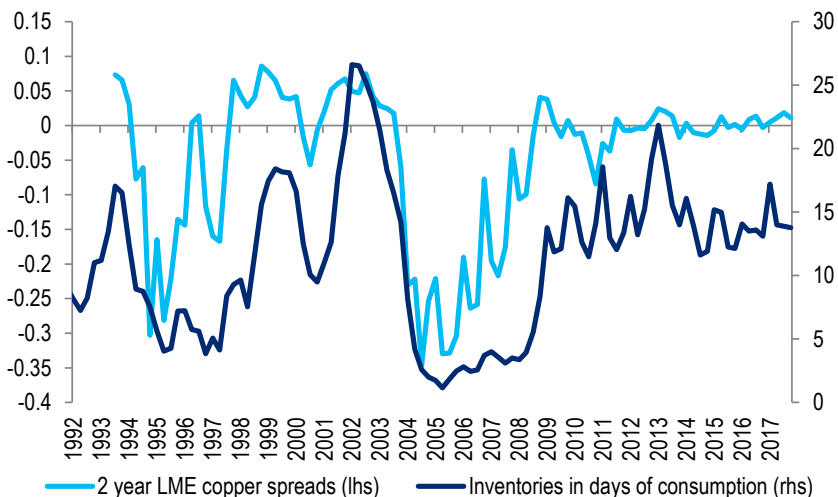
Copper – framing the \$8,000/t copper bull case for 2018

Most bullish commentators pick the ‘round’ \$8,000/t as a target. Here we quantify two scenarios which could see \$8,000/t, the first being an increase in long term demand expectations and the second being a tighter balance and thus tighter copper spreads. A combination of the two scenarios would also suffice.

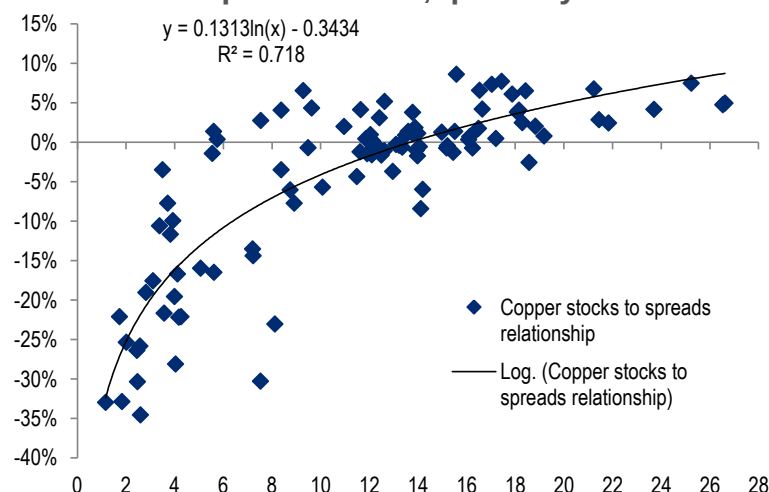
- \$8,000/t Scenario 1 – a higher copper ‘back-end’ (and spreads remain unchanged):** We could see copper move up to \$8,000/t on the back of an extension of the drivers of the current rally – i.e. we could have long term demand expectations continue to shift higher over the coming year, perhaps via more confidence in near term Chinese demand growth. Specifically, for copper to rally to \$8,000/t the market would need to move to price that we need another 1.5mt of copper to be incentivised over the long term, equivalent to an incremental 1%p.a. of growth in copper demand over the next 5 years.
- \$8,000/t Scenario 2 – spreads tighten on a larger than expected copper deficit (and longer term demand and supply expectations – i.e. the back end – remain unchanged):** The second Scenario is where we get \$1,000/t upside from a tightening in spreads on the back of substantial copper deficit. Specifically, based on historical stock to spread relationships we find that copper stocks would need to draw by around 350kt during 2018 and the market would need to price another similar draw during 2019. This would reduce visible and bonded stocks to 0.7 days of consumption, which would historically be consistent with a 14% 2 year backwardation, or a \$1,000/t backwardation over the 3-mo to 27-mo tenor.

Prepared for Rong Lu (research email)

A decline in copper inventories should see spreads tighten



Copper spreads and inventories have a tight historical relationship - 1990-2017, quarterly data

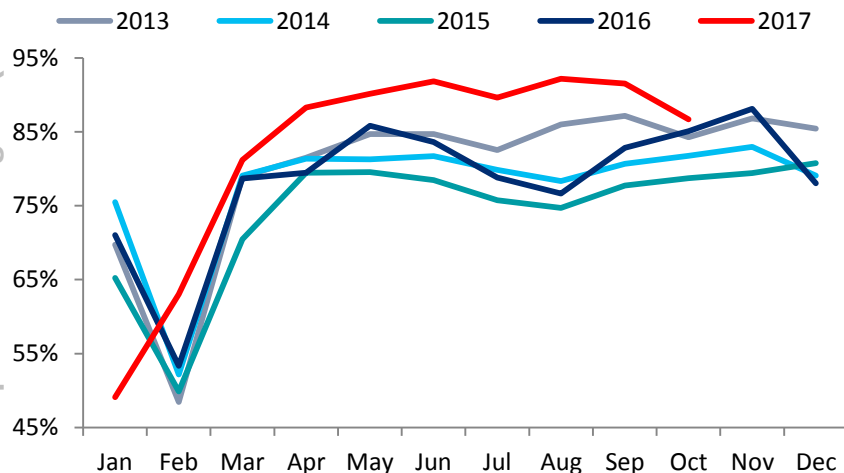


Copper - Chinese copper end-use demand to grow ~2% in 2018

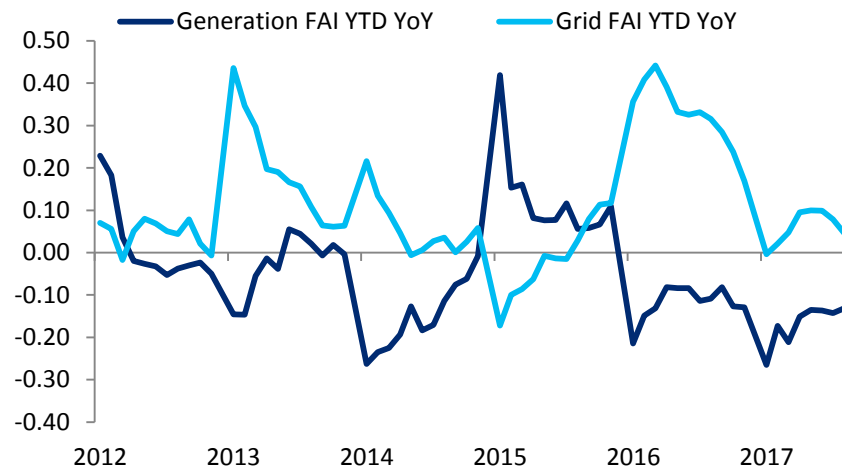
Chinese copper consumption was higher than expected during 2017, as a strong property and export market underpinned growth in air conditioner and power cable demand. We see Chinese copper consumption growth slowing to by 2.3% y/y in 2018, from ~3.3% y/y growth in 2017.

- We expect the construction sector will post growth of **nearly 3% y/y during 2018**, underpinned by rental and social housing, as well as the high level of commercial sales relative to new starts.
- **Consumer goods consumption growth should slow to around 2% y/y in 2018**, from 6% growth in 2017, on the back of a high base and a slowdown in property sales.
- **Copper demand from the automotive sector and NEVs and related infrastructure should remain solid**, and we expect 4-5% growth in demand from this sector in 2018.

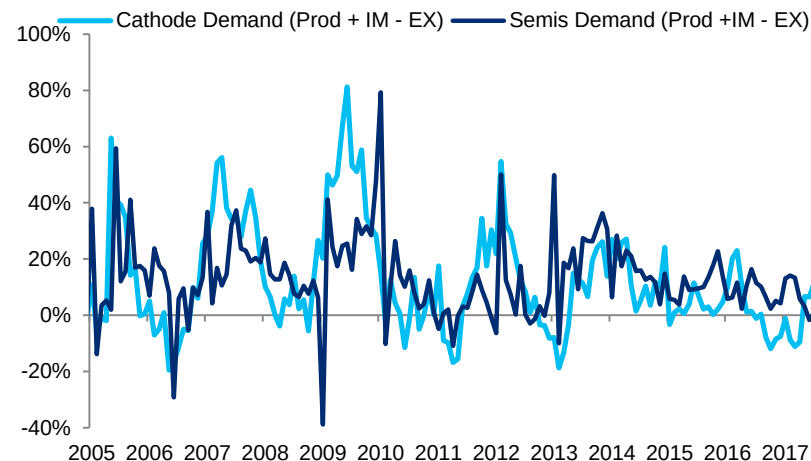
Copper cable operating rates have been well supported



China power grid investment YTD growth has slowed



Copper cathode apparent demand remained strong in 3Q



Source: SMM, NBS, China Customs, Citi Research

Copper – we use an above average disruption allowance of 6%

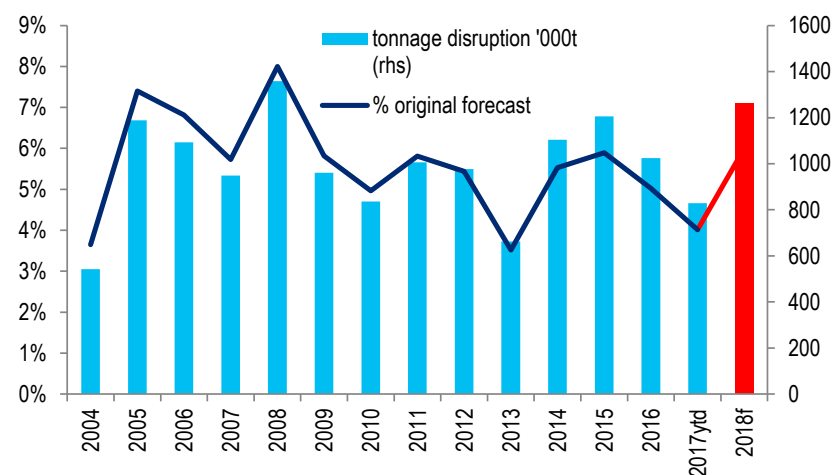
Our base case is that mine supply growth increases only c.250kt during 2018, after taking a 6% disruption allowance for 2018 (this is towards the top end of historical ranges). Supply growth is heavily reliant on Escondida and Glencores Africa operations.

- **Anticipated 2018 copper mine supply additions are quite concentrated** - almost 500kt of the main additions (post disruption allowance) comes from Escondida and Glencores African assets. Other major additions are on paper expected at Grasberg, Olympic Dam, Cobre Panama and Toquepala.
- **To reflect elevated supply risks over the next 12 months we assume a 6.0% disruption allowance for 2018** (or 1.26mt), vs 5.2% average since the financial crisis (or 0.97mt).
- **Overall, post disruption allowance we only expect 1.3% copper mine supply growth in 2018.**

Copper mine supply additions during 2018 – by mine, after disruption

Mine	2017	2018	Change
1 Escondida	893	1208	315
2 KOV Restart / Expansion / Kamoto	0	172	172
3 Ertsberg-Grasberg	490	564	74
4 Olympic Dam	132	188	56
5 Cobre Panama (Petaquilla)	277	314	47
6 Toquepala	0	47	40
7 Bystrinskoe	218	249	38
8 Los Bronces	218	249	36
9 Nkana / Mufulira	25	61	36
Subtotal	2253	3052	814
Others	17474	16925	-564
World total	19727	19977	250

Copper mine supply disruptions – our 2018 forecast is at the top end of historical ranges



Copper – mine supply disruption risks are particular high over the coming year

Supply risks come from the potential impact of the Chinese scrap import ban (there is significant implementation uncertainty here), from labour contract negotiations at operations which account for ¼ of global mine supply, and potential contract and mining issues at Grasberg.

- **There are substantial mine supply risks during 2018.** Most notably, there are over 30 labour contract negotiations associated with c.5mt of mine supply, or 24% of global mine supply. The largest identifiable potential issue concerns the Escondida contract due June, 2018, given the 2 month strike earlier this year. High copper prices may increase the probability of dispute. An end to high grading in the bear market of 2012-1H'16 could also contribute to disruptions.

Labour contract negotiations over the next 12 months

Mine	Location	Company	Expiration	Labour Union/additional information	2018F* Output
1 Quebrada Blanca	Chile	Teck	Nov-17	Sind. Rol General N°1	20
2 Mantos Blancos	Chile	Anglo American	Nov-17		46
3 Collahuasi	Chile	Anglo American/Xstrata	Oct-17	Sindicato N°1 (trabajadores)	493
4 Aurex (Ojos del Salado)	Chile		Dec-17		14
5 Mount Polley	Canada	Imperial Metals	Dec-17		14
6 Escondida	Chile	BHP Billiton / Rio Tinto	Dec-17	Sindicato 2, Staff	1285
7 El Teniente	Chile	Codelco	Dec-17	Sindicato San Lorenzo y Sindicato N°7 (Rajo Sur)	415
8 Quebrada Blanca	Chile	Teck	Dec-17	Sind. Rol General N°2	20
9 Candelaria	Chile		Dec-17		153
10 Lomas Bayas	Chile	Glencore	Jan-18	Sindicato de trabajadores	68
11 Los Pelambres	Chile	Antofagasta Minerals	Feb-18	Sindicato Mina	350
12 Radomiro Tomic	Chile	Codelco	Mar-18	Sindicato de Trabajadores	205
13 Chuquibambilla	Chile	Codelco	Mar-18	Sindicato de Supervisores	427
14 Caserones	Chile	Lumina Copper	Mar-18	Sindicato Trabajadores Planta	155
15 Centinela	Chile	Antofagasta Minerals	May-18	Sindicato Sulfuro Mina	270
16 MM Hales	Chile	Codelco	May-18	Sindicato de Profesionales	186
17 Centinela	Chile	Antofagasta Minerals	Jun-18	Sindicato Sulfuro Planta	270
18 Los Pelambres	Chile	Antofagasta Minerals	Jun-18	Sindicato Planta	350
19 Spence	Chile	BHP	Jun-18	Staff	200
20 Escondida	Chile	BHP	Jun-18	Sindicato N°1 de operadores	1285
21 Caserones	Chile	Lumina Copper	Jul-18	Sindicato Trabajadores Mina	155
22 Andina	Chile	Codelco	Aug-18	Sindicato de Supervisores	215
23 Cerro Colorado	Chile	BHP	Aug-18	Operadores	71
24 Cerro Verde	Peru	Freeport	Sep-18	Agreed 4yr deal effective from Sept 2014	479
25 Centinela	Chile	Antofagasta Minerals	Oct-18	Sindicato Oxido Planta	270
26 Centinela	Chile	Antofagasta Minerals	Oct-18	Sindicato Oxido Mina	270
27 El Teniente	Chile	Codelco	Oct-18	Sindicatos Caletones, El Teniente, N7, N5, Sewell, Mina Unificado	415
28 MM Hales	Chile	Codelco	Nov-18	Sindicato Trabajadores	186
Subtotal potentially impacted mine supply					5066
Pre-disruption global mine supply 2018					21281
% share					24%

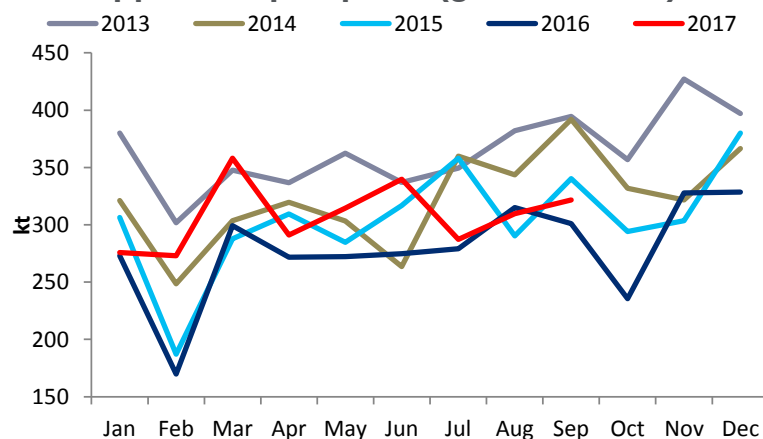
*Italics represents mines with multiple wage negotiations during 2018, woodmac data used for mine by mine and aggregate output estimates
Source: Wood Mackenzie, Company Sources, Reuters

Copper – potential scrap supply disruptions are a significant upside risk

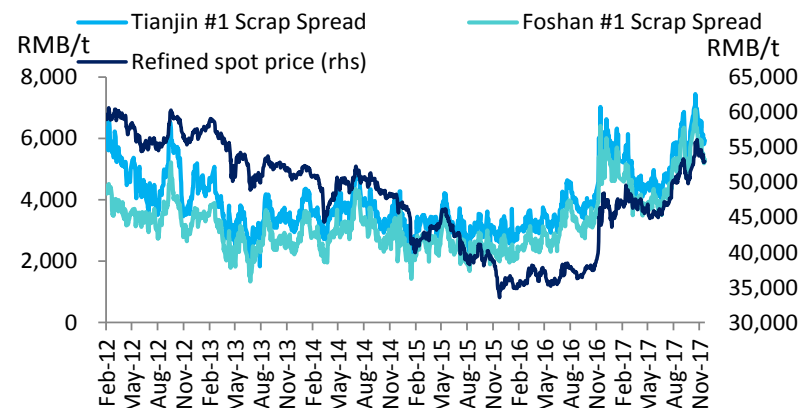
Up to 200-500ktpa (copper content) of copper scrap may be temporarily 'lost' from the market should China enforce it's 'Category 7' scrap ban earlier than the official plan of Jan 2019. If brought forward, there may not be enough time to shift / rebuild processing capacity outside of China. This represents a material upside price risk.

- **Recently, concerns regarding Chinese copper scrap imports have become more prominent, as a few local authorities have reportedly started restricting low-grade copper scrap imports.** The Chinese government is taking a harder stance on copper scrap use, recycling and imports due to environmental concerns. Customs' clearance of imported scrap has reportedly been delayed recently as Beijing mulls over a complete import ban of "category 7" scrap (mainly motors, wire and cable, and machine parts), which has low copper contents, and the enforcement may come earlier than the initially proposed timeline of Jan 2019.
- **Assessing potential impacts of early policy enforcement is imprecise, but under an extreme scenario where all "category 7" scrap is banned, China might lose 40kt/month of copper supply.** Indeed, *BGRIMM* estimates that China's scrap imports might drop 10% y/y in 2018 due to the bans, equivalent to a loss of c.150-200kt of copper content.
- **Should the scrap ban be enforced in 2019, we expect that Chinese scrap processing plants will be relocated to other parts of Asia over the coming year, ultimately mitigating the impact of the ban on the global copper market.** New processing plants outside of China will also likely be constructed over the coming years, owing to the prevailing high level of scrap copper prices, but the scale and precise timing of this will depend on the anticipated likelihood of enforcement.

China copper scrap imports (gross tonnes)



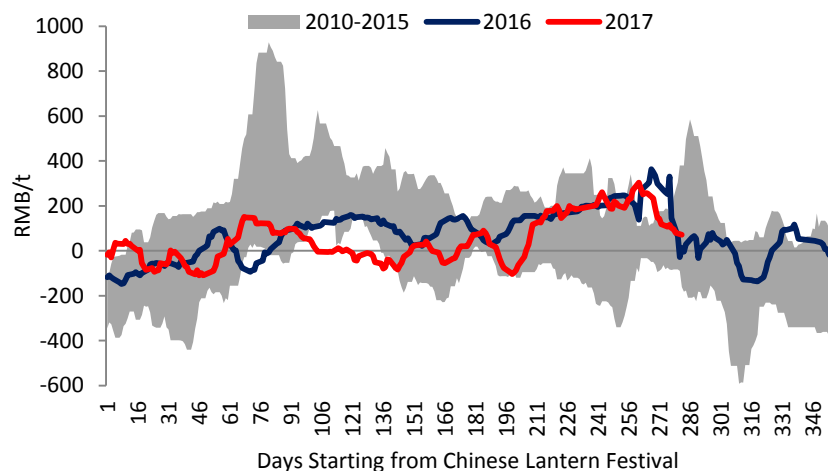
Chinese copper scrap prices and spreads



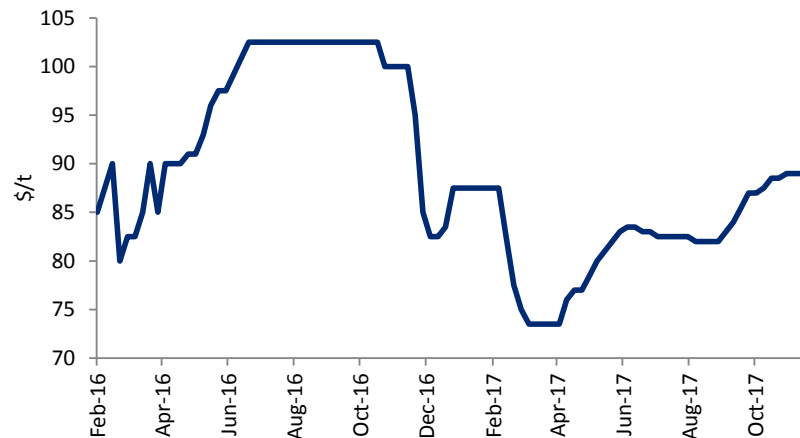
Source: LME, Comex, SHFE, WBMS, SMM, Citi Research

Copper – the 2017 data point to a balanced market, ample concentrate stocks

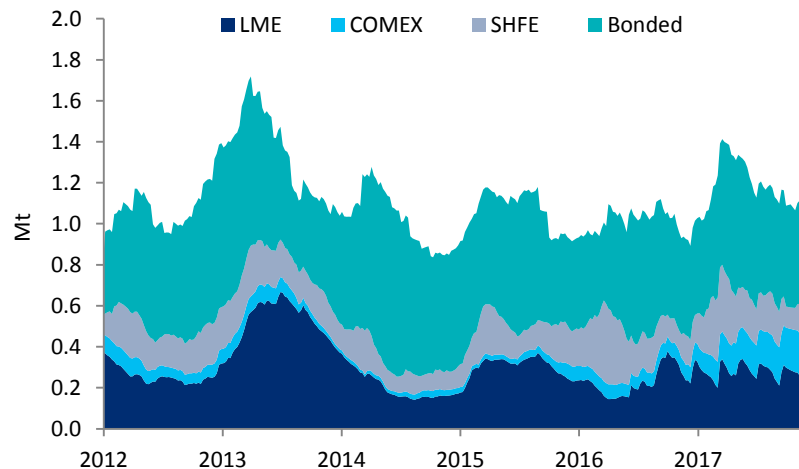
Chinese copper physical premiums around average



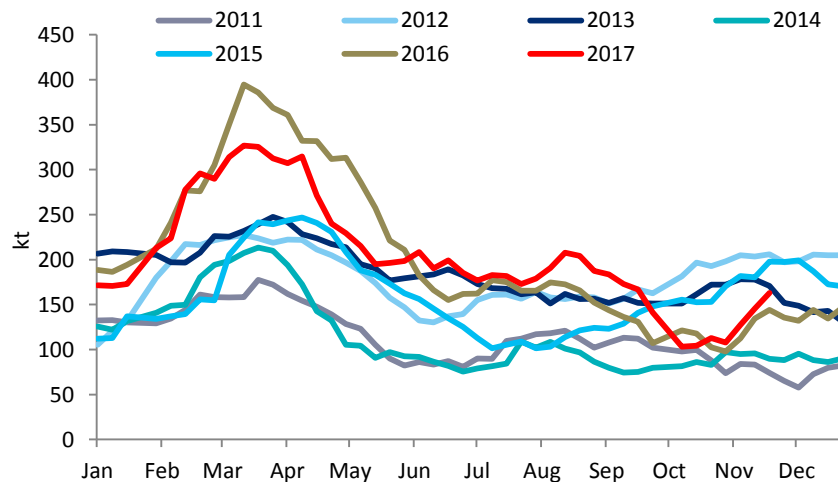
Spot treatment charges (CFR China) are at profitable levels, indicating ample concentrate stocks



'Visible' refined copper stocks have risen over the ytd



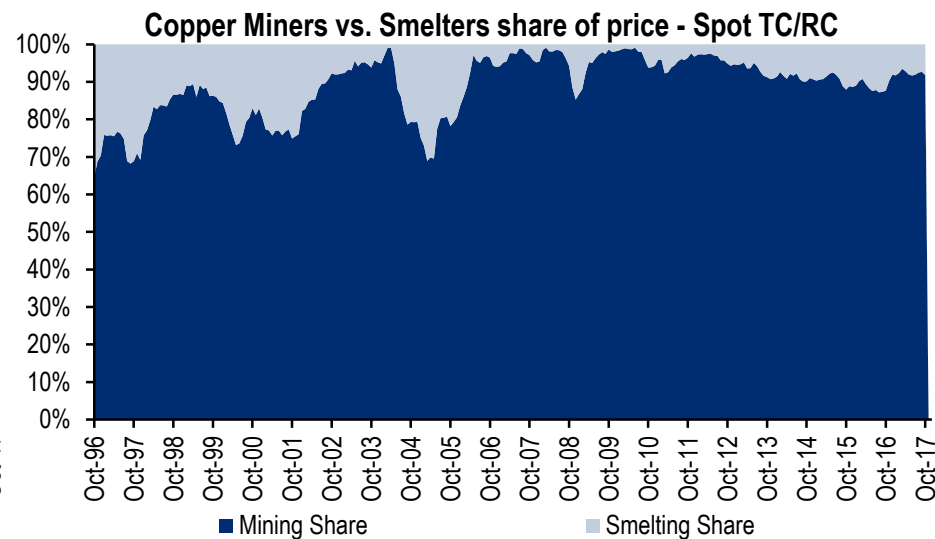
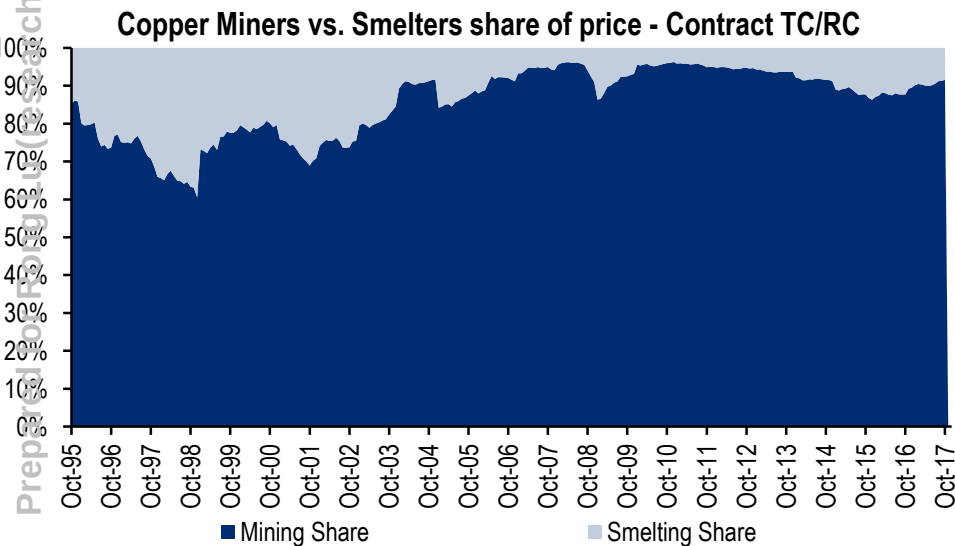
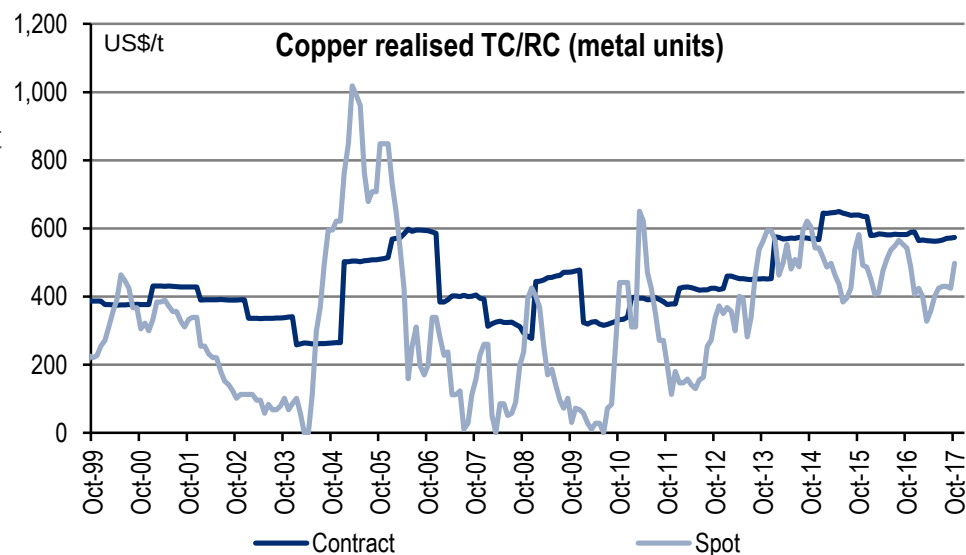
SHFE copper inventories picked up quickly recently



Copper – annual Treatment & Refining Charge (TC/RCs) outlook

2018 benchmark likely to see a moderate decline

- We forecast a 2018 benchmark settlement of TC/RCs of at \$88/t and 8.8c/lb, a ~5% decline vs. 2017 benchmark of \$92.5/9.2c.
- CSPT's floor price of flat-ish YoY settlements appears be a bit aggressive. Divergent starting points for miners and smelters are common as the discussions kick-off but this time the extent of the divergence could lead to a longer than normal negotiation.
- TC/RCs have remained above marginal smelter costs implying copper concentrate stocks remain ample globally.
- Spot TCs have risen in recent months, and were \$88/c8.8 in October after bottoming out at \$58/c5.8 in April.



Aluminium - supply reform a medium to long-term game changer

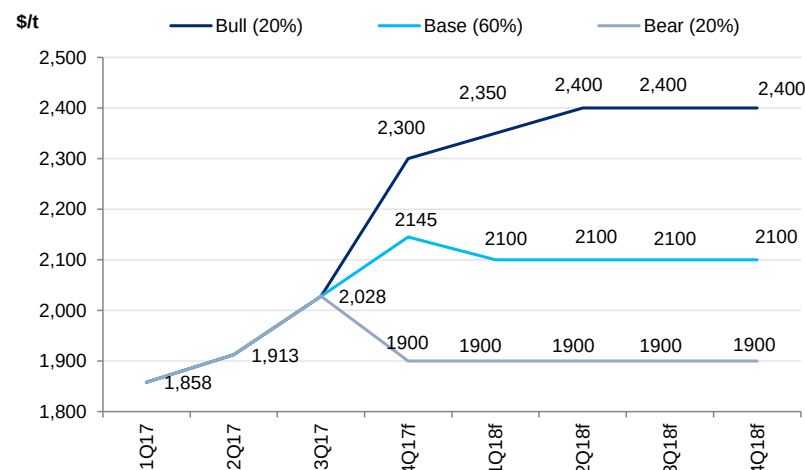
We modestly revise up aluminum price forecasts across through 2020, to reflect our now base case view that Chinese supply side reforms will continue indefinitely. We now forecast that LME aluminium prices will trade well above \$2,000/t over the medium term, in order to incentivise ex-China supply growth.

- **Base Forecast (70%):** LME aluminum prices are expected to consolidate during 2018 around \$2,100/t, with levels below this a buying opportunity in our view. In the near term we are in a 'travel and arrive' situation regarding Chinese capacity cuts – we think these will occur but these have largely been priced. Having said this we see Chinese capacity utilisation rising over the next 1-2 years and see China's supply side reforms as a real long term game changer for aluminium, ultimately meaning ex-China prices need to rise to incentivise ex-China capacity expansion. We forecast global consumption growth of 3-4% y/y for 2018 and 2019.
- **Bull case (20%):** LME aluminum prices reach \$2,400/t by mid 2018, on the back of stronger demand, a lack of Chinese restarts post CNY, and/or tightness in aluminium raw materials (carbon) resulting in ex-China output cuts.
- **Bear case (10%):** LME aluminum prices fall to \$1,900/t during 2018. Chinese supply cut policies reverse in 1H'18 and global aluminium demand growth decelerates rapidly.

Global Refined Aluminum Supply Demand Balance

kt	2016	2017e	2018e	2019e	2020e
Refined Production	59,074	63,463	66,394	68,840	70,773
Supply Incr (%)	3.5%	7.4%	4.6%	3.7%	2.8%
Of Which China	32,200	36,120	37,746	39,443	40,626
Supply Incr (%)	5.3%	12.2%	4.5%	4.5%	3.0%
Refined Consumption	59,610	62,929	66,266	69,352	71,946
Consumption Incr. (%)	5.3%	5.6%	5.3%	4.7%	3.7%
Surplus/Deficit	-536	534	127	-512	-1,173
US\$/t	1,610	1,985	2,100	2,200	2,200

Aluminum price bull-bear scenarios

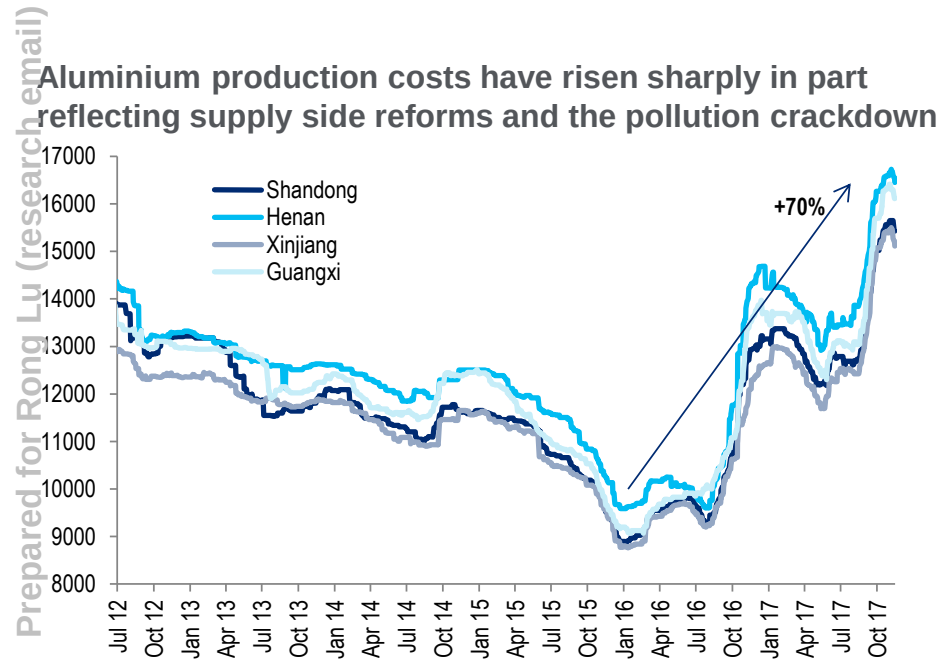


Source: IAI, Wood Mackenzie, NBS, Company data, Bloomberg, LME, Citi Research

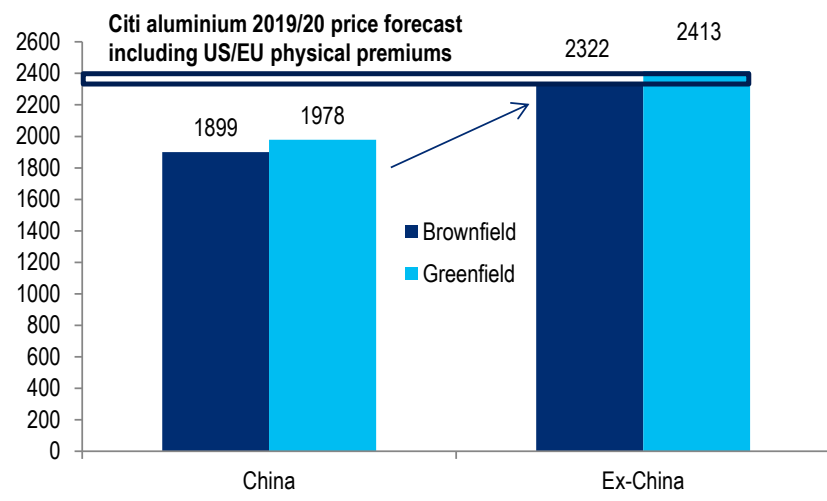
Aluminium production costs are set to remain higher for longer

Beyond the impact of the pollution crackdown on aluminium production costs, Chinese supply side reforms are likely to mean that ex-China will need to build aluminium smelters over the long term. This is set to shift incentive pricing from China to ex-China, where capital costs and hurdle rates are higher.

- China's crackdown on pollution (aimed at reducing pollution by cutting aluminium, alumina, carbon output) and supply side reforms (aimed at reducing overcapacity, affecting coal and alumina output) have contributed to a 70% increase in the cost of producing aluminium over the past 2 years. These policy dynamics are likely to persist indefinitely, and while lower cost production alternatives may be found, a large portion of the increase in costs is likely to persist.
- Perhaps even more critically, Chinese supply side reforms are likely to mean that incentive pricing shifts from China to ex-China, as growth in Chinese exports of aluminium (in all forms) slows down and potentially stops. The reduction in export growth would mean that ex-China would ultimately need to build aluminium smelters. Ex-China incentive prices are much higher than Chinese incentive prices, and this is likely to underpin higher long-dated aluminium prices for longer.



Aluminium smelters are increasingly likely to need to be built outside of China, where costs are higher



Source: Bloomberg, Citi Research

Aluminium - winter cuts slightly disappoint, but the market still set to tighten up

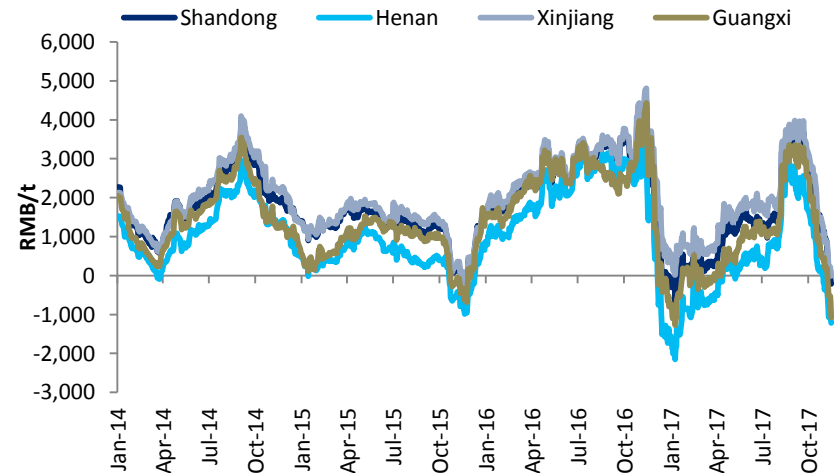
In the near term the market continues to focus on the scale of the winter supply cuts, in particular those at Hongqiao. We expect similar cuts in aluminium, alumina and carbon anode to repeat in the winter of 2018/19, though going forward alumina and carbon anode plants might be able to over-produce ahead of the cuts.

- We estimate the primary aluminium production cuts will result in losses of 1.0Mt (or 9% of 2016/17 winter production), with production risks in our view skewed to the downside. Supply cuts in alumina and carbon anode are triggering price spikes of aluminium raw materials, threatening an unsustainably large portion of Chinese aluminium production. We estimate that planned alumina cuts over the coming months are slightly higher to the aluminium cuts with limited refinery capacity ramp-ups in other regions of China to offset the supply cuts, meaning higher raw materials costs is likely.

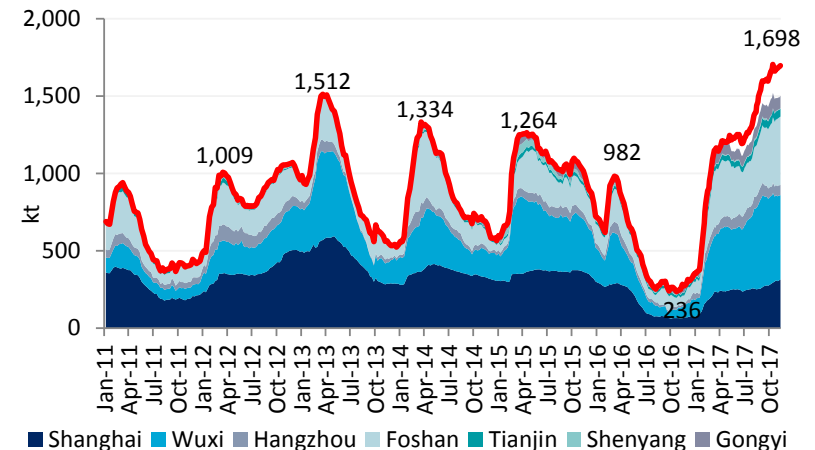
- Chinese smelters are currently running at maximum capacity, leading to consistent physical inventory build despite robust downstream demand. We anticipate inventory should begin to draw during the winter heating season and more so in 1Q'18. This could provide a boost to spot Chinese prices, which are running at a discount to prompt month SHFE prices.

- New Chinese smelting capacity in the pipeline (2-3Mt/y in 2018) should be able to fulfill additional end-user aluminium demand (up 6-7%) during 2018, and we don't see Chinese smelter capacity utilization rising sharply until 2019. A series of unintended consequences of the cuts, including weak local economic growth and unemployment should damp local governments' willingness in cutting more than required.

Chinese smelter margins are unsustainably weak

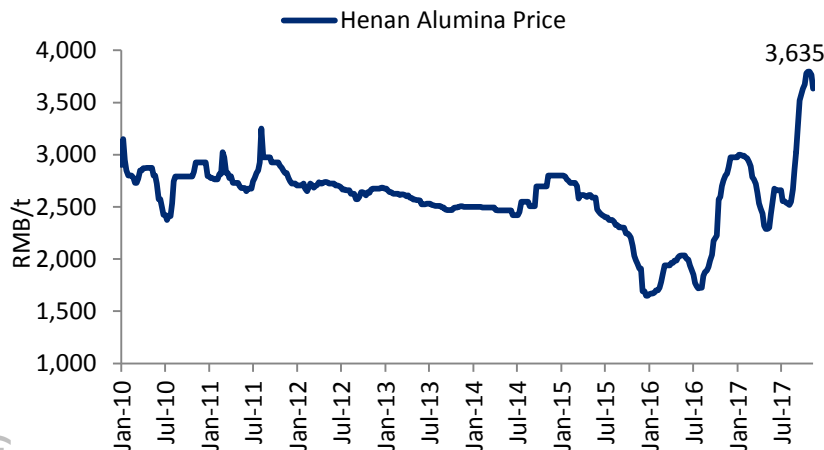


China's aluminium producers have stockpiled ahead of the winter cuts

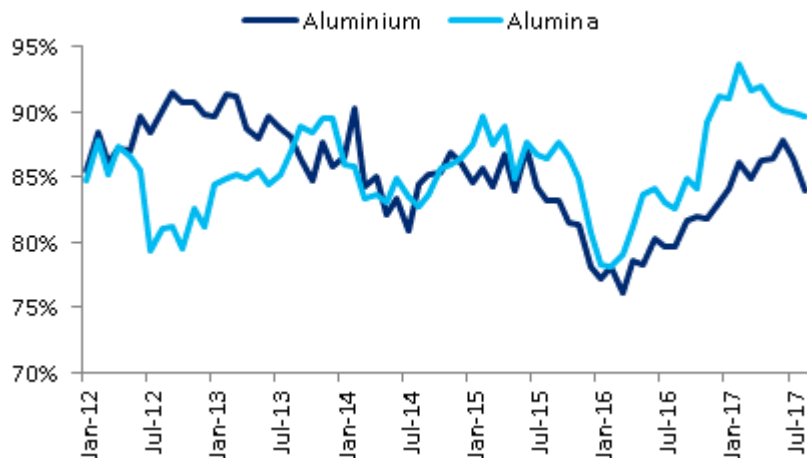


Aluminium – even if alumina unwinds, higher costs are here to stay

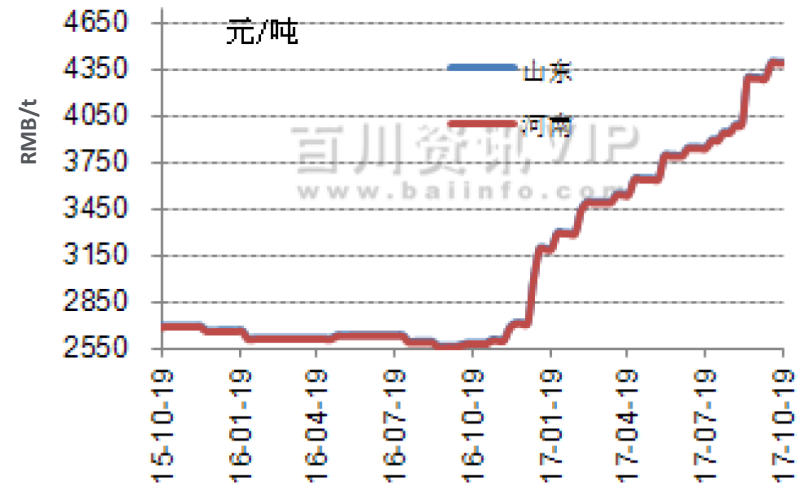
Spot alumina prices have risen sharply since August



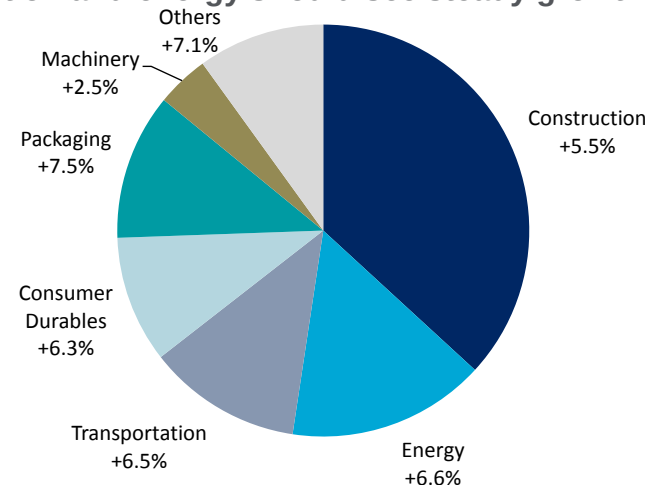
Chinese aluminium and alumina utilization rates have risen to relatively high levels over the past two years



Chinese carbon anode prices have surged 70% ytd



2018 Chinese aluminium demand should continue to benefit from strong growth in consumption-led sectors including transportation and packaging, while construction and energy should see steady growth



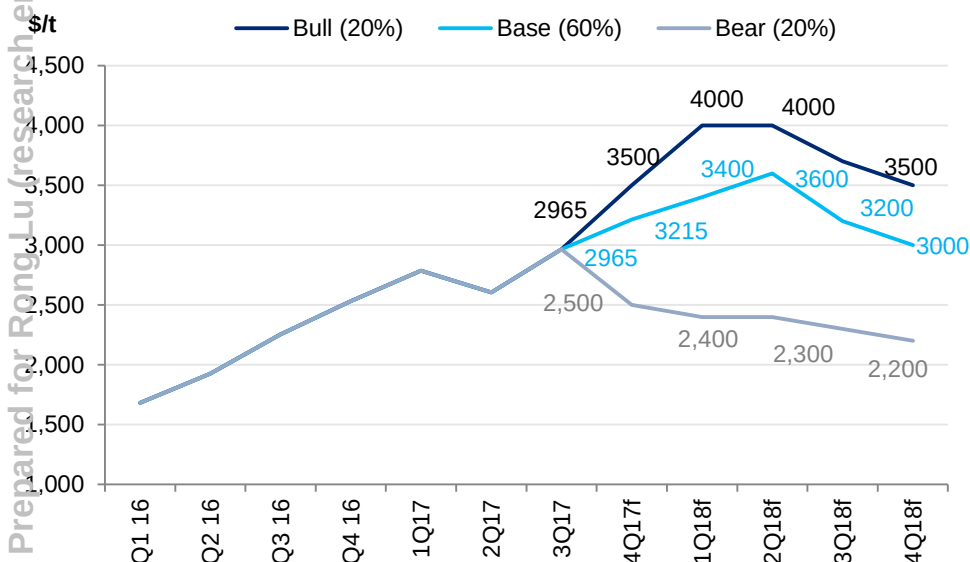
Zinc – another deficit anticipated in 2018, stocks to fall to near critical levels

World Zinc Supply Demand Balance

kt	2016	2017e	2018e	2019e	2020e
Baseline mine production	12,356	13,328	14,053	14,583	15,083
Supply (% chg.)	-7.1%	7.9%	5.4%	3.8%	3.4%
Total Metal Supply	13,788	13,738	14,238	15,038	15,538
Supply (% chg.)	-1.2%	-0.4%	3.6%	5.6%	3.3%
Consumption	14,000	14,266	14,580	14,871	15,139
Consumption (% chg.)	1.4%	1.9%	2.2%	2.0%	1.8%
Surplus/Deficit	-212	-528	-342	167	399
Total reported stocks	1,167	639	298	464	864
Reported Stocks (wks)	30.4	16.4	7.5	11.4	20.8
(US\$/t)	2,099	2,895	3,300	2,650	2,500

- An inventory drawdown and resulting tightening of spreads is expected to drive zinc prices to **average \$3,600/t during 2Q'18**. Our incremental bullishness reflects in part the larger than anticipated stock drawdowns during the year to date, the ~150-200kt of losses of Chinese zinc dust from reduced induction furnace production in China, and our view that Chinese zinc mine supply will continue to disappoint during 2018. However, we expect 2018 will mark the peak in this zinc price cycle, owing to an anticipated weakening in zinc demand growth (as China shifts away from old economy exposed metals like zinc) and strong supply growth set for late 2018 and 2019/2020.

Zinc bull/bear case price forecasts*



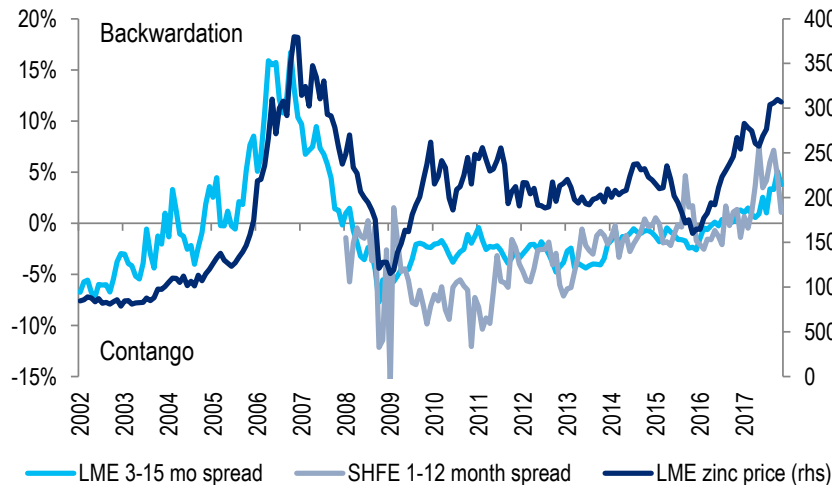
- **Bull case (20%):** Strong global manufacturing activity combines with China policy easing during 1Q'18. Chinese zinc supply cannot respond to stronger demand due to capex cuts and strict environmental conservation measures. A scenario where Glencore extends output cuts beyond our 2H'18 restart base case would be incrementally bullish.
- **Bear case (20%):** An earlier than expected substantial supply response to current prices could ease concentrate tightness. A continued slowdown of infrastructure and construction activity post the winter shutdown period in China is a major downside risk.

Zinc – the historical stock to spread relationship frames our price forecast

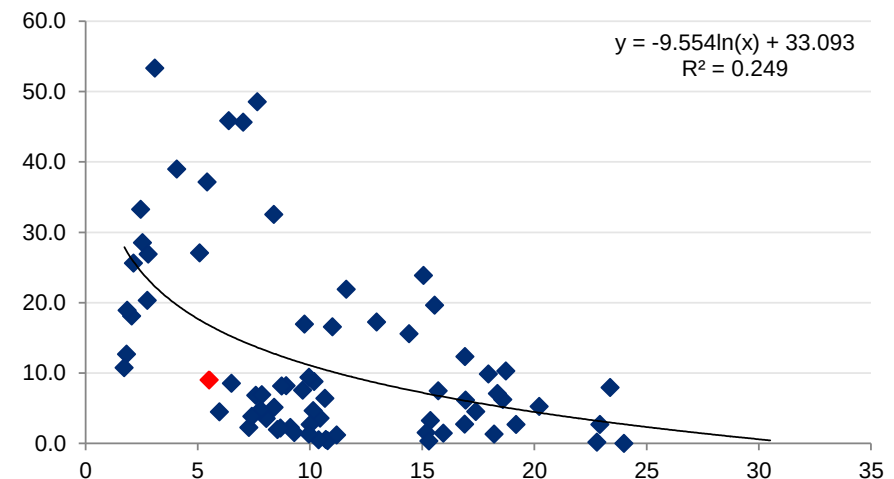
- **Cash to 27 month zinc spreads are currently trading in a 10% backwardation, vs historical relationship of c.18%.** We attribute the lack of a wider backwardation to concerns about off-market stocks. Our view on off-market stocks is that there is around 100kt in China and around 400kt outside of China, or around 12 days of consumption.
- **Our forecast of a c.300kt deficit for the zinc market in 2018 would take total stocks down to extremely low levels.** What will be critical is how much of a drawdown occurs on exchange vs off exchange given there is only c.300kt in exchange stocks. Our base case is that part of the drawdown occurs on exchange, and in particular that stocks decline on the LME to 2-3 days (-100kt), leaving spreads backwardated by 20-25%, and justifying our \$3,600/t zinc price forecast for 2Q'18. Were this deficit to occur completely on-exchange we would see LME and SHFE stocks run out, and spreads would likely widen sharply, with previous peaks possible – indeed our bull case is based on cash-27 month spreads going 35-40% backwardated, driving prices to \$4,000/t. For reference, when inventories were drawing and stocks on the LME were around current levels of 5-6 days of consumption, the backwardation could be as wide as 50%.

The zinc move higher has been both a function of a genuine deficit market and higher global growth expectations

Prepared for Rong Lu (research email)



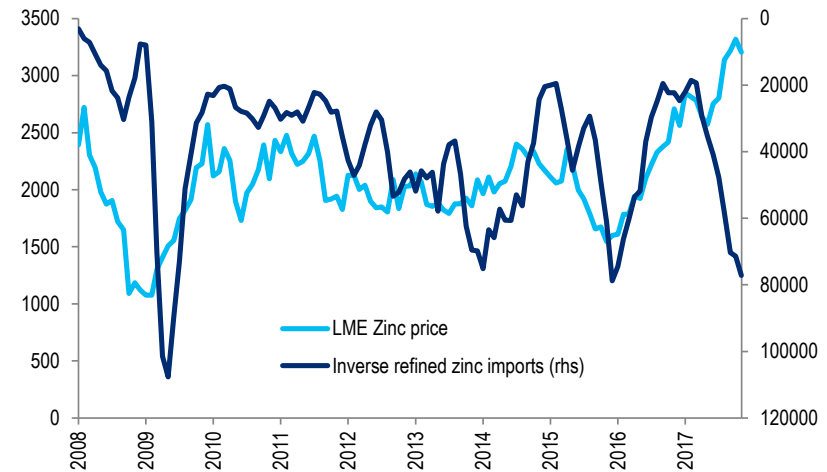
LME zinc inventory and LME zinc spreads (cash-27 month) during periods where inventories were falling



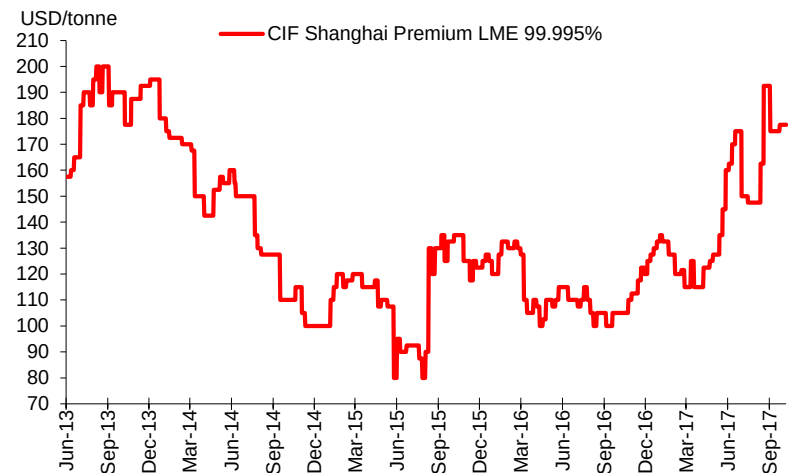
Zinc – Chinese imports suggest China has ‘run out’ of zinc

- China’s rising zinc imports despite high levels suggests that China has run out of zinc.** Chinese zinc imports are usually negatively related to price. This is because China is a net importer and in aggregate tends to de-stock during periods of high / rising prices and re-stock during periods of low / falling prices. After de-stocking during 2H'16 and early 2017, Chinese imports have risen sharply in the face of high/rising prices. This suggests China could be in a domestic deficit to the tune of 500kt. The risk to this view is that Chinese imports could to some degree reflect stock shifting by traders, rather than fundamental domestic market tightness.
- Macroeconomic factors may weigh on the demand profile of zinc due to its exposure to policy-driven sectors, but overall we see modest growth for Chinese end-use zinc demand during 2018.** Typically used through most stages of a development, Chinese housing starts growth, particularly in smaller cities, is set to provide some respite against increasingly cautious market sentiment into the new year. We continue to anticipate solid galvanized steel demand during 2018 driven by the Chinese non-residential sector.
- Zinc demand via galvanized steel is unlikely to face any near-term headwinds despite the fact that market prices have doubled since the end of 2015.** We believe stable end-use demand in a rising broader commodity price environment limits thriving threats for zinc in the galvanized steel sector. Demand destruction is likely to be limited by the fact that the most relevant substitutive metals such as aluminium and nickel have seen similar positive price performances during 2017.

Chinese zinc imports are usually respond to price anti-cyclically, but not this time - China has run out of zinc



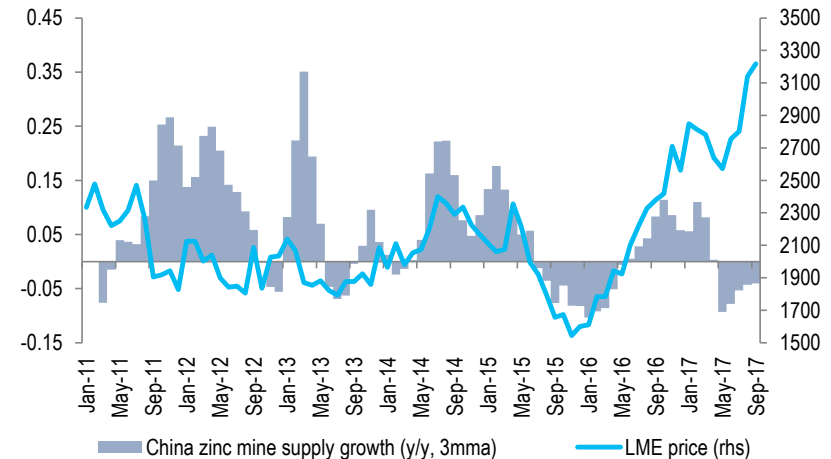
Shanghai premia have gained 34% YTD



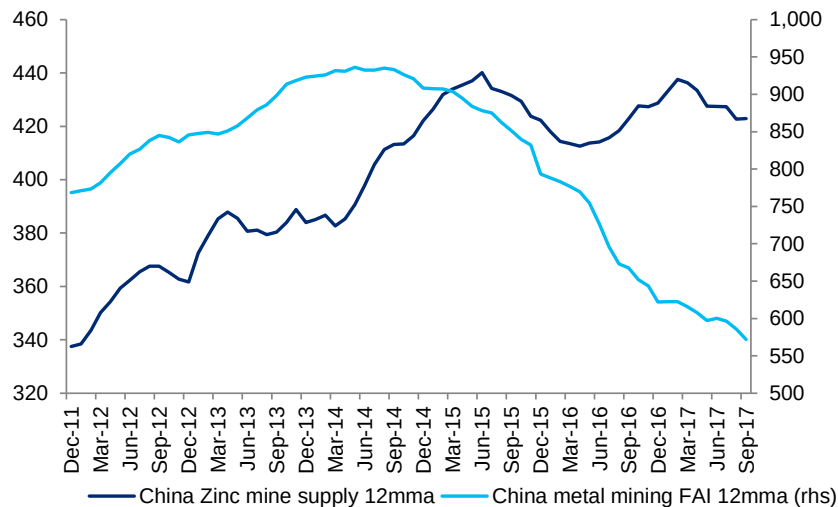
Zinc – Chinese mine supply is expected to grow only modestly during 2018

- Chinese zinc mine supply has fallen 2% y/y ytd September.** Historically Chinese zinc mine supply has been a swing supply source, rising sharply during periods of high prices (rising 20% y/y during much of 2014 for example), and falling sharply following downturns in prices (late 2015/early 2016 -10% y/y). This has not been the case this year, and we believe both **strict environmental inspections and related suspensions, as well as the ongoing collapse in metals mining Fixed Asset investment (-50% over the past 3 years, and still declining y/y) has been impacting China's ability to respond.** Our base case is that this continues, with around 200kt of pre-disruption Chinese mine supply growth forecast for 2018.
- One of the main upside risks to Chinese zinc mine supply and downside risks to our 2018 price forecast is if environmental restrictions eased, or consolidation occurred, enabling restarts.

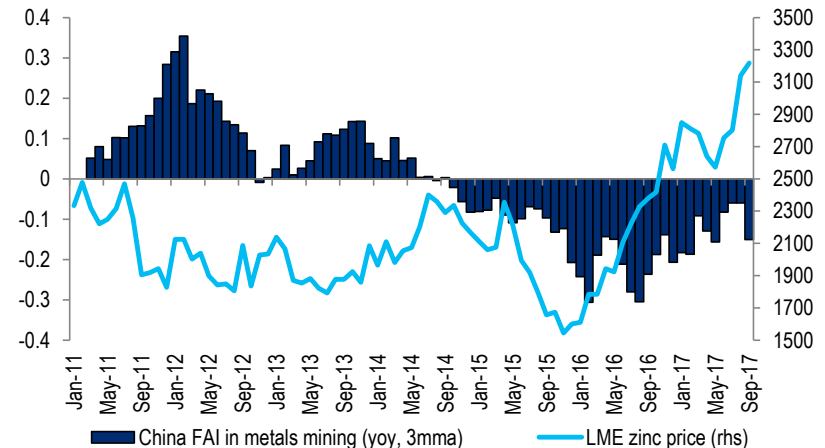
Chinese zinc mine supply usually responds to prices, with the exception of the past 6 months....



Chinese metal mining FAI has collapsed, limiting zinc supply growth



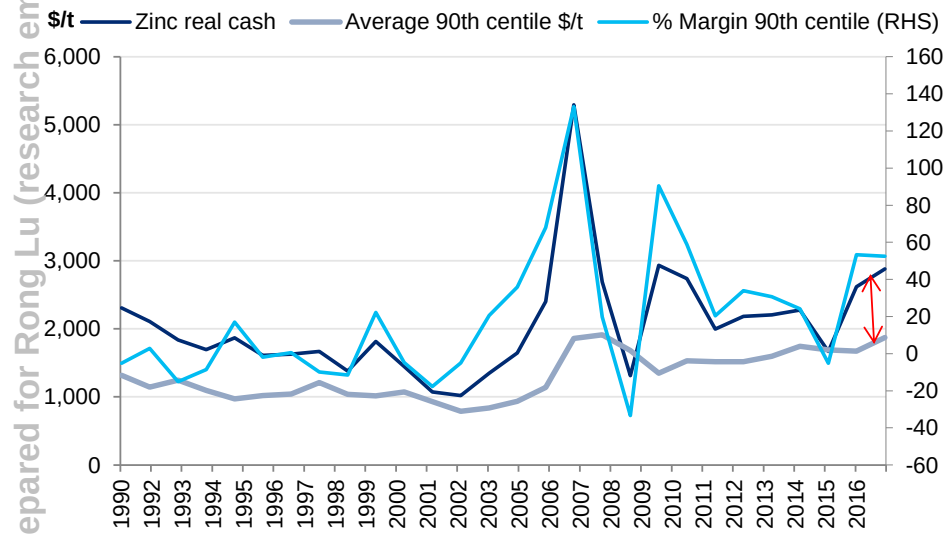
...this may in part reflect declining metals mining FAI as well as environmental inspections/controls



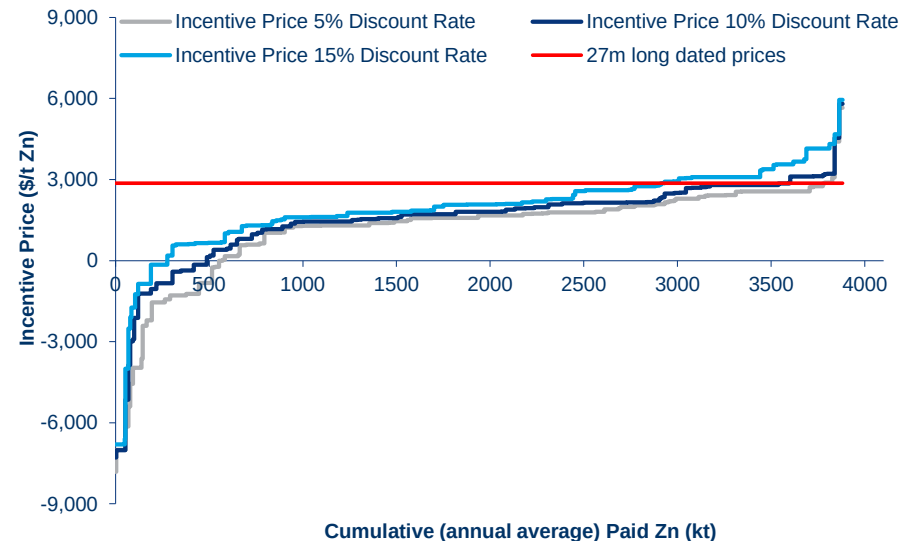
Zinc – producers prepare - 2018 will see the highest zinc price of this decade

- We are strongly of the view that 2018 will see the highest zinc price for this decade (2011-2020). We see prices falling by nearly 30% from spot to \$2,500/t by 2020, for three reasons:
- **First, zinc producer margins are at extremely high levels by historical standards.** Zinc prices are currently more than 50% above the 90%ile of the zinc cost curve. Margins have only been this strong in 4 of the past 27 years. Prices and margins at current levels will incentivise brownfield expansions and capacity restarts.
- **Second, nearly 90% of known/modelled zinc projects, or 3.5mt of paid zinc, is being incentivised to be funded/built at current 2 year forward zinc prices, on paper.** If delivered this would represent ~25% growth in total refined zinc supply, and meet . At trend zinc demand growth, with no growth in scrap or demand substitution, this would cover zinc demand for the next 12 years.
- **Third, zinc is an ‘old economy’ metal with very high exposure to the Chinese construction and infrastructure sectors.** As China shifts to its new era of slower FAI growth, zinc demand growth should slow. This is expected to be offset somewhat by an increase in zinc intensity of use in the Chinese automotive and construction sectors as quality standards are raised.

High cost producer margins exceeds 50% in 2017



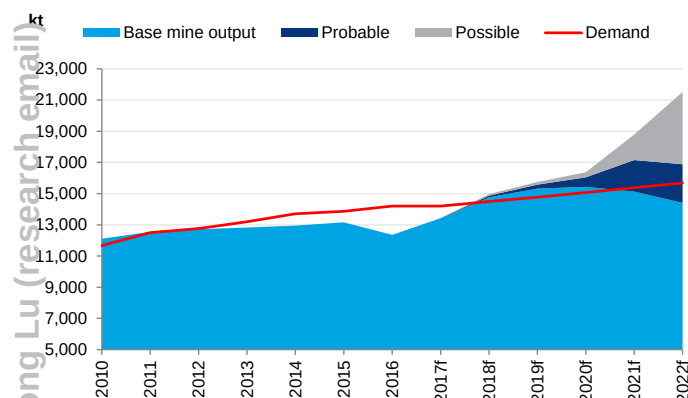
Almost ~3.5mnt of zinc supply incentivised at \$2,850/t



Zinc – ultimately supply will respond, but not enough in 2018 in our base case

- **Zinc mine supply outside of China is responding but we expect the brunt of this will not be felt until 2019.** This is because most projects do not ramp up until late 2018 or 2019. Further, our base case is now that Glencore's will not restart 350kt of production capacity until 2H'18, and we expect this ramp up will be gradual.
- **There are major upside supply risks on a 2019/2020 time horizon.** If Huoshaoyun and New Century reach their potential production targets of 500ktpa and 270ktpa at all, this would represent a bearish supply shock relative to our base case expectations.
- **There is >7mt of 'blue sky' potential supply growth over a 5 year view, representing nearly 50% of 2017 demand.**

Known project potential supply vastly exceeds demand over the long-term



Zinc supply growth is set to moderate during 2018, to around 3% after disruption, before re-accelerating in 2019

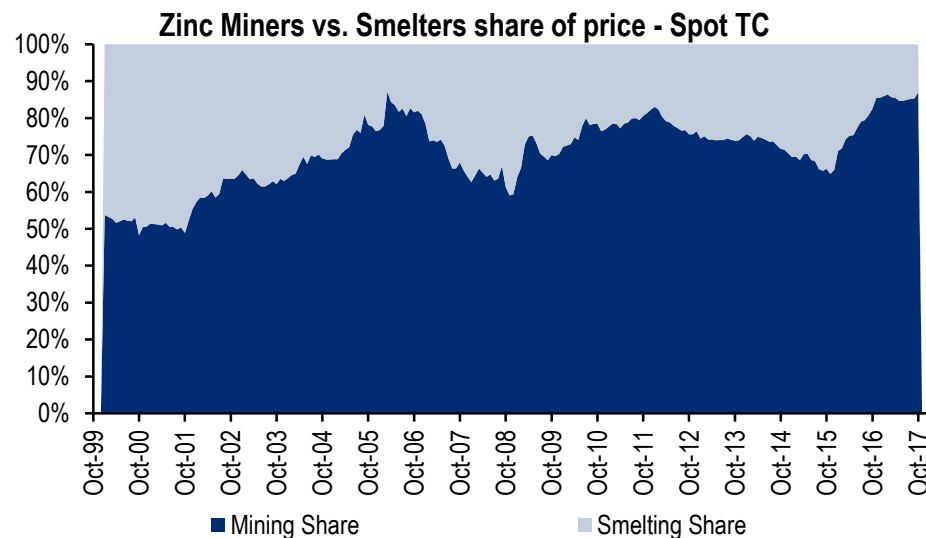
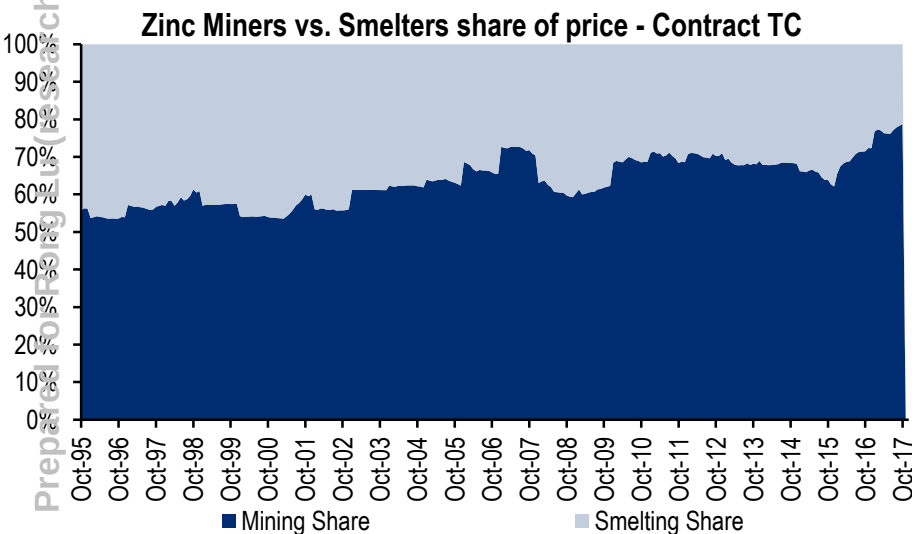
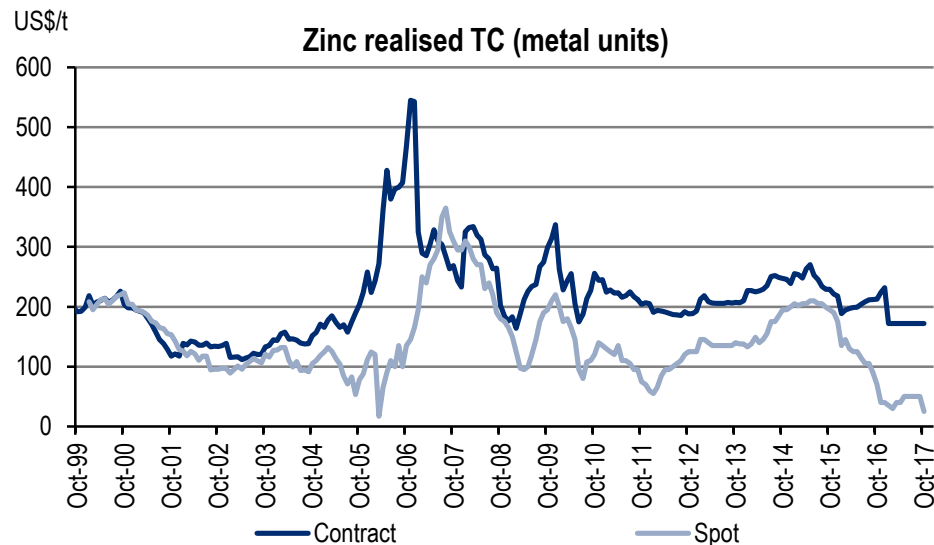
Mine (kt Zn)	Country	2016 Output	2017f Δ	2018f Δ	2019f Δ	2019f Output	2019-2017 Δ
Gamsberg	South Africa	0	0	50	150	200	200
Dugald River	Australia	0	10	110	50	170	170
McArthur River	Australia	200	5	40	125	370	170
Mount Isa Pb/Zn	Australia	288	2	40	120	450	162
Penasquito	Mexico	140	34	-4	120	290	150
Antamina	Peru	198	167	51	-116	300	102
Century Tailings	Australia	0	0	30	70	100	100
Iscaycruz	Peru	0	0	20	60	80	80
Middle Tennessee	USA	0	25	40	15	80	80
Huoshaoyun	China	0	0	0	63	63	63
Sindesar Khurd	India	94	46	0	12	152	58
Zaw ar	India	4	21	30	7	62	58
Rampura-Agucha	India	496	147	-98	5	550	54
Castellanos	Cuba	0	8	25	17	50	50
Changba (Baiyin)	China	62	11	20	7	100	38
Baoshan	China	12	4	23	10	49	37
Yaojialing	China	0	0	25	10	35	35
Taifeng	China	20	7	11	13	51	31
Terrafame	Finland	23	23	4	0	50	27
Caijiaying	China	42	14	10	0	66	24
Top 20 Total		1,580	522	427	737	3,266	1,686
Other		10,776	513	116	79	11,484	708
Total Supply		12,356	1,035	543	816	14,750	2,394

Source: Antaika, Bloomberg, LME, NBS, PBOC, SHFE, WBMS, Wood Mackenzie, Citi Research

Zinc – annual Treatment Charge (TC) outlook

2018 benchmark – market structure facing threat for yet another year

- The outcome of price participation (0% in 2017) and payability (85% currently) negotiations make estimating the realised TC significantly more complex than just simply looking at the benchmark TC negotiations.
- Our base case forecast for 2018 is a \$170/t TC with no price participation and unchanged payability at 85%, which is an unchanged outcome on 2017, albeit much higher refined zinc prices imply a much higher realised.
- The recent spot TC decline to almost negligible levels is not necessarily a reflection of potential benchmark outcomes. However, between lower TC and increased payability, we believe smelters would happily choose the former and prefer to keep the market structure unchanged.

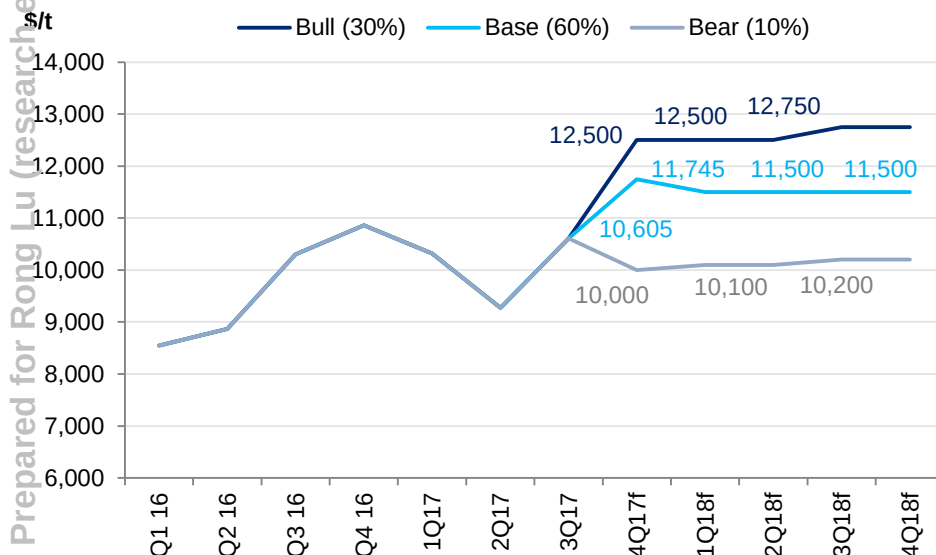


Nickel – neutral view on pricing for 2018

World Nickel Supply Demand Balance

kt	2015	2016	2017e	2018e	2019e	2020e
Mine production	2,145	2,077	2,076	2,080	2,167	2,206
Refined production	1,983	2,023	2,070	2,175	2,275	2,319
- of which: NPI	425	471	570	616	694	749
Supply (%)	-1.0%	2.0%	2.3%	5.1%	4.6%	1.9%
Consumption/Demand	1,880	2,026	2,108	2,178	2,249	2,278
Consumption (%)	1.8%	7.8%	4.0%	3.3%	3.2%	1.3%
Surplus/Deficit	103.2	-3.8	-38.6	-3.7	26.0	41.2
Reported stocks	605.7	594.8	562.2	548.5	574.5	615.7
Reported stock change	80.1	-10.9	-32.6	-13.7	26.0	41.2
Reported stocks (wks)	16.8	15.3	13.9	13.1	13.3	14.1
Implied Global stocks	1023.1	1019.3	980.7	976.9	1002.9	1044.2
- US\$/t	11,886	9,644	10,485	11,500	12,000	13,000

Nickel bull/bear case price forecasts*



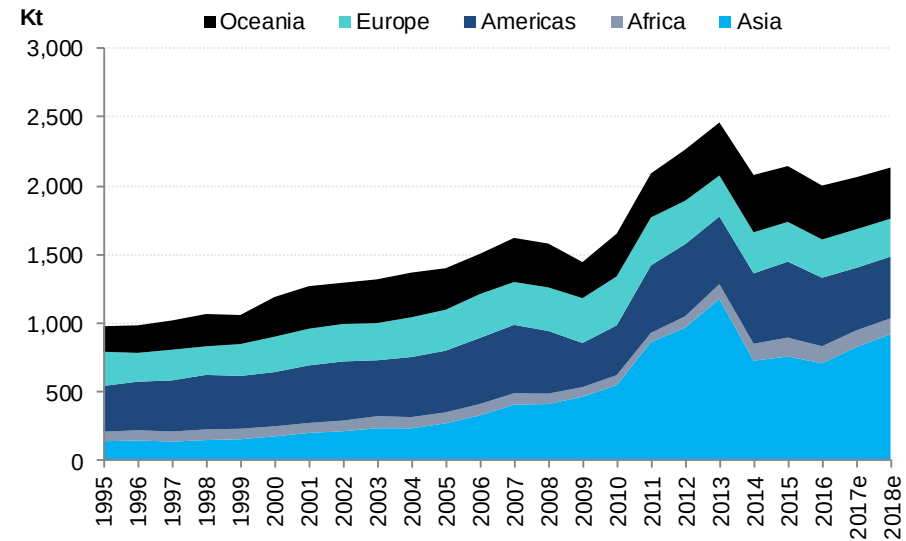
Source: WBMS, Wood Mackenzie, LME, Citi Research. *Subject to revision

- **We hold a neutral view on prices in 2018.** We see nickel supply growth outpacing demand growth during 2018, shifting the market from deficit towards balance.
- **However, we do see nickel prices rising over the medium and longer term.** This reflects our view that nickel is a 'new economy' metal, both from an EV and an income elasticity perspective. Moreover, new high-incentive priced mines will likely need to be built to service future demand for the battery metal. As such, we see the risks surrounding our medium to longer term nickel forecasts are skewed to the upside in our view.
- **To the upside, key risks this year (30%) include:** Better than expected demand from the EV sales could impact prices in 2018 as the market could move to price in the need for future mines to be built. A drawdown of visible stocks (even if simply moved to non-visible locations) could be price supportive especially as Chinese stainless steel mills continue to restock in 1Q'18. Also a change of tune from DENR Secretary Roy Cimatu to resume pressure on active mines deemed erroneous per the audit results conducted earlier in the year, could also be supportive.
- **Key downside risks (10%):** Rising scrap rates could be a headwind. Other downside risks include a potential slump in global stainless steel demand, driven in part by reduced infrastructure investment that typically requires austenitic grade steel in sheet form. EV production ex-China may disappoint and innovation could circumvent nickel as a cathode material in NEV batteries.

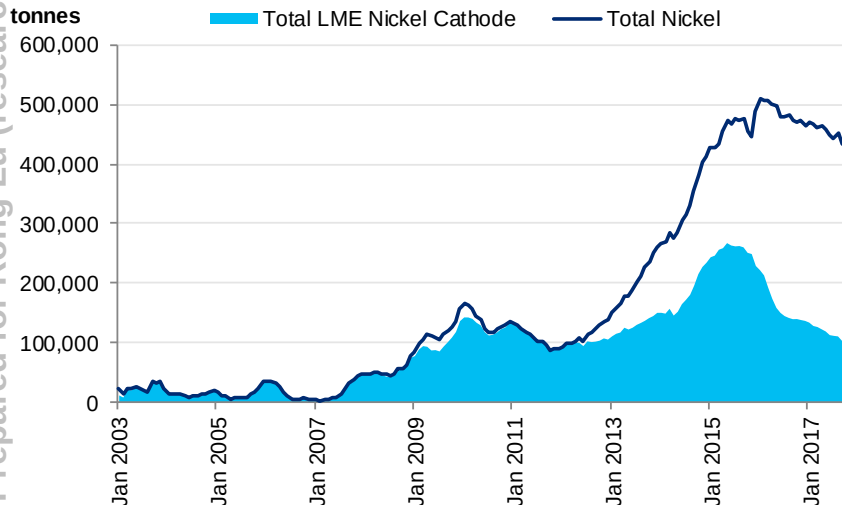
Nickel – capacity cuts have a transitory effect as mine supply should grow

- A round of mine curtailments or closure extensions, despite rising prices and ore exports, supported sentiment **recently**. We note the decision from Vale to put the ~25ktpy Birchtree mine on care and maintenance in September and First Quantum idled the ~20ktpy Ravensthorpe mine in October. Per current available data we expect mine supply to amount to ~2.1mmt in 2018.
- Refined metal stocks have drawn down relative to previous years, though we expect the trend to moderate. LME full plate cathode stocks have fell 35% YTD to 89.4kt in October. Combined LME and SHFE metal stocks have decreased 7.4% over the same period. Moreover, YTD to September exports of Russian nickel are down 25% y/y to 105kt compared to 139kt the year prior.

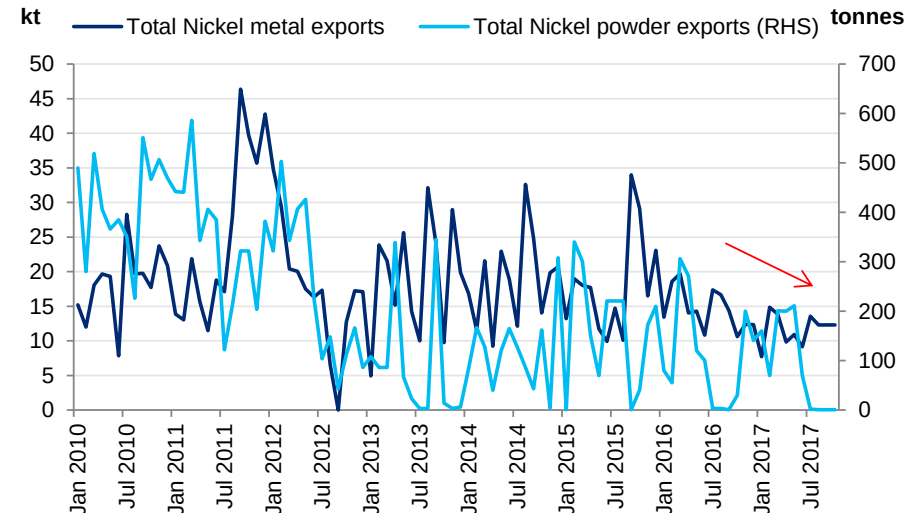
Global mine supply growth expected to pickup



Combined listed stocks have decreased 7.4% YTD



Russian YTD nickel metal exports are down 25% y/y

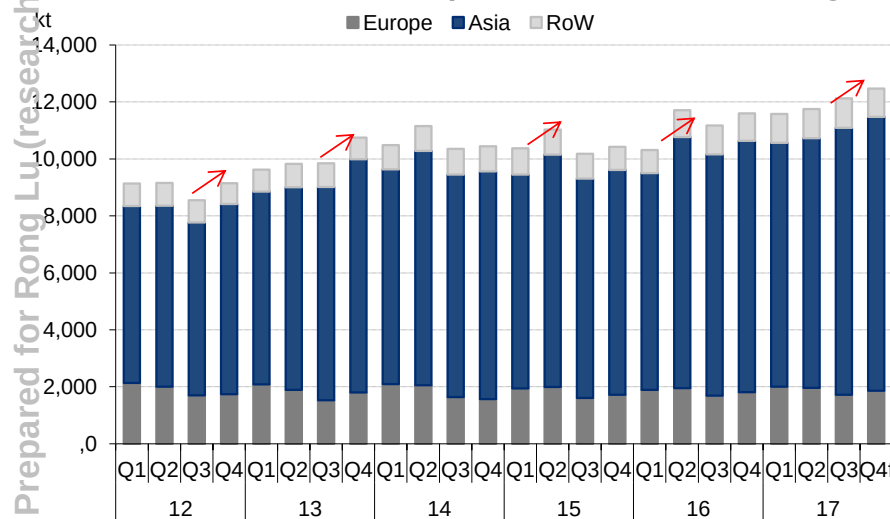


Source: Bloomberg, CRU, INSG, LME, SHFE, Wood Mackenzie, WBMS, Citi Research.

Nickel – demand “renaissance” from EV batteries provides immense upside

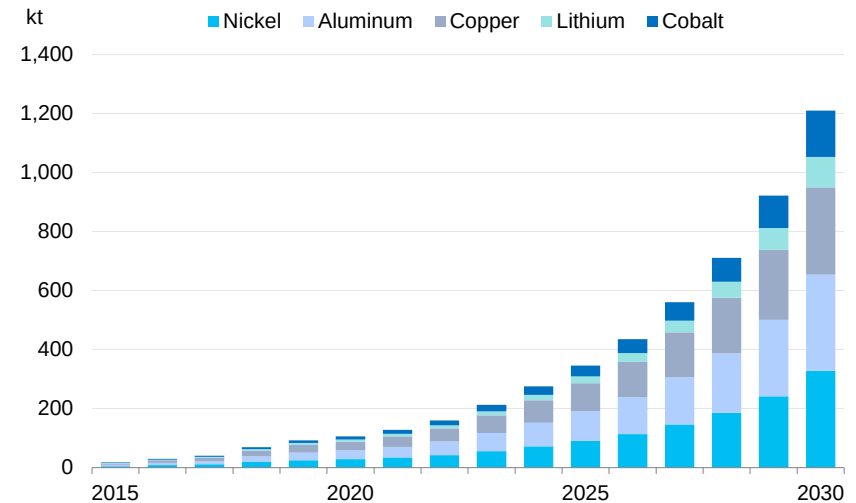
- **Fervent enthusiasm regarding the potential for EV battery demand for nickel has been supportive for nickel pricing.** While we expect nickel demand via the EV sector to amount to ~25kt this year, a supportive policy framework ultimately drives demand growth higher into 2018. We believe risks are skewed to the upside with nickel battery demand potentially doubling next year.
- **Global stainless steel production growth is expected to be supported by rising metal prices and restocking efforts at Chinese mills.** We expect a combination of slack inventory levels from the YTD peak in April and rising alloy surcharge prices to be supportive to both stainless and nickel prices into 1Q'18. However, stainless steel prices may well lose steam should the restocking phase at visible Chinese mills come to a close. Hence, the market deficit in 2018 may narrow should demand for stainless steel soften into 2H'18.

Global stainless steel production continues to grow

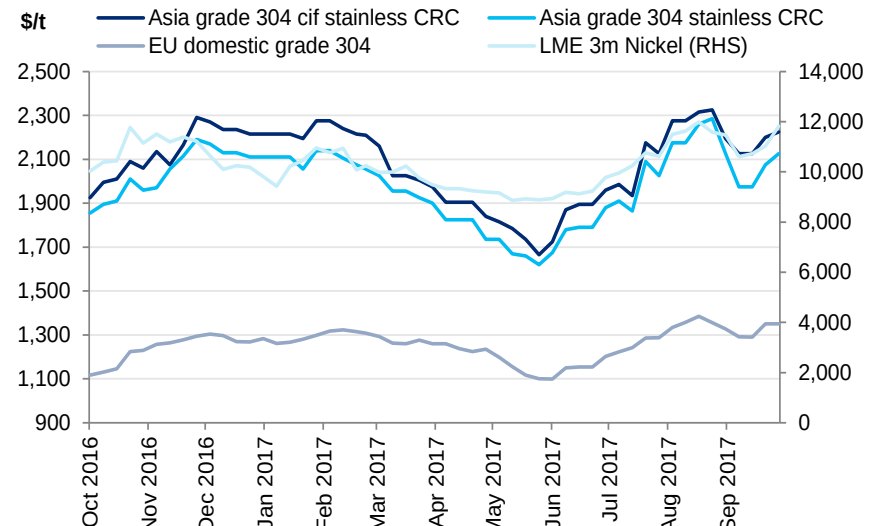


EV battery expected to boost nickel metal demand

Battery manufacturing capacity should double by 2020



Stainless steel prices could soften during 2H'18



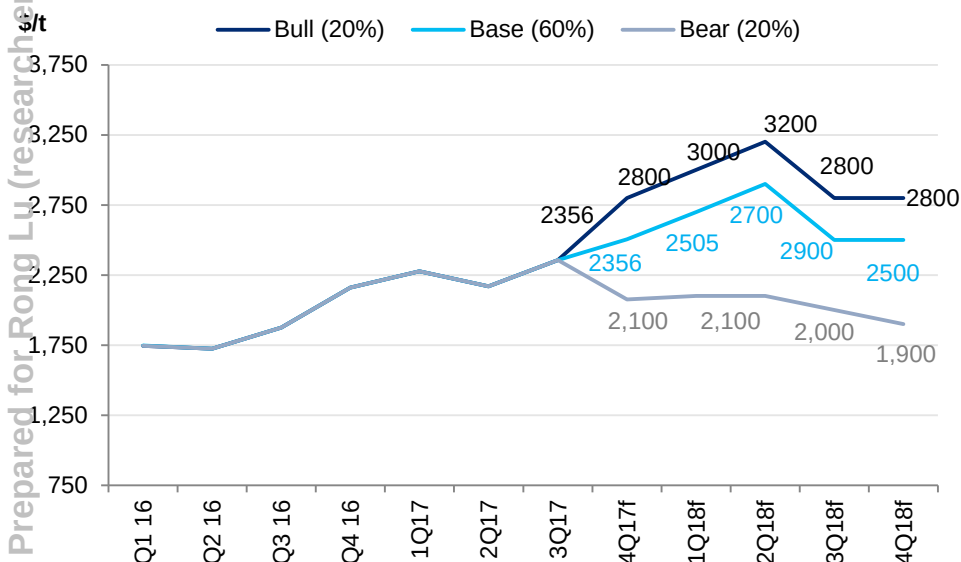
Source: BNEF, CRU, INSG, Wood Mackenzie, Citi Research.

Lead – prices should start the year on the front foot before grinding lower

World Lead Supply Demand Balance

kt	2015	2016	2017e	2018e	2019e	2020e
Mine Production	5,407	5,391	5,103	5,349	5,540	5,578
Primary Production	5,545	5,945	5,697	5,791	5,775	5,865
Secondary Production	6,331	6,391	6,779	7,020	7,271	7,588
Total Lead Supply	11,876	12,336	12,477	12,812	13,047	13,453
Supply (%)	4.0%	3.9%	1.1%	2.7%	1.8%	3.1%
Consumption	11,871	12,259	12,499	12,738	12,978	13,259
Consumption (%)	2.7%	3.3%	2.0%	1.9%	1.9%	2.2%
Surplus/deficit	4.2	77	-23	74	68	194
Reported stocks	616	78	55	129	197	391
Stock change	4.2	-538.0	-22.8	73.7	68.0	194.5
Stocks (wks)	2.7	0.3	0.2	0.5	0.8	1.5
Price (US\$/tonne)	1,795	1,876	2,325	2,650	2,300	2,300

Lead bull/bear case price forecasts*

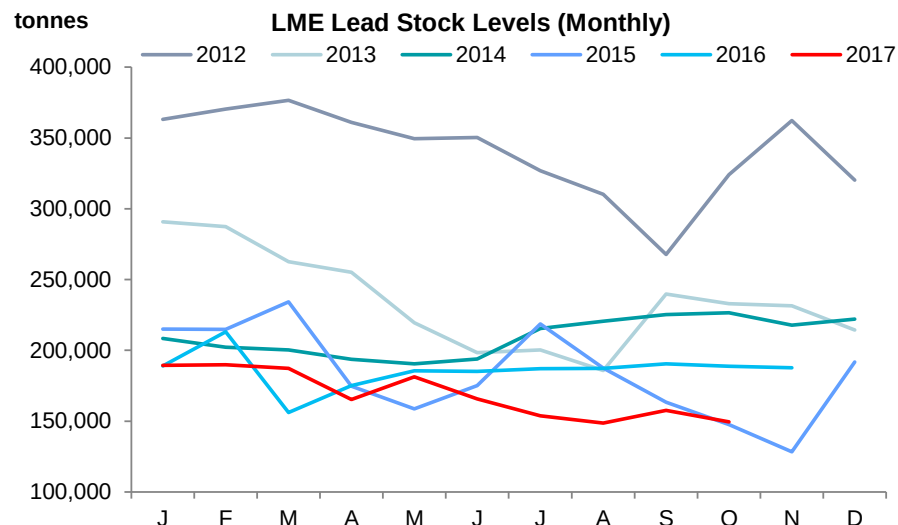


- We upgrade lead prices in 2018 to \$2,650/t (from \$2,255/t) as demand stabilises rather than declines from current levels. We suspect a combination of stable lead-acid battery replacement demand and stable supply to keep prices steady over the mid-term. We see prices tracking zinc prices higher through 1H'18 and lower beyond that, against a backdrop of a moderate surplus lead market.
- **Key downside risks (20%):** A stronger-than-anticipated scrap supply response given the outsized double digit price performance YTD. Should the availability of recycled metal be more than we have forecast in our base case scenario balance, the current deficit of ~27kt could be further reduced. In such an event prices would retrace back closer to ~\$2,000/t levels.
- **To the upside, key risks (20%) include:** Improved macroeconomic conditions in China in the form of better-than-expected GDP and IP data points. In this scenario we would anticipate heightened consumption of automobiles including 2-wheeled motorbikes deepening the deficit for this year, potentially eroding the market surplus for 2018.

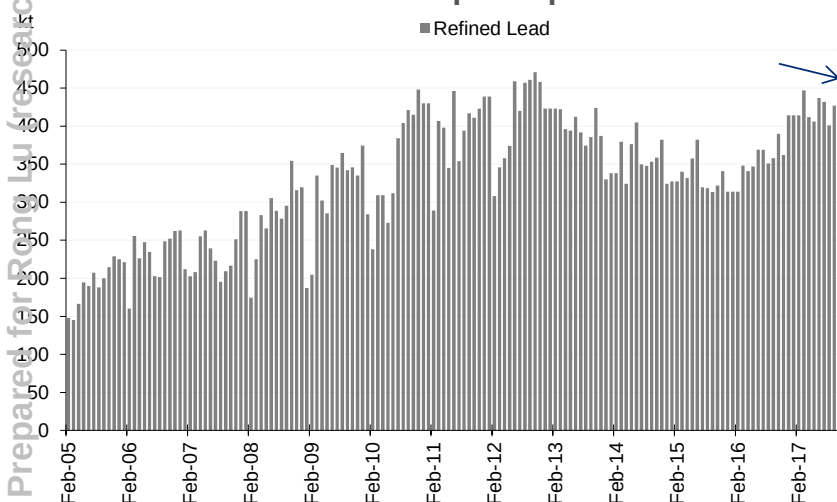
Lead – mixed supply signals

- **A crunch in mine supply has already shown signs of easing.** Negative spot TCs in October reflected severe mine supply tightness. The recent cash/3m backwardation was short-lived as concerns of Chinese zinc availability dissipated following a SHFE stock build of 11% in October to 72kt.
- **LME inventory stocks are very low despite persistent global refined metal output.** We expect global production to grow by 2.7% in 2018 to ~12.8mmt. Chinese refined lead imports have ground lower through 2H'17, largely as a function of the arb-related import glut in Summer, as well as the capacity unwind ahead of the winter shutdowns. A ramp up in domestic scrap feedstock supply was likely incentivized as prices climbed towards \$2,500/t in 4Q'17, but a combination of producer hedging and cooling demand subsequently sent prices lower towards \$2,400/t in mid-November, while physical premia plateaued.

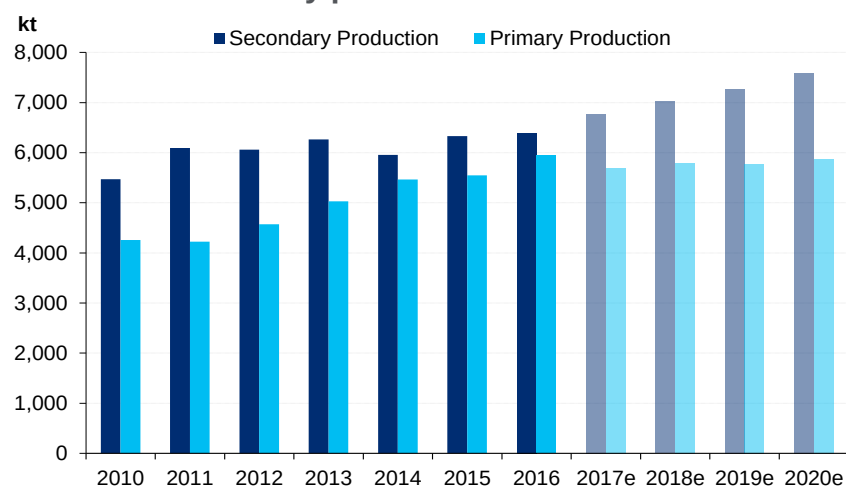
Visible refined inventory is at multi-year lows



Chinese refined lead output expected to flat-line



Global secondary production should continue to rise



Citi base metals price forecasts and bull-bear scenarios

Note: % probabilities apply to the 2018 outlook.

Base metals bull-bear scenarios

Scenarios	3Q17	4Q17f	1Q18f	2Q18f	3Q18f	4Q18f	1Q19f	2Q19f	3Q19f	4Q19f	1Q20f	2Q20f	3Q20f	4Q20f
Copper														
Bull (20%)	6387	7100	7000	7100	7300	7400	7500	7500	7500	7500	8000	8000	8000	8000
Base (70%)	6387	6840	6500	6500	6600	6800	6800	6800	6800	6800	7000	7000	7000	7000
Bear (10%)	6387	5650	5500	5500	5500	5500	5500	5500	5500	5500	5500	5500	5500	5500
Aluminium														
Bull (20%)	2028	2300	2350	2400	2400	2400	2500	2500	2500	2500	2500	2500	2500	2500
Base (60%)	2028	2145	2100	2100	2100	2100	2200	2200	2200	2200	2200	2200	2200	2200
Bear (20%)	2028	1900	1900	1900	1900	1900	1900	1900	1900	1900	1800	1800	1800	1800
Nickel														
Bull (30%)	10605	12500	12500	12500	12750	12750	12750	12750	12750	13000	13000	14000	14500	15000
Base (60%)	10605	11745	11500	11500	11500	11500	12000	12000	12000	12000	13000	13000	13000	13000
Bear (10%)	10605	10000	10100	10100	10200	10200	10250	10250	10500	10500	10500	11000	11500	12000
Zinc														
Bull (20%)	2965	3500	4000	4000	3700	3500	3200	3200	3200	3200	2800	2800	2800	2800
Base (60%)	2965	3215	3400	3600	3200	3000	2800	2800	2500	2500	2500	2500	2500	2500
Bear (20%)	2965	2500	2400	2400	2300	2200	2200	2200	2200	2200	2000	2000	2000	2000
Lead														
Bull (10%)	2356	2800	3000	3200	2800	2800	2600	2600	2600	2600	2600	2600	2600	2600
Base (60%)	2356	2505	2700	2900	2500	2500	2300	2300	2300	2300	2300	2300	2300	2300
Bear (30%)	2356	2075	2100	2100	2000	1900	1900	1900	1900	1900	1700	1700	1700	1700

Prepared for Rong Lu (research email)

Lithium – Supply responding slower than demand

We expect lithium demand to remain strong (2016-2021 CAGR c17%) and supply to continue to disappoint. We upgrade our lithium carbonate price forecasts by 8-25% over medium-term, though we still see price moderating from the current spot (~US\$20k/t).

- Demand Supported by NEV Growth** – We continue to see demand being supported by Chinese NEV growth over medium-term and steady industrial activity globally, which drives modest growth in traditional industrial applications of lithium. On the other hand, supply has continued to disappoint as project timelines slip, commissioning of new projects take longer than expected. Environmental inspections also disrupted the lithium carbonate supply chain in China resulting in further supply side tightness during 2H'17. Overall we expect the market to be largely balanced in early 2018, before moving into surplus in 2019+ as at least 6 new projects commission over next 12 months. Key additions are from Australia, followed by North America.
- Strong Pricing Environment to Continue Near-Term** – We expect strong pricing environment near-term driven by tight supply/demand fundamentals. We upgrade our lithium carbonate price forecasts by 8-25% over medium-term, though we still see prices moderating from the current spot (~US\$20k/t). For the producers that are still not realizing “spot” prices there could also be some catch-up over the next couple of years as older contracts expire, even if the price does moderate from the current levels.

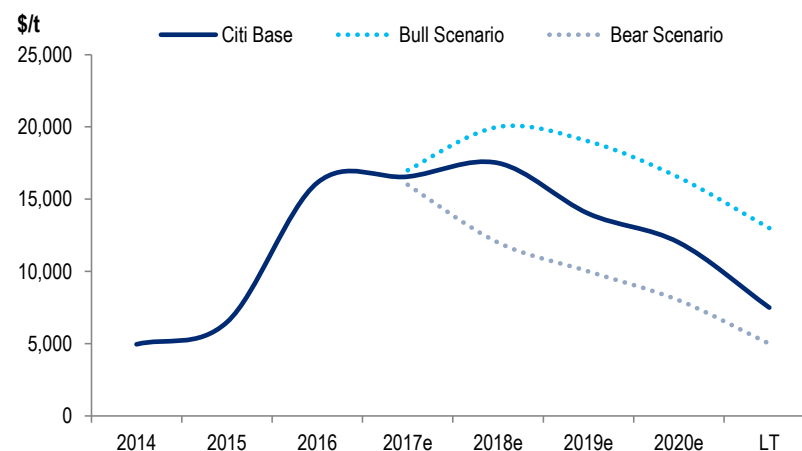
Lithium Supply Demand and Price Forecasts

Summary	2016	2017e	2018e	2019e	2020e
Existing Brine (80% utilisation)	116,050	130,140	149,085	170,660	172,240
Existing Minerals (80% utilisation)	94,950	129,070	138,512	127,664	127,664
Supply under construction (50%)		5,500	29,727	117,644	173,897
Supply under development/evaluation (20%)				240	8,596
Incremental supply lag offset	-	(32,804)	(19,821)	(39,176)	(31,264)
Total Supply	211,000	231,906	297,504	377,033	451,133
YoY %	13%	10%	28%	27%	20%
Total Demand	210,979	233,867	263,222	311,111	379,006
YoY %	12%	11%	13%	18%	22%
Balance	21	(1,961)	34,282	65,922	72,127

Li Carbonate 99% (Industrial Grade) Price (US\$/t)					
New	16,139	16,559	17,500	14,000	12,000
Old	16,139	15,309	14,000	12,000	10,000
% Ch.		8%	25%	17%	20%

Source: Company reports, Roskill, Citi Research Estimates

Lithium Price Forecasts with Bull/Bear Case

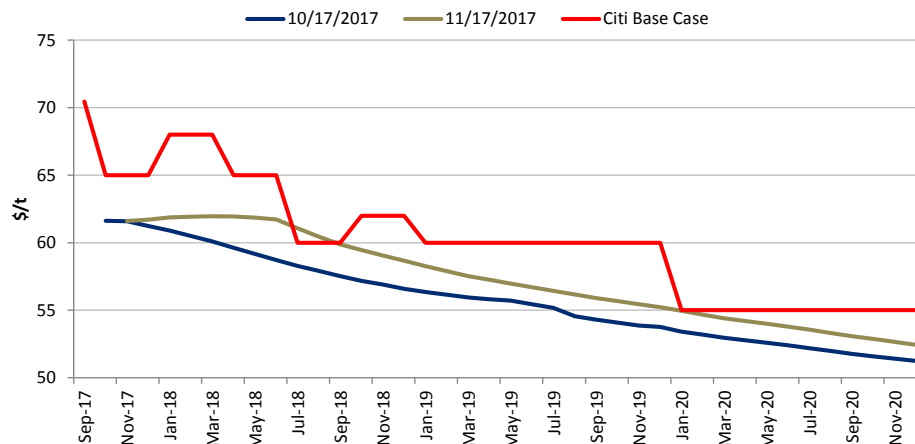


Source: AsianMetal, Citi Research Estimates

5. Bulks: bullish short term iron ore outlook, but bearish the complex medium term

Bulk commodities – Top trade: go long 1Q'18 iron ore for asymmetric upside

We forecast that iron ore prices will rally c.10% to average \$68/t during 1Q'18, with prices trading above \$70/t around Chinese New year, before moderating over the medium term. The risk-reward of going long iron ore over the coming weeks appears the strongest across the metals complex, and we see asymmetric upside risks.



The rally is expected to be driven by a further tightening of the Chinese steel market, Chinese steel mills' active restocking of high-grade iron ore, seasonally weak seaborne supply, and a recognition that iron ore supply growth passes its peak during 1Q'18.

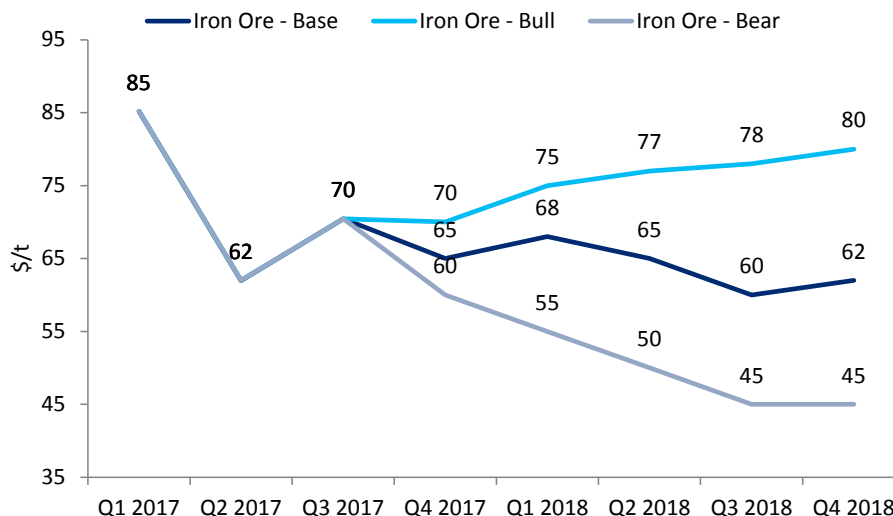
The risk-reward of going long iron ore over the coming weeks is in our view the strongest across the bulk commodities and industrial metals complex. Should Chinese and global growth continue, iron ore could easily overshoot to reach \$75/t during 1Q'18. To get the same return on a zinc, prices would need to rally to \$3800/t and copper, would need to reach \$8300/t.

The upside to iron ore prices is far from being priced in despite iron ore curve shifting to contango recently. Benchmark iron ore prices should see greater upside when a restocking cycle occurs, because current robust steel margins should allow mills to afford much more expensive ores. The restocking cycle post 2018 Chinese New Year could be stronger than historical patterns since blast furnaces with an aggregated ~300Mt/y capacity are expected to return from winter curtailments.

We understand actual restocking activities will unlikely begin until early March, given that the upcoming Chinese New Year will be mid-February. However the iron ore market tends to price in physical buying activity ahead of time, especially since the Dalian contract might have largely been dominated by speculative flows. In 2017, iron ore prices saw rallies in 1Q and 3Q, when actual physical activity was relatively weak.

Iron ore - we revise up 2018 and 2019 prices and expect a rebound in 1Q'18

We revise up short and medium-term iron ore prices to reflect more constructive demand for high-grade iron ore than earlier expected. Our modestly bearish longer term view on prices is informed by the likelihood of slowing demand growth as China shifts in to its new era of quality led economic growth.



Base Case (70%): Iron ore prices rise to \$68/t during 1Q'18 and soften to \$62/t by the end of 2018. At present, the global steel market is tight, as evidenced by solid producer margins. This is particularly the case in China, where steel spreads are tight, margins are strong, and exports are falling. Demand for high-grade iron ore could be higher than expected thanks to steel mills' flight to quality. Our modestly bearish longer term view on prices is also informed by the likelihood of slowing demand growth as China shifts in to its new era of quality-led economic growth, which is likely to see construction activity play less of a role in driving growth domestically.

Bull Case (15%): Iron ore prices march higher to \$70-75/t by the end of 2017 and reach \$80/t by the end of 2018. Chinese steel demand growth accelerates in 2018 thanks to stronger-than-expected China liquidity easing. Winter cuts of steel are enforced more strictly than expected. Seaborne supply growth from Brazil and Australia falls short of expectations.

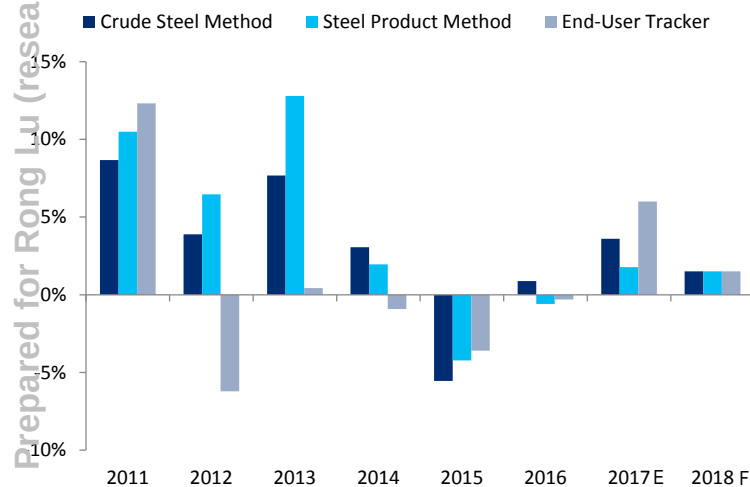
Bear Case (15%): Iron ore prices touch lower to \$55-60/t by the end of 2017 and \$45/t by the end of 2018. Chinese economic growth and real estate investment decelerate more rapidly than expected, and previously closed induction furnaces in China are implicitly allowed to restart operation. Scrap use in global steel production accelerates.

Iron ore - Chinese steel demand growth to remain positive in 2018

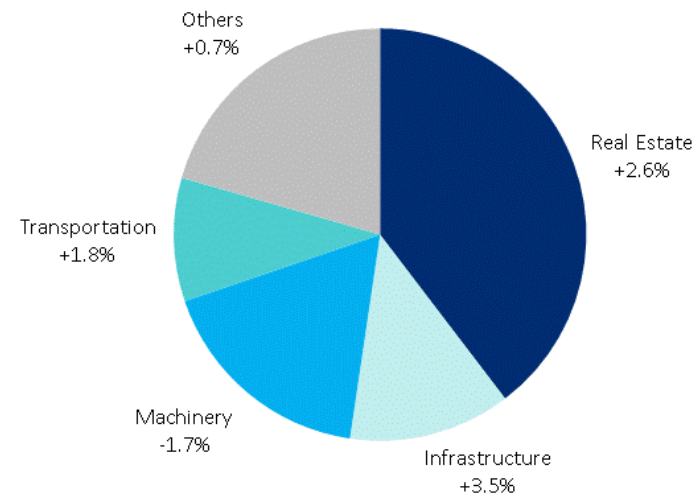
We expect Chinese end-user steel demand to rise by ~1.5% y/y in 2018 compared with ~3% y/y in 2017, with the growth driven primarily by the real estate and infrastructure sectors. Machinery steel demand should taper off partly because construction machinery purchase is likely to have been front-loaded in 2017.

- Given the headline assumption that the Chinese economy should face challenges but will unlikely slow down dramatically in 2018, Citi's China Property Team forecasts real estate investment to decelerate to 5% y/y and new starts growth to slow to 3.7% y/y. As property developers refilled their land bank in 2017 (land sales +6% y/y), the team expects that momentum to start new projects, particularly in tier 3, 4 cities will be solid. Meanwhile, the government's commitment to social housing construction (for rental and purchase) and shantytown upgrading should continue to boost demand for construction steel products. Infrastructure growth is set to slow down, but local government financing conditions are sufficient to allow another year of growth in investment. On the machinery side, a potential slowdown of construction machinery output and tepid investment in general and electric machinery are likely to dampen steel demand.
- Growth in home appliance demand should slow down substantially during 2018, following the trend in property sales, however the slowdown may be mitigated by rising exports and rising penetration rates. Property sales are set to decline 6% y/y in 2018. Growth in Chinese auto sales may also slow down due to high base of 2016 and 2017, and owing to a hike of small-engine car consumption tax from 7.5% to 10% effective in early 2018 (following a hike from 5% to 7.5% in 1Q'17).

China steel output set a record high in mid-Aug



Property and infrastructure drove steel demand growth in 2017

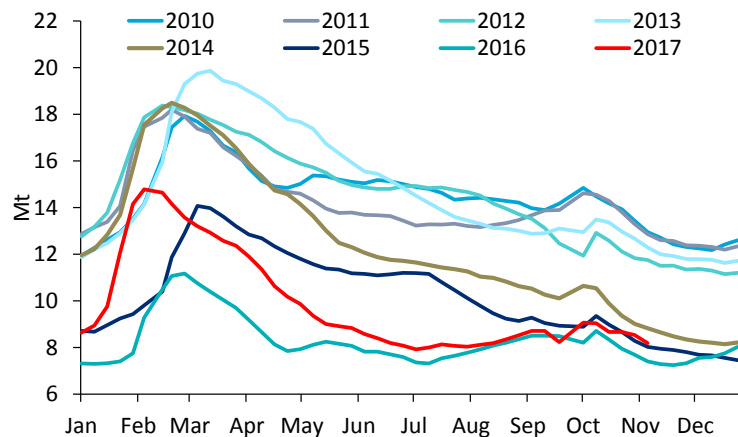


Source: NBS, Wind, Citi Research

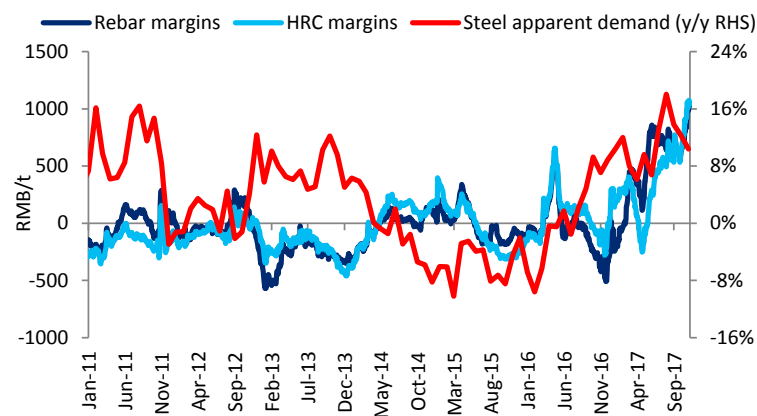
Iron ore - restarts of shuttered steel mills point to a restocking cycle in 1Q'18

We see c.10% upside to iron ore prices by around Chinese New Year. Widened grade differentials could support the operations of Chinese iron ore mines (~30% Fe for raw ore and ~66% Fe for concentrate) for a longer time period than earlier expected.

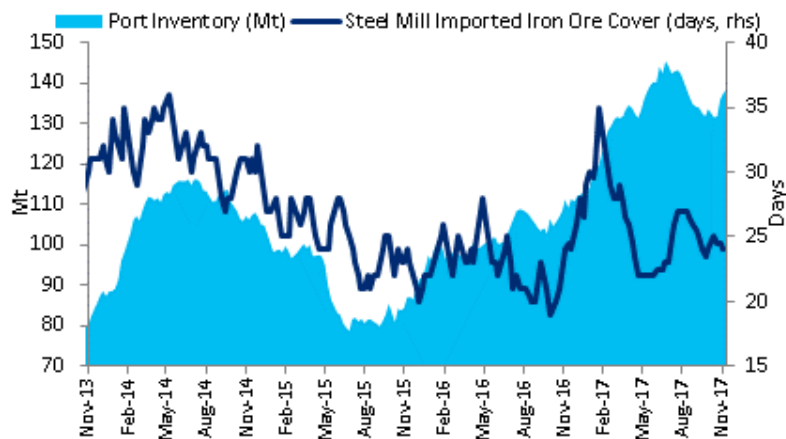
China's steel product inventories point to a tight market



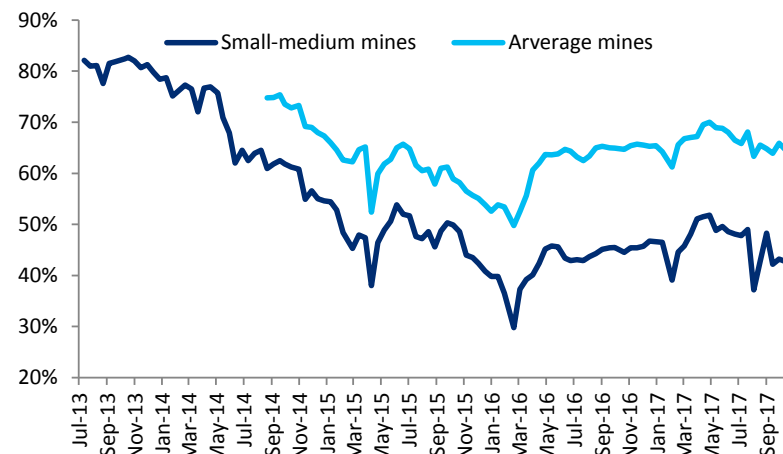
High and rising Chinese steel margins and solid apparent demand growth also point to tightness



Iron ore inventories at Chinese ports are in part WIP inventory, others low grade, and not high vs demand



Chinese iron ore mine operating rates should fall in winter as environmental crackdowns kick in



Iron ore - strong demand for high-grade ores means wide grade differentials

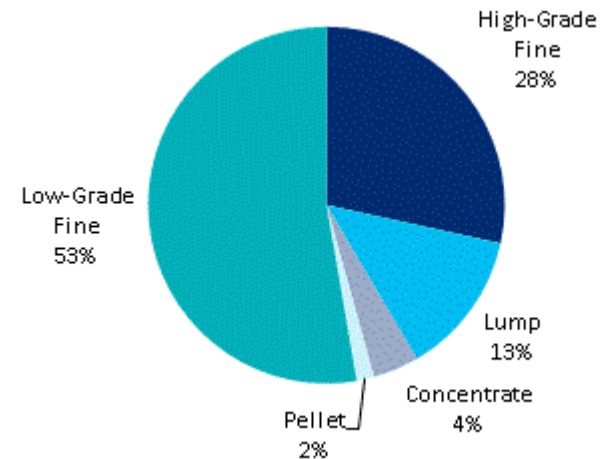
If steel mills' margins stay higher for longer, mills' appetite for mainstream iron ore products will also be higher for longer. The high-grade ore market could shift to deficit because an increasing number of mills are likely to choose to consume a higher share of high-grade ores.

- **The strength in benchmark iron ore prices reflects structural tightness in the high-grade iron ore market**, which was exacerbated in recent months due in part to robust Chinese blast furnace margins and continued environmental crackdowns, which created higher discount for silica contents and other impurities. As long as Chinese steel mills' cash margins are sustained above ~RMB500/t, the oversupply status of the low-grade iron ore market should become increasingly irrelevant to benchmark prices, and the discounts of low-grade products could remain fairly wide.

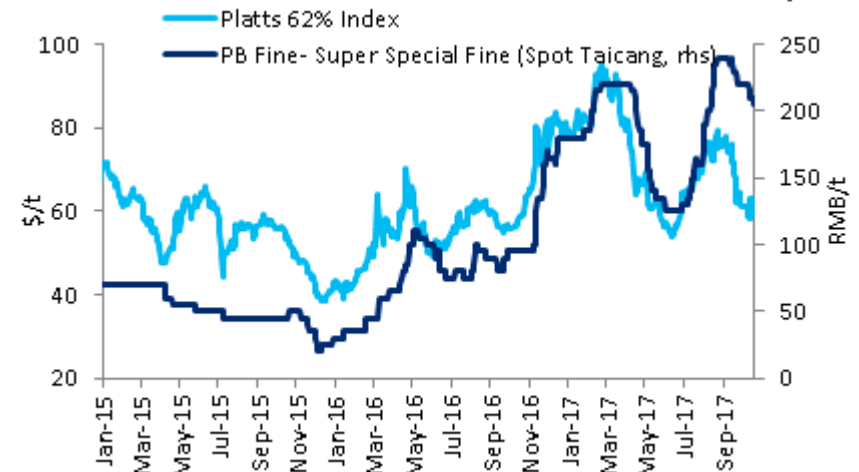
Adding to the market tightness are weak Chinese imports of iron ore with 60% and above Fe content and tepid domestic iron ore output in recent months due to intense environmental inspections in Hebei and Shandong. Iron ore inventories at Chinese ports fell over 7Mt from its peak of 132.6Mt in recent weeks with the majority being low-grade fines.

Higher Chinese steel scrap intake has been challenging iron ore demand, but the bearish impact of this may have peaked. The one-off closure of induction furnaces might have released 60-70Mt/y of scrap supply to the broader steel market, most of which has been inherited directly by blast furnace and new electric arc furnaces in China and abroad. Average scrap use in Chinese blast furnace operations may have risen from 7-8% to 12-13% over the past few months.

Chinese port iron ore inventories have been dominated by low-grade products



Grade differentials move in tandem w/ benchmark prices

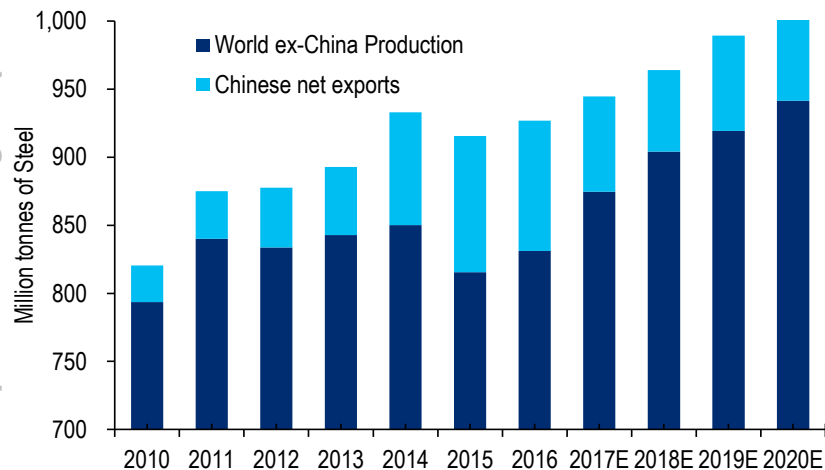


Iron ore - Ex-China steel output sees the highest growth in 6 years

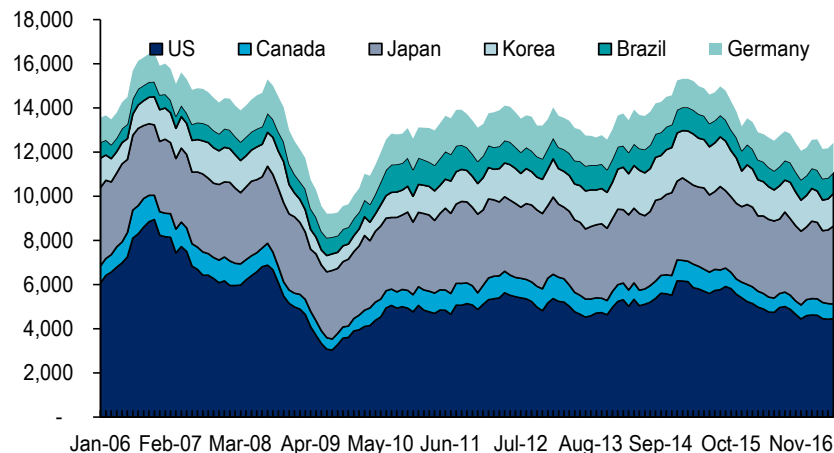
Apparent demand ex-China is effectively flat y/y at the 9M'17 stage; but reduction in Chinese exports by 35mn tonnes/annum annualized is providing more room for output to grow, especially in emerging markets. Our base case is for 1% steel production growth ex-China during 2018.

- **Steel production ex-China has increased by 5.0% per annum in 9M'17 and is on track to increase by c.39mn tonnes, the biggest increase since 2016 and new high for world ex-China steel output.** The renaissance in world ex-China steel production occurred due to lower Chinese steel exports. Steel production ex-China declined from 841mn tonnes in 2011 to 812mn tonnes in 2016, despite apparent demand increasing from 875mn tonnes in 2011 to 927mn tonnes in 2016. This was predominantly due to net exports from China increasing from 35mn tonnes in 2011 to 96mn tonnes in 2016. However with supply side reforms and lower production in China, exports are on track to reduce to below 65mn tonnes in 2017.
- **Apparent demand ex-China has grown 0.5% y/y.** There have been expectations of improving demand in Europe, but it has not come through in the data yet. US demand has disappointed after high expectations last year as the infrastructure spend appears to have been pushed out. Destocking of steel ex-China has been significant with flat steel inventories down 5%/y/y.
- **Our base case is for 1% steel production growth ex-China during 2018, comprising c.5mt of BF output and c.4mt of EAF output.**

Reduction in Chinese net exports benefits ex-China steel production



Ex-China steel inventories down 23% from their Feb-15 peak



Source: Worldsteel, WIND, CRU, Citi Research

Iron ore - supply growth is passing its peak; future growth largely high-grade

We expect seaborne iron ore supply to grow 50-60Mt/y in 2018 and 30-35Mt/y in 2019 (pre disruption allowance), compared with an average of ~80Mt/y during 2015-2017. S11D and Roy Hill are both at or close to their nameplate capacity.

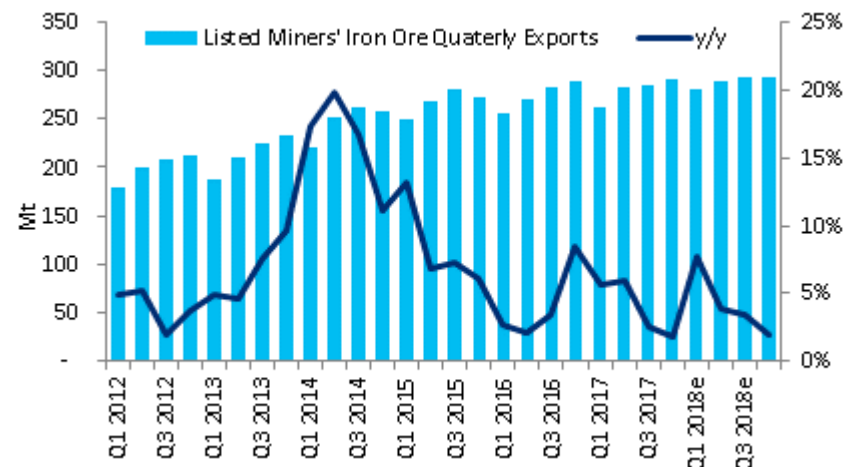
- Chinese iron ore output – traditionally the swing supplier - will be unable to respond to a surprise pick up in demand given a 60% collapse in FAI in iron mining over the past 2 years.
- Future capacity expansion should be focused on high-grade products. S11D continues to advance with mine site now 99% complete and rail duplication 76% complete. Sao Luis pellet plant is still schedule to reopen in 1H'18. Roy Hill is on track to reach capacity soon and we forecast shipments of 55 Mt/y from 2018 onwards. Supply additions may also come from debottlenecks by BHP and Rio Tinto and ramp-up at Sino Iron. Samarco will unlikely return to production until 2019.

Prepared for Rong Lu (research email)

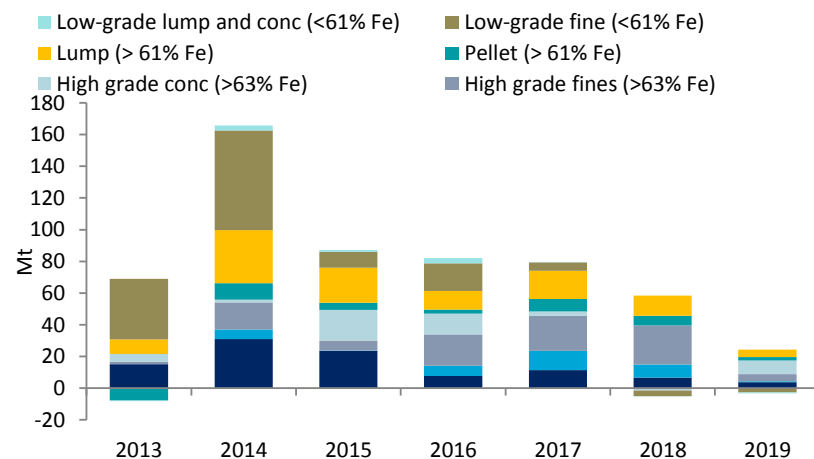
Low-grade iron ore supply is likely to take a hit. Vale opted to curtail high silica production and has announced ~20Mt/y curtailments so far. Indian exports should also face challenges of an oversupplied low-grade market and a lack of competitiveness against some low-cost producers. We anticipate Indian iron ore exports to fall from the current 28Mt/y to ~10Mt/y towards the end of the decade.

That said, capping the upside in the near term, and contributing to our modestly bearish medium term outlook is the likelihood of incrementally higher steel scrap use in steelmaking process, thanks to improvement of steel scrap availability in China and other EM countries, as well as to better economics of running scrap-based mill.

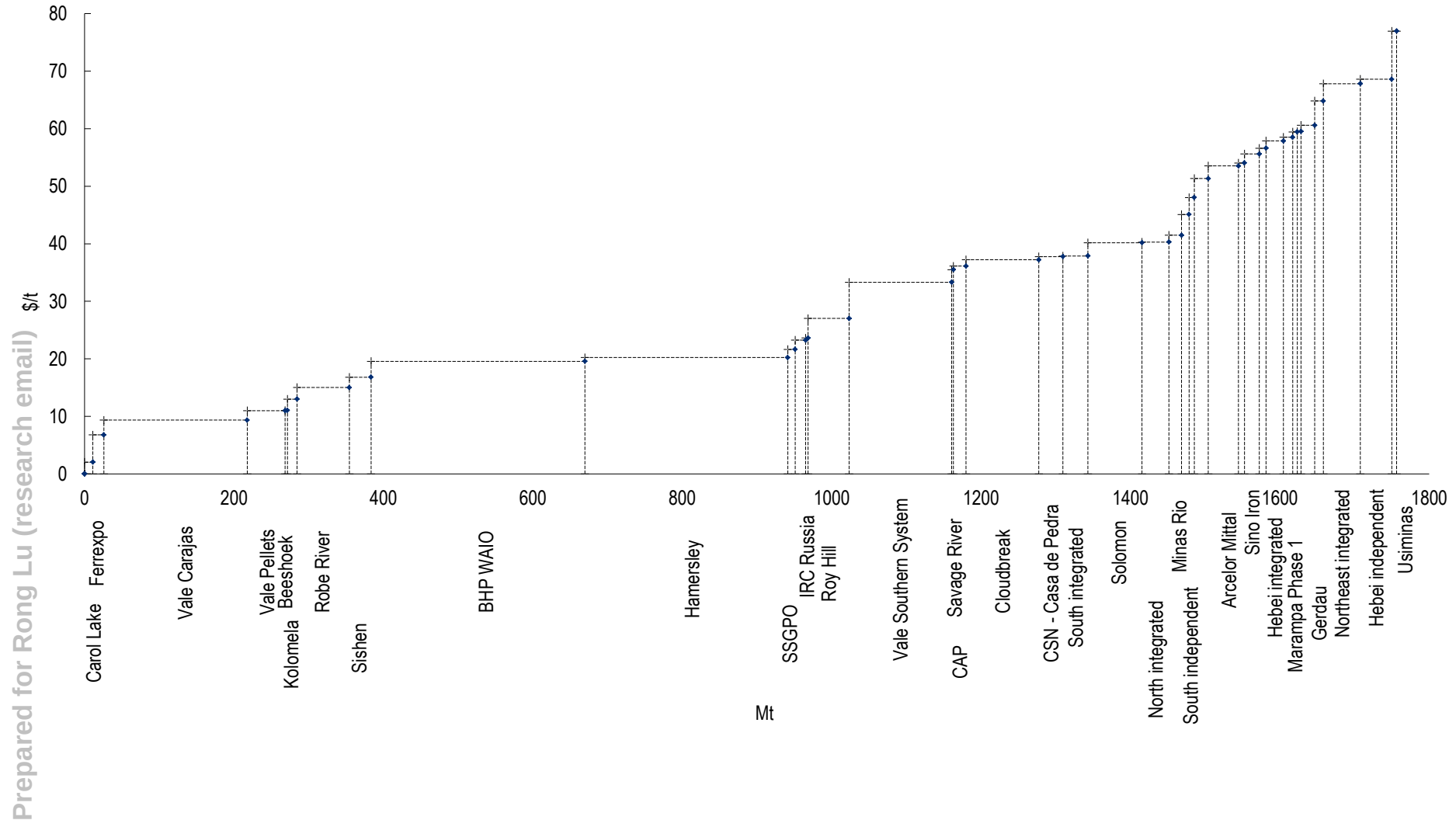
Listed miners' export growth should taper off from 2018 onwards



Global capacity changes by types of products



Iron ore - Current pricing levels are supported by the global cost curve



Source: Company reports, Citi Research, *62% fine normalized, with price adjustments on grade and product type

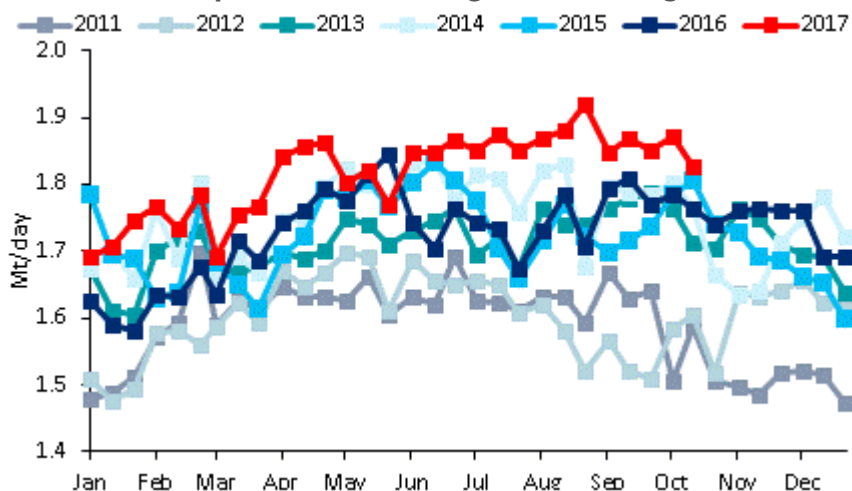
Iron ore - steel winter supply cuts should be in line with expectations

The Chinese flat steel product market is set to experience some tightness towards the year end as supply cuts are being carried out, but the long product market could be more balanced between supply and demand cuts. Capacity shutdowns in the past two years are unlikely to reverse but EAF restarts could help balance the market.

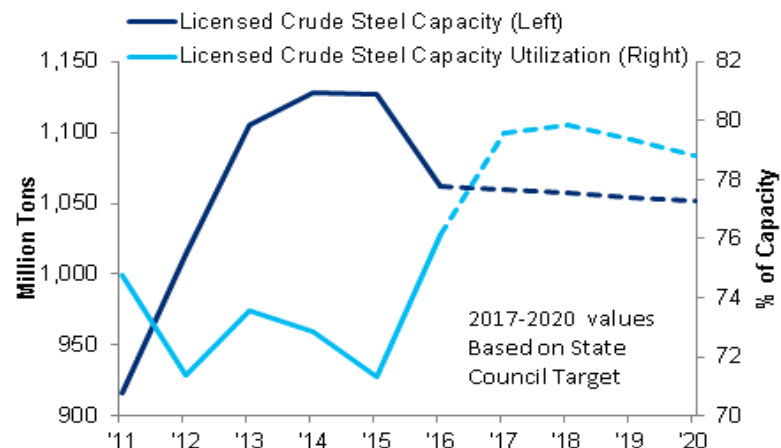
- While the market attention centers around the ongoing winter production cuts at blast furnaces surrounding Beijing, we believe the actual production losses should be in line with expectation, but risks could skew to downside for three reasons. First, steel mills providing steam to urban areas could be exempted from the cuts. At least two large mills in Hebei may have received exemption because of this. Second, instead of requiring all mills to cut 50%, cities including Tangshan have decided output cuts of individual mills based on their track records of environmental compliance. The final scale of the cuts could thus be slightly lower than 50%. Finally, for mills that are only required to cut <30% of their normal output, it appears difficult to monitor actual curtailments because they will likely keep furnaces running while reducing iron feedstock input.
- The government-led steel capacity closures since the beginning of 2016 might be close to an end. So far China may have shuttered over 75Mt/y of blast furnace capacity and over 60Mt/y of induction furnace capacity. The current goal is to keep these operations offline and strictly ban blast furnace restarts. Yet additions of electric-arc furnace (EAF) capacity (in the form of restarts and new construction) has been underway. 20-30Mt/y of EAFs might be brought online by the end of 2017, and another 10Mt/y could be commissioned in 2018. While these additions might not fully offset the earlier cuts, spot steel margins should normalize from the current 5-year high levels but on an average basis stay higher than the past.

Prepared for Rong Lu (research email)

China steel output set record high in mid-Aug



Government-led capacity curtailments close to completion



Iron ore - benchmark prices to rise near term under 2-tier system

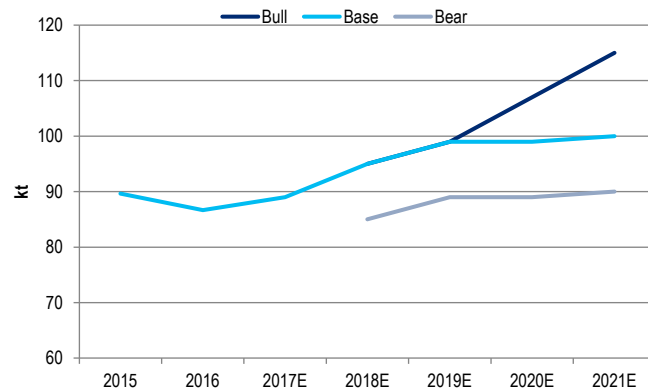
Iron ore prices (62% Fe index) weakened to ~\$62/t very recently compared with ~\$80/t in August thanks to weakened restocking activity by steel mills and rising seaborne supply of high-grade ores. Citi expects benchmark iron ore prices to rebound to an average of \$68/t in 1Q'18 before falling to \$60-62/t in 2H'18.

Mt	Q1 2017	Q2 2017e	Q3 2017	Q4 2017e	Q1 2018e	Q2 2018e	Q3 2018e	Q4 2018e	2014	2015	2016	2017e	2018e	2019e	2020e
Supply (Exports)															
Rio Tinto	77	78	86	90	78	87	90	90	288	319	328	330	345	355	360
BHP	63	71	63	77	65	72	69	69	242	261	261	274	275	293	293
FMG	40	45	44	41	40	45	43	43	153	167	171	169	170	170	170
Australia	192	208	212	227	205	227	224	224	718	767	810	839	880	914	924
Vale	86	92	94	94	86	94	100	100	319	340	349	366	380	386	386
Brazil	91	93	97	101	93	101	107	107	344	366	374	382	409	441	440
India	12	7	3	6	6	6	2	7	10	4	22	28	20	20	10
South Africa	16	16	17	15	14	14	14	14	66	65	65	64	56	56	56
Canada	9	8	9	8	9	8	9	8	31	30	34	34	34	34	34
Others	53	52	51	52	53	53	53	53	213	195	204	208	211	214	212
Additional Price Induced Curtailments					0	-1	-1	-1				0	-3	-38	-124
Total Seaborne Exports	373	384	390	409	380	408	407	412	1,382	1,428	1,508	1,556	1,607	1,642	1,553
Ex-China Demand (Imports)															
Japan	30	33	31	33	33	33	33	33	136	131	130	127	134	136	137
Korea	17	17	19	19	19	19	18	19	74	73	72	73	76	76	76
Taiwan	6	6	6	6	6	6	6	6	23	24	23	25	25	25	25
EU	36	37	36	35	38	39	36	36	153	150	144	145	149	151	153
Others	19	18	19	22	20	18	20	24	47	67	71	78	83	89	94
Total Seaborne Imports	109	112	111	116	116	116	114	119	433	445	440	448	466	476	487
Trade data reconciliation	7	-3	-1	-13	1	-7	-5	-15	-15	-30	-43	-10	-25	-26	-21
Market balance ex-China (62% Fe)	264	262	271	273	259	278	281	271	911	930	1,000	1,071	1,090	1,112	1,019
Chinese Market															
Crude Steel Production (NBS reported)	201	219	219	205	208	217	214	205	823	804	808	843	844	837	829
Scrap Use	19	20	20	22	21	22	23	24	63	70	74	80	89	97	99
Total Demand (62% Fe)	292	319	318	293	300	313	305	289	1,215	1,173	1,175	1,221	1,207	1,183	1,167
Balance Pre-Domestic Production	-28	-57	-47	-19	-41	-35	-24	-18	-304	-243	-175	-150	-118	-71	-148
Domestic Production (62% Fe)	56	64	64	66	60	59	59	56	303	257	245	250	234	206	187
Market Balance	29	7	17	47	20	24	35	38	-1	14	70	100	116	135	39
% seaborne exports +Chinese supply	7%	2%	4%	10%	4%	5%	7%	8%	0%	1%	4%	6%	6%	7%	2%
Iron Ore Price & Forecast	85	62	70	65	68	65	60	62	97	55	58	71	64	60	55

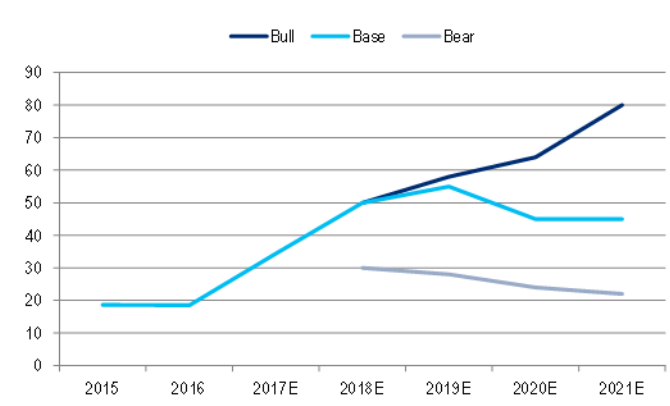
Steel-related Vanadium - outlook moderately bullish near term

We expect vanadium demand to increase by 10kt or 10% in the 2017-2019 period mainly due to increased weightings in Chinese rebar. Demand from vanadium redox battery applications is likely to be challenged by a lack of available vanadium supply coupled with high battery costs.

Vanadium demand scenarios, kt



Vanadium price scenarios, \$/Vkg



Rebar weighting proposal – In July, the China Iron and Steel Research Institute (CISRI) proposed to increase the use of Vanadium in rebar, which would result in an approximate 10-15kt per annum increase in demand, we estimate depending on the range of products captured and the exact standard selected. Our model assumes a positive decision on this recommendation with an increase in demand of 10kt taking place over a 24-month period.

Battery applications challenged – vanadium redox flow batteries possess a number of qualities that make these batteries potentially attractive for use in distributed locations in conjunction with uneven load flow and uneven use over time. However, meeting this potential demand does not look feasible. One scenario of 100Mwh vanadium battery demand (5m units at 20kwh) would require 6 years' mine supply under the current vanadium weightings. CAPEX levels for vanadium resource development are elevated and lead times are long, based on our analysis.

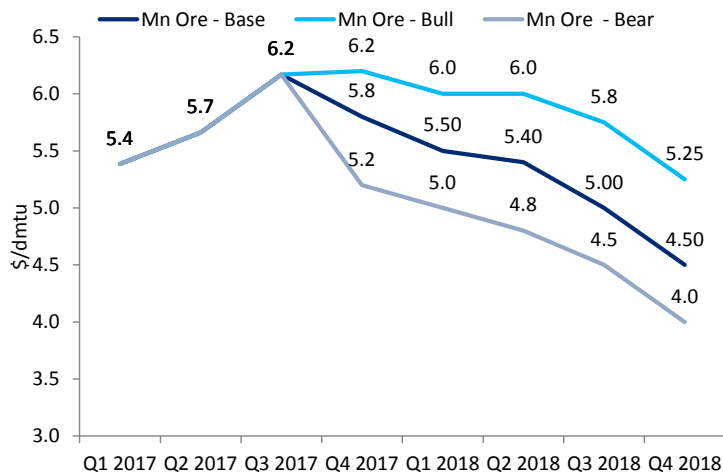
Prices in \$30/Vkg bear case, \$53/Vkg base case – Should China rebar weightings be increased, vanadium prices are likely to be at \$50/Vkg and above for the foreseeable future. Should weightings not be increased, prices could be in the \$20-30/Vkg range.

Steel-related Manganese ore - supply set to catch up during 2018

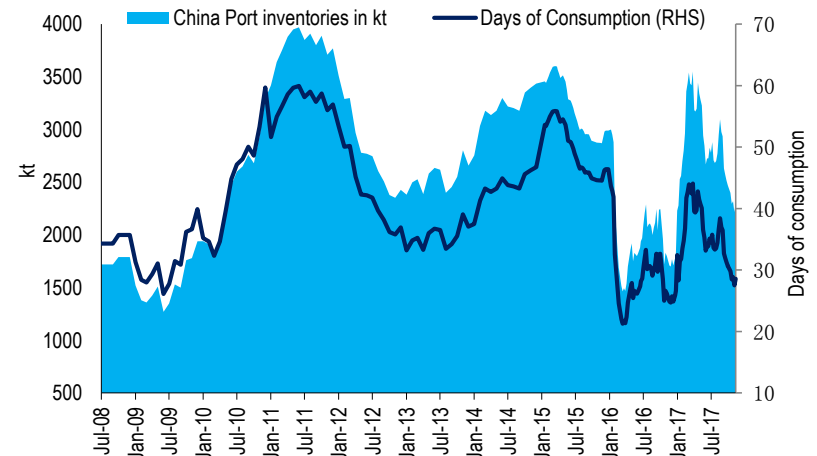
High prices during the last 18 months' are incentivizing capacity restarts and ramp ups resulting in strong supply growth. The market balance is expected to deteriorate further and remain in surplus in the long term.

- **Seaborne prices to fall to \$4.5dmtu by end-2018** – Our base case scenario is for spot ore prices to fall during 2018 as the market surplus grows but prices are not likely to revisit the lows of 2016. Demand from non-steel sectors is expected to remain firm and China's port inventories remain at relatively low levels (30 days of usage) which should support restocking by the end of 1Q'18.
- **Strong supply growth set against seasonal weakness in demand** – Ore production has been very strong so far (YTD production +28% yoy) while real consumption has grown by 16% yoy. This has resulted in market balance improving in Q3 and expected to improve further as we enter the seasonally low demand period of Q4 and Q1. We expect China's steel production growth to moderate from 2018 onwards. Over the medium term we expect global steel demand to remain largely flat but demand from electrolytic metals and energy storage sectors are set to support demand for high grade ore.
- **Capacity restarts to result in market surplus by mid-2018** – We continue to believe that restart and full ramp up of Bootu Creek and Woodie Woodie mines will add significant volumes of high grade ore (~900kt Mn units) to the seaborne market. Other big miners are ramping up production capacity too.

Manganese ore price scenarios



Relatively low China port stocks to support prices



Thermal coal - set to consolidate at higher than previously expected levels

Thermal coal prices have stayed elevated recently, thanks to weaker-than-expected Chinese, Indian and Australian coal production and strong performance of coal-fired power generation. We expect that prices will ultimately normalize in a range of \$70-80/t during 2018.

Mt	Q1 2017	Q2 2017	Q3 2017	Q4 2017	Q1 2018	Q2 2018	Q3 2018	Q4 2018	2013	2014	2015	2016	2017	2018	2019	2020
Supply (exports)																
Indonesia	77.9	77.4	79.8	85.0	82.5	80.7	77.5	82.6	381.2	356.2	327.2	311.2	320.1	323.3	326.5	326.5
Australia	48.5	48.9	52.1	47.9	50.5	48.7	56.0	53.5	188.4	201.0	202.3	201.2	197.4	208.8	213.2	214.2
Russia	36.0	41.2	36.9	39.0	34.8	39.1	39.5	39.7	106.2	119.3	134.2	144.4	153.0	153.0	150.0	147.0
South Africa	19.3	20.9	20.2	20.8	19.5	19.5	19.8	19.8	73.2	76.9	77.4	75.4	81.2	78.6	79.2	77.5
Colombia	22.0	20.7	20.8	20.0	21.0	21.0	22.6	22.6	73.4	85.7	80.5	88.5	83.6	87.1	87.1	87.1
United States	9.1	7.5	7.8	7.8	6.6	8.1	7.8	7.4	47.2	31.2	25.3	17.4	32.1	30.0	30.0	30.0
Other	8.4	5.9	7.4	11.3	8.8	9.6	8.0	7.8	69.0	59.1	43.5	40.6	33.0	34.2	33.8	33.4
Total	221	223	225	232	224	227	231	233	939	929	890	879	900	915	920	916
Demand (imports)																
China	28.6	30.6	33.2	35.7	34.0	23.9	32.0	31.7	192.0	165.6	107.9	124.1	128.2	121.6	79.0	43.1
Japan	38.2	34.8	37.8	38.4	38.2	34.8	37.8	38.4	143.2	144.1	149.7	146.4	149.2	149.2	149.2	149.2
India	37.1	37.0	37.6	41.2	27.2	31.5	37.5	41.1	142.5	178.1	168.8	147.5	152.9	137.3	131.2	126.4
Korea	31.9	28.9	34.0	28.9	31.3	28.3	33.3	28.4	100.2	106.5	109.9	109.7	123.7	121.2	118.8	116.4
Taiwan	16.1	19.1	17.4	16.0	15.3	17.2	16.5	16.0	61.0	67.1	66.9	65.5	68.5	65.0	66.0	67.0
European Union	28.6	25.7	29.5	29.5	27.3	24.6	28.2	28.2	157.9	154.2	138.1	116.9	113.4	108.2	104.6	103.2
Other	49.0	47.9	64.7	75.1	46.8	51.9	72.4	81.2	92.4	111.5	151.2	185.1	236.7	252.4	267.5	288.5
Total	229	224	254	265	220	212	258	265	889	927	892	895	973	955	916	894
Market Balance	-8.4	-1.2	-29.2	-33.3	3.5	14.5	-26.4	-31.5	49.4	2.3	-2.0	-16.3	-72.1	-39.9	3.4	21.8
% of seaborne market	-3.8%	-0.6%	-13.0%	-14.4%	1.6%	6.4%	-11.4%	-13.5%	5.3%	0.2%	-0.2%	-1.9%	-8.0%	-4.4%	0.4%	2.4%
Newcastle Price & Forecast	82	80	93	94	80	75	76	80	84	71	59	66	87	78	75	65

Our forward looking supply & demand balances are based on an assessment of current project pipelines, cost structures, and forecasts of the evolution of demand. Forecast surpluses and deficits should be interpreted as indications of needed market adjustments rather than forecasts of likely actual inventory changes. Thus, a forecast surplus indicates a need for prices low enough to force supply curtailments, project delays, or increased destruction. Similarly, a forecast deficit indicates the need for higher prices to incentivize increased production, new investment, or demand rationing.

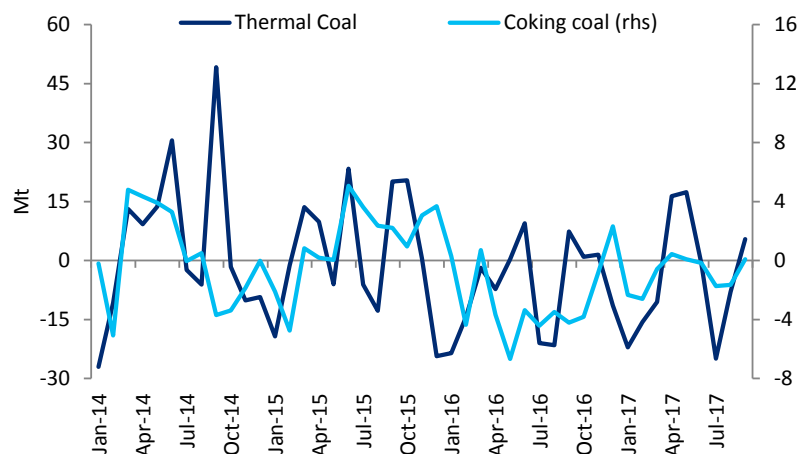
Thermal coal - tightness in the Chinese market may ease during 2018

Thermal coal prices performed strongly this year. Despite rising 5% during 2017, Chinese output remains 3% below 2015 levels, as a result of the 8% cut in output in 2016. Demand rose by 5% during 2017, leaving the market tight. In 2018, the Chinese government is dedicated to lowering prices and demand growth is set to slow.

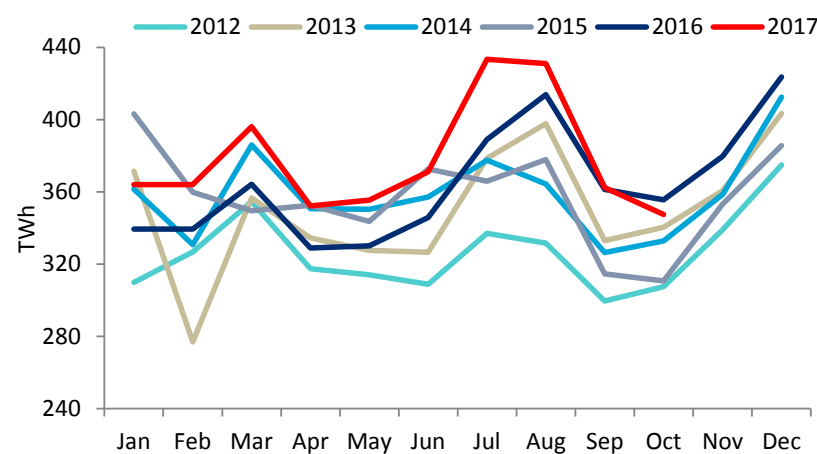
- **Chinese thermal coal prices (QHD 5500 kcal) have traded between RMB560-720/t during 2017**, as domestic production was severely affected by safety controls and environmental inspections, and imports of low-quality ores were dampened by a series of custom controls. Coal-fired power generation outperformed during 2017 owing to hydropower generation issues and robust power consumption by energy-intensive sectors.
- **We expect to see coal prices consolidate during 2018, with the Chinese government dedicated to a lower range of coal prices than current spot.** Though the majority of domestic thermal coal sales are conducted on a long-term contract basis, with prices much lower than current spot, overly high spot prices could put pressure on contract completions and eventually inflation. **We forecast Chinese thermal coal demand to drop -1% y/y in 2018 and production to rise 2.5% y/y.** For the upcoming winter in particular, demand for thermal coal could be hit by curtailments of aluminium, steel and cement production as well as by coal-to-gas switching for household and industrial heating.

• **Upside risks to QHD 5500 kcal index and NEWC 6000 kcal index could lie in continued widening of quality differentials.** Chinese utilities are set to consume coal with higher calorific value, lower ash and sulfur contents over time due to continued environmental crackdowns and stricter emission standards. This is supportive of high-rank coal demand.

China's coal market was tight for most of 2017



Thermal power generation outperformed until recently

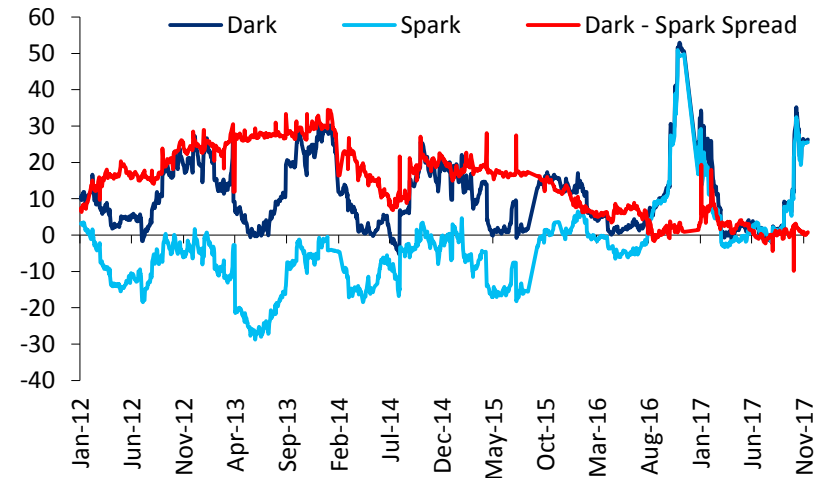


Thermal coal - Ex-China demand growth should moderate in 2018

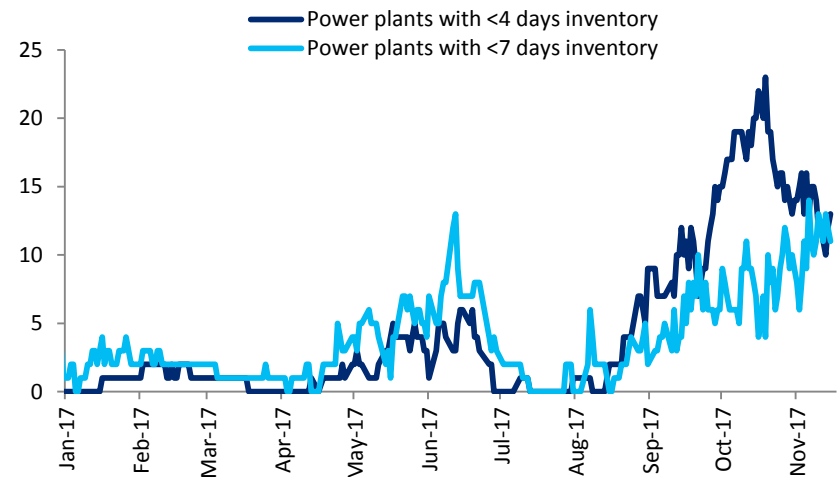
We have revised up our ex-China thermal coal demand estimates to reflect stronger-than-expected economics of coal-fired power generation, particularly in emerging markets where carbon pricing is generally low, and particularly in South and Southeast Asian countries where additional large-scale power capacity is required.

- **EM Asian countries are expected to see strong thermal coal demand growth in the next few years**, thanks to the potential commissioning of new coal-fired power plants, including a total of over 3GW capacity in Malaysia and over 10GW in Vietnam from now to 2020. Pakistan could also see rapid build-out of coal-fired power plants, many of which have received funding from China's Belt and Road Initiative.
- **Indian thermal coal imports are expected to be hit by rising domestic production in the next few years, but demand for imported high-rank coal could stay robust.** In the short run, India could continue its strong buying momentum which emerged since September due to weak Coal India production, rail transportation bottlenecks amid monsoon rains and restocking demand at Indian utilities. *Woodmac* estimates Indian coal imports to stay above 11Mt/month during 4Q'17.
- **European coal demand growth should moderate this winter as low gas prices and elevated coal prices should cause steady coal-to-gas switching.** IHS identifies 10Mt/y of coal demand to be hit by the move in the next 2-3 years. Traditional suppliers for Europe, including South Africa and Russia, are expected to divert more volumes to Asia. In the long run, announcements made by multiple governments to phase out coal-fired plants by 2020/2025 could be a major threat to coal demand.

French base-load front month dark-spark spreads fall



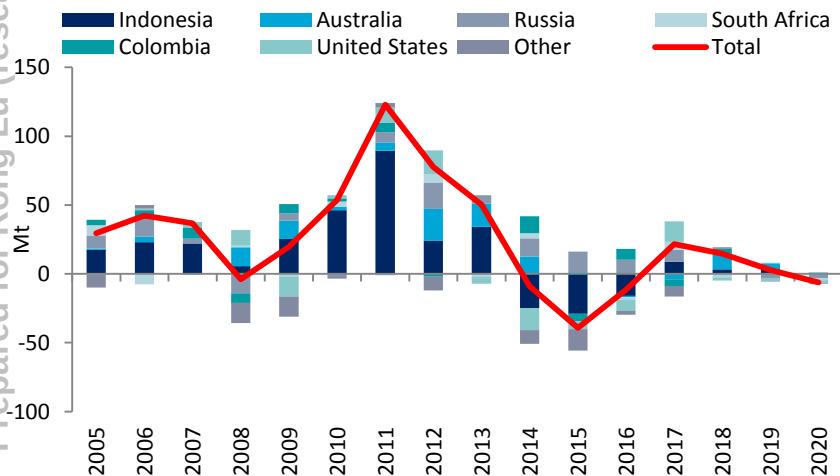
Indian coal power plants began to restock most recently



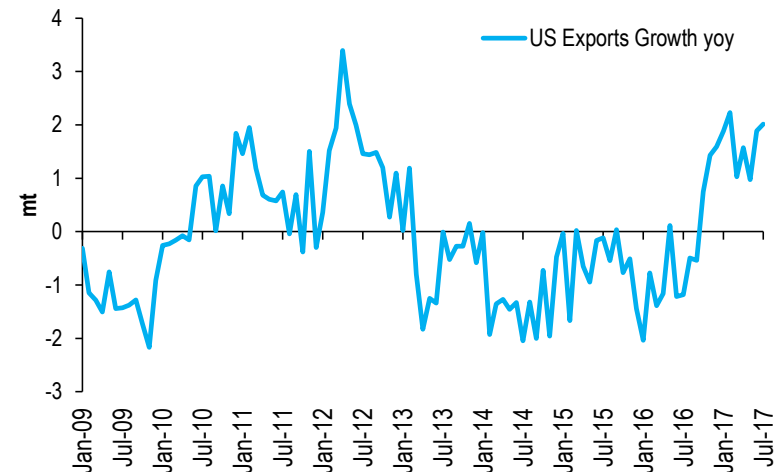
Thermal coal - Global seaborne supply may only rise 2.5% y/y in 2017

- Indonesia:** Output was affected by heavy rains and a lack of sustaining capex despite high coal prices over the past 1-2 years. We expect exports to rebound during Q4 as miners are pushing volumes marginally higher. Domestic demand remains robust and government regulation could continue to be a key risk to the investment sentiment. China's crackdown on low-rank coal imports could hit Indonesian coal directly.
- Australia:** The majority of operations have returned to normal after a series of disruptions earlier in the year. High mine utilization should continue in the medium term, though there remains a lack of spare capacity to combat disruptions. Some high-rank thermal coal has been diverted to the coking coal space due to widened price differentials, and the trend might ease in 2018 as we expect coking coal prices to drop faster.
- Colombia:** Heavy rains hurt output during 3Q, and Colombian exports for the full year might be lower than 2016 levels as a result. Colombian producers could continue to push volumes to the APAC market given favorable pricing.
- United States:** Strong international prices have opened the arbitrage for eastern US operations since June. The ongoing revival of the US coal sector could enable exports to stabilize in the ~30Mt/y territory.
- Russia:** Currency depreciation and steadily improving transportation infrastructure have benefitted Russian coal exports. Key producers are targeting higher export sales, and the country is set to stay competitive in the next few years.

Annual thermal coal export growth (Mt, y/y)

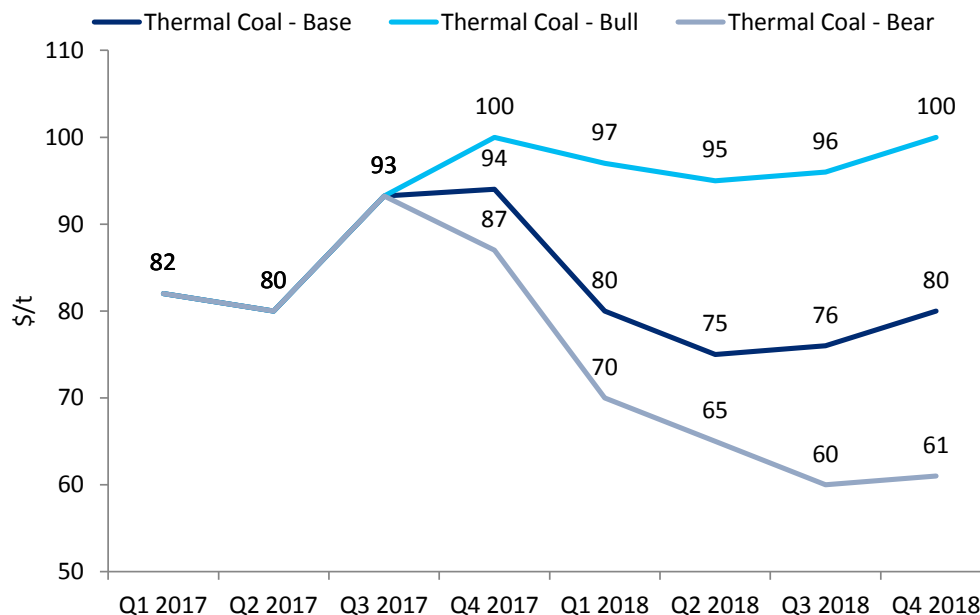


Monthly thermal coal export growth from US



Thermal coal – Revisions, base, bull, and bear scenarios

We revise up medium-term thermal coal prices to reflect tighter-than-expected China balances and cost inflation in Chinese coal operations. A lack of space capacity in the seaborne market and widened quality differentials are also expected to be supportive of benchmark thermal coal prices.



Prepared for Rong Lu (research email)

- **Base case (70%):** Newcastle thermal coal prices are expected to average \$78/t in 2018 and \$75/t in 2019. Global thermal coal prices are expected to consolidate as the Chinese government is dedicated to keeping physical prices in a much lower range than current spot prices, and as improved renewable power generation globally, with a return of hydro, drives a slowing in Chinese and global demand.
- **Bull case (20%):** Thermal coal prices should average \$97/t in 2018, partly because Chinese safety controls and environmental inspections continue to restrain coal output. Price spikes occur frequently due to industrial action and weather disruptions at exporting countries.
- **Bear case (10%):** Prices stay below \$70/t for most of 2018 and average \$64/t for the full year. Conditions of this scenario include major acceleration of Chinese coal production, minimal global disruptions, and rapid growth of coal-to-gas switching and renewable buildout.

Coking coal - prices are expected to moderate over the coming year

Australian hard coking coal prices experienced spikes to \$200-300/t earlier in the year and recently stabilized at ~\$180/t, still much higher than marginal producers' costs of \$100-120/t. We expect 2018 prices to average \$155/t as potential output expansions out of China and other suppliers should bring the market to balance.

Mt	Q1 2017	Q2 2017	Q3 2017	Q4 2017	Q1 2018	Q2 2018	Q3 2018	Q4 2018	2013	2014	2015	2016	2017	2018	2019	2020
Supply (exports)																
Australia	42.9	37.8	46.6	46.8	45.3	43.3	45.7	45.9	169.2	186.4	185.6	189.4	174.0	180.2	178.6	181.4
United States	10.9	12.0	12.0	9.0	10.0	10.0	11.0	11.0	58.6	55.7	40.2	36.1	43.9	42.0	40.0	37.0
Canada	5.8	6.3	6.6	6.0	6.0	6.0	6.0	6.0	31.6	27.2	24.1	24.4	24.7	24.0	23.0	23.0
Russia	6.3	6.1	4.5	5.9	6.3	6.0	4.7	5.9	23.9	21.5	18.5	22.0	22.8	22.9	22.9	22.9
Mongolia	6.4	8.1	5.5	8.1	5.0	7.4	7.4	10.2	15.4	14.8	11.5	23.6	28.1	30.0	31.0	32.0
Other	2.9	3.8	2.6	3.2	5.0	5.6	4.8	5.1	11.2	10.3	8.6	8.4	12.4	20.5	25.7	25.7
Total	75	74	78	79	78	78	80	84	310	316	289	304	306	320	321	322
Demand (imports)																
Japan	11.8	12.7	12.7	13.5	13.5	13.5	13.5	13.5	56.7	55.5	55.5	55.6	50.8	54.1	54.1	54.1
China	16.8	17.3	17.0	12.2	19.5	16.9	18.5	9.3	74.3	61.6	47.0	58.1	63.3	64.1	53.9	47.0
India	10.2	11.3	13.1	12.9	11.1	11.9	13.7	13.5	36.7	44.6	47.5	49.2	47.6	50.1	52.0	53.7
Korea	7.8	7.7	8.1	7.9	7.8	8.2	7.6	8.0	30.2	32.6	33.3	33.0	31.6	31.6	31.7	31.7
Brazil	3.2	3.0	3.3	3.0	3.2	3.2	3.4	3.3	13.3	15.6	15.3	13.6	12.5	13.1	13.8	14.1
Ukraine	2.9	2.9	3.1	2.8	2.8	2.9	3.0	2.8	10.2	8.9	10.7	11.9	11.7	11.5	11.5	11.5
European Union	12.9	13.3	10.2	10.7	13.3	13.8	10.7	11.0	45.0	48.7	42.3	38.0	47.1	48.8	50.0	51.3
Other	6.8	5.8	4.7	5.1	7.5	5.8	5.4	5.9	21.4	22.3	22.2	23.7	22.4	24.6	26.3	28.7
Total	73	74	72	68	79	76	76	67	288	290	274	283	287	298	293	292
Market Balance	2.6	0.1	5.5	10.7	-1.2	2.2	3.9	16.7	22.2	25.9	14.7	20.9	18.9	21.7	27.7	29.8
% of seaborne market	3.5%	0.2%	7.0%	13.5%	-1.5%	2.8%	5.0%	19.9%	7.2%	8.2%	5.1%	6.9%	6.2%	6.8%	8.6%	9.3%
Australia HCC FOB Contract	285	194	175	187	175	165	155	145	159	126	102	114	210	160	133	125
Australia HCC FOB Spot	170	191	189	170	170	160	150	140	148	115	88	145	180	155	128	120

Prepared for Rong

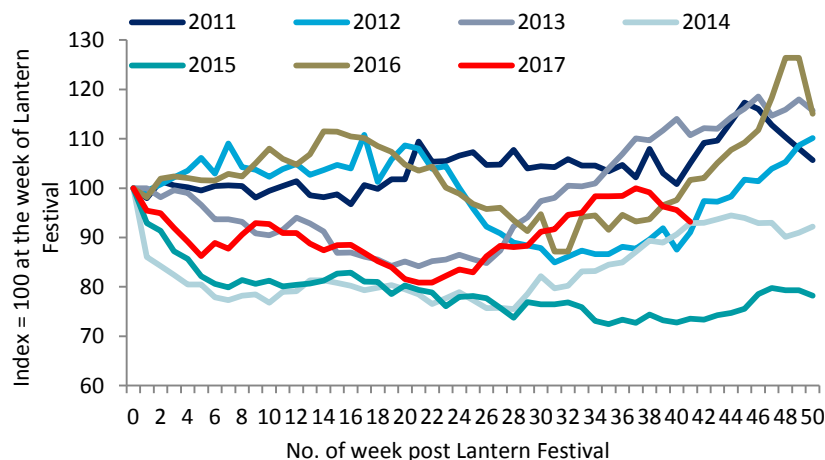
Our forward looking supply & demand balances are based on an assessment of current project pipelines, cost structures, and the forecast evolution of demand. Forecast surpluses and deficits should be interpreted as indicators of needed market adjustments rather than forecasts of likely actual inventory changes. Thus, a forecast surplus indicates a need for prices low enough to force supply curtailments, project delays, or increased destruction. Similarly, a forecast deficit indicates the need for higher prices to incentivize increased production, new investment, or demand rationing. Note also that most countries are shown on a net export or net import basis, and that listed country numbers may not match those reported by customs agencies due to differing coal classifications.

Coking coal – Chinese imports from the seaborne market should slow in 2018

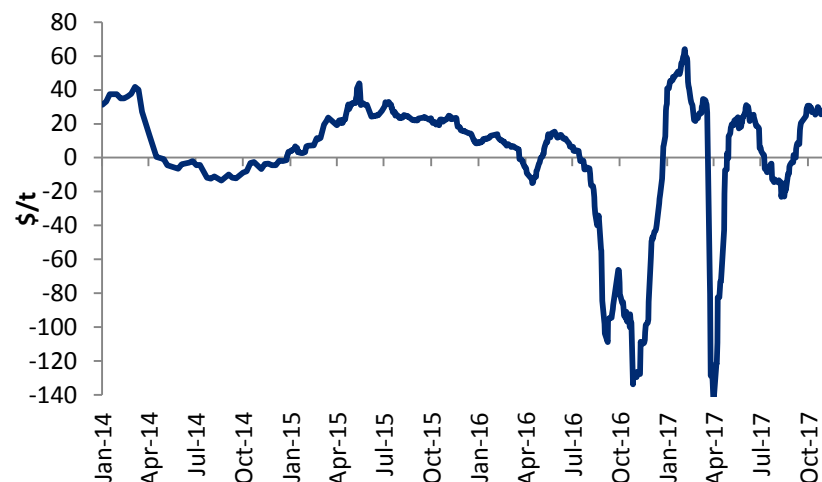
Chinese coking coal imports from the seaborne market should face headwinds from higher scrap intake in the Chinese steel making process, marginally rising Chinese coking coal production and improved imports from Mongolia following completion of transportation infrastructure.

- **The ongoing winter supply curtailments at coke plants and steel mills should hurt demand for coking coal.** We estimate ~22Mt of coking coal demand to be reduced as a result. Similar to the iron ore market, mills' restocking of coking coal could support prices in 1H'18. Yet currently hard coking coal is trading at a ~50% premium to marginal producers' costs while benchmark iron ore is trading at par to some high-grade producers' all-in costs. Hence coking coal should find less price support from potential restocking activities.
- **Demand for premium hard coking coal should receive support from Chinese steel mills' appetite for better quality raw materials given robust margins and strict environmental regulations.**

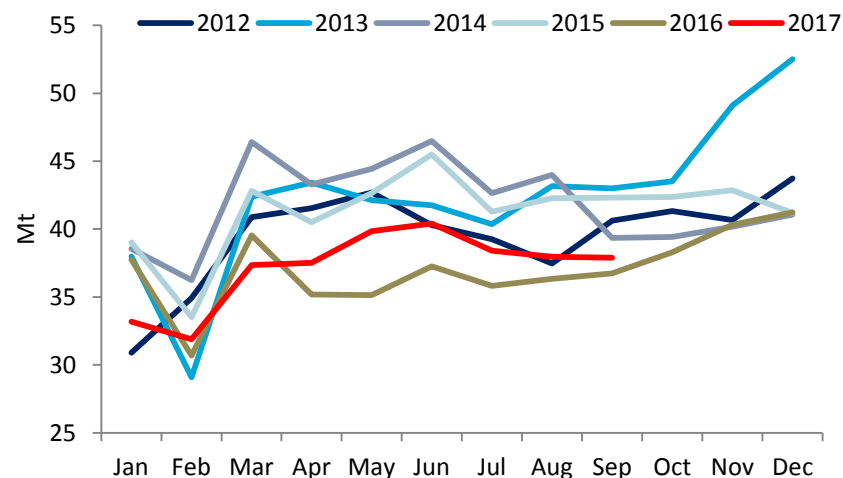
Stocks at coking plants and mills started to draw recently



Chinese coking coal import arbs recovered recently



Chinese coking coal supply remained subdued this year

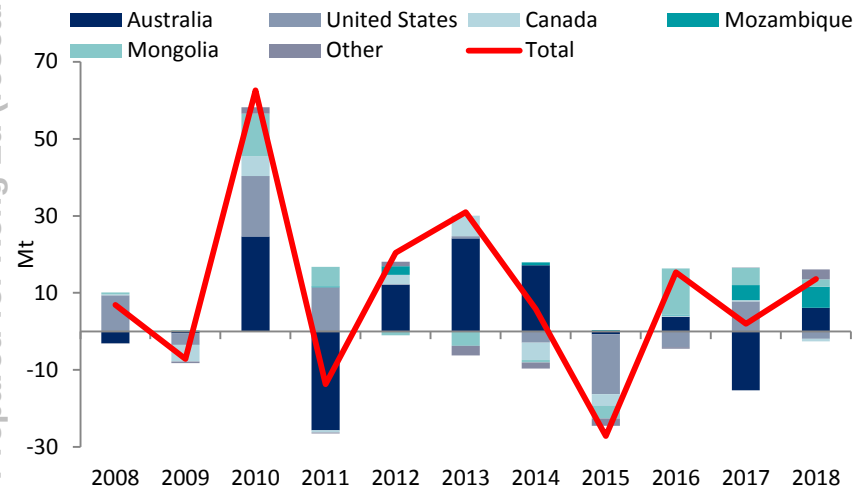


Coking coal – The seaborne supply response looks set to continue

High 2016 coking coal prices triggered a greater supply response from coking coal than thermal coal. Though there have barely been any newly-added greenfield projects, marginal expansions from most suppliers, in particular the US, could potentially kill high prices.

- **September US exports reached an annualized 56Mt in 3Q'17, compared with a peak of annualized 62Mt in 2009.** Industry action in Australia appears to be waning recently, and we expect Australian exports to reach ~180Mt in 2018 compared with 174Mt in 2017. Mozambique reached 3.2Mt production in 3Q'17, among which 60-65% was coking coal. *Woodmac* expect Mozambique to export around 10.5Mt coking coal in 2018, up from about 8Mt this year. Russian and Canadian production has also contributed additional volumes and could stay competitive in the market.
- **That said, miners' appetite for greenfield projects or major investment capex remains subdued, and therefore the market could continue to run at low spare capacity and see high price volatility.** For many of the major coal producers, growth plans remain muted for 2018. A combination of high utilization rates at existing mines, and a cautious approach to capex, will limit the chance of a large output boost in the near term. Adding to the price volatility are potential emergence of collective actions by major miners to boost prices and more labor strikes out of Australia, especially triggered by high coking coal prices.

Seaborne coking coal supply expected to rebound in 2018

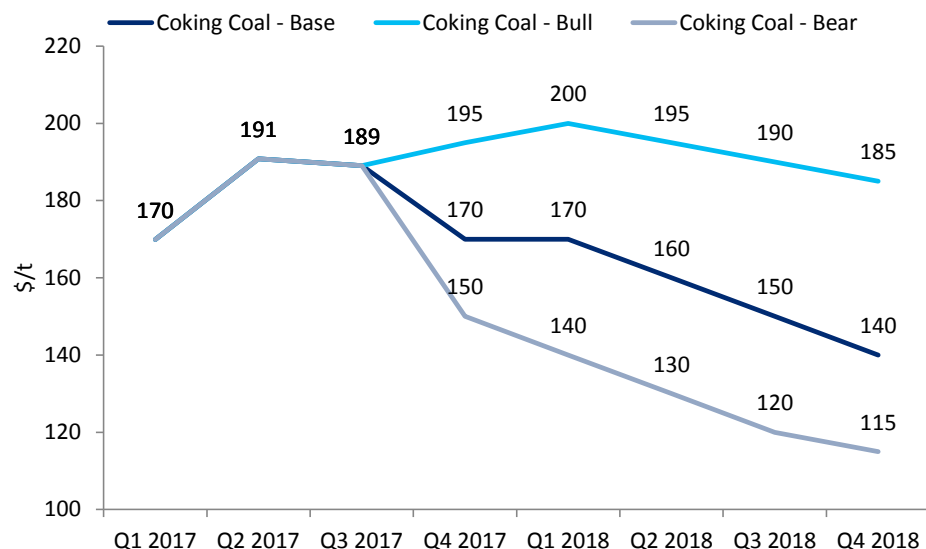


Coking coal prices were revised up on more persistent supply issues



Coking coal – we revise up medium-term prices but stay bearish

Citi expects coking coal prices to fall in the mid-term, from an average of \$180/t in 2017 to \$155/t in 2018, as global steel output growth should decelerate, and marginal seaborne supply expansions are underway. Yet we expect heightened price volatility to continue alongside potential supply disruptions.



- **Base Forecast (60%):** We do not expect prices to drop to marginal producers' cost levels in the near term because coking coal consumption should remain steady in 2018, since we forecast Chinese steel demand to rise ~1.5% y/y and ex-China steel demand to rise ~1% y/y. Elevated steel mills' margins should also allow higher consumption of hard coking coal as supposed to semi-soft and PCI.
- **Bull case (25%):** The base case of our forecast assumes no major disruptions to seaborne coal supply in the next two years, but there is a clear bull case where disruptions could trigger price spikes. Hard coking coal prices may then average \$193/t for 2018.
- **Bear case (15%):** Higher scrap intake in the global steel industry and a larger-than-expected wave of coking coal supply could put pressure on prices. Chinese coking coal output may rise as the government accelerates approvals for new projects. A major slowdown of the Chinese economy and fixed-asset investments could hurt steel and coking coal demand badly.

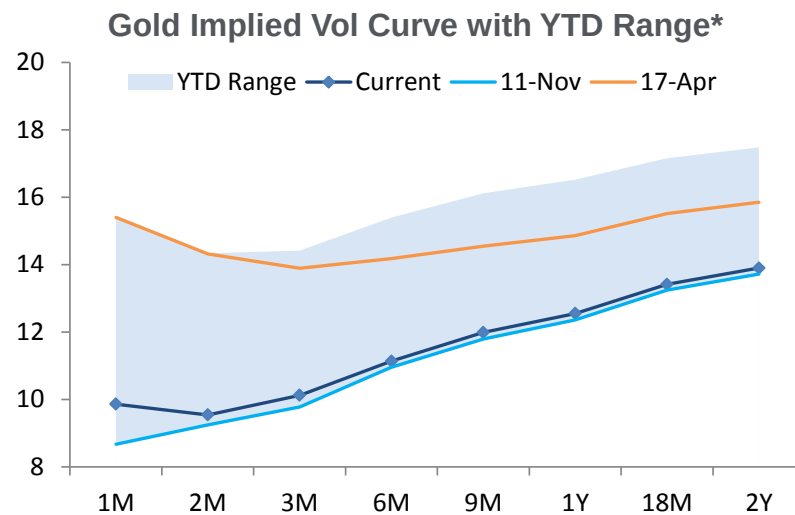
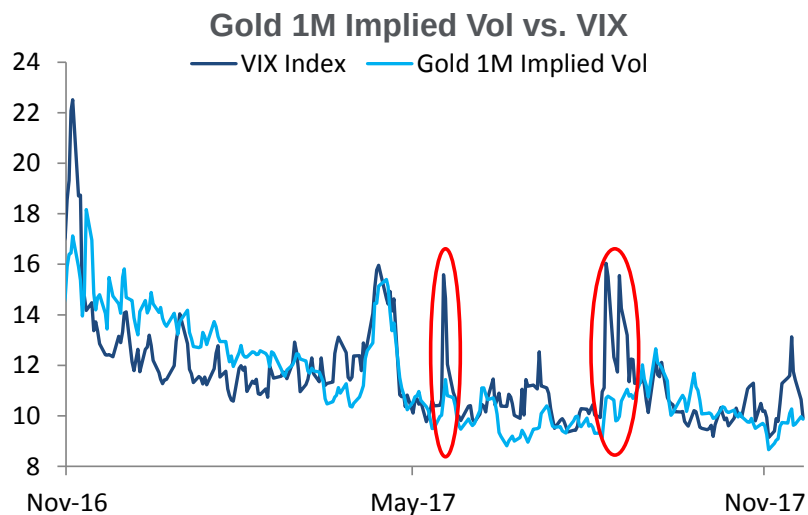
6. Precious Metals: Prices have a neutral bearish tilt in 2018, bar palladium

Precious Metals top calls: Gold still a good hedge on equity unwinds

Gold has been increasingly sensitive to equity market sentiment and geopolitical tail risks. And while our base case gold outlook for 2018 is that COMEX price levels drift lower on higher US rates, we still view the yellow metal as a constructive tail risk hedge and policy play, most notably for equity investors.

- **While we see a moderately lower price trajectory for the yellow metal in 2018 as the base case, we like adding gold to portfolios to hedge against geopolitical and equity drawdown risks.** As discussed in more detail later in this section, the correlation between gold and equity prices has become more negative since the onset of QE, and in that context a risk market drawdown might be a key upside tail risk for gold. While Citi strategists do not expect a US equity market recession in the short-term, we think there is heightened risk for a drawdown in the medium-term. At the same time, we believe event-driven bids for gold are becoming the new normal and should persist in the medium-term, while rates set the baseline.
- **Buying gold upside optionality may be a cheaper hedge than equity downside during risk-off trading, as gold vols are likely to underperform equity vols when stocks sell-off due to different market structures.** Historically, gold's 1M implied vol has traded in a close relationship with the VIX index, as both are volatility measures and are sensitive to risk sentiment. However, we've seen multiple episodes in 2017 where gold vols only edged slightly higher when VIX spiked sharply. This divergence is especially pronounced around unexpected geopolitical events—after the Trump/Comey/Russia story broke out and after the US/North Korea tension escalated. The steep skew in equity vol surface suggests equity vol is likely to rally significantly just by rolling up the skew as prices decline. Also, there are more active vol sellers in equity markets, who reduce their short gamma positioning when vols rally, bidding vols even higher.

Prepared for Rong Lu (research email)

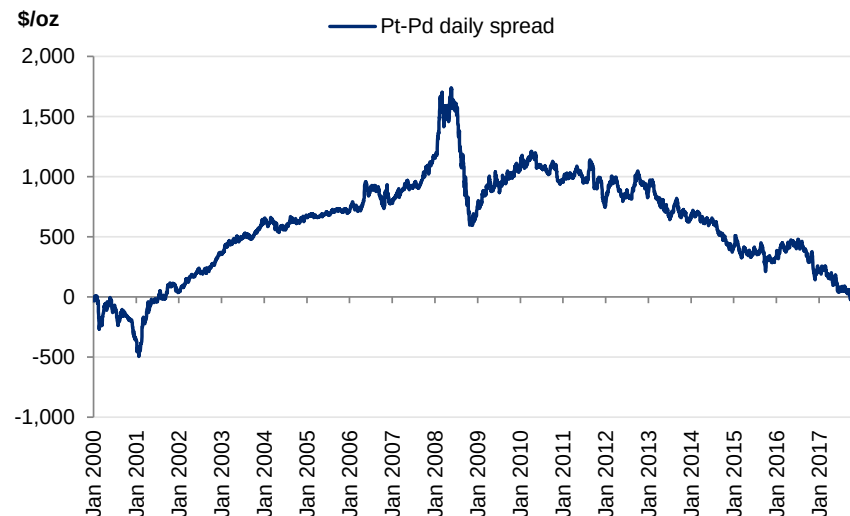


Source: Bloomberg, Citi Research *as of mid-late November.

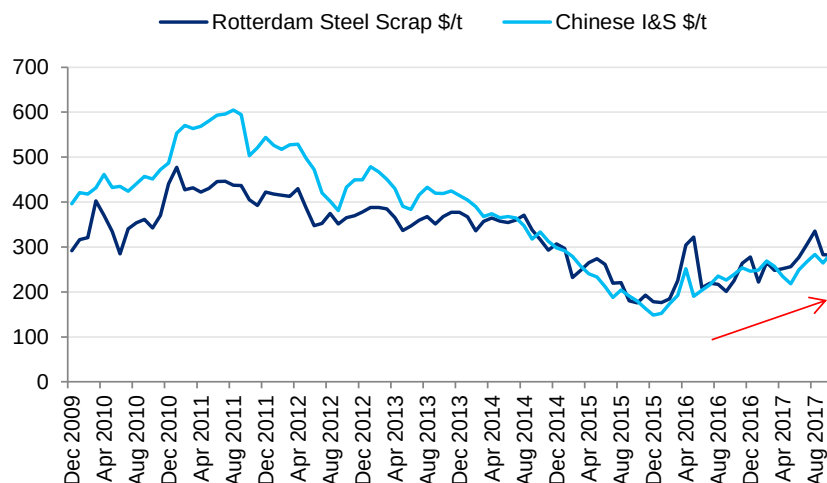
Precious Metals top calls: Negative Pt-Pd spread to widen in 1H'18

- **We expect that palladium will continue to outperform platinum in 1H'18.** Key to this view is our forecast for a sustained palladium deficit, underpinned by increasing gasoline engine penetration in Europe as well as anticipated solid global gasoline vehicle output growth. Prices are expected to come under pressure on rising scrap rates, inventory de-stocking and producer hedging during 2H'18.
- **Scrap represents a headwind for palladium and platinum pricing longer term.** A potential Chinese ban on non-compliant cars in major urban areas, rising scrap steel and related commodity prices, and a rising pool of scrap (owing to the prior boom in Chinese vehicle production), underpin strong anticipated scrap production growth.

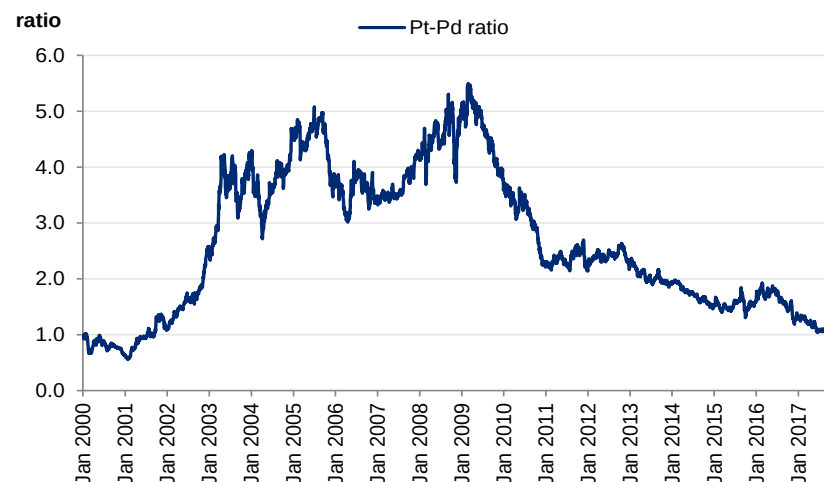
The average negative spread* has doubled in November MTD vs. October



Rising scrap steel prices spurs secondary supply



The ratio* is higher than the QTD average of 0.953x



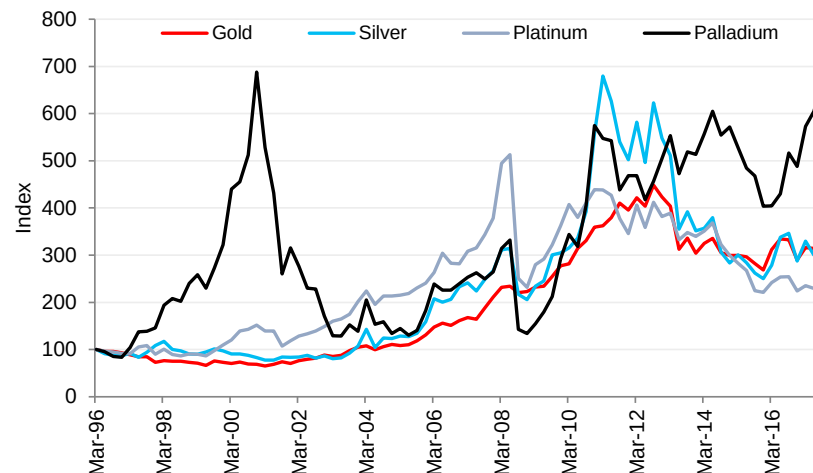
Source: Company Reports, Bloomberg, BNEF, Metals Focus, Citi Research. *Data as of mid-November 2017.

Precious Metals – Prices have a neutral-bearish tilt in 2018, bar palladium

The BCOM precious metals sub-index has gained ~8.6% YTD stabilising near the 1 year average level having cooled from highs in September, primarily driven by gold. But could markets be under-pricing geopolitical risks?

- Gold:** Expect prices to grind lower in 2018, but greater upside risks suggest a soft floor at \$1,200/oz. Increasingly negative correlations between gold and equity prices since the onset of QE, may spur asset allocations into gold in the event of an asset market drawdown, perhaps in the medium term. However, we remain mildly bearish on the price outlook for 2H'18 in line with the short-term bearish forecasts on rates. Balanced investor positioning could allow a price trajectory in either direction into next year.
- Silver:** A muted price trajectory into 2018 due in part to a decline in total physical demand and persistent scrap flow. Net ETF outflows are expected to be ~225t in 2017 vs inflows of 1.43kt last year in line with price softness. The weaker price performance in 2017 did not deter scrap flow from Asian industrial products and this trend is likely to continue next year.
- Platinum:** Markets may be quick to cap any price gains as sour sentiment remains stubborn in 2018. Continued demand destruction in Europe limits positive tailwinds next year. Absent a switch to platinum in gasoline engines, could sputtering South African supply be the saving grace for a structural price recovery in 2H'18? We believe prices may eventually stabilise next year.
- Palladium:** The best performing precious metal for a second consecutive year sees limited downside into 2018. Unlike platinum, prices are expected to remain firm next year albeit to a lesser degree. Tightening emissions policy supports higher loadings. We note fervent EV speculation will only meaningfully erode traditional ICE sales at the margin over the near-term.

Index of historical returns expected to diverge



Key precious metal price ranges*

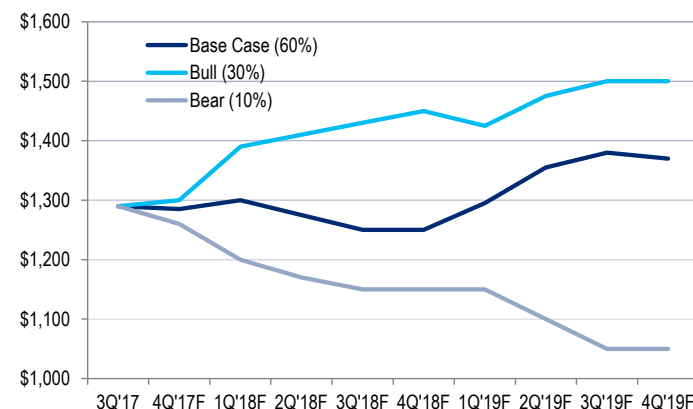
US\$/oz	Current Price	% Return YTD
Gold	1,278	11.4
Silver	17.05	7.1
Platinum	932	3.2
Palladium	987	44.9

Metal	\$ YTD High	\$ YTD Low	\$ Range
Gold	1,349 (Sep)	1,159 (Jan)	190
Silver	18.54 (Apr)	15.62 (Jul)	2.9
Platinum	1,029 (Feb)	897 (May)	132
Palladium	1,016 (Nov)	709 (Jan)	307

Gold – Set to drift lower in 2018, but still a good risk/equity drawdown hedge

- 60% - Base Case:** Our published gold price forecast remains unchanged at \$1,270/oz for 2018 and \$1,350/oz for 2019; slightly bearish the 2H'18 forwards and modestly constructive the deferred strip. The market pricing a slightly more hawkish 2018 Fed, optimism for US tax cuts, sluggish Asian jewelry demand growth. Robust global GDP projections for the next year in both DM and EM economies should underpin a lower price path for the yellow metal. A back-up in US 10s to ~2.6-2.8% per our rates strategists should send 2018 gold trading down to \$1,200-\$1,225/oz in fits and starts. But modest gold price weakness via the rates channel should be at least partially offset by a stable USD outlook and geopolitical tail risk bids which we see as a new normal for gold** amid sizable central bank balance sheets that are only slowly unwinding. Our outlook for inflation and the Fed terminal rate is also less constructive than FOMC projections. Risks of monetary policy errors, asset market drawdowns and more benign 2019-2020F global growth forecasts should then underpin higher baseline gold prices into 2020.

Gold bull/bear case price forecasts*



- 30% - Bull Case:** We assign a higher probability for a bull versus bear case path for gold as the level of optimism priced-in equity markets seems high and overall asset market vols seem too low. If the US tax deal falls apart or a major geopolitical event crystalizes, gold prices should sustain a push above \$1,300/oz. We see several scenarios where gold prices even revert north of \$1,400/oz over the next few years. 2017 has been a great example wherein gold prices averaged higher than consensus forecasts due to ongoing geopolitical bids—in 1H'17 (French elections) and 2H'17 (US/North Korea tensions).

- 10% - Bear Case:** Levels head back towards the post-super cycle lows but remain north of \$1,000/t (~90th percentile of the mining cash cost curve) on the unlikely re-emergence of the 4Q'16 'Trumpflation' trade.

Citi gold price forecast split by scenario: Base case unchanged for 2018/2019/2020*

COMEX Gold (\$/t)	0-3m	6-12m	4Q'17F	2017F	1Q'18F	2Q'18F	3Q'18F	4Q'18F	2018F	1Q'19F	2Q'19F	3Q'19F	4Q'19F	2019F	2020F
Old Base Case			1,310	1,270	1,300	1,275	1,250	1,250	1,270	N/A	N/A	N/A	N/A	1,350	1,370
Gold Base Case (60%)	1,300	1,250	1,285	1,265	1,300	1,275	1,250	1,250	1,270	1,295	1,355	1,380	1,370	1,350	1,370
Revision (%)			-2%	0%	0%	0%	0%	0%	0%	N/A	N/A	N/A	N/A	0%	0%
Scenarios:															
Bull (30%)			1,300		1,390	1,410	1,430	1,450	1,420	1,425	1,475	1,500	1,500	1,475	N/A
Bear (10%)			1,260		1,200	1,170	1,150	1,150	1,170	1,150	1,100	1,050	1,050	1,090	N/A

Source: Bloomberg, Citi Research, *subject to revision; forecasts published in Must C: Global Gold Book (October 2017), **see: Why the Geopolitics of Gold Matters (November 2017)

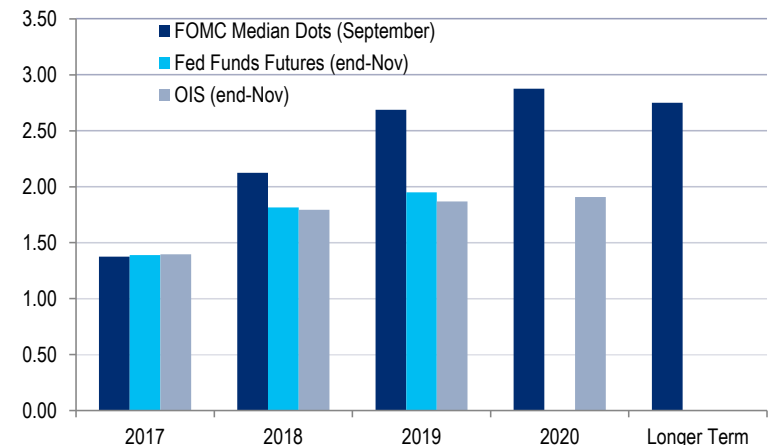
Gold – What asset markets giveth they may taketh away, aiding gold hedges

- The correlation between gold and equity prices has become more negative since the onset of QE and in that context a risk market drawdown might be a key upside tail risk for gold.** Examining historical rolling quarterly returns for spot gold and US equities since 1975, we identify about a half-dozen periods of sustained and steep retracement in the US stock market >15% (on average -22%) and note that gold mean returns were +4% during those periods. Interestingly, gold market returns were on average 0% in the quarter immediately following a sharp equity market correction, suggesting the gold price move is coincidental with a risk unwind but then typically loses steam as markets normalize. *But the elevated price environment for gold typically lasts for a not inconsequential four to six months.*
- While Citi strategists do not expect a US equity market recession in the short-term, we think there is heightened risk for a drawdown in the medium-term.** Geopolitical risks and macro uncertainty appear more sticky given ‘vox populi’ election trends, lower-for-longer oil prices hitting OPEC revenues and general instability due to monetary and fiscal policy gaps (e.g. unfunded liabilities, low wages). In fact, our macro equity colleagues note that a major geopolitical uprising is probably the one element that could derail the current global stock market rally.
- Fed projections suggest a long-term neutral rate of 2.75% but we believe this is closer to ~2.0-2.25% while money markets are pricing ~1.90%. All-else equal, gold price downside seems limited in this policy environment.** If the median dots eventually shifted down it would further support the investment case for gold. Indeed, the Fed terminal rate has been coming-in for some time now, including at the recent September meeting, capping the long-end of the curve as inflation has undershot committee projections.

Gold spot versus S&P 500 rolling 1Y weekly correlation



FOMC dot plot and market-implied expectations



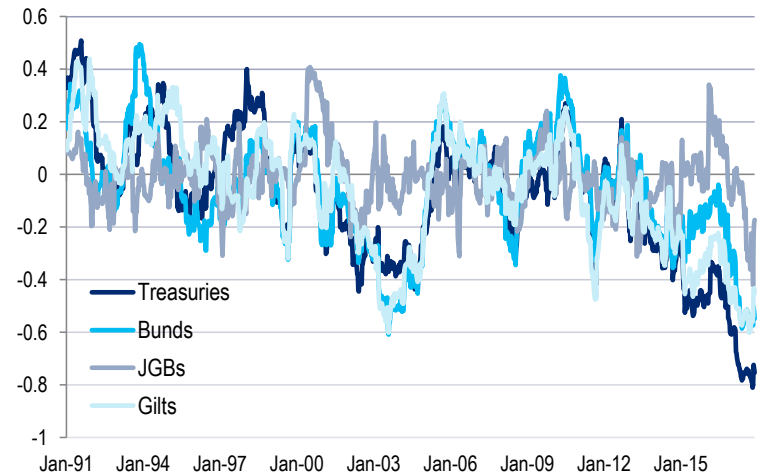
Gold – US and core G3 rates may have bottomed, but remain structurally low

- **We are mildly bearish 2H'18 gold prices as our short-term rates view remains bearish, particularly the long end.** Tax cuts are likely coming in the US this winter and while the efficacy of fiscal stimulus measures should be modest, we think animal spirits and the impact on Treasury supply (higher deficits) can push rates higher, capping gold upside. In our gold-rates-inflation model, 50-75% of the COMEX gold price move is explained by shifts in real medium-term US yields, given the assumption gold behaves like a long duration zero-coupon bond*. Our US rates strategists are targeting 10s in a 2.45-2.50% area for 2Q'18, backing-up to ~2.75% in 4Q'18. Examining rolling inflation-adjusted rates betas across the UST yield curve, this outlook suggests gold forwards are fairly valued around ~\$1,260-\$1,310/oz in 1H'18 and ~\$1,200-1,250/oz in 2H'18, all-else equal. However, there are other factors that influence spot gold pricing and trading flows, such as investor bids for tail hedges as well as risk-on/risk-off safe-haven flows. We think many of these exogenous factors, discussed later in this section, could provide structural support for gold prices from falling too hard on higher rates. Accounting for term premiums, 10Y USTs may even *top out* around ~3.0-3.25% in the medium-term rates cycle; suggesting aggressive gold dips may find steady value buyers.
- **While the Fed seems confident in its short-term policy outlook, based on its own projections and statements—as well as those of the ECB, Bank of Japan and Bank of England—short-to-mid duration real yields should remain very low or negative across core G10 for some years to come.** Such a historically low rate environment should buttress gold markets above \$1,175-1,200/oz, reducing the opportunity cost of holding a hard asset that does not distribute cash. The US dollar is also off the cycle peaks and appears highly unlikely to rebound to TTM highs in our baseline forecast horizon.

Real Yield betas in US rates markets across tenors**



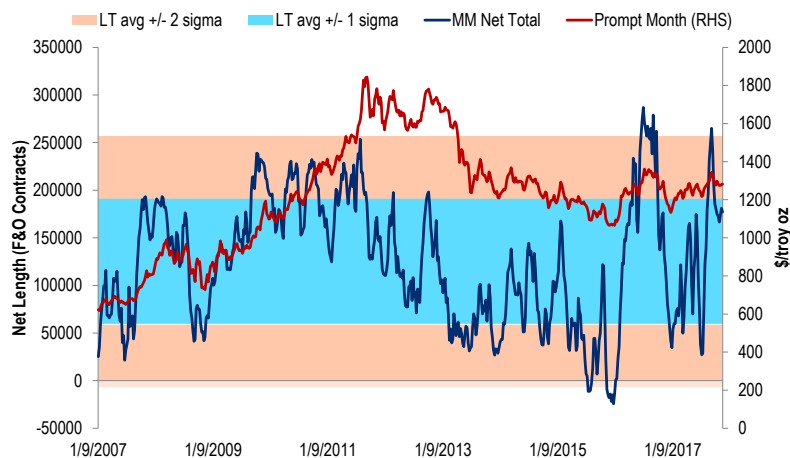
Rolling 1Y correlation between weekly shifts in G4 bond yields and weekly changes in spot gold price



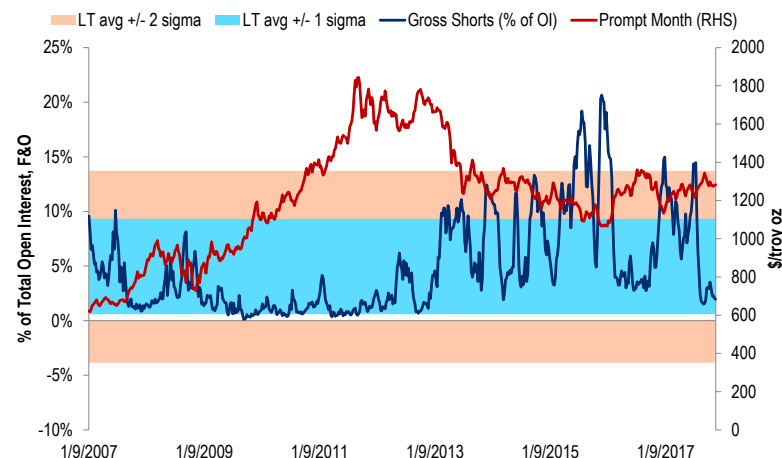
Gold – Positioning* balanced heading into 2018, allowing either/or price moves

Money manager positioning in gold has stabilized to end 2017 down ~100k lots from YTD peaks but up ~35% from the long-term average. Normalized gross longs and gross shorts are better-balanced today versus 4Q'16. Gold ETF holdings grew 200t in 2017 to ~2,150t and should have scope to grow further on institutional buying.

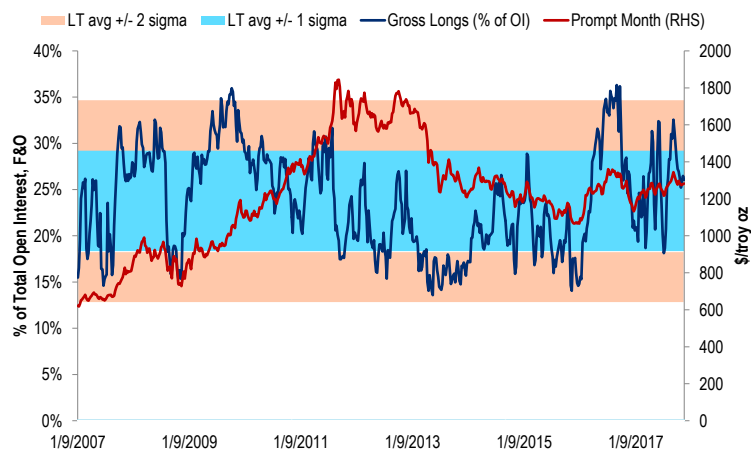
Money Manager Net Length and Prompt Prices, +/- 1,2 sigmas – COMEX gold



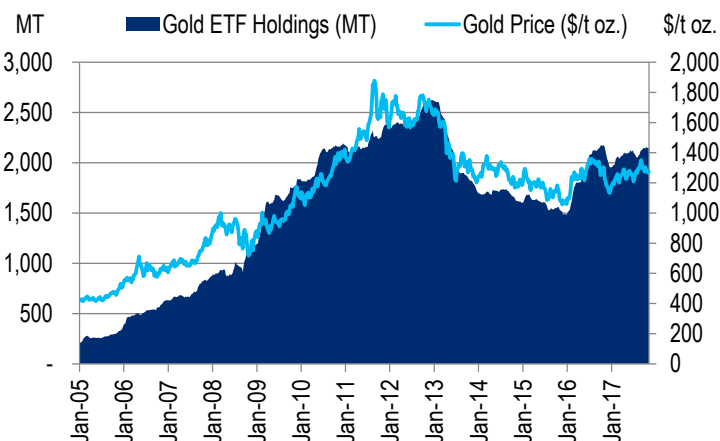
Money Manager Normalized Gross Short and Prompt Prices, +/- 1,2 sigmas – COMEX gold



Money Manager Normalized Gross Long and Prompt Prices, +/- 1,2 sigmas – COMEX gold



Physically-backed Bullion ETF holdings (kt)



Source: Bloomberg, Citi Research, *as of mid-late November

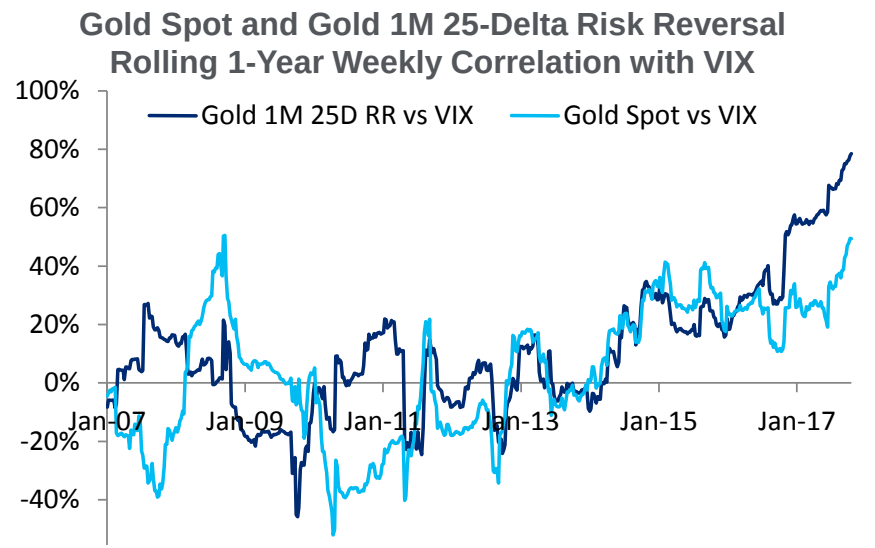
Geopolitical risks are becoming increasingly impactful on gold prices

While the basis of the short and medium-term Citi gold price outlook is mostly a function of US interest rates, inflation expectations and foreign exchange, bullion prices have been trading increasingly sensitive to geopolitical events and the market's risk sentiment over the past two years. We see this as supportive for gold.

- **Event-driven bids for gold are becoming the new normal and should persist in the medium-term, while rates set the baseline.** In a world of 'lowflation' and record stock market valuations, geopolitical risks have become more relevant for gold investors. From a macro investor's perspective, geopolitical risks can be viewed as risk and exposure to uncertainty via the political, military and financial channels. Thus, focusing on the post-Bretton Woods modern era, we broadly categorize three geopolitical areas: (1) Elections/political outcomes; (2) Military attacks/strikes and; (3) Macroeconomic crises/asset market unwinds. Heightened geopolitical risks should thus generally be positive for gold prices while low geopolitical risks may be construed as negative.

Though the constitution of a geopolitical event is a qualitative exercise, the gold price reaction function to the perceived level of geopolitical risk can be quantified by co-movements in asset markets. During periods of elevated geopolitical tensions and fat tail risks, gold price returns show an inverse relationship to core G4 rates, particularly UST yields, as well as to equity valuation. This provides a basic proof of concept for gold as a safe-haven product and a "risk-off" portfolio hedge.

Geopolitically-fueled gold price rallies seem to be occurring more frequently. Just as seen in 2017, bullion prices ran up sharply in advance of Round 1 of the French elections in 1H and on North Korea missile tests in 2H. Indeed, both gold prices and implied vol skews have been trading in a tighter relationship with the market's risk sentiment, marked by their rising correlations with the VIX index since 2015.



Expected price reaction function for risk assets and gold markets

Geopolitical Risks	Risk Assets	Gold Markets
High	Negative	Positive
Low	Positive	Negative
Neutral	?	?

Broad categories of geopolitical events that might influence the gold price

Geopolitical Categories
Elections and political votes
Military attacks/strikes
Economic and financial crises

Prepared for Rong Lu (research email)

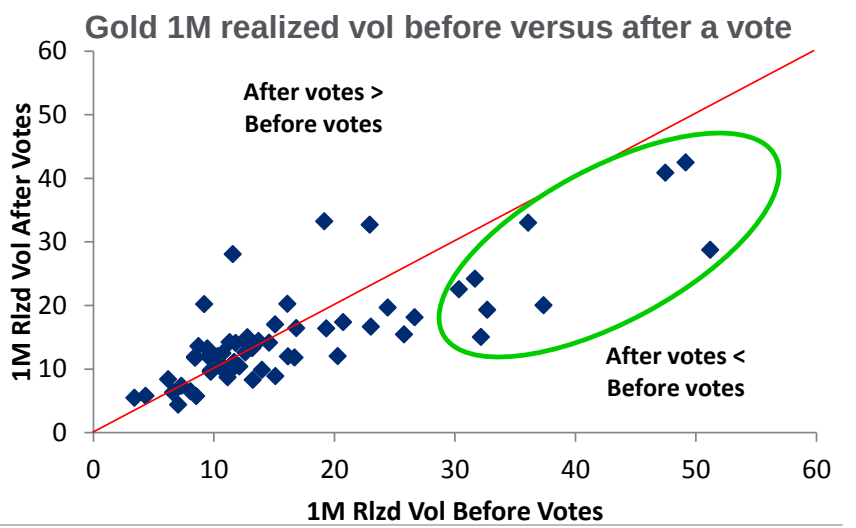
Long gold outright, not calls, may be a good hedge against election tail risks

Changes in gold prices and volatility around major elections and referendums in the post-Bretton era suggests that investors should considering taking profits from their gold longs shortly after the vote and that long gold outright may be more efficient than buying call options as a hedge against election and referendum tail risks.

- While there is not a consistent pattern for gold price performance around past elections, gold prices have rallied more frequently as opposed to selling-off immediately after a vote, although these rallies tended to fade quickly. Studying gold's price returns around the 62 elections in world's major economies (the US, Japan, the UK, Germany, and France) since 1975 as well as two referendums (e.g. the 2016 "Brexit" and the 2015 "Grexit" referendums), we find ~60% of the time gold prices traded higher on the following day and within the following week. But a month after an election, the likelihood of positive returns drops to just 44%, and the average return retreats. This erosion in returns is seen across all three core economic regions.
- Gold prices tend to trade choppy immediately after an election, but the volatility has typically subsided rather quickly after the results were unveiled. The realized volatility the month following an election oftentimes has been lower than during the month running up to it. Notably, volatility fell after all nine votes where the pre-event volatility was higher than 30 vol points. In these cases, the factors that kept volatility elevated before the event—either the vote itself or other macro drivers—might have calmed soon after the vote took place.
- Owning optionality on gold ahead of an important political vote seems logical for investors who look to profit from enhanced price volatility or to add portfolio hedges, but such strategies had not worked as well as expected in the past due to elevated implied vol levels.

Gold spot price returns before and after major political votes since 1975 by core economic region*

Region	Before Votes		After Votes		
	1W	1D	1D	1W	1M
Gold Spot Average Returns					
US	1.21%	0.58%	0.26%	-0.33%	-0.09%
Europe	-0.64%	-0.07%	0.22%	0.27%	0.25%
Japan	-0.56%	-0.15%	-0.14%	1.82%	0.93%
Total	-0.31%	0.02%	0.14%	0.54%	0.34%
% of Positive Returns as of Total					
US	55%	45%	73%	64%	45%
Europe	42%	50%	53%	55%	42%
Japan	33%	40%	67%	73%	47%
Total	42%	47%	59%	61%	44%



Source: Bloomberg, Citi Research *When a vote took place on a trading day, which has typically been the case in the US and UK, the gold price move on the voting day is considered as happening before the event in our calculation, since votes are usually tallied and counted after the settlement.

Gold shines brighter during economic turmoil than military/terrorist attacks

Market timing has proven challenging and long gold could be a pain trade for investors during wartime. But during an economic crisis, it is not too late to add gold to the portfolio even after the market crash has started.

- **Military strikes usually have far-reaching impacts on regional and sometimes global economies, but the gold price rally driven by increasing haven demand has tended to be benign and short-lived historically.** Wars and terrorist attacks are typically accompanied by heightened risk aversion among investors who are prone to sell risk assets such as stocks and buy safe havens including gold to protect portfolios from losses. Sometimes, just the threat of military action could trigger risk-off, sending gold prices higher as well. A war could also dent the economic growth and lead to a rapid currency depreciation in the region under military attack, increasing local gold demand from investors or consumers in that region.

Meanwhile, bullion prices have tended to get a bigger and more sustained boost during economic crises, as discussed in more detail in the macro section. We believe this is due to the fact that there is little uncertainty during a real market crisis in terms of its negative impact on the economy and the knock-on impact on risk assets, and this influence is direct, almost instantaneous and usually systemic. By contrast, the impact from military exercises and terrorism are unpredictable and oftentimes predominated by impacts from other factors, such as a change in monetary policy, a shift in the inflation outlook or a surprise in economic data. For example, it may have been the disappointing China GDP data (7.7% versus 8% consensus estimates) rather than diminished risk appetite that was driving asset price moves on April 15, 2013, when the Boston Marathon Bombing took place, including the ~9% plunge in gold prices.

Major military strikes and terror attacks and gold price returns*

Event	Date	Gold Price Returns			
		-1M	1D	1W	1M
Falkland Islands War	4/2/1982	-9.5%	0.2%	7.5%	10.5%
Beirut Barracks Bombing	10/23/1983	-4.1%	0.1%	-1.6%	-5.0%
US Invades Grenada	10/25/1983	-4.9%	0.9%	-3.2%	-4.8%
US Bombs Libya	4/15/1986	-1.8%	0.0%	-0.1%	0.2%
Invasion of Panama	12/20/1989	4.4%	0.0%	-1.6%	-0.9%
Iraq Invades Kuwait	8/2/1990	3.7%	2.7%	3.8%	4.3%
Gulf War	1/17/1991	7.1%	-6.0%	-6.1%	-9.7%
WTC Bombing	2/26/1993	0.2%	-0.5%	-0.8%	0.9%
Oklahoma City Bombing	4/19/1995	3.2%	-0.1%	-1.3%	-2.9%
US Embassy Bombing in Africa	8/7/1998	-2.0%	-0.2%	-0.7%	-0.5%
USS Cole Bombing	10/12/2000	-0.8%	2.0%	-0.2%	-2.5%
9/11	9/11/2001	-1.0%	6.5%	6.5%	4.8%
War in Afghanistan	10/7/2001	7.0%	0.3%	-2.3%	-4.1%
Iraq War	3/20/2003	-3.7%	-1.2%	-1.8%	-1.0%
Madrid Train Bombing	3/11/2004	-1.5%	0.7%	1.8%	1.6%
London Train Bombing	7/7/2005	-0.5%	0.1%	0.1%	2.7%
Lebanon War	7/12/2006	6.2%	1.6%	-1.3%	-1.4%
Russo-Georgian War	8/7/2008	-5.0%	-0.8%	-6.0%	-8.8%
Arab Spring/Libyan Civil War	2/15/2011	0.0%	0.9%	2.7%	4.7%
Boston Marathon Bombing	4/15/2013	-6.9%	-9.1%	-5.3%	-3.5%
Median Returns		-0.9%	0.1%	-1.0%	-1.0%
Average Returns		-0.5%	-0.1%	-0.5%	-0.8%
Positive Returns as % of Total		35%	60%	30%	40%

Source: Bloomberg, LPL Research, Google search, Citi Research *-1M represents gold price returns during the month before the attack.

Gold – India: Gold serves as multi-pronged asset to a diverse demographic

India has a huge and culturally varied population and each distinct subsection of the demographic have equally unique perspectives on gold. Changes in the socioeconomic profile of the country are meaningful to demand.

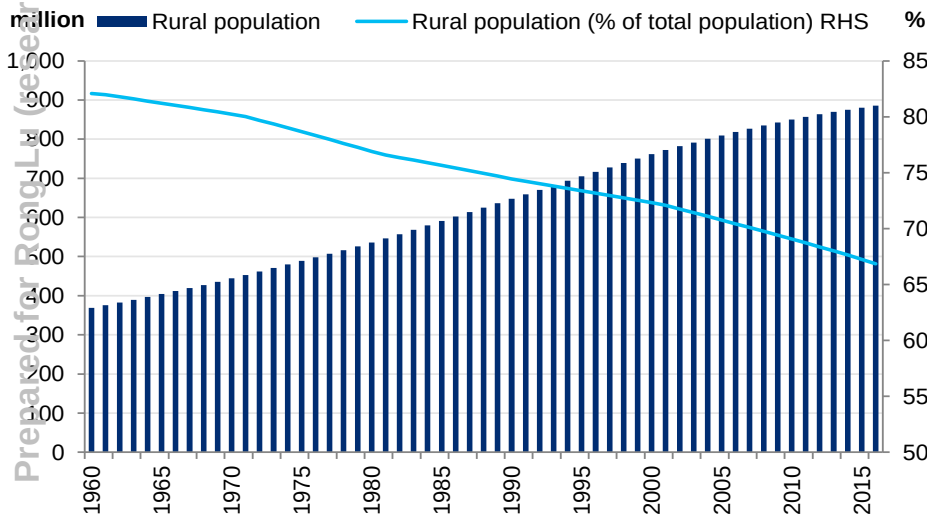
- As India's population matures and becomes wealthier, physical gold demand should shift from jewelry to bars and coins.** The vast majority of the Indian demographic live in rural areas amounting to ~885m or 67% of the population, while an additional ~380m live in urban areas, per census statistics. India's educated middle class may double in the next half-decade to ~547m. These changes in the socioeconomic profile of the country are meaningful to gold demand over the mid-term.
- Despite continued population growth, the overall share of middle income inhabitants should increase and surpass that of the rural population.** In this scenario, it is likely that high purity bars and coins may be favoured over jewelry items.

Rural: Gold is seen as a store of wealth and insurance: Population has no access to banking, limited literacy. There is high, seasonal demand for coins, biscuits and jewelry. Population has limited geopolitical awareness.

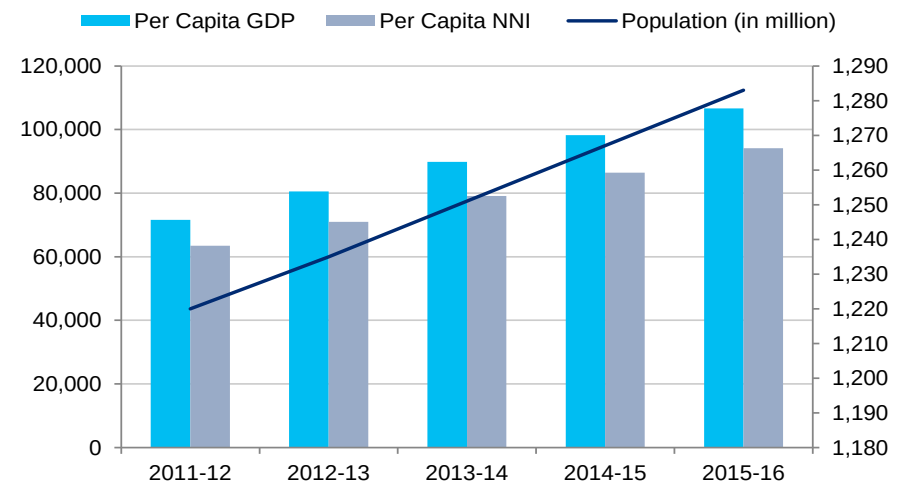
Middle income: Gold as a store of wealth, jewelry or status symbol: Access to high-street banking, moderate to average literacy. Less seasonal demand for coins, biscuits and jewellery. Moderate geopolitical awareness.

Top quintile: Gold as a status symbol and portfolio diversifier: Access to private wealth management and online management platforms, high literacy. Moderate-to-low demand for physical gold. Sensitive to geopolitical events.

The majority of India's population is rural: ~67% at 885m



Per capita GDP has on average increased 5.5% y/y since 2011



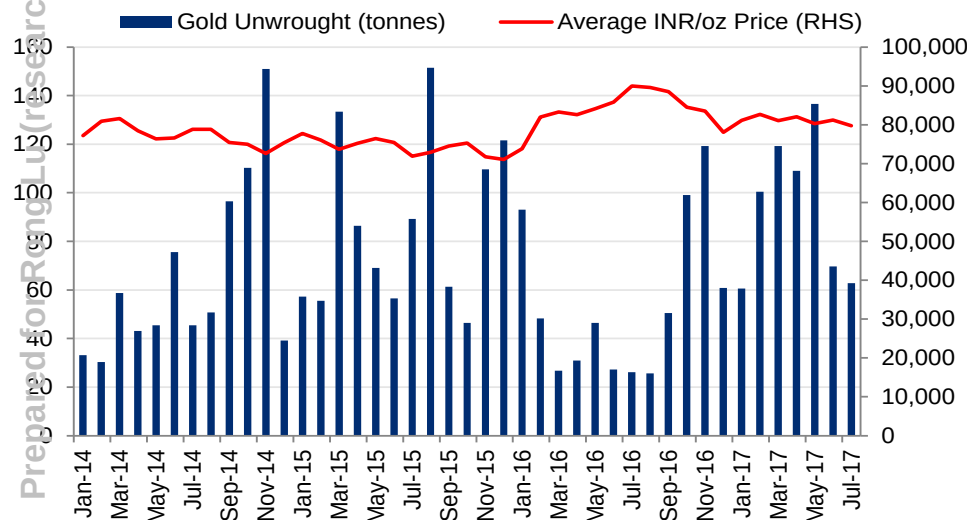
Source: Bloomberg, GFMS, Metals Focus, Ministry of Statistics, WGC, World Bank Data, Citi Research

Gold – India: Seasonal physical demand trends unlikely to change in 2018

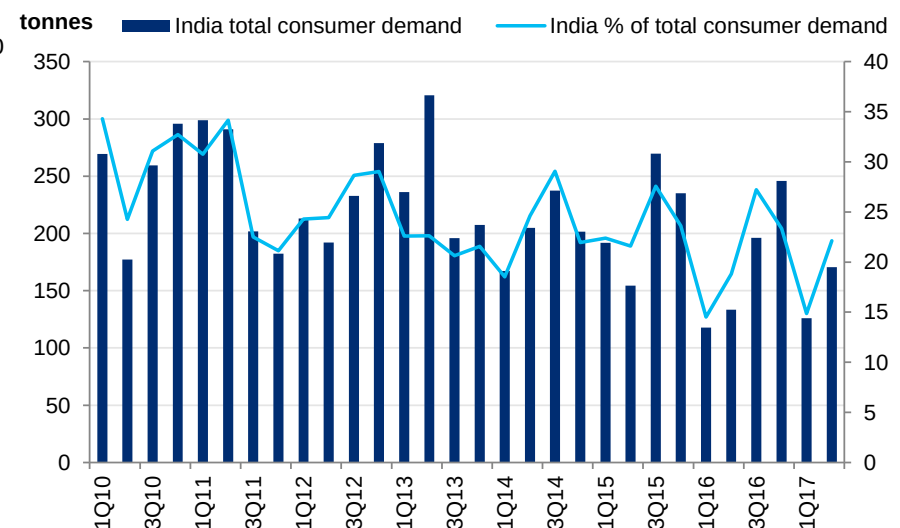
Indian geopolitics have rarely impacted consumer gold demand. Seasonal factors that currently drive the majority of gold demand such as religious events and weddings are deeply rooted into the Indian culture.

- **Both low-to-middle income socioeconomic tiers see high-to-moderate seasonality in their gold buying trends.** Indian geopolitics has rarely impacted consumer gold demand. Instead, a combination of INR denominated gold price, “carrot vs. stick” national and state-level policy measures, as well as changes to disposable income typically define the trends seen in bullion consumption. The spring and autumn harvest seasons occur either side of the 4-month Indian monsoon between June and September. As such, Akshaya Tritiya and Dhanteras are notable festivals within the spring and autumn harvest seasons that are typically associated with gold buying for auspicious reasons. Moreover, in conjunction to the 4Q’17 festival season, favourable winter weather prompts a spike in weddings that include a dowry largely made of gold items. Hence, the combination of weather-dependent agricultural profits and festive gift-giving are critical in terms of gold retail demand.
- **Annual Indian gold demand is likely to plateau near the 3 year average to ~650t in 2018.** Gold in the form of jewelry still comprises the lion’s share of consumer demand in India. Looking further ahead, the development of sophisticated derivative products may spur a slow and steady adoption of paper gold exposure as opposed to physical bars and coins. Indeed, we could see greater traded volumes ahead of and during key festivals. Hence, the seasonality of Indian gold consumption is unlikely to dissipate as quickly as it has outside of Asia.

Gold imports spiked in May ahead of GST implementation



India's global share of gold demand has averaged ~25%



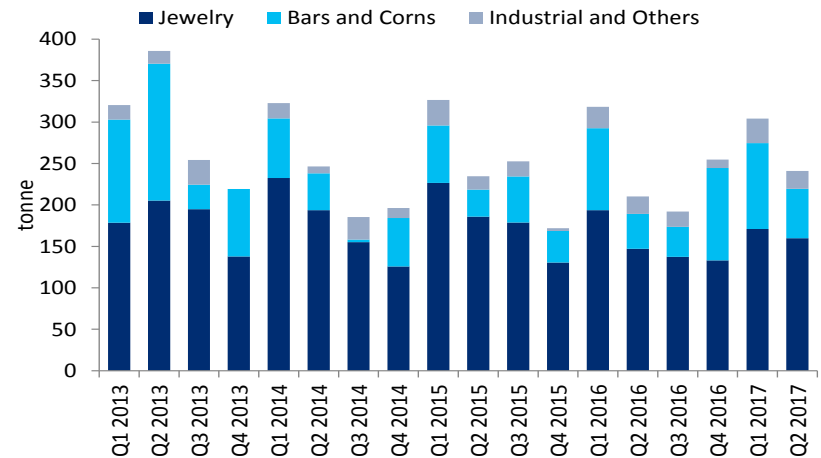
Source: Bloomberg, GFMS, India Department of Commerce, Metals Focus, WGC, Citi Research

Gold – China: Favored by Chinese retail and political authorities alike

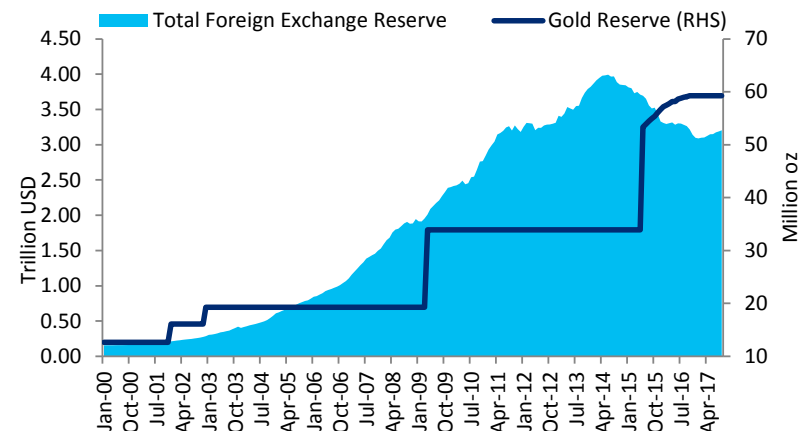
Physical gold allows Chinese investors access to a true 'foreign currency' despite local capital controls. Chinese demand for gold benefits from rising household income and savings. PBoC holdings are set to grow as they diversify holdings and as US-Sino relations under Trump have cooled.

- **China is now the largest global gold consumer, producer and importer.** Robust local demand for gold is benefiting from rising personal income and savings rates, limitations on existing investment options and Chinese retail investors' cultural appetite for gold, which also is a form of capital flight.
- **Chinese gold jewelry consumption grew rapidly in the past decade until 2015.** While we expect a potential slowdown of jewelry demand, gold demand for investment could be buttressed by the country's closed capital account and limited domestic investment options.
- **PBoC gold holdings are also set to grow.** China's official gold holdings totaled 1,842 tonnes by the end of 3Q'17 and constituted 5% of total global official gold reserves. The exponential surge of Chinese official gold reserves suggests the government's strategic move to diversify its FX reserves, in particular away from the US dollar, which currently consists 60-70% of Chinese FX reserves, yet gold only constitutes ~2-3% of total Chinese official reserves.
- **Though the PBoC seems to have eased back on gold purchase since November 2016, they might resume and accelerate the purchase in the long run,** not only due to heightened geopolitical concerns and challenges in Sino-US relations and associated higher safe-haven demand on a national level, but also because gold could help preserve value of China's FX reserves, which could marginally enhance the credibility of RMB in international trade and settlement.

China has become the world's largest bullion consumer



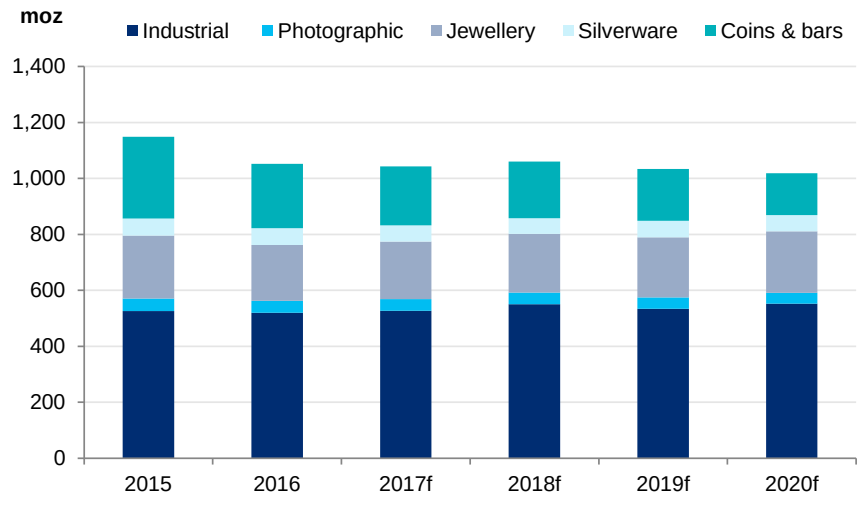
PBoC's gold reserve should continue growing



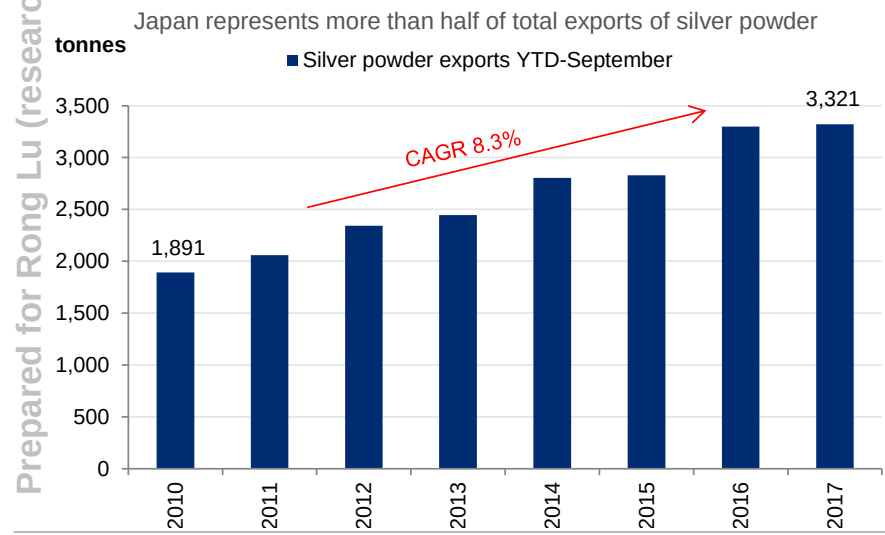
Silver – Industrial demand expected to pick-up in 2018 with limited price impact

- Industrial silver demand continues to grow, while other end-use sectors shrink in 2018.** Despite ongoing concerns for thrifting in the PV sector, *BNEF* estimates point to installed capacity this year of 95GW amounting to ~90moz of silver in 2017. Japan exported 3.3kt of silver powder YTD to September, akin to levels last year. As such, we expect industrial silver demand to continue to grow by ~5% y/y in 2018. However, a resurgence of industrial scrap in 3Q'17 could pose some headwinds. While we do not expect scrap rates to accelerate in 2018, it is likely that a combination of thrifting and improved recovery of industrial silver scrap would compensate for any incremental demand increase into 1H'18.
- As renewable energy targets accelerate, so too will silver paste demand.** As leaders in PV installation, China and Taiwan should boost global new build capacity in 2018 to over 100GW, potentially increasing silver demand by another 5moz.

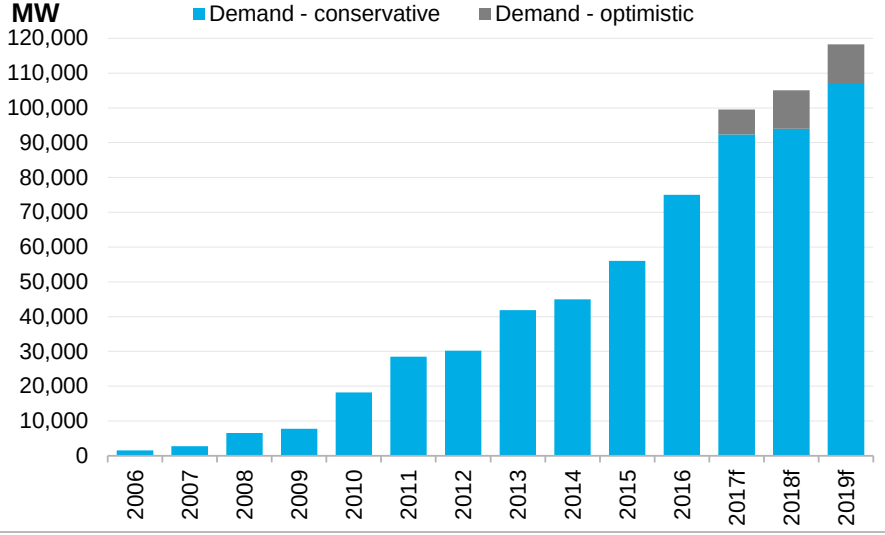
Industrial demand makes up 51% of total consumption



Robust silver powder exports suggest stable demand



Global PV new build may breach 100GW in 2018



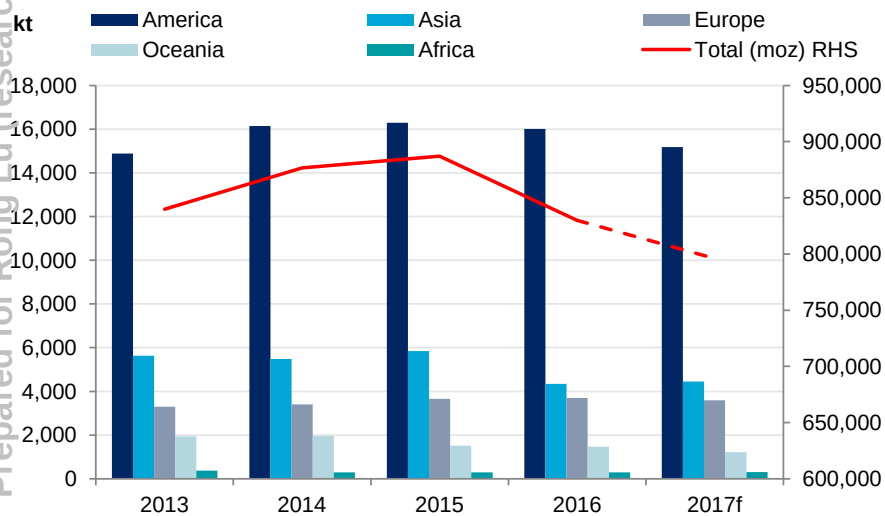
Source: BNEF, Metals Focus, WGC, Bloomberg, Citi Research. Where silver powder is defined by HS Code 7106.10.

Silver – Mine output misses may reverse next year, softening fundamentals

- A recovery in by-product supply should see total silver mine output stabilise over the mid-term.** We suspect the effects of weak silver mine production may be short-lived. Improved output of Chilean copper and Mexican zinc/lead mines should boost silver supply within the next 12 months. In terms of by-product ranking, lead/zinc mines contributed 35% of silver output in 2016, followed by copper mines at 23% and gold mining at 12%, per *GFMS* estimates. As such, we expect primary silver and by-product copper mines should curb output misses seen in 2017 and lift y/y supply growth by 1.5% to 885moz in 2018 from ~872moz in 2017.
- A combination of rising copper price expectations, increased copper mine investment and improved Mexican silver output may soften prices in 2H'18.** An extra ~13moz may be derived next year, particularly if Goldcorp's Penasquito output recovers from 2017 levels.

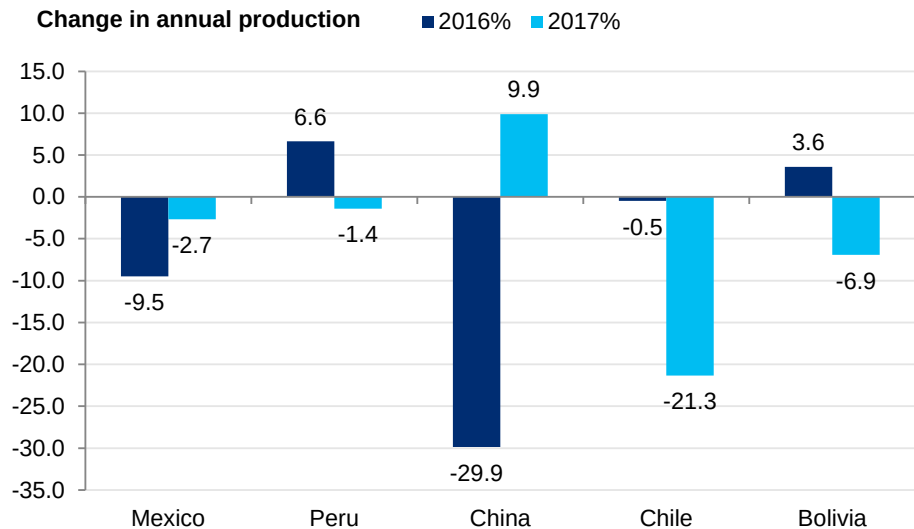
Prepared for Rong Lu (research email)

Mine output misses should recover over the mid-term

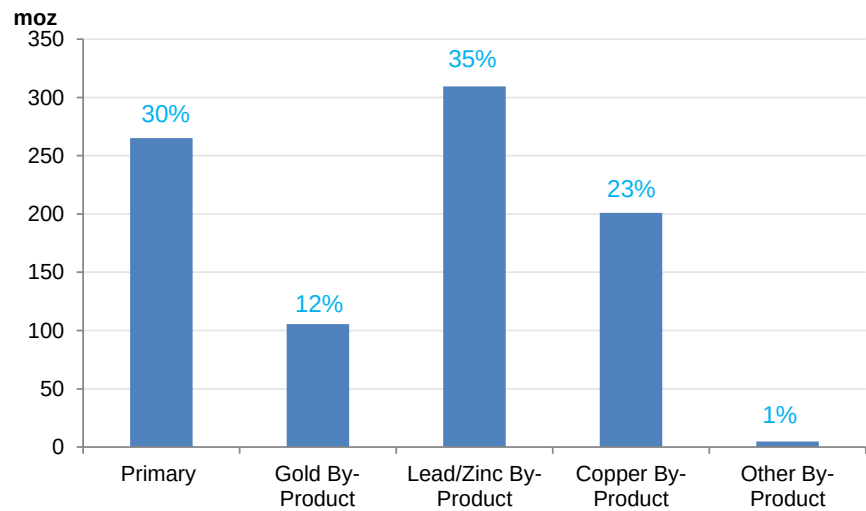


Source: Bloomberg, GFMS, Reports Web 'Precious Metals Mining', WBMS, Citi Research.

Mixed results from top silver producing countries



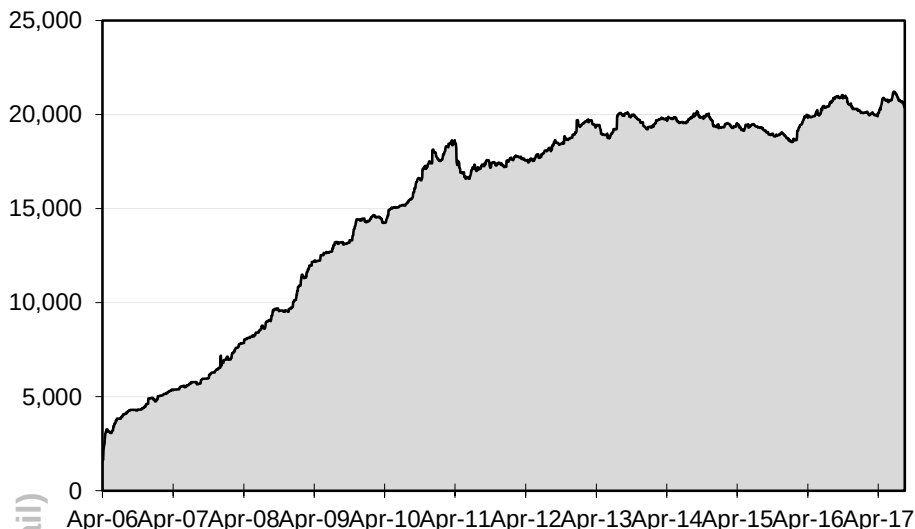
Source of silver supply is skewed to by-products



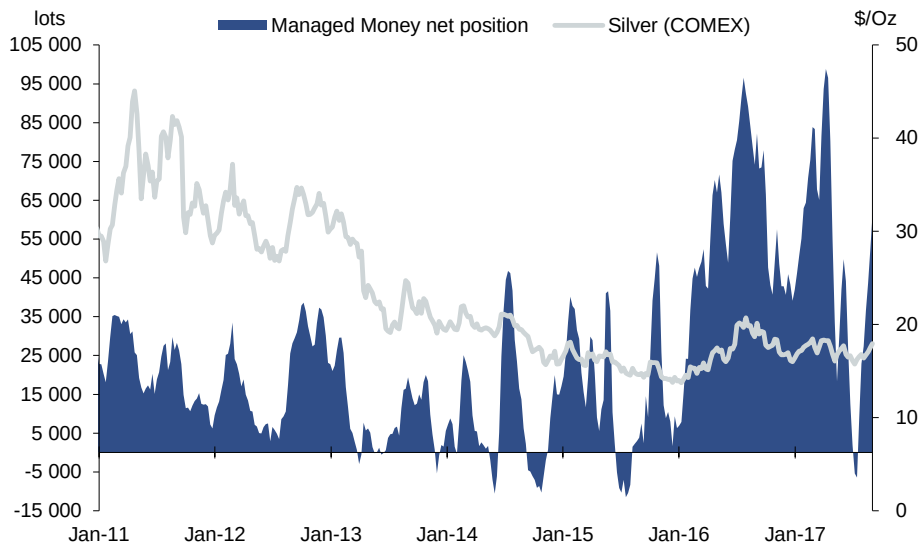
Silver – Price stability has not been enough to reverse persistent ETF outflows

ETF net outflows slow QTD, but remain sizeable at 248t

tonnes

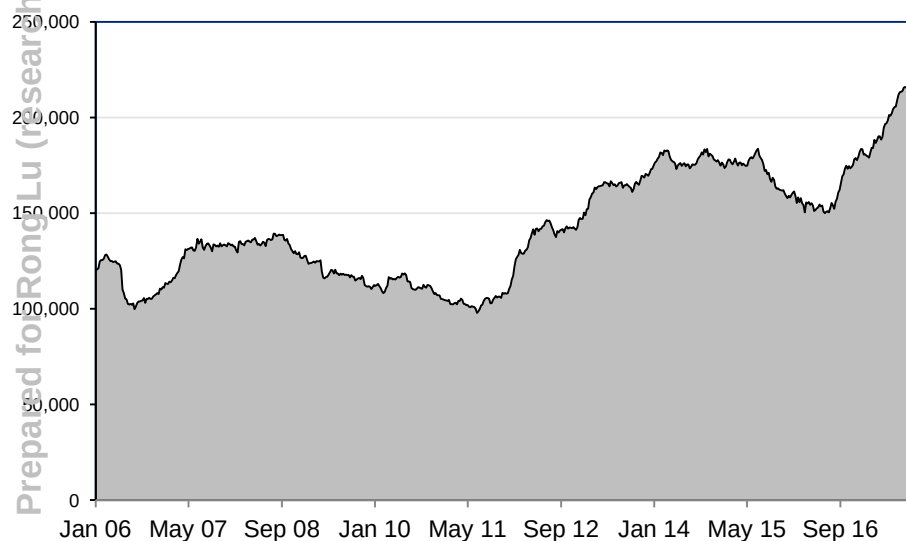


Money manager net length recovers to from the July nadir

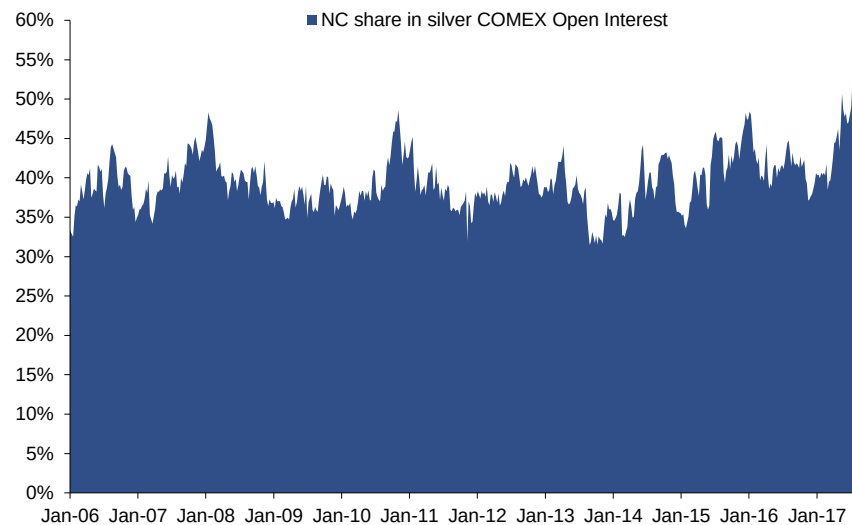


COMEX inventory levels rise 26.3% YTD to 232koz

ounces



Spec share of OI cools to 42% from multi-year highs



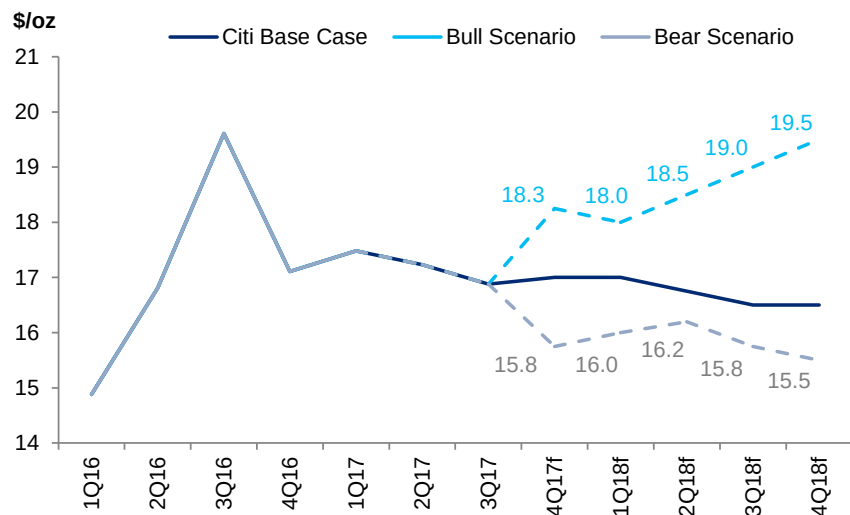
Source: Bloomberg, CFTC, Citi Research. Data as of mid-November.

Silver – Market prices face challenges into 2018 as positive sentiment wanes

Silver Price Chart in USD and EUR terms



Silver bull/bear case price forecasts*



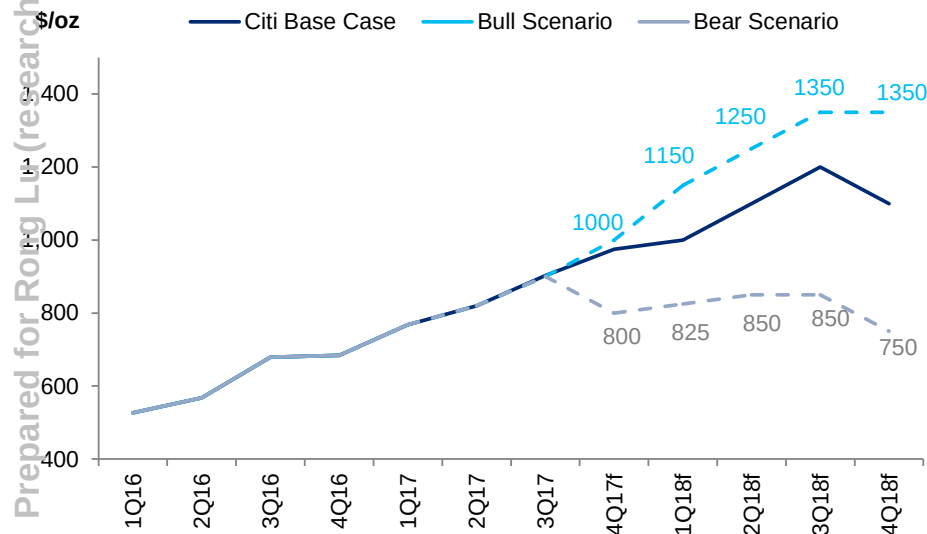
- We moderately lower 4Q'17 prices to \$17.00/oz (from \$17.70/oz) and expect 1H'18 prices to grind lower.**
 Following the price correction in September, we were right to expect a short-term price recovery in 4Q'17 as spot silver has thus far gained ~4%. However, we do not envisage the price gain to extend beyond \$17.50/oz. The gold:silver ratio has reversed out of YTD lows in April of ~68.4x to stabilise to ~74.5x at the time of writing. Consistent with the declining silver price trajectory in 1H'18 we expect the ratio to maintain levels above 75x. In light of these price dynamics we still maintain a bias to upside price risks into the new year.
- Upside price risks (20%):** Our bullish scenario points to prices potentially breaching \$18.5/oz in the latter part of the year. Geopolitical risks may be currently under-priced and may still persist translating into increased ETF or retail bar and coin investor buying. Hence, we still maintain a marginally higher probability weighting to our silver bull case scenario in light of an undercurrent of safe-haven support that is still at play.
- Downside price risks (15%):** Continued producer hedging activity may cap future price increases. A collapse of industrial demand would quickly reverse any notional market tightness and prices could easily slip into our bear-case price scenario near \$16.0/oz. Outside of seasonal trends a quicker than expected decline in jewellery demand from Asia would result in fabricator inventory backlogs, heightened product competition driving sales of more 'entry-level' luxury goods and ultimately lower prices.

Palladium – expecting \$1,200/oz by 3Q'18 on sustained deficits

World Palladium Supply Demand Balance

Pd 000 oz	2015	2016	2017f	2018f	2019f	2020f
Autocatalysts	6240	6660	7812	8182	8228	8188
Jewellery	335	318	328	336	341	338
Investment	-625	-410	-235	-160	-175	-250
Industrial	2305	2263	2277	2292	2307	2322
Total demand	8255	8830	10183	10651	10701	10598
Mining, of which:						
Southern Africa	3004	2966	3236	3283	3350	3393
Russia	2434	2773	2725	2650	2500	2500
North America	874	894	890	890	915	915
Rest of World	142	129	180	180	180	180
Total Mine Supply	6454	6762	7032	7003	6945	6988
Autocatalyst Scrap	1891	1990	2206	2321	2454	2486
Jewellery & Other Scrap	466	461	470	492	515	526
Total Supply	8811	9213	9708	9816	9914	10000
Surplus/(Deficit)	557	383	-475	-835	-787	-598
Palladium Price (US\$/oz)	692	614	865	1100	1000	900

Palladium bull/bear case price forecasts*



- Our new market balance and bullish 2018 price forecasts for palladium reflect our expectation of sustained solid global growth and a deficit market of ~835koz. Our updated forecast methodology incorporates palladium loading estimates per region, which should see autocatalyst demand continue to increase, breaching 8moz by 2018, though to some degree this should be offset by sizeable increases in autocatalyst scrap.
- Looking further we forecast that the tightness in palladium relative to platinum and resulting relative price performance will ultimately prove self-defeating. Indeed, by 2019/20 we assume that the strong forecast palladium to platinum price ratio in 2018 ultimately results in substitution of palladium for platinum in gasoline engines over the longer term, mitigating palladium deficits.
- **Key downside risks (20%):** Weaker than anticipated global income and automotive sales growth and a stronger than expected scrap recovery represent key downside risks.
- **To the upside, key risks (20%) include:** A significant easing in Chinese tightening policies would boost Chinese industrial output and would be bullish for palladium. An extension of the small engine tax-rebate later next year might prompt higher than expected sales into the latter part of 2018. Alternatively, limited to no substitution of platinum in gasoline engines would be bullish for palladium.

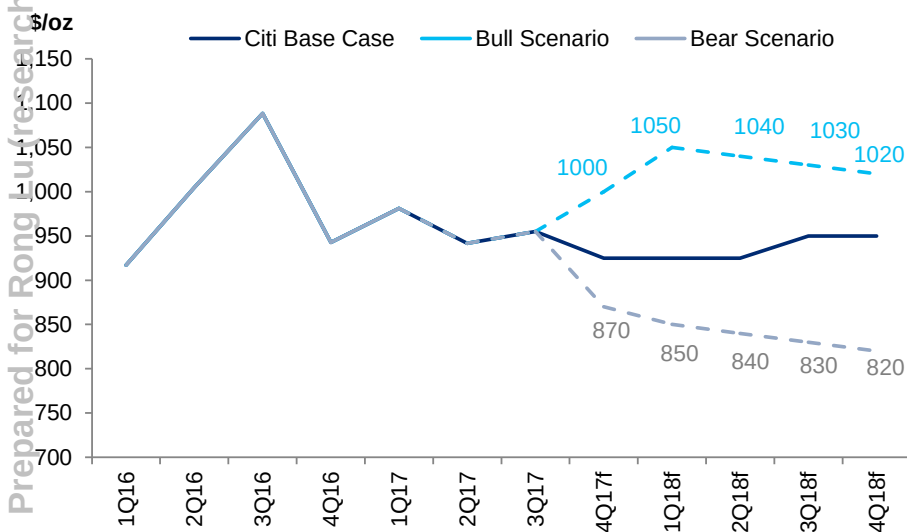
Source: Bloomberg, Thomson Reuters GFMS, CPM, Citi Research. *Subject to revision

Platinum – surplus environment persists in 2018 keeping prices range bound

World Platinum Supply Demand Balance

Pt 000 oz	2015	2016	2017f	2018f	2019f	2020f
Autocatalysts	2970	3117	3341	3529	3562	3587
Jewellery	2860	2518	2550	2574	2606	2762
Investment	-320	-135	95	105	325	450
Industrial	1463	1480	1490	1500	1510	1531
Total demand	6973	6979	7475	7708	8003	8330
Mining, of which:						
Southern Africa	4972	4881	5079	5133	5268	5354
Russia	670	723	824	800	750	750
North America	316	338	366	367	375	382
Rest of World	151	161	120	110	110	115
Total Mine Supply	6109	6103	6389	6410	6503	6602
Autocatalyst Scrap	1191	1197	1243	1276	1292	1290
Jewellery & Other Scrap	384	467	404	404	405	405
Total Supply	7683	7767	8036	8091	8200	8297
Surplus/(Deficit)	710	788	561	383	197	-33
Platinum Price (US\$/oz)	1056	988	950	940	1000	1150

Platinum bull/bear case price forecasts*



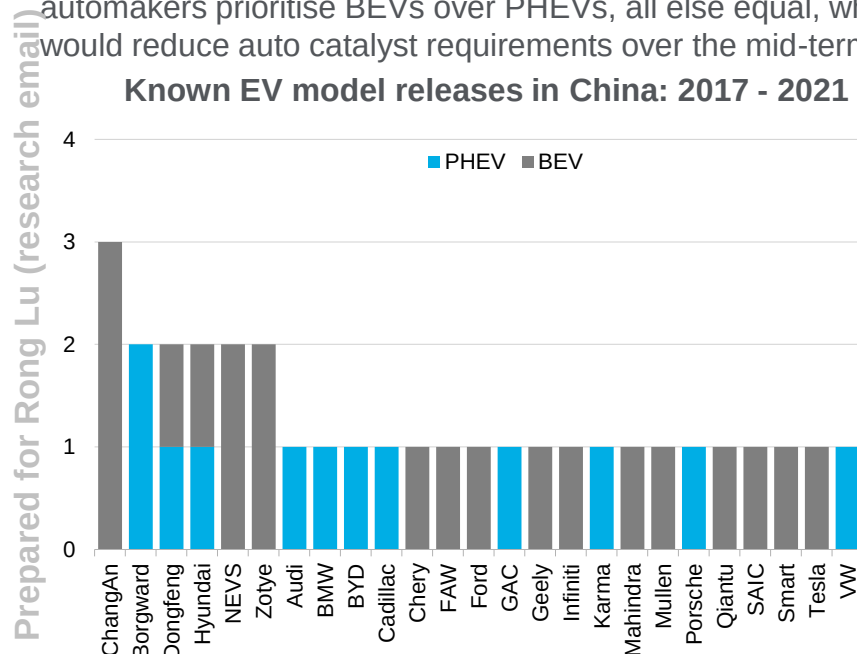
- Neutral outlook for platinum.** We continue to see the platinum market in moderate surplus during 2018. Looking further ahead we now believe that sustained palladium strength relative to platinum will eventually result in substitution to platinum in gasoline engines. As such, our base case is that platinum surpluses decrease over the longer term.
- We maintain probability weightings of 25% for the bear case and 15% for the bull case due to the transformative changes underway within the automotive demand segment (NEVs, shifts from diesel to gasoline engines).
- Key downside risks (25%):** A quicker than expected decline of diesel-engine market share in Europe, particularly in urban areas, from 52% in 2016 to less than 45% in 2020. Such a scenario would certainly push platinum surpluses higher post 2018. Shifting spending patterns to 'experiences' away from 'luxury goods' may also slow platinum-based jewellery demand.
- To the upside, key risks (15%) include:** Improved diesel-engine based vehicle sales (most hybrids for example) in Europe could counter-act or indeed boost platinum demand. Secondly, if PGM parity or a palladium-platinum premium were to persist for several months, it would incentivise substitution of platinum into gasoline-engine vehicles, with the highest risk of the autocatalyst thrifting expected by 2H'18. We note that further penetration or diversification of passenger vehicle powertrains from zero-emission contenders is a longer-term (post 2020) consideration.

Source: Bloomberg, Thomson Reuters GFMS, WPIC, Citi Research. *Subject to revision

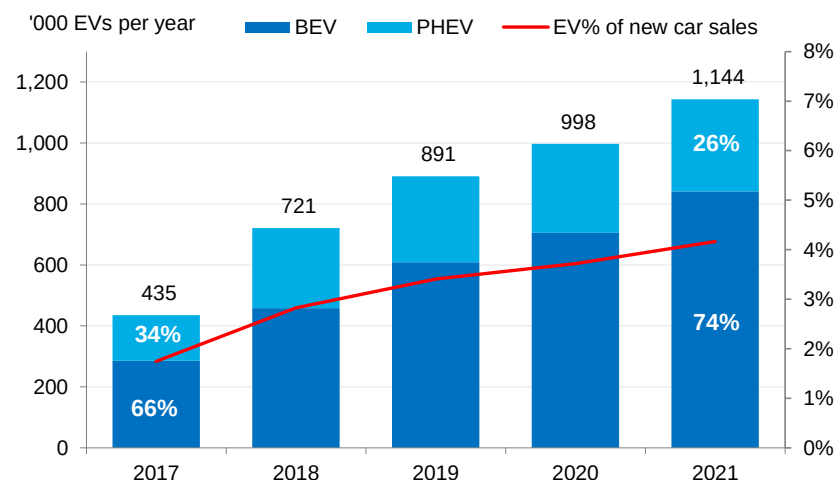
PGMs – China's NEV quota system has a limited impact on 2018 car sales...

- Commencing in 2019, a new system of Chinese automaker 'credits' will be imposed, where the 'credits' are a function of the average driving range of NEVs, supporting the use of larger, more efficient batteries. 'Credits' form part of a new 'NEV score' for large automakers under the 'cap-and-trade' policy. Per the recent specifications laid out by the MIIT and MoF, the policy demands automakers sell EVs to generate credits equivalent to 10% of total passenger vehicle sales by 2019 and 12% by 2020.
- PHEVs can in general achieve less than half the credits of a BEV, which could change the powertrain mix in China versus other regions. Each plug-in hybrid electric vehicle can obtain 2 credits, while pure battery vehicles can reach a maximum of 5 credits. Once implemented in 2019 we could see Chinese automakers prioritise BEVs over PHEVs, all else equal, which would reduce auto catalyst requirements over the mid-term.

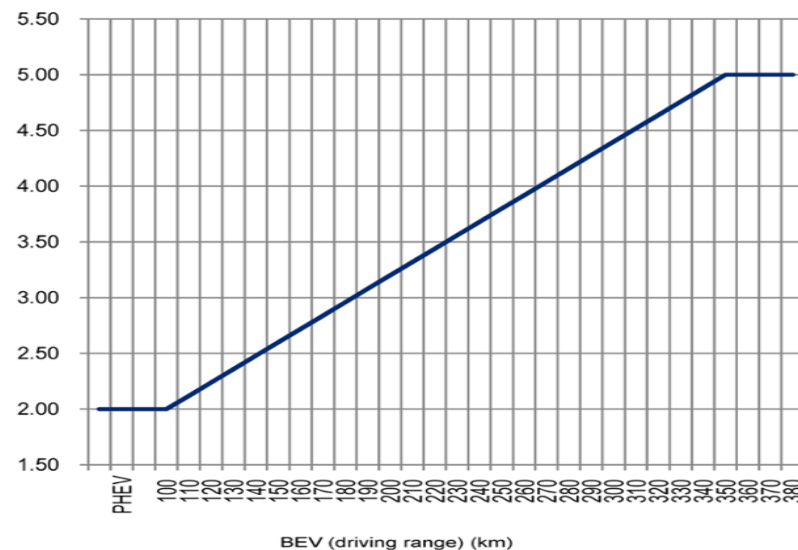
Known EV model releases in China: 2017 - 2021



PHEV share of sales will first grow then decline



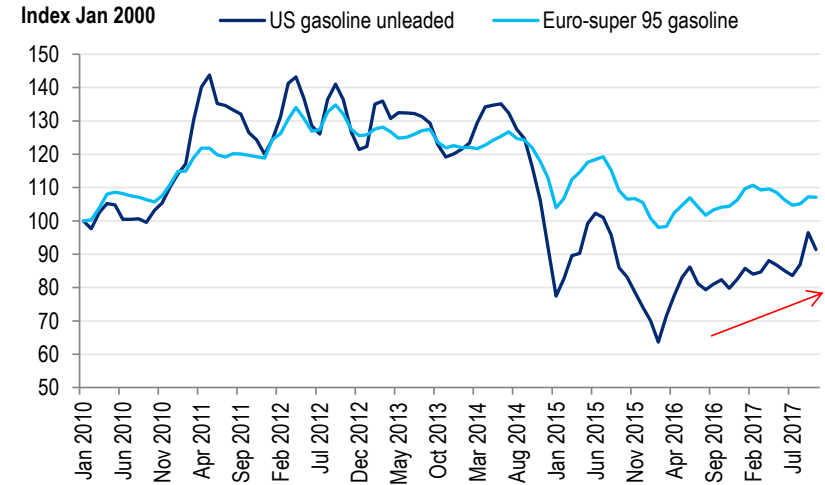
Chinese automaker credits earned by NEV type



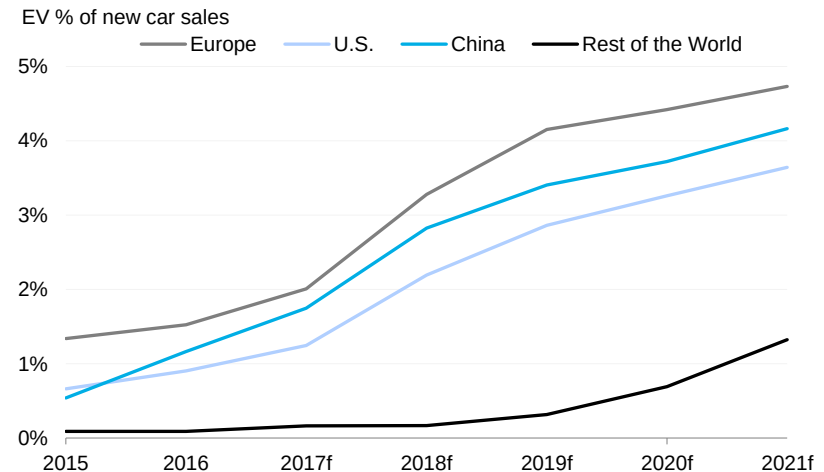
PGMs – ...however PGMs demand looks challenged over the very long term

- **Over the long term we expect pure NEVs will dominate global vehicle production.** In particular, it is likely there will be changes to the favoured battery chemistry within BEVs that make this battery technology more competitive over the long term.
- **It is highly likely that new BEV battery technology will not require expensive platinum and palladium as catalysts.** We therefore assume PGM autocatalyst demand destruction will become irreversible in an environment where BEVs rather than PHEVs are bought instead of traditional ICE (internal combustion engine) vehicles.
- **The overarching policy framework and associated incentive should ultimately define the power train mix.** Europe and to a larger extent, the wider Western world has effectively embraced hybrids, while battery cars still face infrastructure and life-time cost related hurdles. Conversely in China, hybrids have not been incentivised as much as pure battery vehicles. Hence we expect the battery vehicle share of the overall fleet to be larger in China.
- **Following two years of stagnant fuel prices, rising bills at the pump in 2018 could help push new car owners to alternative vehicles.** European gasoline pump prices (with tax) have been rising, notably, prices have risen 4.1% since the start of 2H'17 alone. The average US gasoline unleaded price has risen 14.9% over the same time frame. As such, new car owners may begin to consider hybrids battery cars.

Western world pump prices are on an upward trend



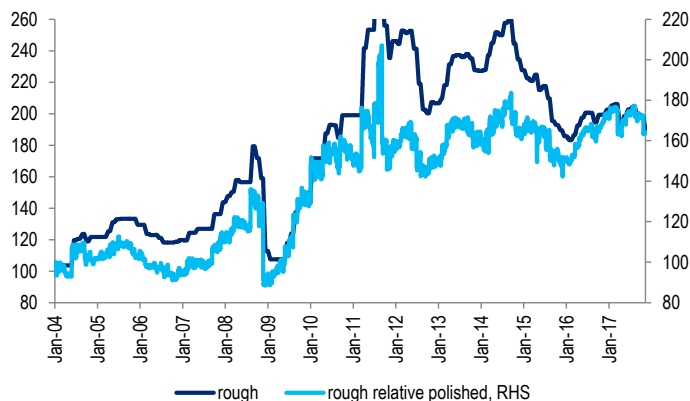
EVs will be set to account for a growing share of new car sales



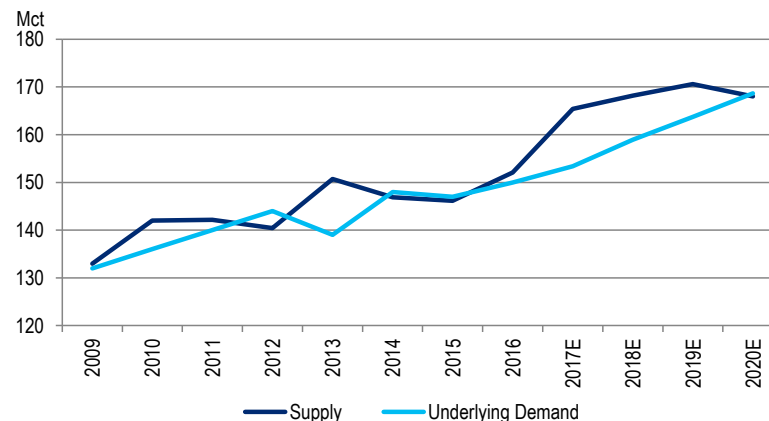
Diamonds - Destocking downturn imminent

We continue to believe that the diamond industry is moving into a destocking cycle. The midstream segment is in the second consecutive year of significant restocking, that will in our view lead to a destocking cycle starting later in 2017 or 2018. Prices have remained broadly stable YTD however we expect 6% decline in 2018.

Rough prices and rough relative polished



Rough diamond supply/demand



Prepared for Rong Lu (research email)

• **2018 repeat of 2015?** - The diamond industry is moving into a destocking cycle in our view. We think that 2018 is shaping up to be an extremely difficult year for the diamond industry. We expect industry sales to fall sharply in late 2017 and 2018 as:

- 1) retail diamond demand growth is likely to be only 2%
- 2) against 2% underlying end demand (at retail), we forecast 7% increase in rough sales volumes representing a second consecutive year of midstream restocking;
- 3) rough sales growth remains high despite weak end demand because diamond miners expand production by 9% and will seek to move these products into the channel which is likely to lead to higher inventories in the midstream and a subsequent destocking for rough diamonds.

• **Alarm bells are ringing** - The rough to polished price ratio remains at an elevated level, supporting our view that retail (polished) demand is not as strong as channel (rough) demand. This leads us to view that diamond prices could decline 6% in 2018.

• **Lab-grown threat and millennial behaviour** - The medium to longer term outlook remains at risk from growth in lab grown supply and changes in millennial behavior.

7. The US farm belt is to cereals what US shale is to global oil; CBOT prices lower for longer? Bullish ICE sugar still in play...

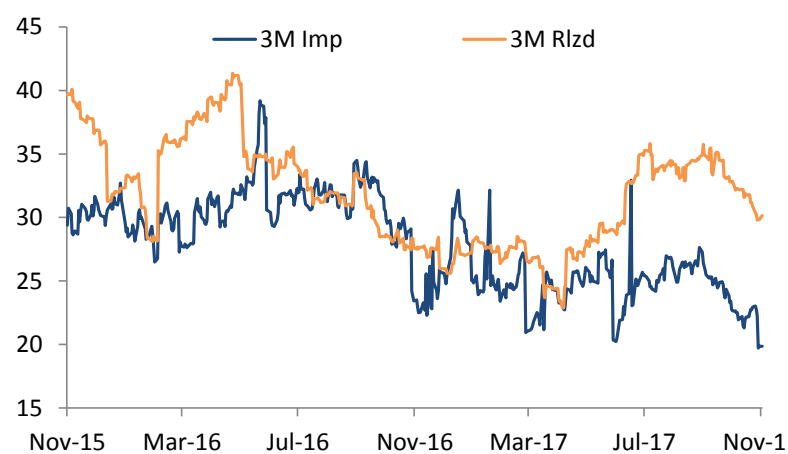
Agriculture Sector Conviction Call – We are still long 2H'18 ICE Sugar

We expect ICE sugar prices to trade 10-15% higher versus the NY curve in 2H'18* and recommend positioning for upside. Cheap volatility and flat skew make outright calls and bullish risk reversals look the most attractive.

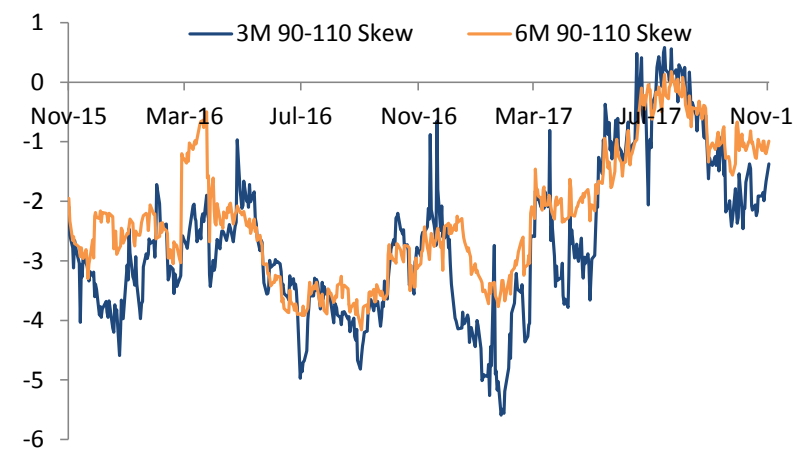
- **ICE sugar prices look biased to move higher through 2018 in our view, given supportive Brazilian ethanol dynamics and tightening balances.** Since mid-August, sugar has been trading at a discount to ethanol in CS Brazil and if these pricing dynamics persist, the CS Brazil sugar mix could come in as low as 40%, which could swing 2018/19 balances back into deficit. Thus, we expect sugar prices would need to move ~\$0.02/lb higher relative to ethanol to incentivize sufficient sugar production in CS Brazil. Current ethanol prices are ~\$0.15/lb in sugar-equivalent terms, and with Brent expect to trade at \$54/bbl average price in 2018, we think these levels could be maintained. As such, we expect ICE sugar prices to trade up to the \$0.17/lb range in 2H'18, some ~12% above current forwards* and we think consumers and funds will be keen to position for upside in 2H'18.
- **Given the cheap implied volatility and flat skew, we find outright calls and bullish risk reversals the most attractive ways to position for the upside in sugar prices going into 2018.** The 3M implied volatility is trading below 20 vol points, the lowest level in more than two years and much lower than the trailing realized vol. To be sure, while implied volatility is subdued across the commodity complex, only a few commodities are seeing a negative vol risk premium, most of which are in the grains sector driven by a seasonal dip in volatility in 4Q. At the same time, although call skews have re-priced higher recently with the bearish investor sentiment abating, put vols remain rich versus calls historically, making selling puts to fund upside exposure more attractive.

Prepared for Rong Lu (research email)

ICE Sugar 3M Implied versus Realized Volatility



ICE Sugar 3M and 6M 90-110% Moneyness Skews



Source: Bloomberg, Citi Research, *as of mid-November 2017 although we have published this outlook and left our ICE outlook unchanged since early October 2017.

Productivity gains, record yields, low prices...US shale or the US farm-belt?

The 2017/18 campaign might mark a turning point in the long-term grains cycle whereby market assumptions for 'trend crop yields' are structurally tilted upwards, with bearish implications for baseline CBOT prices absent a major weather event, and headline negative for food-at-home inflation. On the other hand, our forecast is slightly more constructive for the equatorial softs, especially 2018/19 ICE sugar, but for idiosyncratic one-off factors.

- **US row crop supply growth over the past half-decade has consistently surprised to the upside. The Midwest farmer is transforming the global staple cereals landscape in a similar way that US tight oil has impacted the energy sector:** structurally depressing prices, allowing stockpiles to build-up domestically and abroad and providing trade competition, just as the likes of Brazil, Russia and other key producers have started to ramp-up their own harvest profiles.
- **Precision farming techniques, weather-resistant seed technology and farmer input cost deflation across the North American farm belt have underpinned back-to-back bumper row crop harvests and record corn and soybean yields.** Since nominal cereal trading levels hit an all-time high during the late stages of the 2011/12 crop year a half-decade ago, average CBOT marketing year price declines have been 10-12% per annum on both a crop year and calendar year basis. Farmer gross incomes have similarly declined for the longest stretch since the early/mid-1970s at four consecutive crop cycles. And if the record-setting trend for higher annual crop yields continues apace, then we are hard-pressed to call for a significant price recovery in the staple cereals complex, especially given a relatively healthy level of global importer stocks and relatively robust output levels from Brazil and Argentina in the Southern Hemisphere.

• **While significant retrenchment for the CBOT grains complex appears more limited at this juncture, price upside also seems challenged without deeper supply-side rebalancing effects (assuming normal weather).** To be sure, mid-4Q'17 corn and wheat prices at \$3.30-3.50/bu and \$4.20-4.40/bu are not much above where variable cost support would suggest, historically. Additionally, hedge funds are holding near record short positions across the complex and there should be some demand impulse from price-elastic consumers. But as a baseline, we modestly cut our outlook for grains prices over the next year or two, struggling to see a sustained push in corn, wheat or soy prices above \$4/bu, \$5/bu and \$10.5/bu, respectively.

• **On the other hand, the Brazilian softs that trade on ICE such as coffee and especially sugar, should see price recoveries next year on one-off factors.** For raw sugar in particular, aging cane, higher oil prices and under hedged mills in Brazil should lead to a sharp decline in 2018/19 output. Our previously bullish ICE cocoa market outlook is now more neutral the curve, as prices have rose, per Citi, above \$2,100/t.

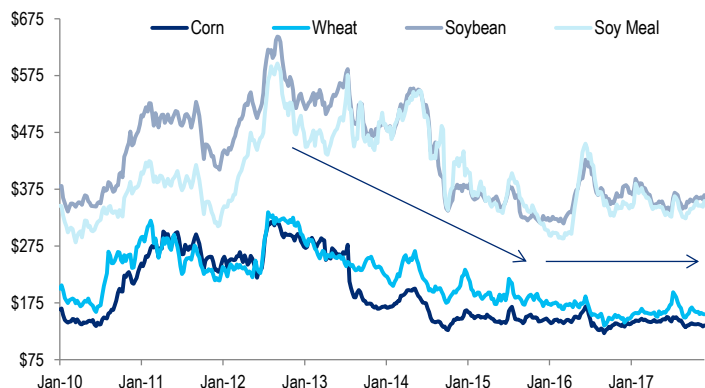
6-12m forward outlook and directional bias (benchmark CBOT/ICE) versus spot levels/curve baseline*

Bullish	Neutral-Bullish	Neutral	Neutral-Bearish	Bearish
Sugar	Coffee, Ethanol	Corn, Wheat	Soybeans	
		Cocoa		

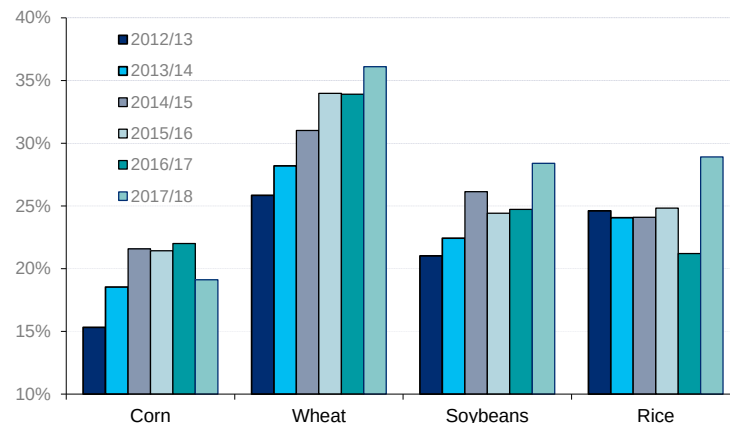
Source: Citi Research, *as of mid-November 2017 and subject to revision; assumes normal weather

CBOT prices downgraded versus 4Q'17 update; ICE price outlook unchanged

Benchmark feed grain prices 2010 – November 2017(\$/t):
Lower for longer without deeper rebalancing effects



Global grains stocks-to-use ratios (%): Inventories still loom large and well-above 2013/14 levels across markets



Citi Commodities Agriculture market price outlook and changes versus 4Q'17 outlook*

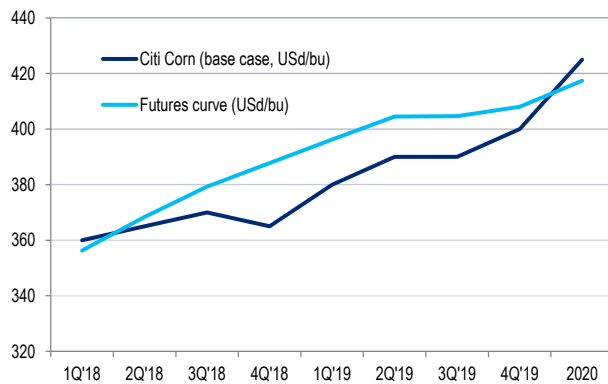
Contract*	Unit	0-3m	6-12m	4Q'17F	1Q'18F	2Q'18F	3Q'18F	4Q'18F	1Q'19F	2Q'19F	3Q'19F	4Q'19F	2014A	2015A	2016A	2017F	2018F	2019F	2020F
Corn	USD/bu	360	370	345	360	365	370	365	380	390	390	400	415	377	358	360	365	390	425
Soybean	USD/bu	945	955	975	945	930	955	975	990	1015	1010	1025	1245	945	989	975	951	1010	1050
Wheat	USD/bu	468	485	445	468	480	485	505	510	505	500	525	588	507	436	442	485	510	550
Cocoa	USD/t	2070	2200	2100	2070	2150	2200	2175	N/A	N/A	N/A	N/A	3010	3093	2880	2005	2150	2200	N/A
Coffee	USD/lb	134	138	126	134	140	135	130	N/A	N/A	N/A	N/A	178	132	138	135	135	134	N/A
Sugar	USD/lb	15.0	17.0	15.0	15.0	16.0	17.0	17.0	N/A	N/A	N/A	N/A	16.0	13.1	18.4	16.0	16.5	16.0	N/A
Ethanol	USD/gal	1.45	1.50										2.06	1.51	1.51	1.50	1.60	1.55	N/A
Changes																			
Contract*	Unit	0-3m	6-12 m	4Q'17F	1Q'18F	2Q'18F	3Q'18F	4Q'18F	1Q'19F	2Q'19F	3Q'19F	4Q'19F	2014A	2015A	2016A	2017F	2018F	2019F	2020F
Corn	USD/bu			-10	-15	-20	-25	-50	N/A	N/A	N/A	N/A	415	377	358	362	-28	-60	N/A
Soybean	USD/bu			33	0	-60	-45	5	N/A	N/A	N/A	N/A	1245	945	989	968	-24	-90	N/A
Wheat	USD/bu			-5	3	0	0	20	N/A	N/A	N/A	N/A	588	507	436	445	5	-65	N/A
Cocoa	USD/t			175	0	0	0	0	N/A	N/A	N/A	N/A	3010	3093	2880	2005	0	N/A	N/A
Coffee	USD/lb			-9	-5	-5	-10	-20	N/A	N/A	N/A	N/A	178	132	138	135	-10	N/A	N/A
Sugar	USD/lb			0	0.0	0.0	0.0	0.0	N/A	N/A	N/A	N/A	16.0	13.1	18.4	16.0	0.0	N/A	N/A
Ethanol	USD/gal												2.06	1.51	1.51	1.50	-0.20	N/A	N/A

Upgrades: CBOT wheat by less than 1% for 2018F. We remain bullish ICE sugar versus 2H'18 forwards.

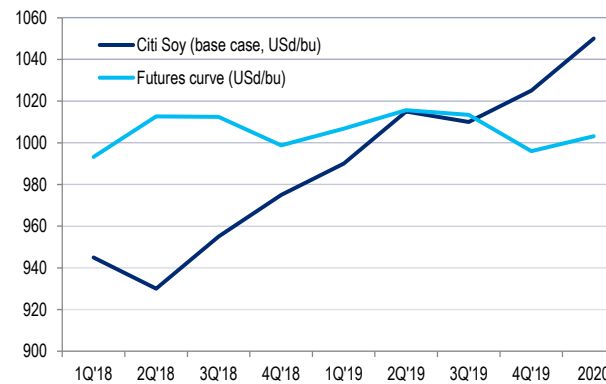
Downgrades: CBOT corn and soybeans 3-5% for 2018F and 5-10% for 2019F assuming normal weather.

Risks for CBOT grains complex – upside and downside considerations

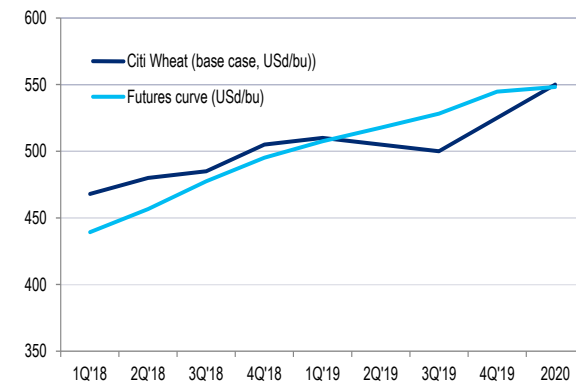
Citi corn price outlook versus futures*



Citi soy price outlook versus futures*



Citi wheat price outlook versus futures*



- CBOT corn:** Our base case is more neutral the curve in cal'18 albeit bearish cal'19, given the high carry in the corn market. Our bear case scenario for 2018 CBOT corn prices is \$3.10-3.15/bu on average from 2Q-4Q'18, down 15% from our baseline cal'18 average and near variable cost support levels. We see this as a low probability event, requiring US stocks-to-use for 2018/19 to be pegged by the market at 22-23% (levels not seen since the early 1990s) and for Brazil, Ukraine and Argentina to all concurrently post near record harvests. Our bull case outlook would be for prices to average \$4.20/bu or 15% above our current cal'18 forecast, should the US carryout ratio dip to the T5Y average of 11.5% on a weather event and lower plantings. **Citi corn risk profile: base case (70%); bear case (15%); bull case (15%)**

- CBOT soybean:** Our base case is slightly bearish the forwards in cal'18 at \$9.51/bu on average and mixed for cal'19/cal'20. Our bear case scenario of \$8.50/bu could materialize on US soy acres surging above 92mn with trend/high yields, Brazil posting a new-crop record harvest closer to 114-115mmt as opposed to our baseline y/y cut and for China to push out imports. A bull case scenario of \$10.50-11.00/bu would likely require an ENSO weather event in our view, and US stocks-to-use falling towards 7% for 2018/19 on better export demand. **Citi soy risk profile: base case (60%); bear case (15%); bull case (25%).**

- CBOT wheat:** Our base case is neutral the cal'18 curve and parts of the long-dated curve at \$4.85/bu for cal'18 and \$5.50/bu for cal'20. With global wheat supply outpacing demand growth for a fifth consecutive cycle, a bear case scenario could see CBOT levels average ~\$4.15/bu in 2018 on robust USD strength and a plunge in USGC export volumes below 900mn bu and a lack of US producer discipline to cut all-wheat area, as Russia and EU take more foreign market share. The bullish flipside would be sharper all-wheat acreage cuts, tightness in high-protein feed markets and the implementation of a Black Sea export ban similar to what Russia tried in 2010/11. This could see wheat prices rally and stay above \$5.25-5.50/bu in the next year. **Citi wheat risk profile: base case (65%); bear case (10%); bull case (25%).**

Source: CBOT, Citi Research *as of mid-November 2017; Chicago benchmark contracts trading on CBOT

The current crop environment stands out over the decades

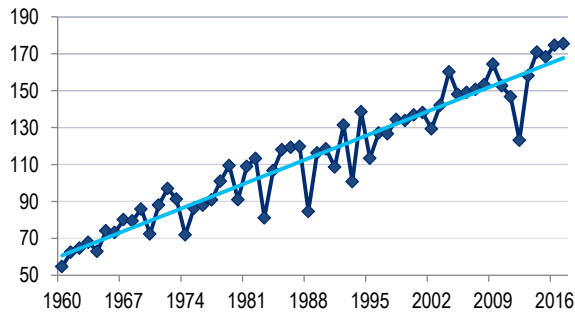
Corp Year Return					Calendar Year Return				
	Corn	Soy	Wheat	Sector		Corn	Soy	Wheat	Sector
70/71	-23%	16%	17%	3%	1970	28%	19%	14%	21%
71/72	10%	7%	-7%	3%	1971	-21%	6%	-3%	-6%
72/73	86%	108%	93%	96%	1972	26%	37%	62%	41%
73/74	38%	2%	24%	21%	1973	73%	36%	107%	72%
74/75	-12%	-24%	-13%	-16%	1974	27%	20%	-16%	11%
75/76	-7%	18%	16%	9%	1975	-24%	-36%	-27%	-29%
76/77	-36%	-23%	-33%	-31%	1976	-2%	57%	-17%	13%
77/78	17%	23%	34%	25%	1977	-13%	-16%	1%	-9%
78/79	34%	10%	12%	19%	1978	4%	14%	23%	13%
79/80	25%	10%	9%	14%	1979	25%	-5%	32%	17%
80/81	-18%	-16%	2%	-11%	1980	31%	23%	10%	21%
81/82	-22%	-17%	-13%	-17%	1981	-28%	-23%	-22%	-24%
82/83	57%	62%	-1%	39%	1982	-10%	-8%	-16%	-11%
83/84	-16%	-27%	4%	-13%	1983	38%	44%	10%	31%
84/85	-22%	-19%	-11%	-17%	1984	-20%	-30%	-4%	-18%
85/86	-33%	-6%	-21%	-20%	1985	-8%	-7%	-1%	-5%
86/87	3%	5%	10%	6%	1986	-36%	-8%	-20%	-21%
87/88	81%	68%	28%	59%	1987	15%	24%	13%	17%
88/89	-19%	-32%	11%	-13%	1988	54%	33%	42%	43%
89/90	5%	3%	-14%	-2%	1989	-16%	-29%	-7%	-17%
90/91	2%	-4%	-13%	-5%	1990	-3%	-1%	-36%	-14%
91/92	-12%	-5%	22%	2%	1991	9%	-1%	55%	21%
92/93	4%	19%	-19%	2%	1992	-14%	3%	-13%	-8%
93/94	-2%	-11%	16%	1%	1993	41%	24%	7%	24%
94/95	33%	6%	13%	17%	1994	-25%	-22%	6%	-13%
95/96	25%	32%	38%	32%	1995	60%	33%	28%	40%
96/97	-28%	-18%	-32%	-26%	1996	-30%	-6%	-26%	-21%
97/98	-30%	-23%	-21%	-24%	1997	3%	-3%	-15%	-5%
98/99	8%	-9%	-11%	-4%	1998	-19%	-20%	-15%	-18%
99/00	-11%	2%	9%	0%	1999	-4%	-14%	-10%	-9%
00/01	20%	-3%	-1%	5%	2000	13%	8%	12%	11%
01/02	18%	17%	4%	13%	2001	-10%	-16%	3%	-7%
02/03	-10%	6%	15%	4%	2002	13%	35%	12%	20%
03/04	-2%	5%	12%	5%	2003	4%	39%	16%	20%
04/05	-14%	-8%	-15%	-12%	2004	-17%	-31%	-18%	-22%
05/06	12%	-9%	19%	7%	2005	5%	10%	10%	9%
06/07	41%	61%	32%	45%	2006	81%	14%	48%	47%
07/08	75%	53%	46%	58%	2007	17%	75%	77%	56%
08/09	-43%	-17%	-16%	-25%	2008	-11%	-19%	-31%	-20%
09/10	36%	-1%	-32%	1%	2009	2%	7%	-11%	-1%
10/11	75%	44%	74%	64%	2010	52%	34%	47%	44%
11/12	10%	24%	-15%	6%	2011	3%	-14%	-18%	-10%
12/13	-38%	-19%	15%	-14%	2012	8%	18%	19%	15%
13/14	-27%	-23%	-11%	-21%	2013	-40%	-7%	-22%	-23%
14/15	1%	-18%	-24%	-13%	2014	-6%	-22%	-3%	-10%
15/16	-15%	9%	-6%	-4%	2015	-10%	-15%	-20%	-15%
16/17	10%	-2%	-9%	-1%	2016	-2%	14%	-13%	0%
Average	6%	6%	5%		2017 YTD	-4%	-2%	3%	-1%
Std. Dev	32%	29%	26%		Average 70-16	6%	6%	6%	
					Std. Dev 70-16	28%	25%	30%	

- The staple cereal sell-off since the record 2012 drought has been unprecedented, at least looking back to 1970, both in its scale and consistency. Crop year and calendar year* price declines since the 2012 peak have been 10-12% vs. a long-term mean of 5-6%. Implications abound:
 - World and US food CPI have been on a structural decline, with the deflationary trend only reversing briefly this year. US farm income has declined sequentially as LatAm continues to expand its footprint and US farmers have not cutback enough acres.
 - Low levels of US and global grain stocks (2010-2013), exacerbated by 'export bans' from the Black Sea, have grown 50-100% .
 - Speculative trading in CBOT cereals seems to have become more tactical and opportunistic short-term vs. a long-term passive play. Weaker curve term structure, especially for CBOT wheat, has supported this trend. From 2007-June 2013, hedge funds held a net long ~290k lots in CBOT corn/soy/wheat. Since then, the funds have averaged a net short over ~80k lots.
- Absent a major weather shock, we expect these trends to hold in 2018/19 (even if only a repeat of 2017 weather, which was less than ideal versus 2014-2016 during July and late August/September).

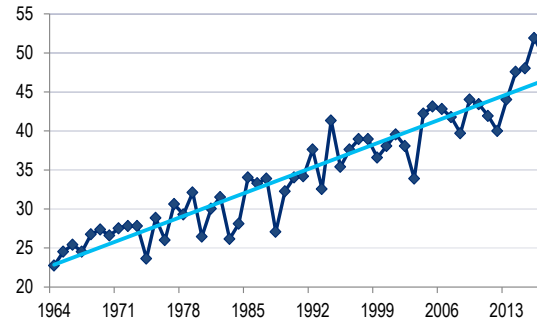
Source: Bloomberg, Citi Research, *as of mid November 2017; Northern Hemisphere crop year = (Sep/Aug for corn and soy, Jun/May for wheat)

US row crop yields should be pegged at a higher baseline on *average* weather

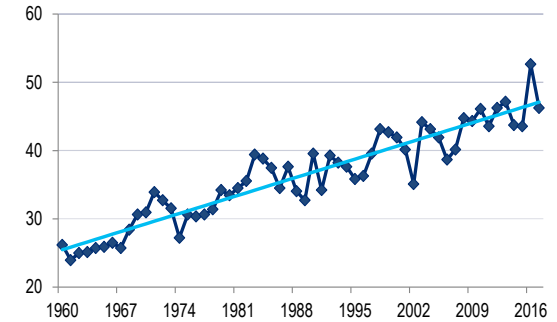
US Corn yield versus trend (bu/acre)



US Soy yield versus trend (bu/acre)



US Wheat yield versus trend (bu/acre)



- US row crop yields have posted a CAGR of 2.3% for corn and 1.5% for soybeans over the past half-century.** Yields plunged 10% below trend during the devastating 2012 drought so perhaps more remarkable is the strengthening of the upward yield momentum for soybeans since 2013 to ~2.4% per annum and US corn yields sustaining a 2.1% growth rate. To be sure, US corn and soy yields each hit three fresh records over the past four growing cycles. At face value this is not unusual given the long-term upward trajectory for yields. But that these yield gains are regularly coming-in north of consensus expectations and despite less-than-ideal growing conditions (e.g. 2017/18 is a clear example of this), suggests a breakdown in the conventional USDA crop-condition yield models (both at the national and individual state levels). For example, Citi's own weekly US yield model as well as Bloomberg and Reuters monthly surveys regularly pointed to downside yield risks following the USDA July crop report this year. Yet, the August, September and October WASDE reports continued to scale yields above expectations. US corn yields, for example, have posted 10-bu/acre or 6% higher than our own August estimate. We do not think this is a one-off issue, and we think the past half-decade represents a new higher trend for crop yields in the absence of a sustained adverse weather shock. We thus peg 2018/19 US corn and soy yields at ~173-bu/acre and ~49-bu/acre*.

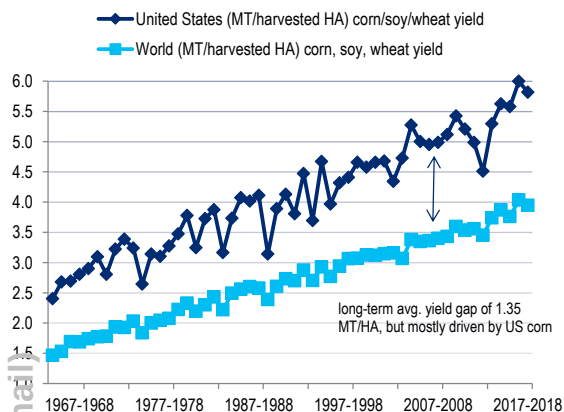
The application of biotechnology and emergence of drought, disease and heat resistant genetic seeds, precision farming with satellite technology and improved fertilizer use are boosting corn and soy yield prospects, just as key pockets of demand growth (e.g. biofuel processing) start to plateau. This means without adequate acreage cuts or a major weather event, surplus inventory levels can remain stickier. Consider that since 2014/15, CBOT corn and soy prices have averaged a stable \$3.65/bu and \$9.75/bu, respectively, with very little price volatility for mean seasonal levels. Indeed, CBOT corn prices have only sustained a push above \$4/bu on three brief occasions since 3Q'14. While wheat is a non-GMO crop across most of the world, with a more modest 1% US yield CAGR, there the bigger issue is that the 'rest-of-world' has caught up with the US in terms of wheat farming productivity, prompting significant trade competition, and keeping prices low.

Source: Bloomberg, Industry reports, USDA, Citi Research, *this is still conservative to what some investors or consultants might price for 2018/19.

The US plugs the row crop yield gap, but the world can grow wheat...

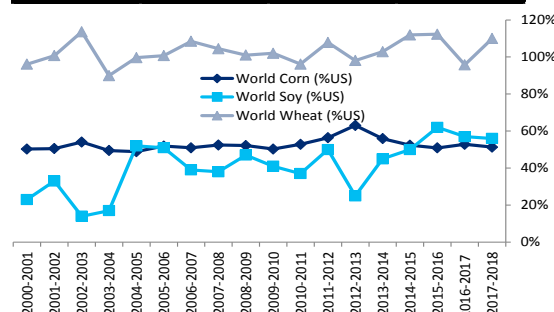
The rest-of-world (RoW) has closed-in on the US with respect to crop yields and in many regions are ahead of the US for wheat. Wheat being a primary foodstuff cereal and less weather-sensitive than its row crop brethren has seen both Southern and Northern Hemisphere producers sharply increase productivity in recent years.

US and World grain yields LT trend*

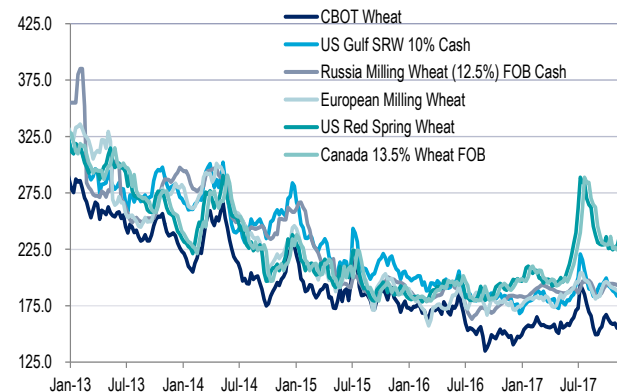


Long-term CAGR of crop yields and US versus World yield comps

	Corn	Soy	Wheat
US	1.90%	1.50%	1.10%
World	1.90%	1.70%	1.90%



Various wheat prices/benchmarks (\$/t)

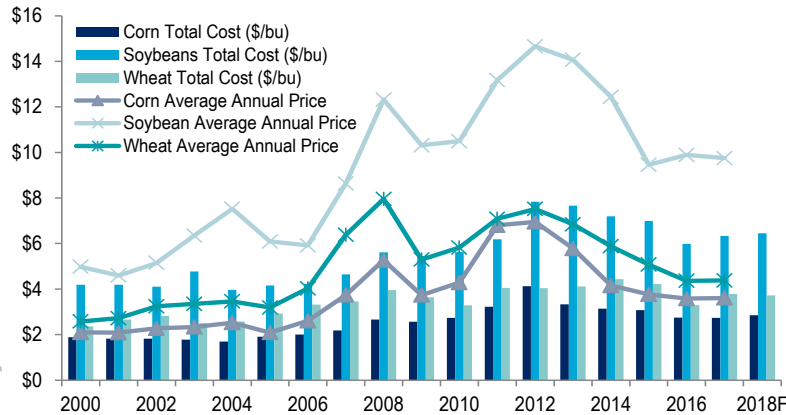


- **The US remains the leading global 'ag economy' with the largest and most productive farms, although increasingly the RoW is catching-up.** The long-term staple cereals yield gap* between the US and RoW remains a measurable 1.35MT/HA but most of this is due to corn, which is more weather-sensitive and fert/chem intensive. Interestingly, while the CAGR for both US and world maize yields are at ~1.9%, the RoW growth rate in soy and wheat yields are eclipsing the US by 0.2% and 0.8%, respectively. Productivity and area gains in the Southern Hemisphere, especially in Brazil and Argentina, have been notable in damping cereal price spikes in recent years and providing export flows to the Northern Hemisphere (e.g. China) as local stocks seasonally draw. This has exacerbated the weak cereal price environment amid record US yields. This impact has been most material for wheat, which by design is an easier and cheaper crop to cultivate and is grown year-round in many regions—allowing producers in Black Sea, Canada, Argentina and Australia to provide stiff competition against USGC cargoes.
- **The Southern Hemisphere has a nitrogen deficit versus the Northern Hemisphere, and the long-term CAGR of farm area expansion below the equator is about double the rate of combined US/Europe/Asia.** With more arable land and a greater delta to improve crop efficiency and productivity, we could see Brazil continue to challenge the US pole position as the global breadbasket for grains in the coming years, and a reduced supplier dependence on crop imports in Africa. However, we would note that the Southern Hemisphere remains more exposed to weather risk vis-à-vis the US for the time being.

Total farmer costs/acre up 80-100% since 2000, but record yields mute impact

Producer costs are front-and-back loaded but efficiency gains can be exponential in between at almost no cost for grain output, thanks to Mother Nature. Indeed, while total farmer costs per acre are up 80-100% for corn, soy and wheat since 2000, on a per bushel basis they are only up 50-57% across the board, deflating since 2012.

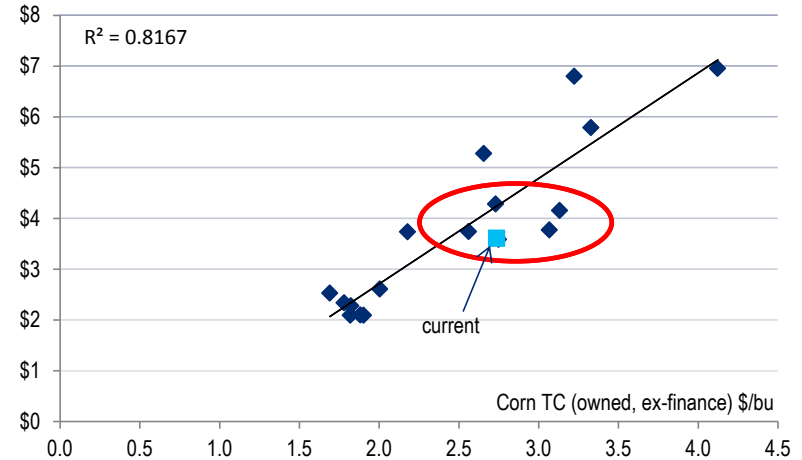
Corn, soy and wheat production costs (2000-2018F)



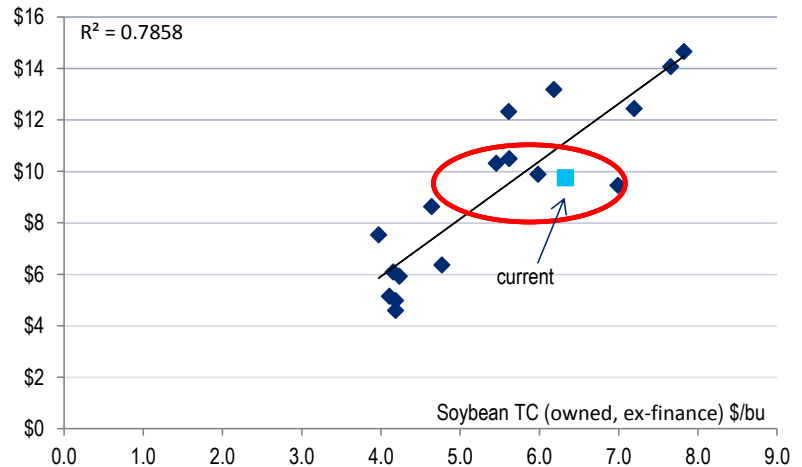
All-else equal, lower farmer costs are a negative for crop prices, as they buttress producer margins upstream in an otherwise low cash flow environment downstream. Higher costs tend to correlate with higher pass-through to wholesale prices. Reductions in expenses can disincentivize austerity (e.g. acreage cuts, reduced inputs) required to rebalance large inventory overhangs. This has been amplified in the record row crop yield environment after 2013/14 – essentially akin to productivity gains for the industry – since farmers are able to capture more bushels per harvested acre.

We expect 1.3-1.4% total cost increases per acre for corn and wheat and 0.8% for soy in 2018/19. This is modestly price supportive. Albeit if high yields persist, upside may be limited; the per-unit cost may stay flat or can even deflate.

Total corn farmer costs versus average annual corn price

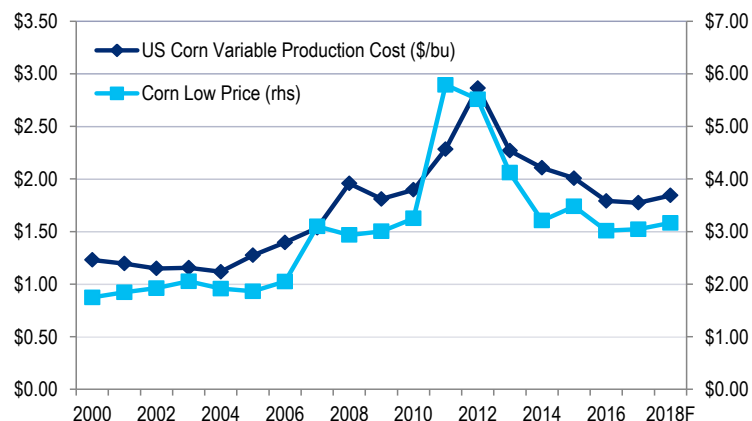


Total soy farmer costs versus average annual soy price

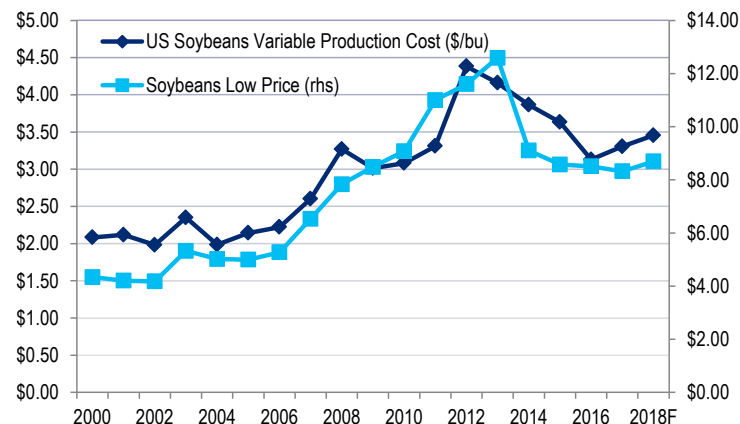


Variable expenses help set a CBOT price floor but have flat-lined at low levels

US Corn Variable CoP vs Annual Low Price on CBOT

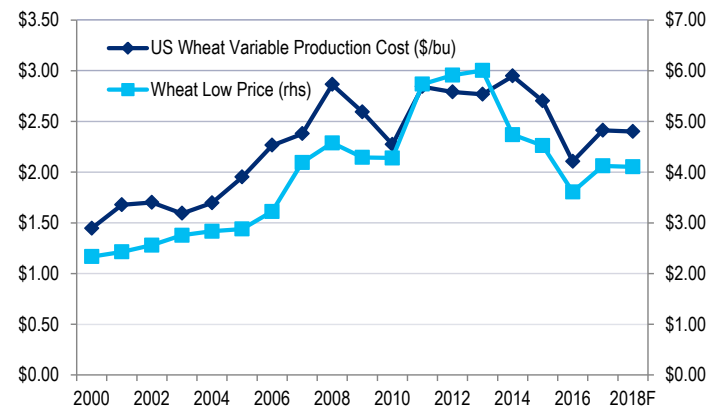


US Soy Variable CoP vs Annual Low Price on CBOT



- Seeds, fertilizers and fuel constitute over 60% of the non-fixed cost (CoP) for a typical staple cereal farm but inflationary pressures seem unlikely to measurably boost CBOT trading. Having deflated since 2014, we pencil-in 2.4-2.7% increases in variable costs for 2018/19 versus a long-run CAGR of 3.8-4.6%. Since 2000, both CBOT corn and wheat prices have traded at a floor ~70% above CoP, adjusted for yields. For CBOT soy, the relationship has been less clear but beans also trade at a higher premium since exportable surplus is smaller, there are less meal substitutes and Chinese demand has grown 20x since the 1990s.
- To the extent the historical relationship between CBOT trading and farmer CoP holds, then we anticipate fairly modest cost support in the next year (assuming trend yields) at \$3.15-3.20/bu corn and \$4.10-4.15/bu wheat. The good news is that this relationship has been fairly consistent and reliable for grains before, during and after the commodities super cycle. And with spot trading for corn and wheat only marginally above these levels during peak harvest, downside could be limited from here.

US Wheat Variable CoP vs Annual Low Price on CBOT



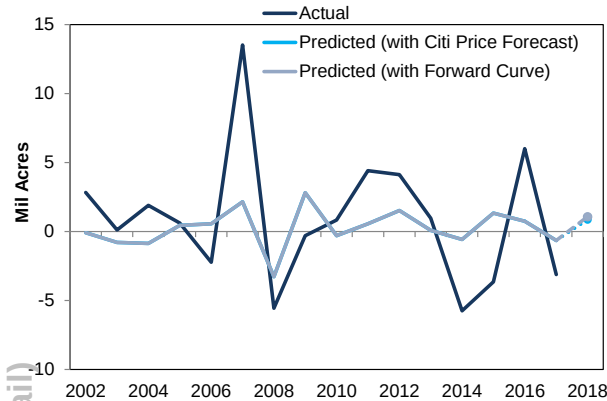
Source: IE, USDA, Citi Research, *assuming normal weather; 2018 yields used are per Citi balances. Oil/fuel/nat gas prices, unit hired labor costs and seed prices est. higher y/y.

US all-wheat area likely to keep declining but soybeans may hit new record

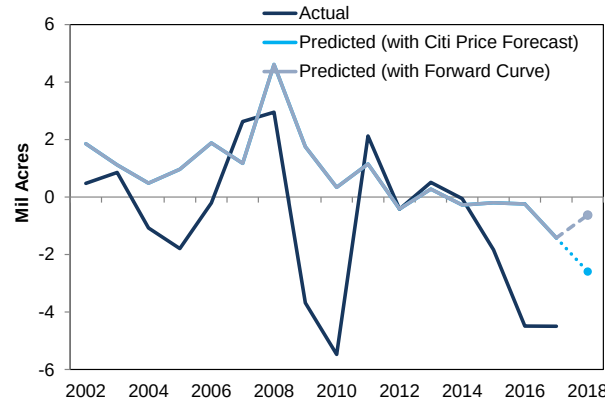
Lower CBOT grain (corn, wheat) harvest prices and soybean relative outperformance QTD has prompted us to cut US corn and wheat plantings for 2018/19 versus our 4Q'17 outlook while we upgrade new-crop soy sowing to a record 90.3mn. We still expect corn area to grow modestly y/y but for all-wheat plantings to decline 5+% y/y.

Out-of-Sample Predicted Acreage versus Actual Acreage*

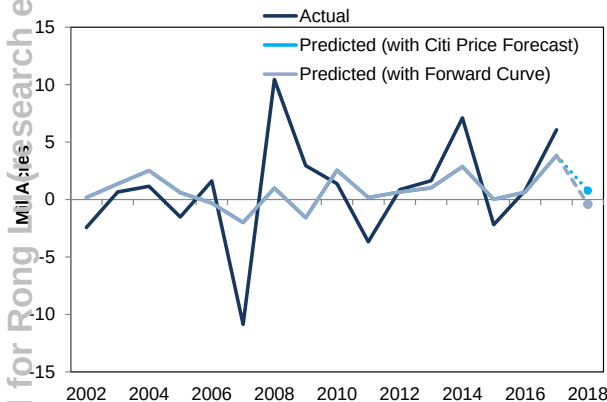
Corn



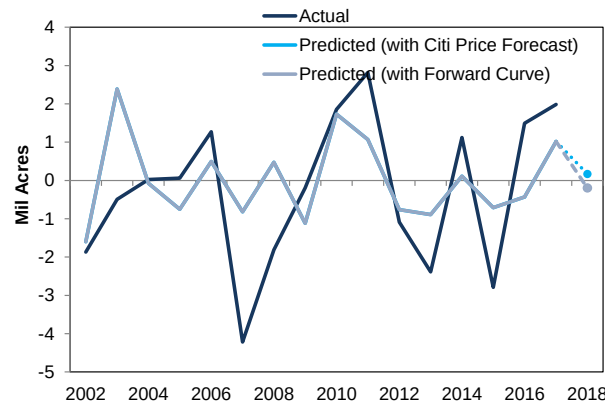
Wheat



Soybean



Cotton***



US Grains Planting Acreage Forecast for 2018 (Million Acres)

Year	Forecast	Corn	Soybean	Wheat	Cotton	Total
2018	Citi	91.7	90.3	43.1	12.2	237.3
	Futures Curve**	92.0	89.1	45.0	11.9	238.0
2017		90.9	89.5	45.7	12.1	238.1
y/y Change	Citi	0.9%	0.9%	-5.7%	1.4%	-0.3%

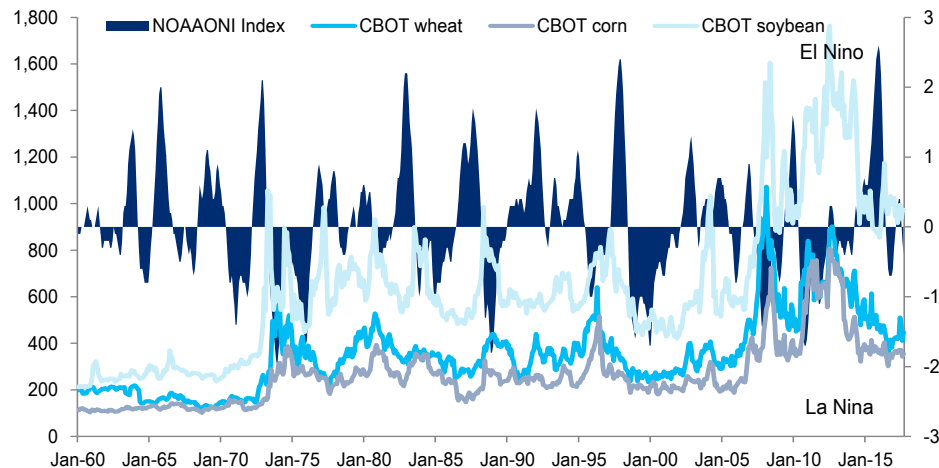
- Some factors that influence planted area include crop yields, futures prices and price ratios among substitute crops. We use an elastic net regression to select relevant factors and generate US planted area forecasts for 2018*. Corn area may expand slightly at the expense of wheat which provides the weakest farmer returns, to 91.7m. A higher soy-corn ratio and bean price outperformance could see US sowing hit a record above 90mn in 2018/19, in our view. Total sowing area** may decline by 0.3% y/y in 2018 driven by a steep all-wheat drop in area if low spot prices persist.

- All four crops picked different factors and models were selected by minimizing the out-of-sample errors. That said, results indicating 5.7% lower wheat area y/y and 0.9% gain in corn are subject to revision as economics might change this winter, but the collapse in wheat prices has weighed on our plantings outlook. A 0.9% y/y increase in soy is consistent with a neutral (~\$9.7) pricing outlook for soybeans in December 2018 amid the 4Q'17 collapse in CBOT grain prices.

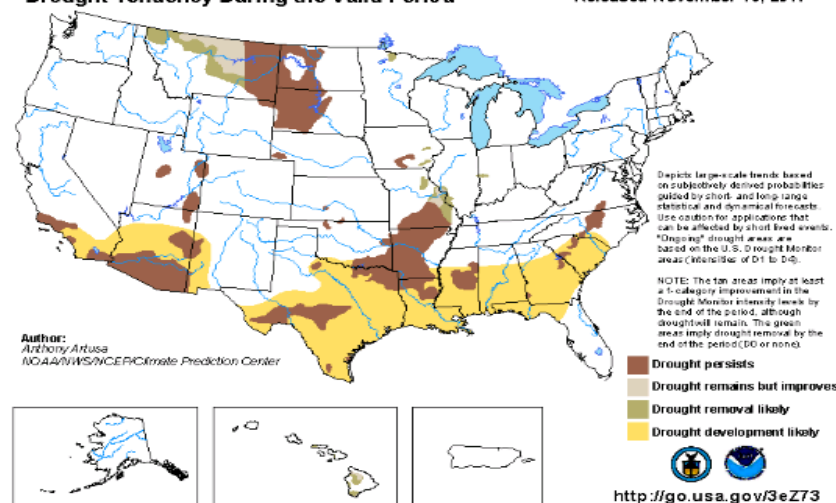
Source: Bloomberg, USDA, Citi Research, *subject to revision; futures prices as of end-October; Citi published prices used as of Sept 2017, acreage as of September WASDE. USDA Ag Outlook forum data available February 2018 and prospective plantings survey work will be completed and published in March 2018, so these figures could deviate based on prices, crop ratios, LatAm supply, etc. **For simplicity, we focus on corn, soybean, wheat and cotton as those are the top four US field crops. ***Citi Research does not cover US or global cotton.

The weather situation for now does not look disruptive for the grains complex

ENSO cycles and benchmark cereal prices (1960-present)



U.S. Seasonal Drought Outlook Valid for November 16 - February 28, 2018 Drought Tendency During the Valid Period Released November 16, 2017



- The US Climate Prediction Center and the Australia Bureau of Meteorology both project *mild* La Niña conditions will continue during the Northern Hemisphere winter (the South American planting and growing cycle). However, a weak ENSO cycle pegged by agencies at 65-75% probability through end-1Q'18 seems unlikely to materially damage LatAm supplies for now. Sizable inventory buffers in the US and globally should further damp mild weather impacts, especially compared to the La Niña in 2011 which was strong and also arrived at a time when US and world stockpiles were already declining. Indeed, a La Niña might even bring above average snowpack and precipitation for the US farm-belt over the next 3-5 months. For the southern part of the US, a La Niña historically brings about warmer temps and below-median precipitation. However, this can benefit the harvest speed and most farmland in the US sits idle between December-February, ahead of new-crop plantings.

- July/August weather matter more for the Northern Hemisphere row crop harvests. And as the 2017 crop progression implies, the US farmer with better seed technology is now able to withstand short bouts of weather disruptions.

Correlation matrix – Crop yields versus precipitation and temperature*

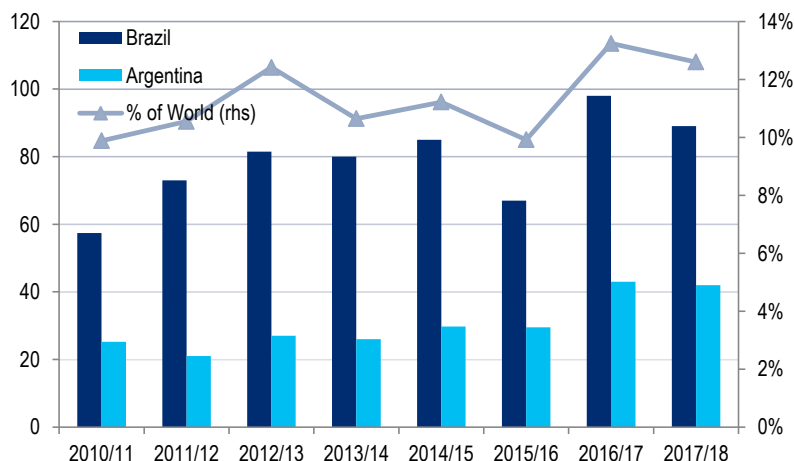
		Corn Yield	Wheat Yield	Soybean Yield
Precipitation	June	12%	24%	21%
	July	10%	44%	27%
	August	10%	4%	56%
Max Temperature	June	-42%	-22%	20%
	July	-79%	-11%	-44%
	August	-55%	2%	-42%
Min Temperature	June	-33%	-23%	21%
	July	-69%	0%	-39%
	August	-31%	-7%	-18%

Source: CPC, NOAA, Citi Research, * We highlight significant correlations with p-value less than 5% as red bold text

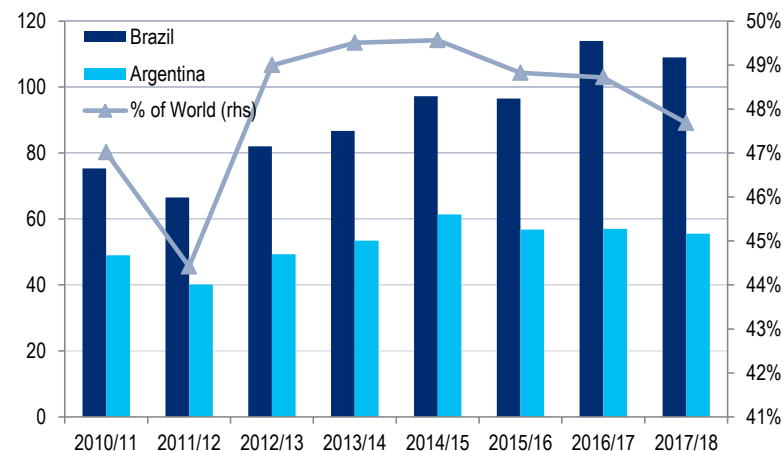
Latin America row crop supply prospects down y/y but look 'good enough'

- **With the ideal soybean planting window closed in Brazil, excellent rainfall in mid-November is a positive sign for the crop which we peg at 108.5mmt, down from 114mmt in the previous campaign but still historically high, especially in proportion to robust US stocks-to-use.** Soy sowing was behind schedule in Brazil but has completely caught up, and we see total acreage 2% higher y/y at over 85mn. On the corn side, we estimate a total Brazilian harvest of 89mmt with 'safrinha' second maize about 75% of this figure. Corn plantings/prices are struggling in certain states with high costs and lower area. CONAB is still optimistic, pegging a 92mmt Brazilian corn crop with the USDA at 95mmt. We think this figure could come down in the coming months and thus are already quite conservative in our forecasts.
- **In Argentina, a favorable tax regime for corn versus soy exports is favoring a switch in planted area towards maize, even as the Macri government might start lowering bean export tariffs 0.5%/mo next year.** 36% of the Argentina corn crop has been sowed as of mid-November while 34% of the beans are in the ground, mostly across the central region. Our ARS production forecasts for corn and soy are 42mmt and 55.5mmt, respectively, up 2% y/y for corn and down 4% y/y for beans. We expect acreage growth and physical flow to shift back towards beans in 2019, as the soy tax regime drifts toward parity and since soy products like biodiesel face external tariff pressures from US trade disputes*.
- **Row crop production in Brazil and Argentina should decline y/y but this is probably understood across the trade.** The BRL/ARS new-crop corn production profile in a global context is also perfectly in-line with recent seasons and is even has a smaller weighting for soybeans in the short-term, given a sizable 2017/18 US harvest.

Brazil and Argentina corn production profiles (est. mmt)



Brazil and Argentina soy production profiles (est. mmt)

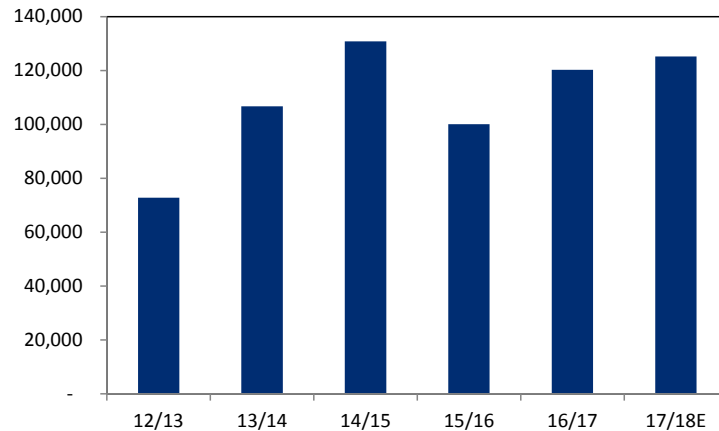


Source: AgRural, Bloomberg, CSA, USDA, Citi Research, *see: RINS notes on citivelocity.com

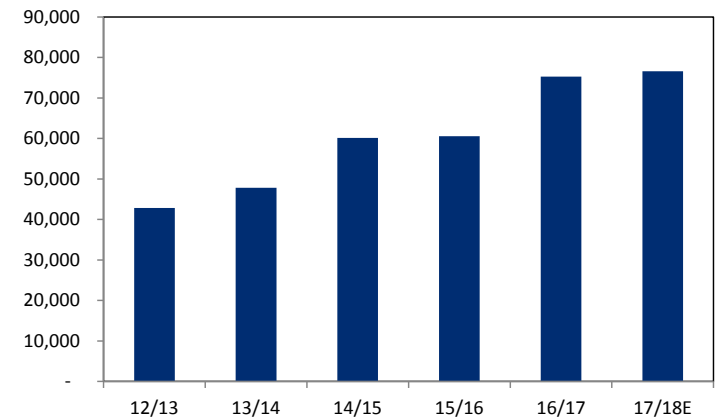
World cereal stocks ex-China are no longer building albeit not drawing either

China runs a net trade deficit in grains/oilseeds and we can assume crops produced locally or entering the country via imports are unlikely to 'exit'; strategic policy for the second largest global economy and the largest global population. So it is important to note that staple cereal inventories should remain steady into 2018/19.

Global corn stocks/reserves ex-China (kt)



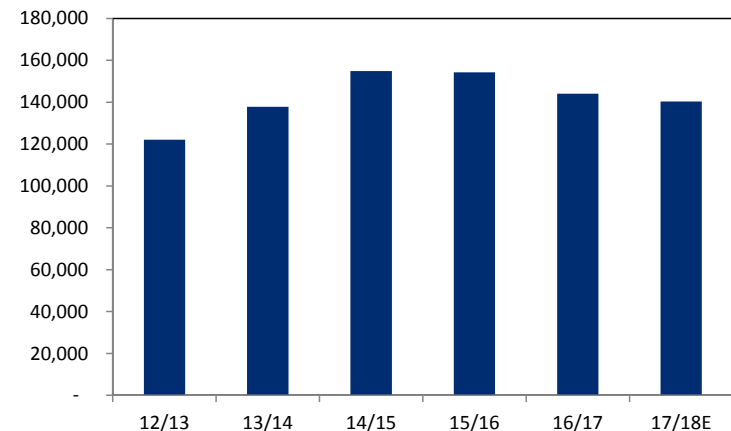
Global soy stocks/reserves ex-China/East Asia (kt)



Global corn and soy stocks may even build slightly outside of China on the back of a larger-than-expected US harvest and sizable old-crop carryover in Brazil. USDA data suggest that combined global row crop reserves might even hit a record in the next year, although the pace of growth is clearly declining. We view 100mmt as a 'comfortable' global stock level for corn ex-China which would be ~10% of world output, and 2018/19 carry-in ex-China should be around 120mmt. For global soybeans, we think ~65mmt would be adequate and expect this figure to post between 75-77mmt.

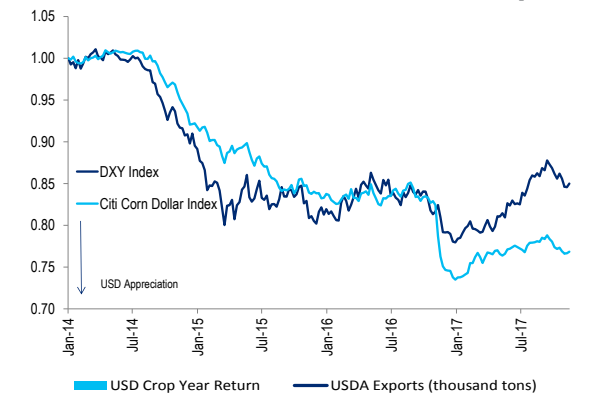
For wheat, despite record Russian supplies, global stockpiles are likely to plateau as US harvests have fallen and farmers are culling acreage there versus other crops. Still, overall inventories should be larger than in 2012-2014.

Global wheat stocks/reserves ex-China (kt)

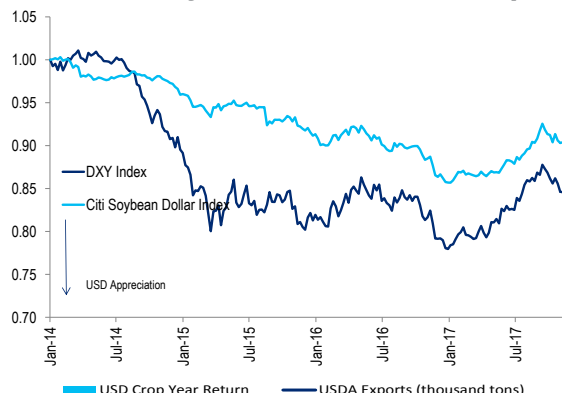


US grain exports: US\$ could be a tailwind into 2019, but hold your cargoes...

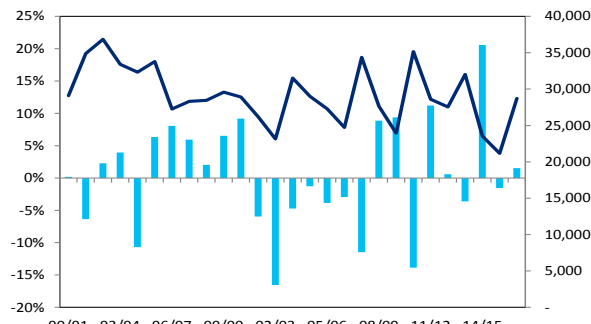
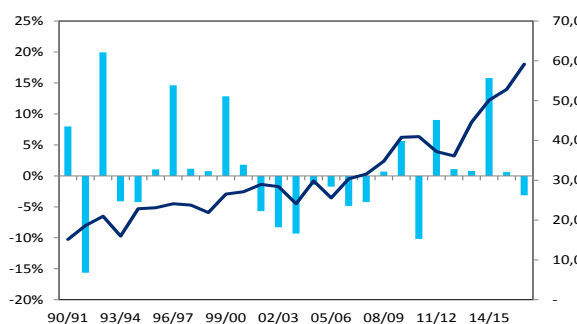
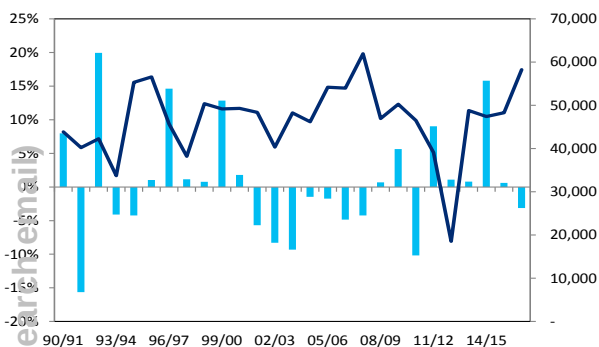
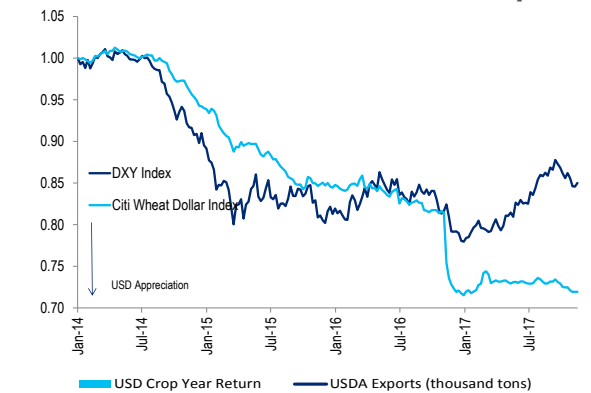
Citi US Corn-Dollar index and exports



Citi US Soy-Dollar index and exports



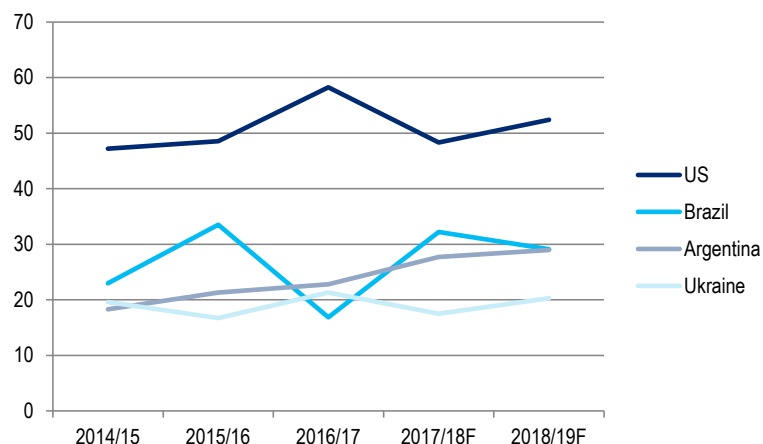
Citi US Wheat-Dollar index and exports



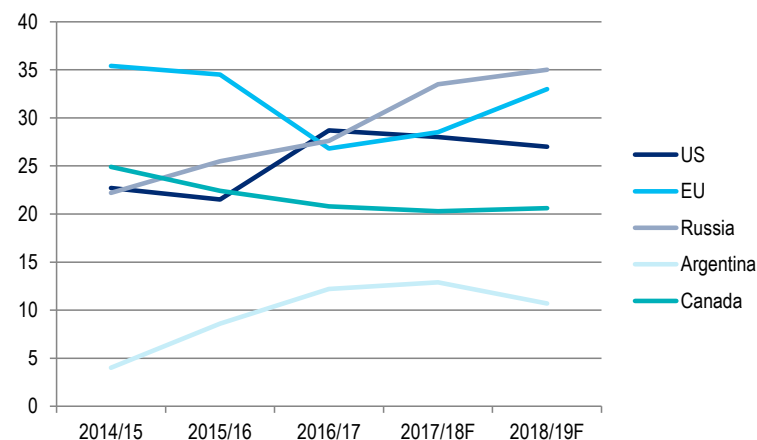
A weak US dollar should boost nominal CBOT cereal prices via denomination effects and US export competitiveness, and Citi expects ~5% depreciation in the USD index by end-2018. But this should not impact all crops equally so a soft 'big dollar' outlook is not necessarily bullish US grain shipments per se. While most market participants track the DXY (US dollar) index as a proxy for USD strength, that measure is skewed towards core G3 trading partners such as EUR. Citi grain-USD indexes instead weight importer pairs such as Mexico, Egypt, Algeria, Iran, Nigeria and South Korea; providing a more direct trade measure. The Citi crop-USD indexes tend to correlate to the broader US dollar index but can also diverge. With China representing two-thirds of the global soybean trade, for example, we have seen steeper 2016-2017 soy-index USD depreciation versus corn and wheat. High USD-EM can also persist away from EUR and JPY depreciation for large importer economies that are less connected to the OECD, such as Egypt and Nigeria. Additionally, CBOT prices do not always track US export levels, especially when the forward supply outlook appears constructive. The Citi-crop currency indexes overall appear less positive for US grains export growth than the 'big dollar' DXY outlook would suggest, in our view.

Global staple cereal export balances not so tight; the US is not alone in trade

**Corn Exports of Major Exporters
(mmt) (Sep/Aug)**



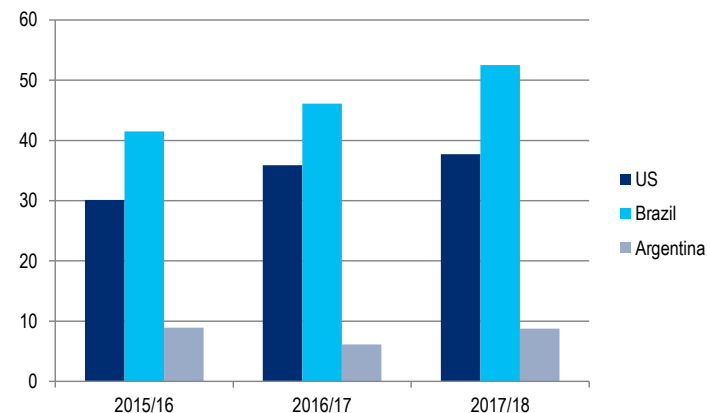
**Wheat Exports of Major Exporters
(mmt) (Jul/Jun)**



US corn exports in both 2017/18 and 2018/19 should remain below 2016/17 levels of ~58mmt while Argentina grows volumes given the local tax regime favoring corn sales. For wheat, US exports are likely to continue falling over the next few years as both Russia and Europe grow trade share at the expense of USGC markets. All-in-all, we expect combined Russia/EU wheat exports by 2018/19 to hit multi-year highs of 68mmt+, above 60% of global trade, versus the US at 26-27mmt. In short, wheat trade balances remain in surplus ex-US while for corn the market looks stable, with the US holding more than adequate supply to pick-up any slack.

China represents 2/3 of the global physical soy trade, and may import a record 97mmt in 2017/18 and 100mmt in 2018/19. Notably, Brazil bean volumes have surpassed US soy sales to China both on aggregate and in terms of y/y growth. We expect this trend to continue next year; capping US CBOT soy prices absent a weather event in Latin America.

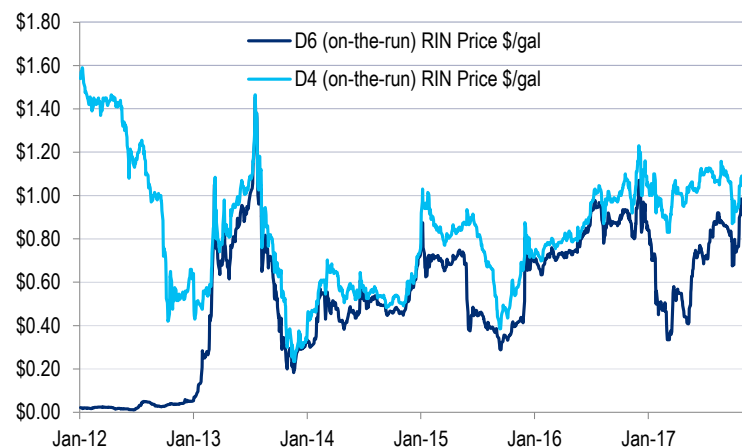
**US and key LatAm Soy exports to China
(mmt) (Sep/Aug)**



Renewable fuel policy* and positive for biofuel usage, though growth plateaus

- **High RIN prices and a binding Renewable Fuels Standard*, amid steep US trade tariffs on soy oil-based biodiesel from Argentina and palm-based from Indonesia, should continue to support the demand side of the ledger for corn and bean use for biofuel production.** Indeed, a key development in 2H'17 was the Trump administration telling the EPA to affirm its support for the US farmer lobby/biofuel industry at the expense of merchant refiners. Elevated RIN credit prices further support these feedstocks by incentivizing racks to sell higher level blends.
- **Though biofuel production remains a key source of consumption for row crop balance sheets, the pace of growth is set to moderate alongside US transport fuel demand growth.** To that end, export markets should remain key for the US industry going forward, particularly China and EM in general; economies planning to expand their gasoline-biofuel blend rates. But we think growth will peak by 2020 and that China's targets for E10 may prove unrealistic to achieve so quickly.

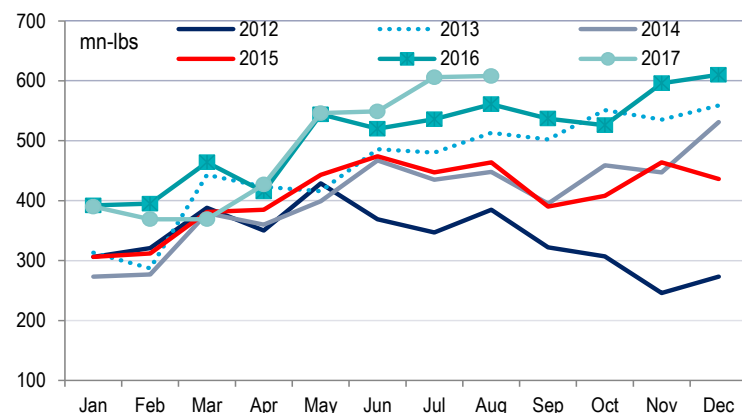
Benchmark on-the-run RIN prices (\$/gal)



US ethanol capacity utilization (% of nameplate): 95-100% likely the new normal, but plant expansions limited



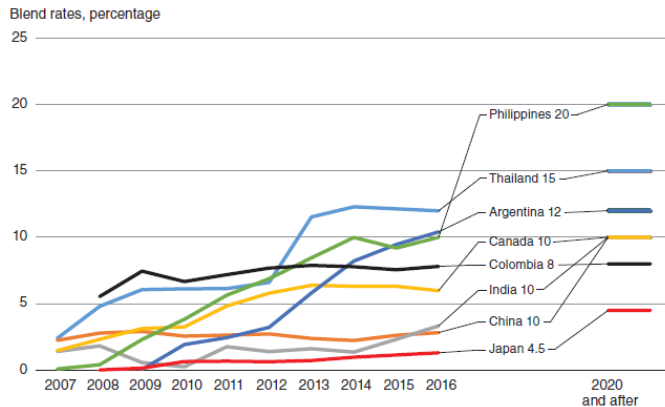
Feedstock/veg oil/soy oil into US biodiesel output (mn lbs) a record in 2H'17 and likely to grow in 2018 due to tariffs*



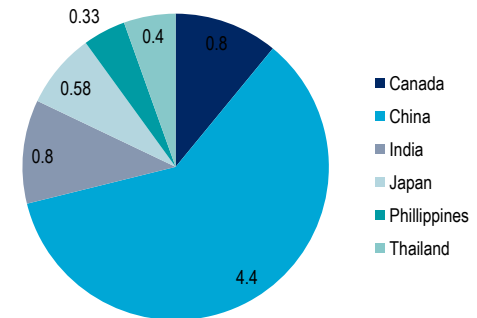
Ethanol blending poised to increase, particularly in emerging markets...

The growth of ethanol blending standards are poised to increase ethanol exports post-2020 by 7-8bn gallons. For reference, Citi estimates 2017 US ethanol exports (gross) will be a record ~1.3 billion gallons. China should see the largest incremental gain as its gasoline demand grows from ~38 to ~50 billion gallons. It seems both the US and Brazil ag producers can benefit from this scenario...however electric vehicles could displace this growth.

Historical ethanol blend rates and medium-term targets



Estimated net import gap (bn gal pa) post-2020 for select EMs + Japan



- **Electric vehicles displace not only gasoline but also ethanol. Fuel economy also weighs on overall motor transportation fuel consumption.**
 - Post-2020 concern, but significant post-2025. Policy impulse seems stronger for EV versus biofuels for now.
- **Political concerns and the ‘food-for-fuel’ debate is likely to gain traction again if a ‘2012-type’ Northern Hemisphere drought situation were to reemerge.**
 - Is California LCFS an example of a more complex but complete fuel standard?
- **E15 waiver possibility in the US?**
 - Will refiners—particularly merchants—give-up market share?
- **Biofuel trade wars are heating up again.**
 - US biodiesel versus Argentina and Indonesia; Brazil ethanol versus US, EU tariffs, China (e.g. DDGs), etc. Could the Asian Dragon change its stance on ethanol policy? Provincial versus federal concerns still loom.

US Corn Balances*

Millions of Acres Millions of Bushels		USDA				Citi Estimates	
	13/14	14/15	15/16	16/17	17/18	17/18	18/19
Area Planted	95.4	90.6	88.0	94.0	90.4	90.4	91.7
Area Harvested	87.4	83.1	80.8	86.7	83.1	83.1	84.1
Percent Harvested	92%	92%	92%	92%	92%	92%	92%
Yield per Harvested Acre	158.2	171.0	168.4	174.6	175.4	175.4	173.0
Beginning Stocks	821	1,231	1,731	1,737	2,295	2,295	2,500
Production	13,829	14,216	13,602	15,148	14,578	14,577	14,555
Imports	36	32	68	57	50	45	40
Total Supply	14,686	15,479	15,401	16,942	16,923	16,917	17,095
Feed and Residual	5,002	5,284	5,114	5,463	5,575	5,600	5,625
Ethanol Usage	5,124	5,200	5,224	5,206	5,475	5,460	5,475
Food, Seed and Industrial	6,531	6,597	6,648	6,891	6,935	6,915	6,945
Total Domestic Usage	11,533	11,881	11,763	12,354	12,510	12,515	12,570
Exports	1,921	1,867	1,901	2,293	1,925	1,902	2,075
Total Demand	13,454	13,748	13,664	14,647	14,435	14,417	14,645
Ending Stocks	1,232	1,731	1,737	2,295	2,488	2,500	2,450
Stocks to Usage	9.2%	12.6%	12.7%	15.7%	17.2%	17.3%	16.7%

- Bottom Line:** CBOT corn prices pegged to average ~\$3.60/bu for 2017/18 and ~\$3.80/bu for 2018/19 – a ~3-5% downgrade from our 4Q'17 report. We project record 2017/18 maize carryout of 2.5bn bu, with new-crop stocks-to-use growing 1.6 points y/y to 17.3%, more than double 2012/13 levels. Nevertheless, our regression model indicates a slight increase in US corn sowing next year to 91.7mn. Assuming 'trend' yields of 173-bu/acre, which would be down both versus 2016/17 and 2017/18, the 2018/19 harvest is still penciled-in only 20-25mn bu smaller y/y, with deferred carryout remaining at a hefty 2.45bn bu and stocks-to-use of 16.7% still 3.2 points above the T5Y average, keeping our mean forecasts sub-\$4/bu.
- US demand may take a hit in 2017/18, down slightly to 14.42bn bu as export flows should struggle on South American competition** (Brazil has posted record monthly volumes in 2H'17 and Argentina is expanding its new-crop maize). Corn-grind for ethanol should rise over the next few years, but we still see a ceiling below 5.5bn bu, given capacity constraints and substitution. We boost feed/resid use for 2017/18 to 5.6bn bu and to 5.625bn bu in 2018/19, correlating with higher supply.
- US corn balances since 2013/14 are trending to the loosest environment in about a quarter century. Absent a major weather event during the Northern Hemisphere planting cycle, we think it is unlikely corn prices can sustain a rally.**

Source: USDA, Citi Research, *subject to revision

US Soybean Balances*

Millions of Acres Millions of Bushels		USDA				Citi Estimates	
	13/14	14/15	15/16	16/17	17/18	17/18	18/19
Area Planted	76.8	83.3	82.7	83.4	90.2	90.2	90.3
Area Harvested	76.3	82.6	81.7	82.7	89.5	89.5	89.6
Percent Harvested	99%	99%	99%	99%	99%	99%	99%
Yield per Harvested Acre	44.0	47.6	48.0	51.9	49.5	49.5	49.0
Beginning Stocks	140	92	190	196	301	301	430
Production	3,358	3,927	3,926	4,296	4,425	4,425	4,388
Imports	72	33	24	22	25	20	20
Total Supply	3,570	4,052	4,140	4,515	4,752	4,746	4,838
Crush	1,734	1,873	1,886	1,899	1,940	1,945	1,950
Seed	97	96	97	105	101	101	105
Residual	10	50	18	36	35	25	25
Total Domestic	1,840	2,019	2,001	2,040	2,076	2,071	2,080
Exports	1,639	1,842	1,942	2,174	2,250	2,245	2,345
Total Demand	3,479	3,861	3,943	4,214	4,326	4,316	4,425
Ending Stocks	92	191	197	301	425	430	413
Stocks to Usage	2.6%	4.9%	5.0%	7.1%	9.8%	10.0%	9.3%

- Bottom Line:** CBOT soy prices expected to average ~\$9.50/bu in current season which is bearish the forwards. In 2018/19, we expect a modest recovery to just shy of \$10/bu on average – both are downgrades versus our previous projections. New-crop US soy carryout of 430mn bu would be up 43% y/y and the stocks-to-use ratio of 10% would be more than 3x 2013/14 levels and 2x 2014-2016 levels. In our view, this should contain CBOT soy trading well-below \$10.50/bu on normal weather and implies pricing should not scale above \$10/bu on a sustained basis until perhaps the 2018/19 harvest.
- Given the better profitability for soy versus the grains (versus both total and variable farmer costs) and the ongoing growth in Chinese imports to utilize local crush capacity, we model US area to expand marginally in 2018 and for LatAm supply growth to continue apace into 2020.** This should lead to a modest draw in 2018/19 US carryout, hitting just above 410mn bu, while stocks-to-use might fall to the single digits again to a still high 9.3%. Indeed, the soybean export story continues to remain constructive, with China imports likely to hit a record 100mnt in 2018/19, although Brazil has taken top share away from the US in recent years and that seems likely to stay. We peg US export growth of ~70mn bu y/y in 2017/18 and another ~100mn bu y/y in 2018/19 to a record 2.345bn bu. With higher carry-in and robust planted acreage, US soy supply may scale a record 4.8bn bu by 4Q'18. We see stronger use for soy oil versus meal, due to US biodiesel blending/trade policy.

Source: USDA, Citi Research, *subject to revision

US All-Wheat Balances*

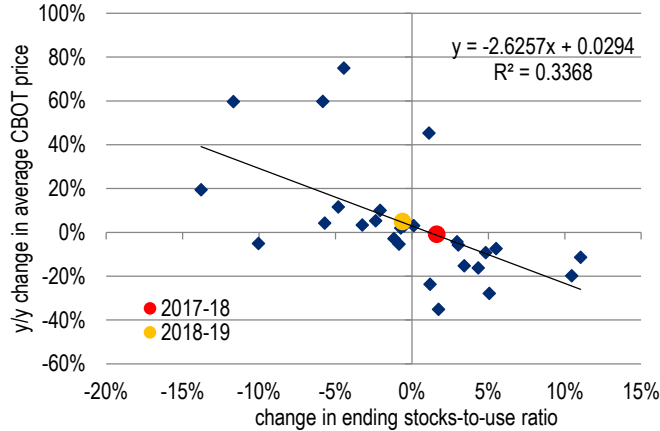
Millions of Acres Millions of Bushels	USDA					Citi Estimates	
	13/14	14/15	15/16	16/17	17/18	17/18	18/19
Area Planted	56.2	56.8	55.0	50.1	46.0	46.0	43.1
Area Harvested	45.3	46.4	47.3	43.9	37.6	37.6	37.0
Percent Harvested	81%	82%	86%	88%	82%	82%	86%
Yield per Harvested Acre	47.1	43.7	43.6	52.6	46.2	46.2	47.5
Beginning Stocks	717	590	752	976	1,181	1,181	959
Production	2,135	2,026	2,062	2,309	1,741	1,741	1,758
Imports	172	151	113	118	150	150	160
Total Supply	3,025	2,768	2,927	3,402	3,071	3,072	2,877
Food	955	958	957	949	949	950	948
Seed	77	79	67	61	66	63	55
Feed and Residual	228	113	149	156	120	125	130
Total Domestic Usage	1,259	1,151	1,174	1,167	1,136	1,138	1,133
Exports	1,176	864	778	1,055	1,000	975	950
Total Demand	2,435	2,015	1,952	2,222	2,136	2,113	2,083
Ending Stocks	590	752	976	1,181	935	959	794
Stocks to Usage	24.2%	37.3%	50.0%	53.2%	43.8%	45.4%	38.1%

- Bottom line:** We see the US all-wheat balance tightening ~8 points in 2017/18 to 45.4% and to just above ~38% in 2018/19, as steep acreage cuts (to century-long lows) continue given the current low margin environment for wheat versus row crops. However, 2018/19 US ending stocks at ~800mn bu would still be north of the 2013-2016 average and we still view wheat markets as relatively loose on any US stocks-to-use figure above 30-32%. Indeed, we peg all-wheat production slightly higher in 2018/19 on the back of a reversion to trend yields. The EU and Russia should also keep taking wheat export market share away from the US, in our view. And while low wheat prices could spur a demand response, global inventories are also projected at a record ~267mnt as wheat harvest growth outpaces demand for the fifth straight marketing campaign.
- From a physical balancing perspective, we think local producer discipline, which has already materialized for wheat versus the row crops in the past few years, can continue to limit downside corrections.** Indeed, model-based prices on the US stocks-to-use decline for wheat would suggest a CBOT push above \$5/bu in 1H'18, although the picture is more benign globally. We think a tight ~\$0.75/bu wheat premium to corn over the short-run could encourage some feed demand but that overall this is not too material. We are also bearish USGC wheat exports and total demand versus the USDA and consensus.

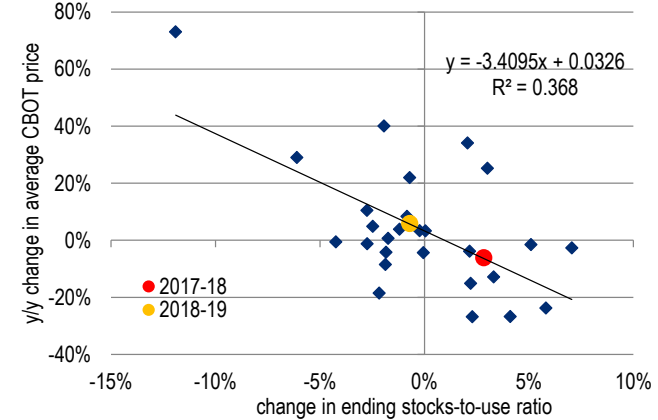
Source: USDA, Citi Research, *subject to revision

High and stable stocks-to-use ratios for row crops in particular should persist

Y/Y (%) corn price vs Y/Y US stocks-to-use ratio



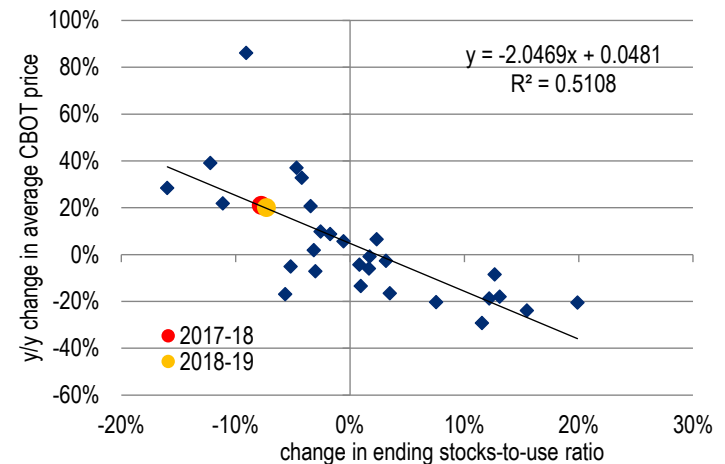
Y/Y (%) soybean price vs Y/Y US stocks-to-use ratio



- Just as we utilize total farm costs and variable expense ratios to help peg CBOT prices, so too is the measure of physical inventories versus total demand. The historical relationship between the (seasonal) average CBOT contract price and changes in the ratio of stocks-to-use, using USDA data since 1990, show an especially strong inverse relationship for wheat but also both the row crops.

We utilize model outputs for average seasonal price ranges between \$3.55-3.85/bu for corn and \$9.35-9.95/bu for beans. For wheat, the sharp reduction in stocks-to-use ratios projected in 2017/18 and also 2018/19, off a very high base, suggest CBOT prices could get to \$5/bu+ in 1H'18, but we are conservative on our price outlook given record Russian supplies, expensive importer costs due to rising freight rates (e.g. see Baltic Dry Index) and strong Northern Hemisphere export competition. However, on a base case scenario, we think CBOT wheat has more upside versus corn and beans.

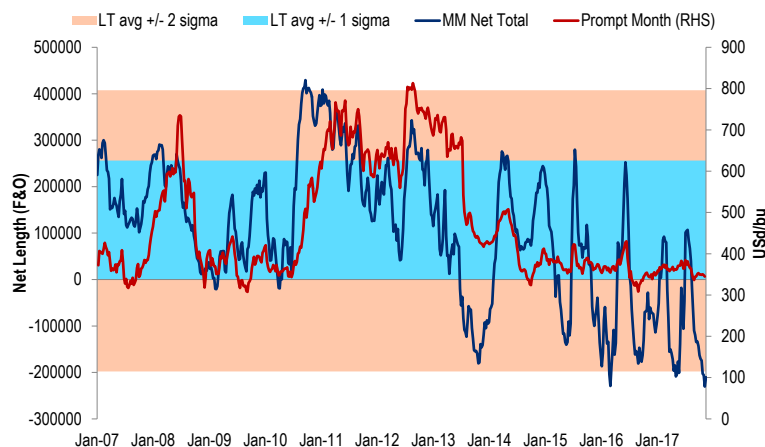
Y/Y (%) wheat price vs Y/Y US stocks-to-use ratio



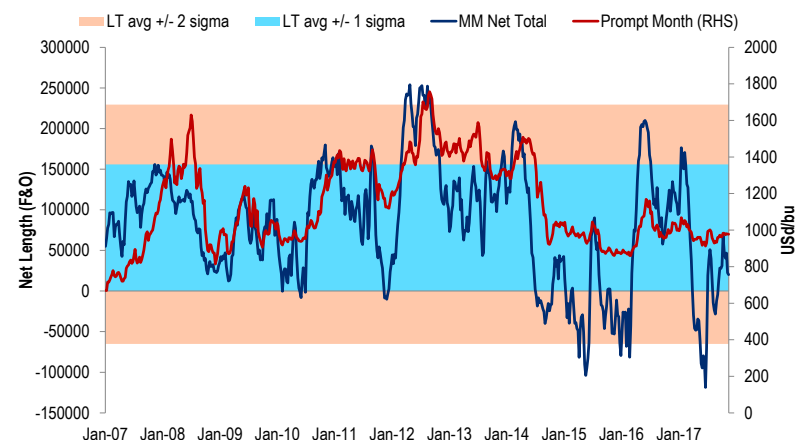
CBOT positioning may be stretched too short, limiting steeper price declines

The managed money corn net short is near a record while the combined CBOT fund length in the market for corn, soy and wheat is historically short on an aggregate, seasonal and normalized basis. With spreads also very weak for wheat and corn, beyond exchange storage costs, price downside might be limited here.

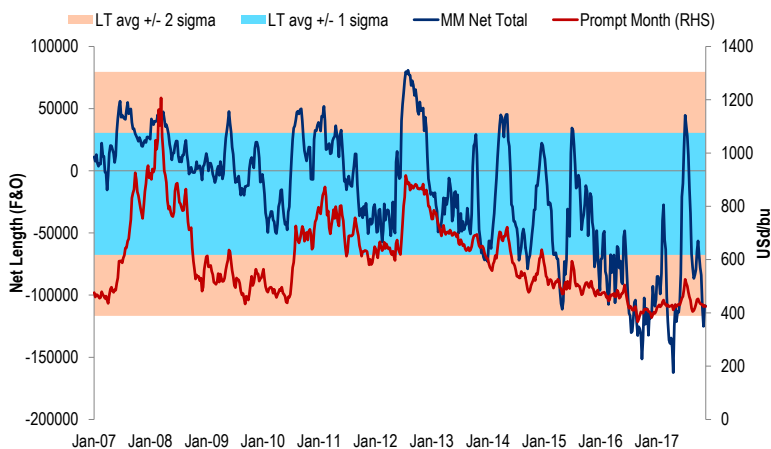
Money Manager Net Length and Prompt Prices, +/- 1,2 sigmas – CBOT Corn



Money Manager Net Length and Prompt Prices, +/- 1,2 sigmas – CBOT Soybeans



Money Manager Net Length and Prompt Prices, +/- 1,2 sigmas – CBOT Wheat



Normalized money manager net length (% of F&O OI) – CBOT Corn, Soybeans and Wheat

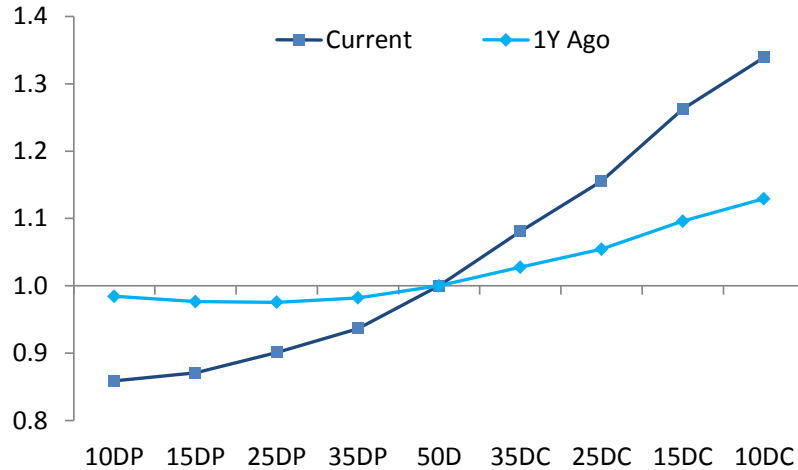


Source: Bloomberg, Citi Research, note data are through mid-late November 2017

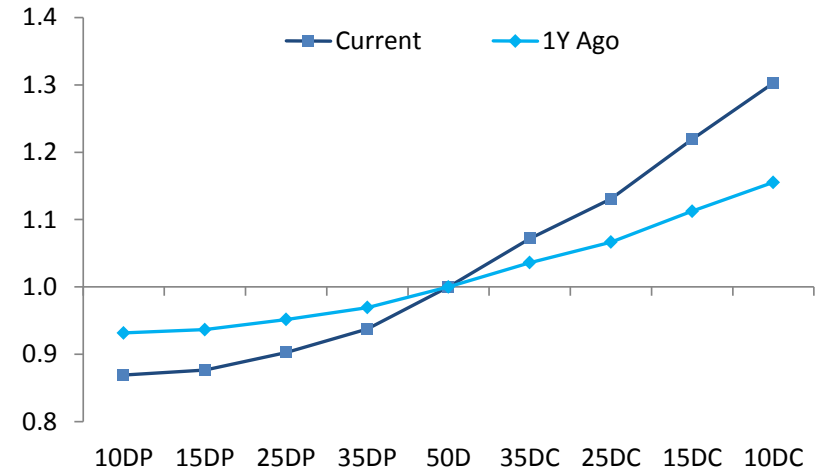
Steeper long-dated skews indicative of positioning for topside CBOT tail risks

Longer-dated skews are trading a lot steeper than the same time last year for soybean and corn, while prompt skews have declined markedly for corn and wheat recently, reflective of the bearish investor positioning. Volatility should remain contained in 1Q before edging higher in 2Q when the planting season kicks off.

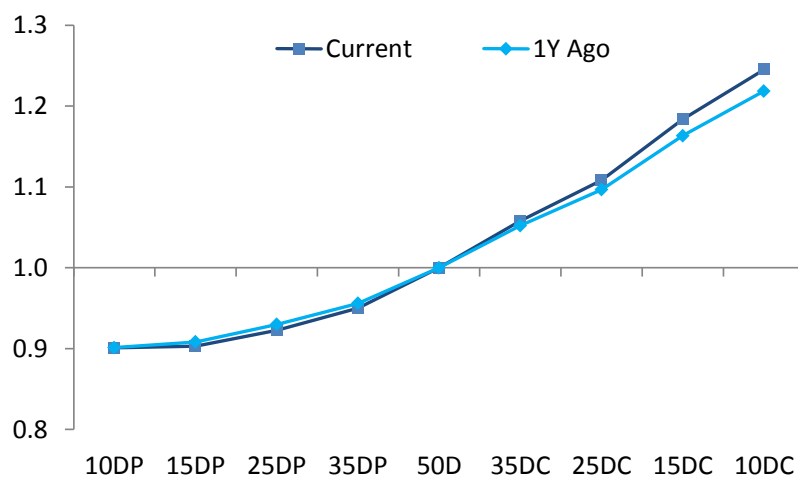
CBOT Soybean Normalized 6M Implied Vol Skew*



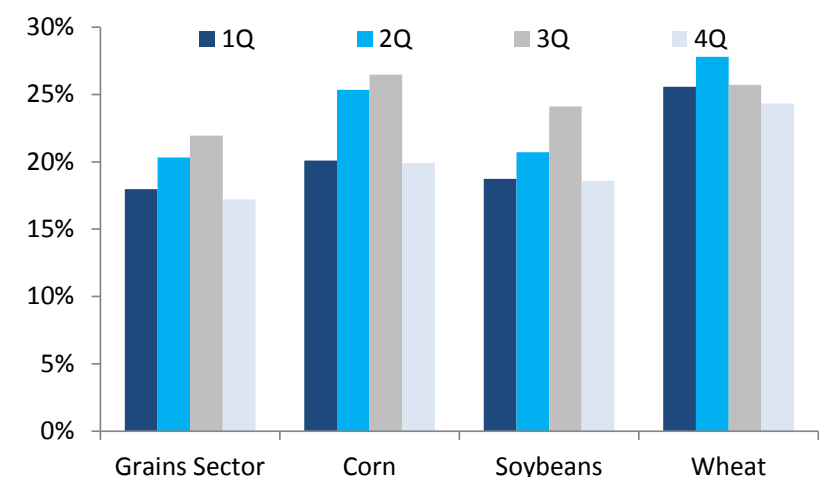
CBOT Corn Normalized 6M Implied Vol Skew



CBOT Wheat Normalized 6M Implied Vol Skew



Grains LT Average Quarterly Realized Volatility**

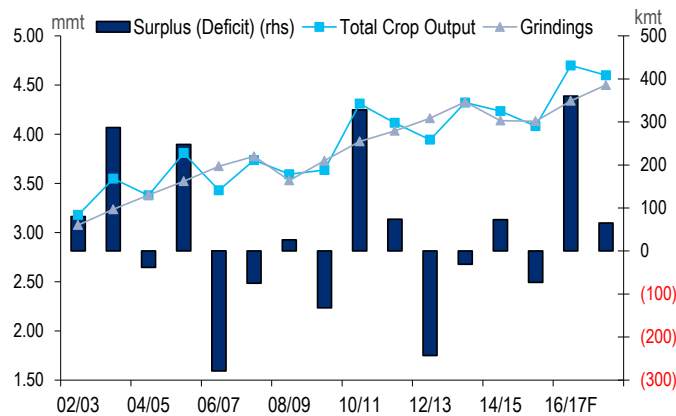


Source: Bloomberg, Citi Research *All skews are normalized by ATM implied vol. **Average quarterly volatility over the past 20 years.

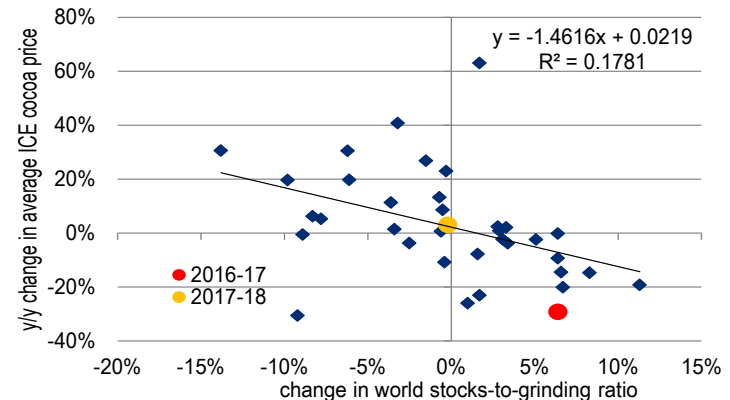
ICE cocoa prices look neutral, as prices have rallied off the lows to \$2,000/t+

We maintain our cal'18 NY cocoa price average at \$2,150/t as terminal prices and forwards rallied from \$1,900/t to \$2,100/t since mid-3Q'17, exactly in-line with the Citi published outlook. Overall, we see a large 2016/17 market surplus swinging closer to balance in 2017/18, on lower West African supply and improved grindings.

World aggregate Cocoa Balances



Y/Y (%) cocoa price versus Y/Y stocks-to-grinding ratio change



Prepared for Rong Lu (research email)

The 2016/17 global cocoa surplus posted a bit lower than our prior update at 0.36mmt versus 0.49mmt albeit we maintain a better-balanced new-crop surplus at ~50-75kt. Ivory Coast and Ghana account for ~2/3 of the world harvest and production for both countries may drop ~5-7% y/y to 1.9mmt and 0.9mmt, respectively. Arrivals are a bit behind schedule but good rainfalls and pod counts suggest that the main crop in 2018 could be solid. Ghana faces more issues that could stunt its cocoa production growth going forward: aging trees, terminal prices below farmer break-evens and a challenging fiscal situation. Combined with downside risk to Indonesian output, which we pencil-in below 280kt and a 20% drop from 2015/16 levels, world cocoa production may decline 2% y/y to 4.6mmt amid a robust Northern Hemisphere near-term grindings outlook.

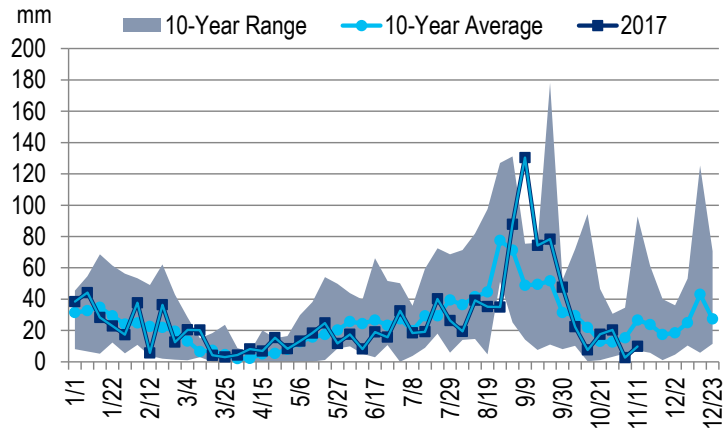
After negative grindings growth in both 2014/15 and 2015/16, cocoa demand grew ~5% in 2016/17 and may grow 3.6% during the new-crop campaign. Though headwinds of declining OECD candy sales and smaller packaging remain, the ~50% peak-to-trough drop in cocoa prices in the past few years to \$1,800/t levels in mid-2017 has clearly incentivized cocoa processing growth based on regional data from North America, Asia and the EU industry groups. Indeed, we see the global cocoa stocks-to-grinding ratio falling to 44% in the next year versus our previous assessments closer to 47%. Based on the tight inverse relationship between y/y changes in the mean seasonal ICE cocoa price and the ratio over the past 35 years, we expect ICE trading to maintain a bid north of \$2,000/t on average in the next year, mostly hovering in a \$2,100-\$2,200/t range, before rising to \$2,200-2,300/t in 2019. The price outlook is unchanged since July 2017, as the ICE curve rallied to our forecast.

Source: CAA, ECA, ICCO, Industry Reports, NCA, Citi Research, *see: Global Commodities Focus – NY cocoa prices looking firmer for cal'18 but still low for now. Citi: August 2017.

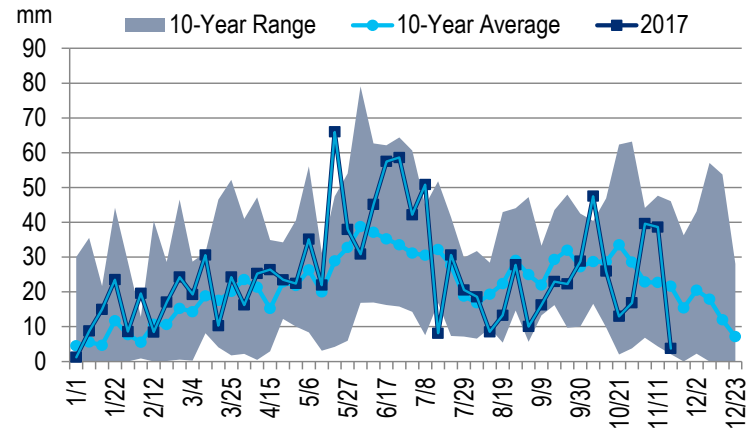
Rainfall average in key cocoa producers, more erratic in Ghana this year

Precipitation has been average this year across key cocoa growers Ivory Coast and Indonesia and a bit more erratic in Ghana. The largest global producer, Ivory Coast, should see its harvest fall only slightly y/y in 2017/18. We maintain a world harvest forecast of 4.6mmt for 2017/18, with modest declines in West Africa offset by modest upticks in Brazil and elsewhere. Supply discipline on low farm gate prices could accelerate rebalancing.

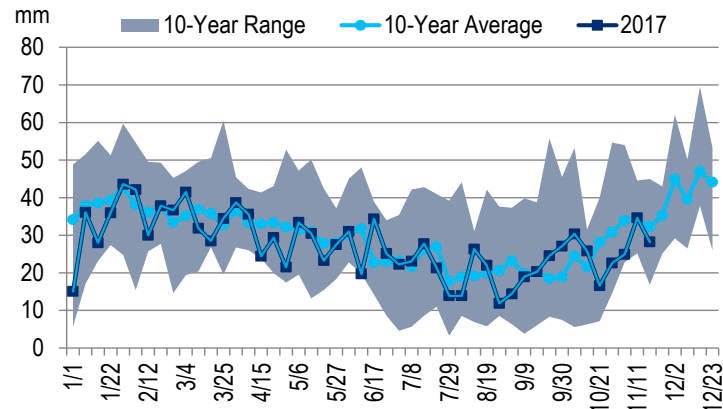
Weekly precipitation in Ivory Coast



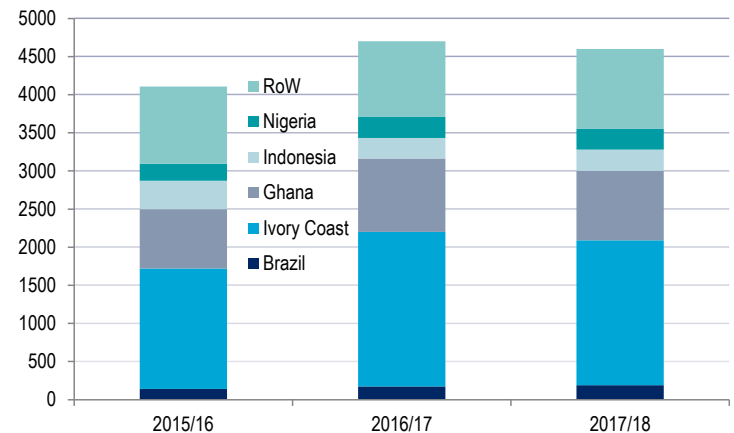
Weekly precipitation in Ghana



Weekly precipitation in Indonesia



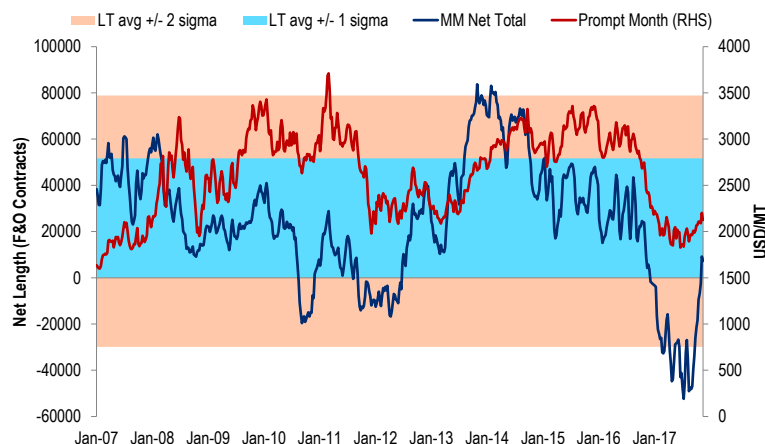
Crop forecasts by country (kt)



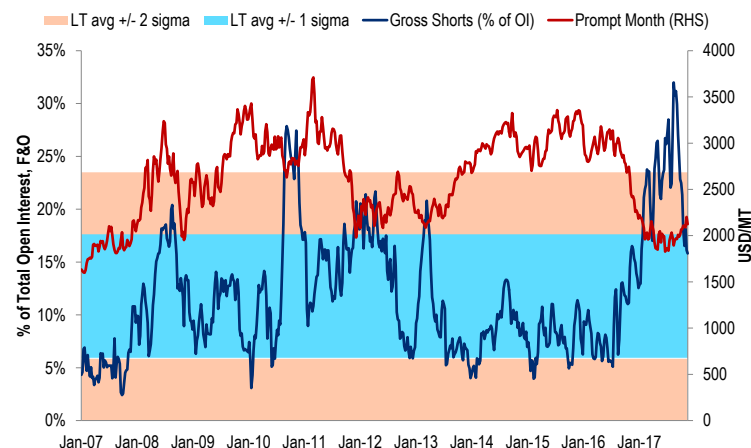
Grindings should trend positive but funds have bid cocoa prices in anticipation

The record ICE cocoa fund short held as recently as 3Q'17 has been fully covered as prices jumped ~10% to \$2,100/t. This came both in anticipation and realization of stronger cocoa processor demand on the back of the ~50% peak-to-trough drop in terminal prices from 4Q'15 to 2Q'17. We expect grindings growth to support ICE prices above \$2,000/t, but think fund buying will slow without a weather disruption in West Africa.

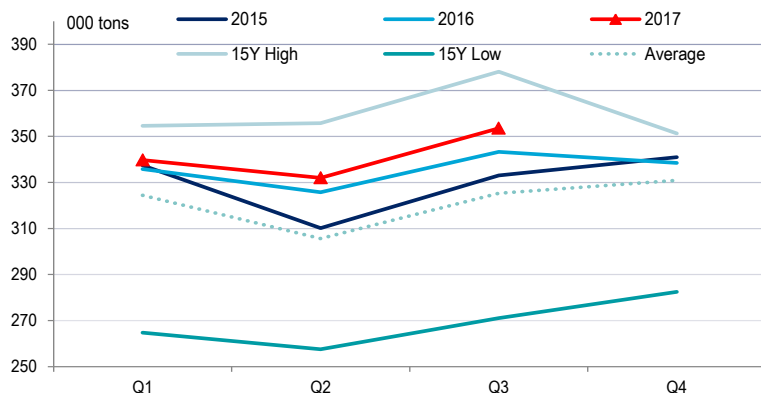
Money Manager Net Length and Prompt Prices, +/- 1,2 sigmas – ICE Cocoa



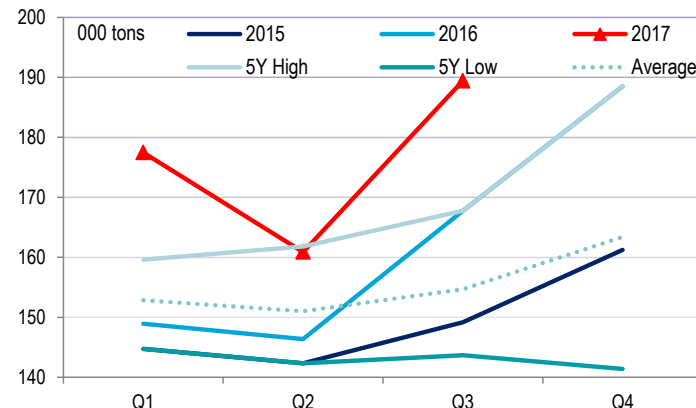
Money Manager Normalized Gross Shorts, +/- 1, 2 sigmas – ICE Cocoa



European Cocoa Grindings



Asia Cocoa Grindings



World coffee supply/demand balances point to stable prices of \$1.30-1.40/lb*

million 60-kg bags

	2012/13	2013/14	2014/15	2015/16	2016/17	2017/18
Beginning Stocks						
Exporters	21.3	27.8	30.2	28.0	21.6	15.9
Importers	60.1	61.0	65.2	67.1	73.0	78.6
Total Beginning Stocks	81.4	88.9	95.4	95.1	94.6	94.4
Production						
Brazil	55.4	54.7	52.3	50.4	55.0	50.7
Colombia	9.9	12.2	13.3	14.0	14.5	14.5
Vietnam	23.4	27.6	26.5	28.7	25.5	27.5
Indonesia	11.5	11.3	11.4	12.3	11.5	10.6
Total Production	149.6	152.2	148.7	151.6	153.9	152.5
Total Supply	231.0	241.1	244.1	246.6	248.5	246.9
Consumption						
European Union	41.4	42.0	41.9	41.7	42.0	42.3
United States	22.8	23.6	24.1	25.2	25.6	26.0
Japan	7.3	7.5	7.6	7.8	8.1	8.2
Other Importers	25.3	26.5	28.2	29.0	30.0	30.8
Total Importer Consumption	96.8	99.6	101.8	103.8	105.7	107.3
Exporter Consumption	45.3	46.1	47.2	48.2	48.3	48.8
Total Consumption	142.1	145.7	149.1	152.0	154.0	156.1
Imports						
European Union	41.5	42.2	42.4	42.2	43.0	42.9
United States	23.2	24.0	24.5	25.5	26.3	26.6
Japan	7.6	7.9	7.7	8.0	8.1	8.3
Other Importers	25.4	29.7	29.0	34.1	33.8	30.9
Exporters	12.2	11.2	8.3	9.9	10.2	7.8
Total Imports	110.0	115.0	111.9	119.6	121.5	116.6
Exports						
Brazil	29.3	32.8	36.9	36.9	36.9	36.9
Colombia	8.8	10.8	12.3	12.3	12.3	12.8
Vietnam	20.7	24.9	21.3	26.4	26.4	25.4
Indonesia	11.3	10.0	6.7	8.0	8.0	8.0
Total Exports	110.0	115.0	111.9	119.6	121.5	116.6
Ending Stocks						
Exporters	27.8	30.2	28.0	21.6	15.9	10.9
Importers	61.0	65.2	67.1	73.0	78.6	80.0
Total Ending Stocks	88.9	95.4	95.1	94.6	94.4	90.9

- We expect global production to post around 152.5m bags, down by 1.4m y/y driven by Brazil and partly offset by gains in Vietnam and rest of the world. The ratio between Arabica and Robusta should back-up slightly from last year to about 60/40. Demand growth is expected to remain stable at 1.3-1.5% p.a. evenly distributed between importers and exporters on average. Brazil and Vietnam should maintain their status as the top exporters for Arabica and Robusta, respectively. Exports in Colombia have been near historical highs in 2016/17 and are expected to grow another 0.5mn bags in 2017/18. Importer stocks look significantly higher for 2017/18 versus 2013/14, when coffee traded \$2+.
- Dryness concern for Brazil is not yet completely relieved, as rainfall in Minas Gerais is still slightly below average and dryness in Espirito Santo could last through December. However, as coffee trees in Brazil are entering the high-yielding half of the biannual cycle, we would still expect a decent production in 2018/19. Without taking growth in the rest of the world into account, as long as weather co-operates in Brazil, the world could potentially see a pause from the five-year long coffee deficit cycle and with near perfect weather, we could even potentially see global balances lifted to surplus.

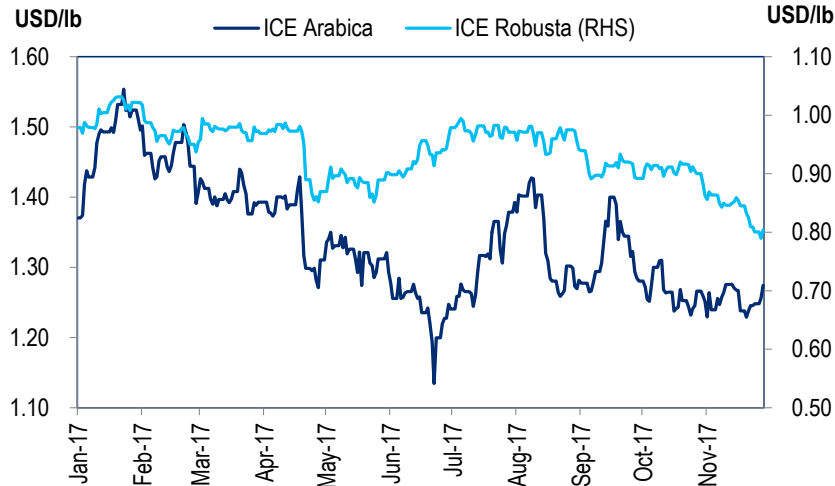
*Beginning stocks are for the crop year for each respective exporter, and adjusted for a Oct-Sept crop year basis for all importers.

Exporter nation imports were derived according to industry reports and balances. Rest of World estimates are derived from USDA and ICO global balances and major exporter/importer balances.

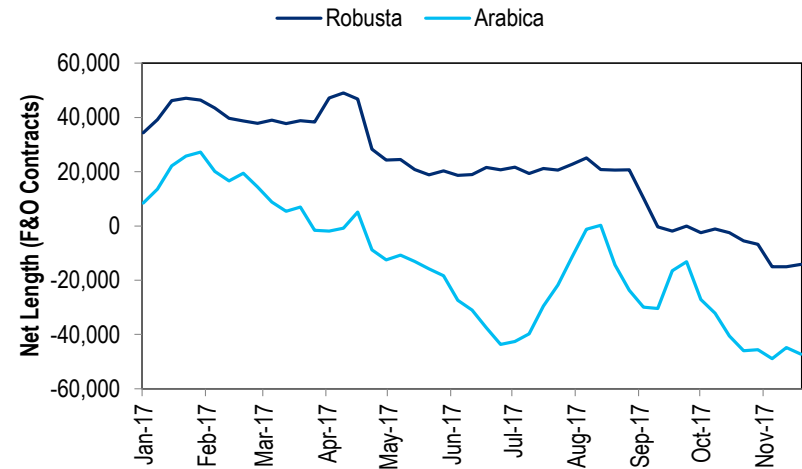
Source: ICO, USDA, Industry reports, Citi Research, *average ICE (Arabica) price

ICE coffee: Prices are supported by S/D deficit and tight market in winter

Arabica and Robusta Coffee Prices



Arabica and Robusta Money Manager Net Positions



- Timely precipitation relieved some concerns for 2018/19 Brazil production and Arabica coffee prices dropped 11% from the highest point hitting \$1.30/lb. Meanwhile, Arabica money manager net short positions hit a record high at ~49k lots in mid-November. Nevertheless, as we are still expecting a ~3.6mn bag deficit globally in 2017/18, we still have a neutral-to-bullish outlook for 2018. Short-term, early harvest supply is expected to be slow this year in Vietnam and Colombia due to rainfall, which could signal a tight coffee market balance towards the end of this year although the actual supply could be well-supported by inventories. Together with the likelihood of money manager short covering, we are expecting to see prices recover through 1Q'18. For the full year, we set our 2018 annual price outlook for ICE coffee at \$1.35/lb, slightly above the forwards as of November 2017, expecting stable prices in 2019.
- Robusta prices also dropped this quarter and the current money manager net short position is 14-15k lots there, which is also the highest since early 2016. However, the Vietnam harvest is expected to come in good volume overall and we expect to see Robusta coffee inventories start to rise as the new harvest posts.

Citi ICE Arabica Price Outlook*

	1Q'17	2Q'17	3Q'17	4Q'17	1Q'18	2Q'18	3Q'18	4Q'18	2017F	2018F	2019F
ICE Arabica (c/lb)	145	131	132	126	134	140	135	130	133	135	134

Source: Bloomberg, CFTC, Citi Research, *subject to revision

Sugar: 2017/18 surplus remains hefty but 2018/19 could see a rebalancing

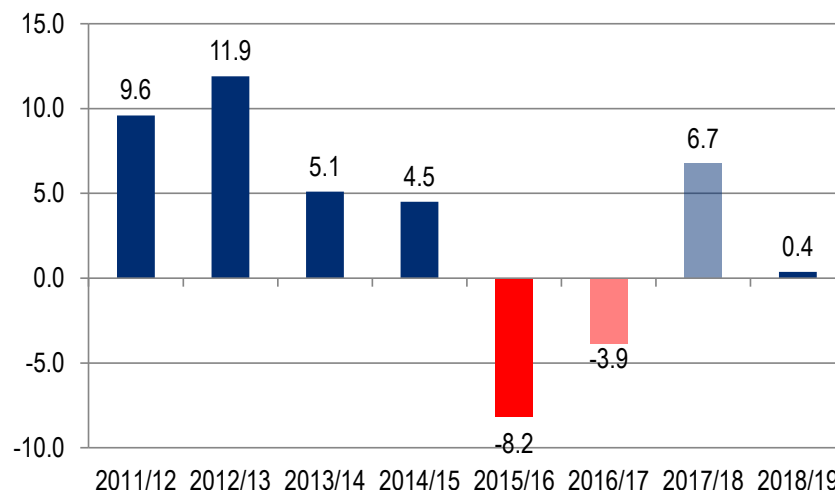
Despite recent oil-driven price strength, we continue to expect the hefty 2017/18 surplus to cap prices through 1H'18. But looking ahead to 2018/19, the sugar market may become more balanced, assuming Brazilian mills react to weak sugar economics by reducing their sugar mix, which should promote higher prices in 2H'18.

Global Sugar surplus/deficit (mn mttq)*

(in million mt, Local Crop year, Tel Quel)

Production - Key Players	2017/18e	2018/19e	Change (mt)	Change (%)
Brazil CS (May/Apr)	35.7	32.9	-2.8	-8%
India (Oct/Sep)	25.0	26.0	1.0	4%
EU 28 (Jan/Dec)	19.9	18.7	-1.2	-6%
China (Oct/Sep)	10.2	10.0	-0.2	-2%
Thailand (Oct/Sep)	11.8	11.6	-0.2	-2%
Russia (Oct/Sep)	5.9	5.2	-0.7	-12%
USA (Sep/Aug)	7.5	7.7	0.2	3%
Mexico (Oct/Sep)	6.0	6.1	0.1	2%
Australia (Jul/Jun)	4.5	4.6	0.1	2%
Others	50.1	49.7	-0.4	-1%
Global Production	176.6	172.5	-4.1	-2.32%
Global Consumption	169.9	172.1	2.3	1.43%
World Surplus/Deficit	6.7	0.4		

Global Sugar supply/demand balance*



Stronger ethanol pricing, on the back of rising global crude benchmarks, helped to lift ICE sugar prices through 4Q'17, but we expect further price strength to remain subdued through 1H'18 as harvests in Asia and Europe peak. But in 2018/19, the market may begin to look more balanced, with the major balancing factor being Brazilian production. We expect the sugar mix in CS Brazil to fall to 44% next year, with downside risks should current sugar/ethanol spreads persist, which could mean a nearly 3-mn mttq decrease y/y in raws output. Elsewhere, the production increases seen in 2017/18 should not be sustained in 2018/19, as they have been largely driven by one-time factors (recovery from drought in Asia and deregulation in the EU). Though we expect Indian production to continue to expand, the y/y increase may not match 2017/18. Consumption growth, however, should continue. All things considered, we expect only a modest surplus in 2018/19 which should support prices through 2H'18 and 2019, though falling crude prices during 2019/20 may damp relative price cheer.

	1Q'17	2Q'17	3Q'17	4Q'17F	1Q'18F	2Q'18F	3Q'18F	4Q'18F	2016	2017F	2018F	2019F
ICE Sugar (c/lb)	19.6	15.2	14.1	15.0	15.0	16.0	17.0	17.0	18.3	16.0	16.5	16.0

Source: GP, Platts, Citi Research, *subject to revision; some figures may round, Crop years are approximated per local country or USDA reporting. Citi raw sugar price outlook unchanged since September 2017 and maintained in this report.

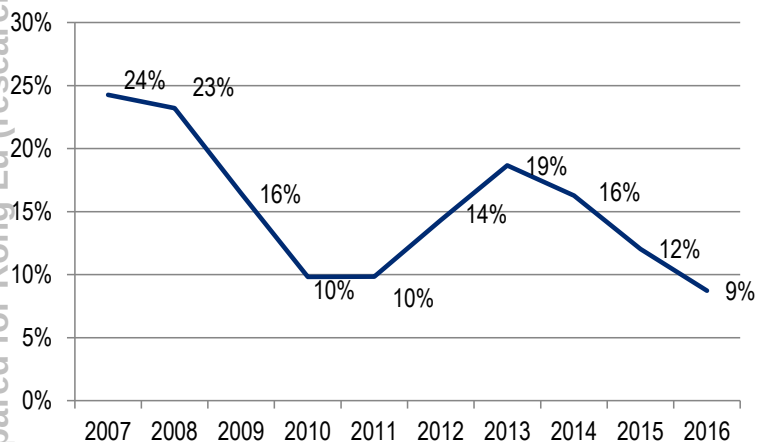
CS Brazil: Strong 2017/18 harvest, but 2018/19 could see output curtailed

A sharp switch towards ethanol late in the 2017/18 harvest and slightly disappointing cane crush have prompted us to revise our production expectations lower to 35.7-mn mttq for this season, though still record high. In 2018/19, we expect lower cane crush and a swing towards ethanol to cut sugar output by nearly 3-mn mttq y/y.

Centre South Brazil Production Forecast

Item ('000 mt/mn litres)	2017/18	2018/19	YOY Change (c/a)
Cane	593,000	585,000	-1.35%
TRS	0.1362	0.1340	-1.62%
Sugar Mix (%)	46.4%	44.0%	
Sugar Production	35,708	32,865	-7.96%
Ethanol Production	25,386	25,742	1.40%

Centre South Brazil Renovation Rates

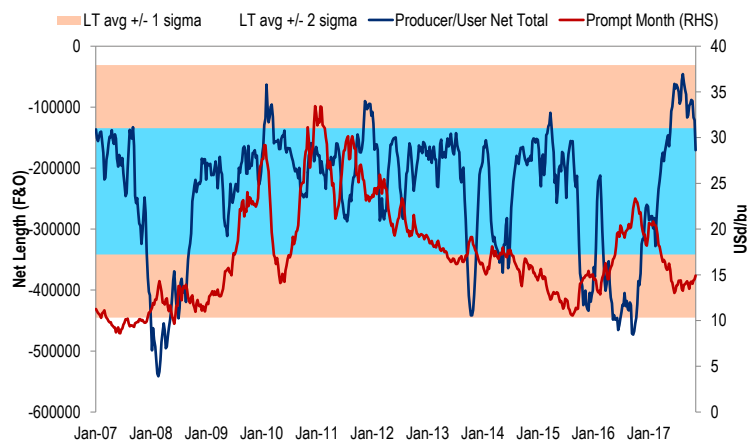


- As the CS (centre-south) Brazil harvest has started winding down, our production expectations for 2017/18 have been tempered somewhat on the back of slightly lower expectations for cane crush through harvest-end as well as a significantly lower sugar mix. Biweekly cane crush in the last four fortnights has averaged ~0.6-mmt lower than during the same four fortnights last year after averaging 2.3-mmt higher during the first half of the harvest (excluding the first two weeks when 2016/17 crush was inflated due to the large amount of cana bisada*). If crush through the remainder of the season were to continue to average ~0.5-mmt below last year's crush, crush would come in at 593-mmt, but there may even further downside risk if biweekly cane crush continues to fall y/y. We have upgraded our expected TRS yield to 136.2-kg/mt, but the impact is offset by a significantly reduced sugar mix which we now peg closer to our pre-harvest forecast at 46.4% as Brazilian mills have reacted to the negative sugar to ethanol premium. All in, we expect production to come in at 35.7-mn mttq in 2017/18.
- Looking ahead to 2018/19, we are expecting both lower cane crush and a strong switch away from sugar production at Brazilian mills to reduce sugar output by nearly 3-mn mttq y/y. Since the beginning of September, sugar has traded at a discount to ethanol averaging ~60 cent points below. If sugar prices remain this low through 1Q'18, which looks likely given the size of the trade flow surplus this winter, mills should react and structurally reduce their seasonal sugar mix. At the same time, persistently low renovation rates over the last few years have led to a significant ageing of the cane, which we expect to weigh on crush in 2017/18.

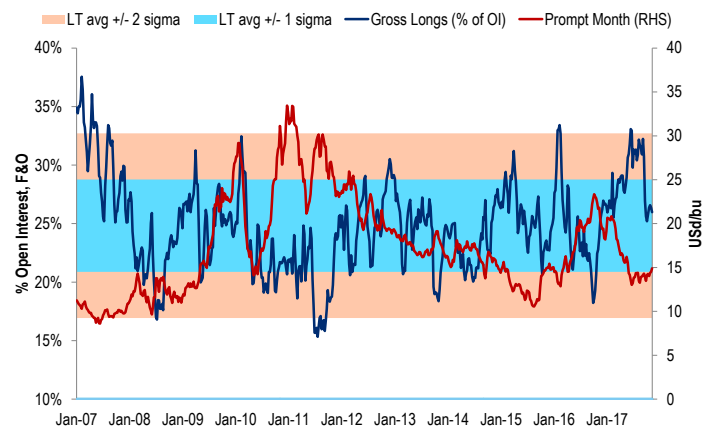
Brazilian mills have flexibility in 2018/19 as specs shorts may get squeezed

Financial flows can further our bullish new-crop sugar price outlook. Though ticking-up in recent weeks as the cal'18 forward curve drifted above \$0.15/lb, producer shorts in 4Q'17 are down dramatically y/y, allowing CS mills flexibility in the sugar mix. Meanwhile, the spec short on ICE remains very large historically and seasonally.

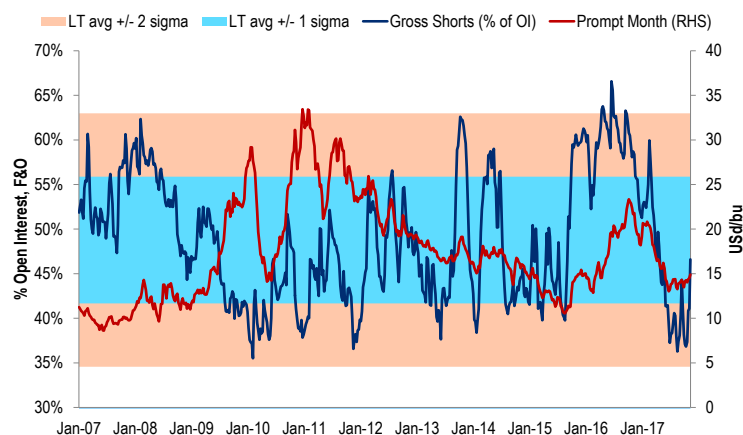
Producer Net Short and Prompt Prices, +/- 1,2 sigmas – ICE Sugar



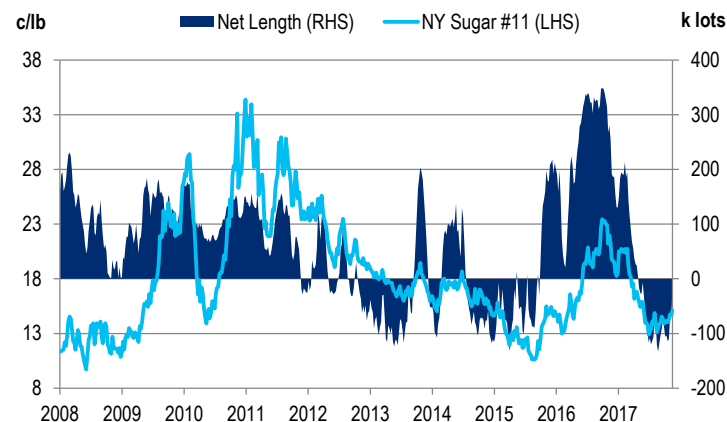
Normalized Consumer Gross Long and Prompt Prices, +/- 1,2 sigmas – ICE Sugar



Normalized Producer Gross Short and Prompt Prices, +/- 1,2 sigmas – ICE Sugar



Non-commercial (ex-index) speculative net length – ICE Sugar



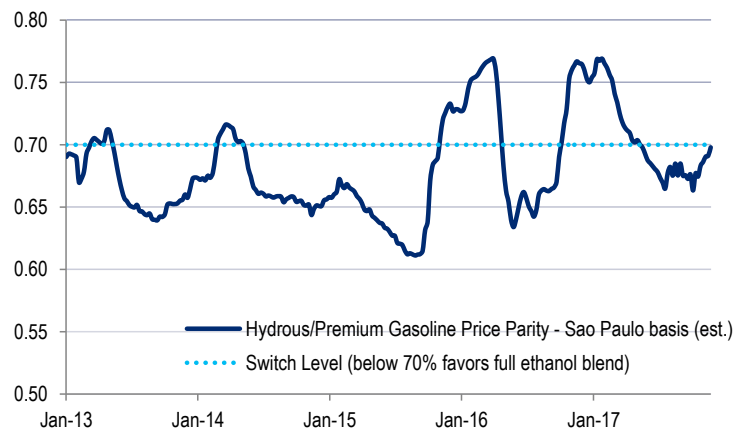
Prepared for Rong Lu (research email)

Source: Bloomberg, CFTC, Citi Research, note data are through mid-November 2017

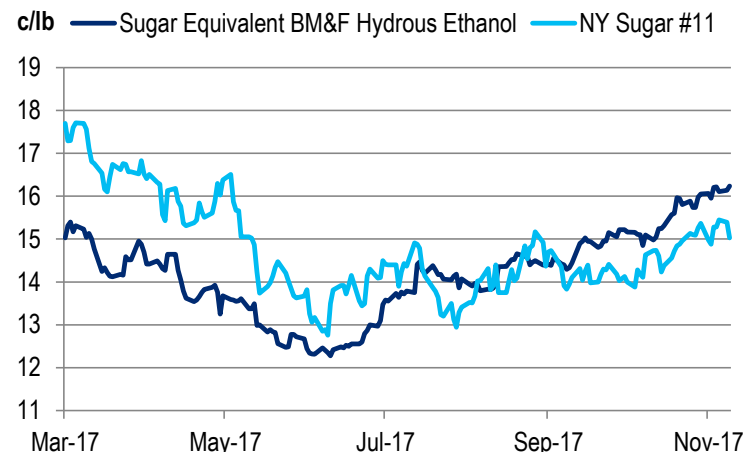
Local Brazil ethanol/sugar pricing dynamics also do not look sustainable

Several factors lifted domestic ethanol pricing in Brazil through 2H'17 including rising global crude benchmarks which have increased ethanol competitiveness, with hydrous ethanol now trading slightly cheaper than anhydrous gasoline blend. While this is supportive of sugar pricing, sugar continues to trade below ethanol.

Hydrous / Gasoline Price Parity



ICE Sugar versus Brazil Hydrous Ethanol (sugar-equivalent)



Higher domestic ethanol prices in 2H'17 on the back of rallying global crude benchmarks and changes to Brazil's domestic ethanol policies, including the imposition of an import tariff and the increasing of the PIS/Cofins tax on gasoline, helped to lift sugar prices despite looming supplies from the Northern Hemisphere. Amid higher domestic gasoline pricing, demand for ethanol has picked up with hydrous now trading slightly cheaper than anhydrous gasoline blend. But while these dynamics have been supportive of sugar, sugar prices have continued to trade below the ethanol "floor" since mid-August when it was first breached. Through 2H'17, sugar has traded at an average discount of ~60 cent points to ethanol.

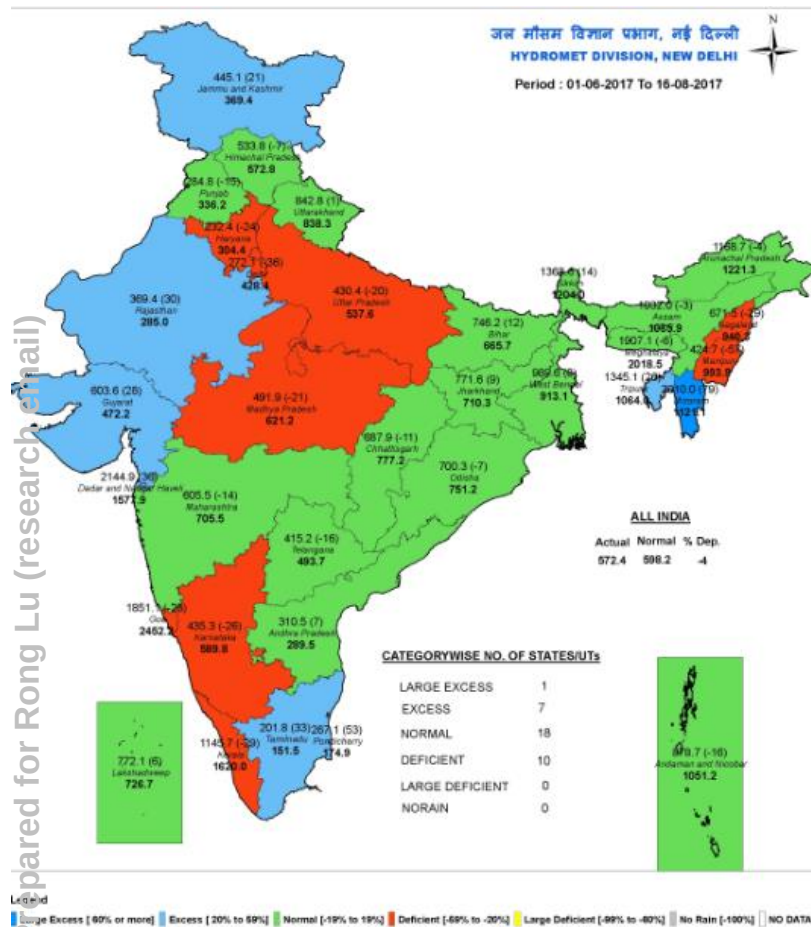
These trading dynamics do not look sustainable in our view. Typically, sugar must trade at a slight premium to ethanol in order to incentivize production because ethanol offers a cash flow advantage over sugar. For this reason, we expect sugar prices to move above the ethanol floor to trade at a slight premium as we get closer to the beginning of the 2018/19 harvest, though for the spread to remain relatively narrow, and thus tentatively peg the sugar mix next year at 44%. But if current depressed trading levels persist, the sugar mix could be as low as 40% or even lower which could cut sugar production in CS Brazil by a further 3-mn mttq versus our current forecast, likely swinging 2018/19 balances back into a deficit.

Source: ANP, Bloomberg, Citi Research,. *in neat gasoline energy equivalent terms

India – 2017/18 cane harvest off to positive start, strong y/y growth expected

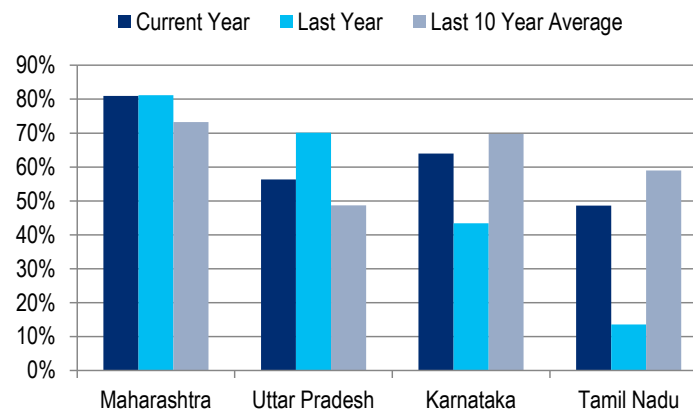
Mills have begun crushing in India with positive results already reported out of Uttar Pradesh and Maharashtra. The delayed withdrawal of the monsoon brought extra rains which should be beneficial to both the 2017/18 and 2018/19 crops. We slightly upgrade our 2017/18 forecast to 25.0-mn mttq and peg 2018/19 at 26.0-mn mttq.

India 2017 Monsoon Rainfall through mid-August (Red=Deficient, Blue=Excess, Green=Normal)



- Rainfall during the 2017 monsoon varied considerably across regions, with Maharashtra and Tamil Nadu receiving rainfall well above their LPAs*, while parts of Uttar Pradesh and Karnataka saw more mixed rains. Reservoir levels in Karnataka were already low after an erratic monsoon in 2016 following the 2015 drought, so we expect only a modest rebound in production there. But Maharashtra, which saw excellent rains this year, should see a strong rebound with acreage estimated up almost ~300-k HA y/y to 902-k HA. Despite the drier weather in Uttar Pradesh, the increasing adoption of a high yielding cane variety should support new record production.
- We forecast sugar production of 25.0-mn mttq in 2017/18, slightly upgraded from our previous estimate given late monsoon rains. Looking ahead, assuming good weather continues in 2018, we expect 2018/19 production could expand even further to 26-mn mttq.

Nov 2017 Reservoir Levels in Key States

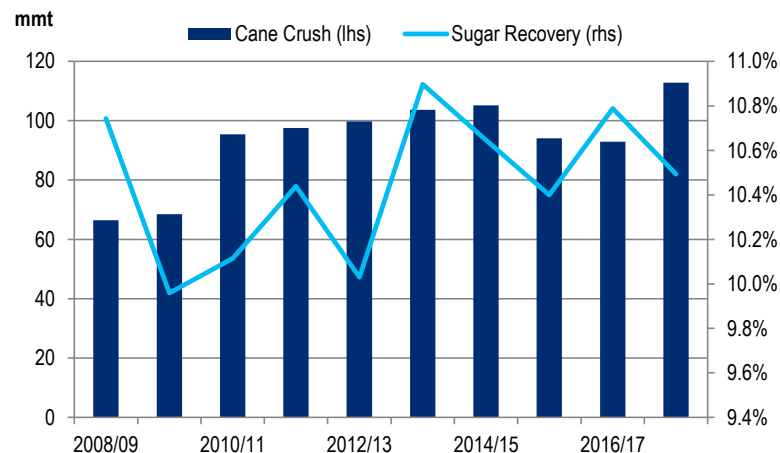


Other Asia sugar – 2017/18 output to rise on weather, strong domestic pricing

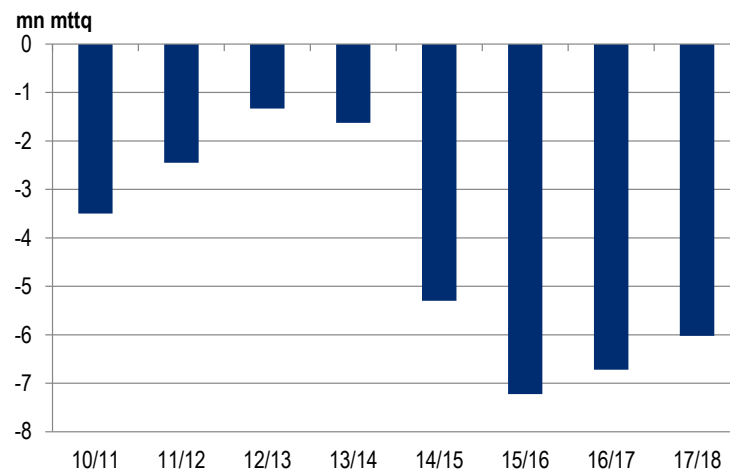
China's 2017/18 harvest has begun, while the start of the Thai harvest is fast approaching, with Thailand likely kicking off harvesting imminently. We expect production to rise in both China and Thailand in 2017/18; in Thailand on an ongoing rebound from the 2015 drought and in China on domestic price support measures.

- Thailand saw excellent rains through the 2017 monsoon season, and we expect this to drive record cane crush in 2017/18 of 110.1-mmt, up 18% versus the previous year. We also expect sugar recovery rate of 107-kg/mt, which is slightly above the 5-year average, but still slightly lower than last year.
- On the back of our cane crush and recovery forecasts, we are expecting production of ~11.8-mn mttq in 2017/18. This would represent record production for Thailand, exceeding the previous record production hit in 2013/14 by 0.5-mn mttq..
- In China, stronger domestic pricing, propped up by government price support measures, has been paying dividends, with output increasing last year and expected to rise again this year by ~1-mn mttq. Chinese sugar production rebounded slightly in 2016/17 to 9.2-mn mttq after hitting a low point in 2015/16 at 8.8-mn mttq. We expect Chinese sugar production to grow again in 2017/18 to ~10.2-mn mttq, in line with the forecast from the China Sugar Association announced in early November. This should narrow the domestic deficit to ~6-mn mttq.
- Measures restricting imports, both legitimate and smuggled, have been instrumental in supporting the domestic sugar price and driving acreage expansion in China. Besides cracking down on smuggling, the Chinese government also imposed steep tariff increases in early 2017, increasing out-of-quota (OOQ) import tariffs over the next three years by 45%, 40% and 35%, respectively. These measures have driven up domestic sugar prices, helping to support the production expansion.

Thai Annual Cane Crush and Yields



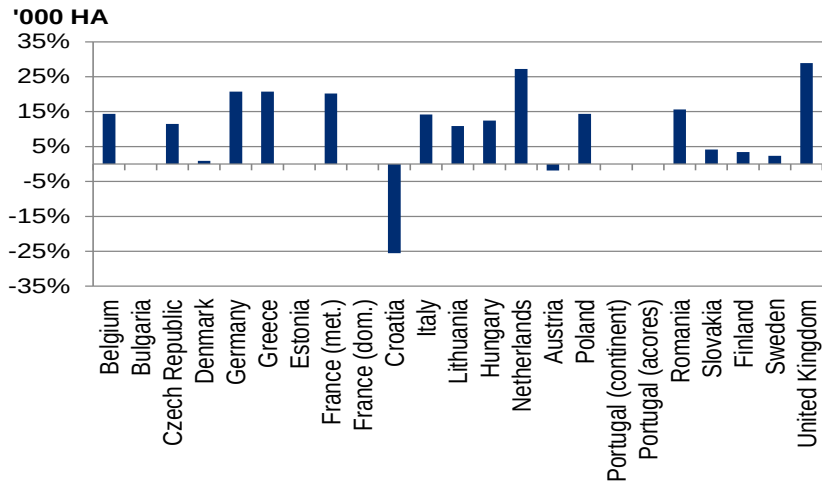
China Sugar Surplus/Deficit



EU – Sugar production surprising to the upside but exports face bottlenecks

The EU 2017/18 harvest, the first of the new deregulated era, began in October with positive early results on beet yields prompting us to upgrade our production forecast to 19.9-mn mttq. But while production has surprised to the upside, exports have been weaker than expected amid a choppy transition to net exporter status.

EU Sugar Beet Area Planted YoY Change By Country

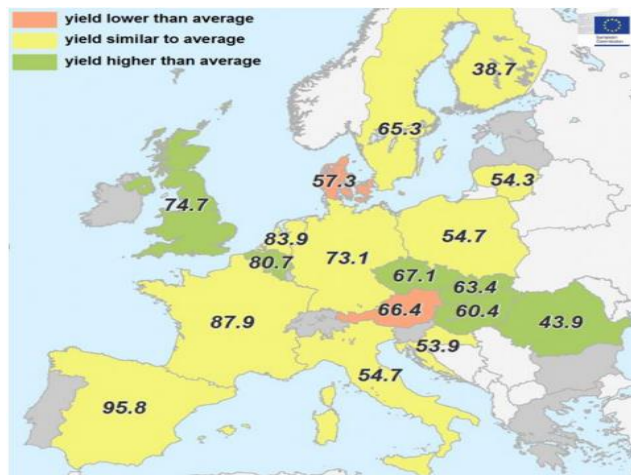


- 2017/18 beet harvesting has begun in Europe, marking the beginning of a new, deregulated era for EU sugar production. On the back of the deregulation, beet acreage in Europe was up significantly y/y, with our estimates pegging the increase in total acreage at 17%. The largest increases in acreage were observed amongst the low cost western beet producers, with acreage in France, Germany, and Greece all up ~20%, while the UK and the Netherlands increased acreage by nearly 30%.

- In the context of this sharp acreage increase, the positive early results of the EU harvest, with beet yields coming in favorably, point to potentially a very large crop this year. Early results from the harvest indicate that positive weather conditions this summer, with a mix of rain and sunshine, have indeed helped to support strong crop growth. We expect above normal yields across a host of major producing countries, and upgrade our forecast for 2017/18 production to 19.9-mn mttq.

- But logistical issues associated with the sudden swing to net exporter status amid the industry deregulation has led to disappointing exports so far this season. Reportedly, EU exports have faced delays in loading of breakbulk cargoes at some ports. This has led to weaker than expected whites exports during the early harvest, though logistical issues may well work themselves out over the course of the 2017/18 campaign.

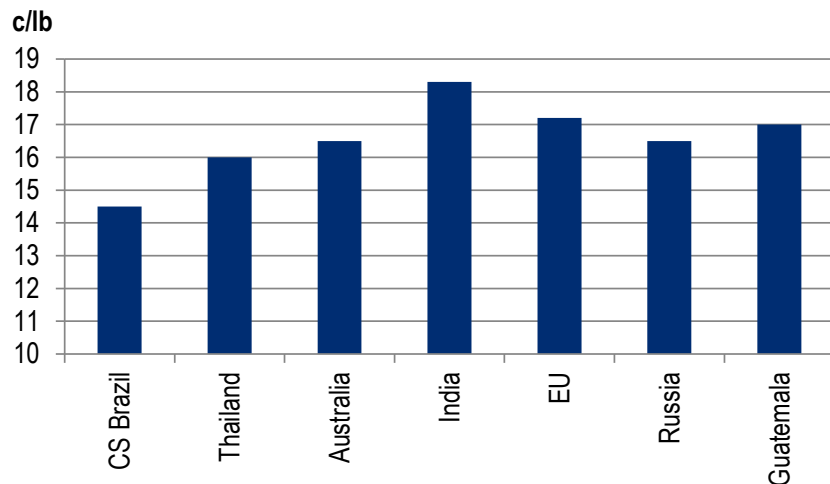
EU 2017/18 Sugar Beet Yield Forecast



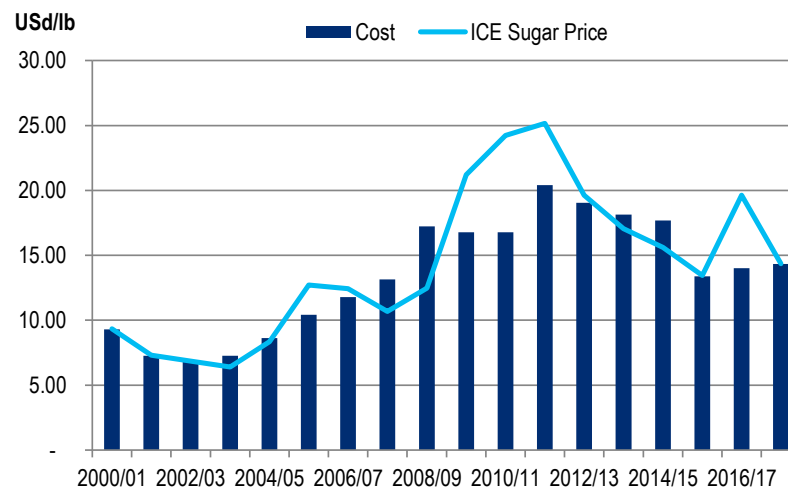
In the medium-term, sugar prices should eventually cover costs of production

Prolonged periods of prices trading below costs of production should eventually lead to acreage cuts, crop switching and lower levels of production. Sub-\$0.15/lb ICE sugar prices do not look sustainable in the medium term, given that they are below costs even in CS Brazil, the lowest cost global producer.

Global Sugar Costs of Production



CS Brazil Sugar Costs of Production versus Average raw sugar price



Brazilian costs of production fell dramatically over the course of 2014-2016, largely due to a substantial depreciation in the currency. Costs for Brazilian producers hit as low as \$0.13/lb in 2015/16 and remain relatively depressed at an estimated \$0.14-0.15/lb range for 2017/18. ICE sugar prices historically have displayed a close relationship with Brazilian costs of production, with major divergences typically disappearing within one to three years.

Most global producers are not as efficient as the Brazilian mills, nor did they see as significant currency depreciation over the last two years, and current costs of production are typically \$0.02-03/lb higher in the other major producing countries. Granted, domestic pricing in several of these countries is not necessarily tied to global pricing due to government intervention, for example in India and China. But even so, it appears likely in the long-term for global pricing to revert higher, at least towards average blended costs as several are large producers of high quality raws (e.g. Thailand, Guatemala). This relationship in-turn will hinge on tightness of physical trade flows, on a deferred basis.

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